LOGIC INTERFACING

CHAPTER 12

Although sheer "number crunching" is an important application of digital electronics, the real power of digital techniques is seen when digital methods are applied to analog (or "linear") signals and processes. In this chapter we begin with a brief chronology of the rise and fall of digital logic families and a review of the input and output properties of the surviving (mostly CMOS) families that you are likely to use in circuit design. This is essential for understanding how to interface logic families to each other and to digital-input devices (switches, rotary encoders, keypads, comparators, etc.) and output devices (indicator LEDs, relays, power MOSFETs, etc.). We continue with the important subject of bringing digital signals on and off circuit boards, in and out of instruments, and through cables. We discuss optoelectronic devices (fiber-optic drivers and receivers, optocouplers, LCD and LED displays, and solid-state relays). And in the next chapter we discuss the major subject of conversion between analog and digital signals. Finally, with an understanding of these techniques, we look at a number of applications in which combined analog and digital techniques provide powerful solutions to interesting problems. Much of this material is applicable not only to "discrete" digital logic, but also to the PLDs and FPGAs of Chapter 11 and to the microcomputers and microcontrollers of Chapters 14 and 15.

12.1 CMOS and TTL logic interfacing

12.1.1 Logic family chronology – a brief history

In the prehistoric early 1960s, adventurous people who didn't want to build their logic from discrete transistors struggled with RTL (resistor–transistor logic), a simple logic family introduced by Fairchild and characterized by poor fan-out and poor noise immunity. Figure 12.1 shows the problem, namely, a logic threshold at one V_{BE} above ground and miserable fan-out (in some cases one output could drive only one input!) caused by passive pullup and a low-impedance current-sinking load. Those were the days of small integration, and the most complicated function you could get was a dual flip-flop, which would operate to 4 MHz (the MC790P, in case you want to look it up). We bravely built circuits with RTL; they sometimes malfunctioned when someone switched on a soldering iron in the same room.

The death knell for RTL came a few years later with the introduction by Signetics of DTL (diode–transistor logic) and, soon thereafter, Sylvania's SUHL – "Sylvania Universal High Speed Logic" – subsequently called TTL (transistor–transistor logic). Signetics had a popular mixture of the two, called 8000-series DCL Utilogic ("Designer's Choice Logic"). TTL caught on quickly, particularly in the "74*xx*" numbering system originated by Texas Instruments. (Although bipolar TTL is pretty much history now, the 74*xx* designations have remained healthy to this day, in contemporary CMOS incarnations.) These families used current-*sourcing* inputs with logic threshold at two V_{BE}'s and (usually) push–pull "totem-pole" outputs (Figure 12.1). DTL and TTL began the era of +5 V logic (RTL used +3.6 V), and offered speeds of 25 MHz and fan-outs of 10 (i.e., an output could drive 10 inputs). Designers rejoiced at the speed, reliability, and complex functions (divide-by-10 counters, for example) of these families. It seemed to us that you couldn't ask for more; TTL would live forever.

People are greedy, though. They wanted more speed. They wanted less power consumption. They soon got both, sort of. In the high-speed arena, a souped-up TTL (74H series; "High-speed" TTL) delivered roughly twice the speed, at twice the power! (It accomplished this underwhelming feat by cutting all resistor values in half.) Another family, ECL (emitter-coupled logic), delivered real speed (30 MHz in its original version), using a negative power supply and rather closely spaced logic levels (−0.9 V and −1.75 V); it consumed lots of power (30 mW/gate) and came only in small integration. For low power there was a souped-down TTL (74L series; "Low-power" TTL) with 1/4 the speed at 1/10 the power of the corresponding 7400 "standard" TTL.

Back at RCA, the first of the MOSFET logic families was developed, 4000-series CMOS. It had zero quiescent power consumption and a wide supply range (+3 V to

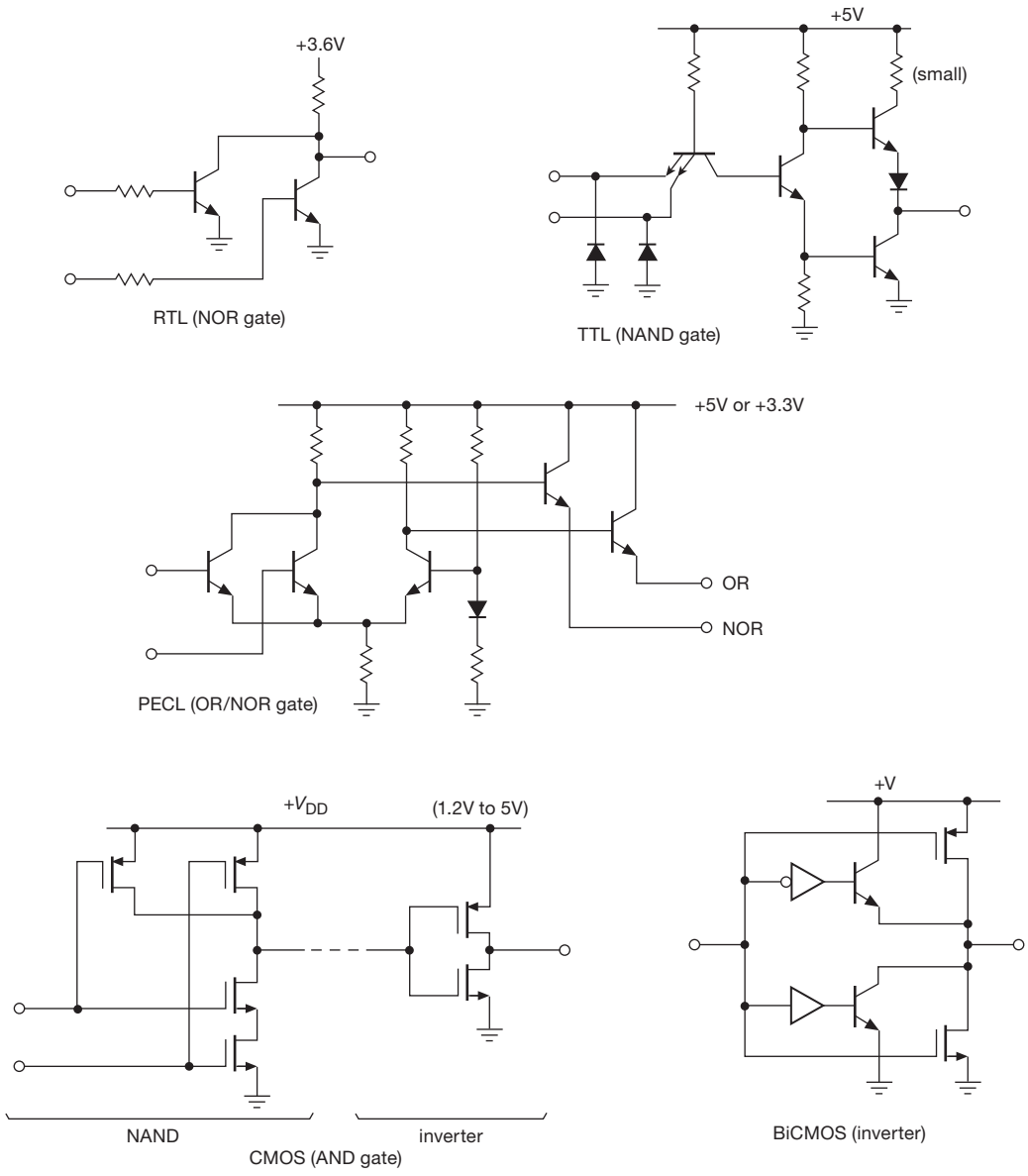


Figure 12.1. Simplified diagrams of several logic families.

+ 12V). The outputs swung rail-to-rail, and the inputs drew no current. That was the good news. The bad news was the speed (1 MHz at 5 V supply) and the price (about $20 for a package of four gates). In spite of the price, a whole generation of battery-powered instrument designers grew up on the micropower CMOS simply because there was no alternative. They learned the true meaning of static electricity as they worked with the easily damaged inputs.

This, then, was the situation at the beginning of the 1970s – two main lines of bipolar logic (TTL and ECL) and the extraordinary CMOS. The TTL variants were essentially compatible, except that 74L TTL had feeble output drive (3.6 mA sink) and could drive only two standard (74-series) TTL loads (whose inputs sourced 1.6 mA when held LOW). There was almost no compatibility between the major families (though a pulled-up TTL could drive CMOS, and 5 V CMOS could just barely drive a single 74L TTL load).

During the 1970s there were steady improvements on all fronts. TTL sprouted the nonsaturating "Schottky-

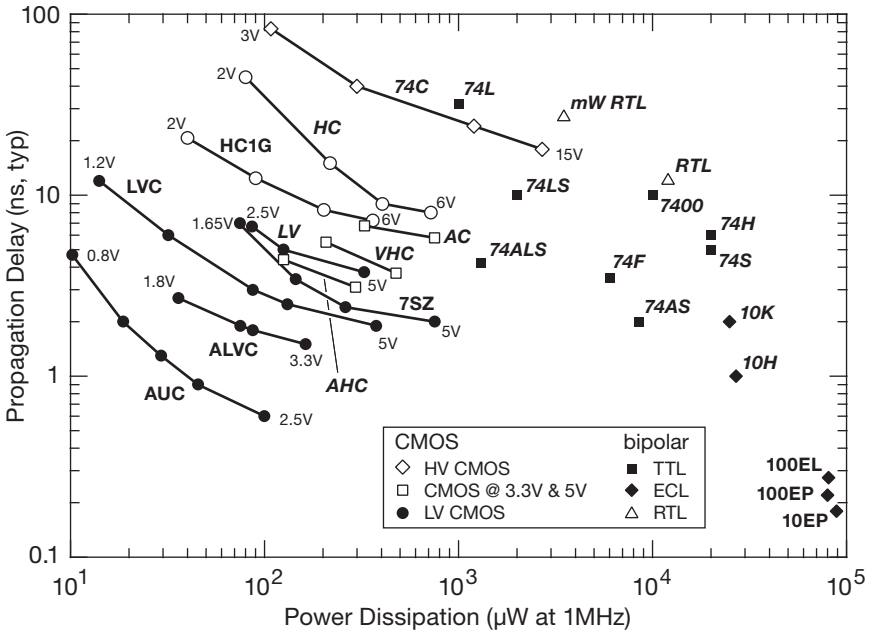


Figure 12.2. Gate delay versus power for various logic families. Power dissipation depends on supply voltage and frequency of logic switching: $P = C_{\text{pd}} V_{\text{cc}}^2 f$, where C_{pd} is called the "power dissipation capacitance." (C_{pd} does not include external wiring and load capacitances.) CMOS families permit operation over a range of supply voltages, whereas bipolar logic runs at a defined voltage (either 3 V or 5 V). Families in *italics* are available in DIP (through-hole packages). See also Table 10.1 and Figure 12.3.

clamped" families: first the 74S (Schottky) series, which made 74H obsolete by giving three times the speed at twice the power; then the 74LS ("Low-power Schottky"), which made the "standard" 74-series TTL obsolete by delivering slightly improved speed at 1/5 the power. Life with 74LS and 74S was nice, but then Fairchild came up with 74F (F for FAST: "Fairchild Advanced Schottky TTL"), which is up to 50% faster than 74S at 1/3 the power; it also has other improved properties that make it extremely nice to design with. Texas Instruments, the originators of many 74*xx* lines, came up with a pair of advanced Schottky families, 74AS ("Advanced Schottky") and 74ALS ("Advanced Low-power Schottky"); the former was intended to replace 74S, and the latter was intended to replace 74LS. All these TTL families have the same logic levels and plenty of output drive, and so they can be mixed within a circuit. Tables 10.1 and Figure 12.2 illustrate the speed and power of these families.

Meanwhile, the 4000-series CMOS evolved into the improved 4000B series, with wider supply range (3 V to 18 V), better input protection, and higher speed (3.5 MHz at 5 V). The 74C-series was essentially the same, with 74-family functions and pinouts to take advantage of the tremendous success of the 74-family bipolar logic. ECL sprouted the ECL II, ECL III, ECL 10,000, and ECL 100,000 series, with speeds to 500 MHz.

The situation in 1980, then, was this: most design was done with 74LS, with 74F (or 74AS) mixed in where higher speed was needed. This same TTL was used as "glue" to hold together nMOS microprocessor circuits, whose inputs and outputs are TTL compatible. Micropower design was always done with 4000B or 74C CMOS, equivalent and compatible with each other. And for the highest speeds (100–500 MHz), ECL was the only game in town. There wasn't much mixing between families, except occasional combination of CMOS and TTL, or perhaps TTL interfaces to an ECL high-speed circuit.

During the 1980s came the remarkable development of CMOS logic with the speed and output drive of TTL: first 74HC ("High-speed CMOS"), with the same speed as 74LS, and, of course, zero quiescent current; then 74AC ("Advanced CMOS"), with the same speed as 74F or 74AS. With rail-to-rail output swings, and input threshold at half the supply voltage, this logic combined the best features of previous TTL and CMOS, and it essentially replaced bipolar TTL. However, there was an incompatibility when both kinds of logic were used within a circuit, because the +2.4 V (minimum guaranteed) logic HIGH output of

traditional TTL standard logic (and also of more complex functions that were being done in nMOS) was insufficient to drive an HC or AC input (with their requirement of +3.5 V, min).

To address this problem, each CMOS family offered a variant with the lower input threshold. These are named 74HCT and 74ACT ("High-speed CMOS with TTL threshold"). During the 1980s, also, complex large-scale integrated (LSI) and very large-scaled integrated (VLSI) devices (microprocessors, memory, etc.) evolved from nMOS to CMOS (with consequent low power and, usually, full-swing output CMOS compatibility), at the same time increasing speed and complexity. And at the extreme high-speed end, there was some development of GaAs (gallium arsenide) logic devices (by companies like GigaBit Logic and Vitesse) to reach speeds of ∼3 GHz.

Predictably, things just got better and better in the following two decades. The most important development was the improvement in CMOS performance, brought about by reduced feature size on the silicon chip ("scaling"). First, as the length scales were shrunk, it became possible to put many more transistors on a reasonable-sized chip – this led to the development of massive processor, memory, and other complex function (e.g., video) chips, with transistor counts going from millions to billions. Second, and perhaps as important, the scaling increased the speed and also reduced the operating voltage and the power per gate.[1] The result is new families of low-voltage CMOS logic parts (74LVC, 74AUC, and so on; see Figure 12.3) with pin-to-pin speeds in the several-hundred megahertz range (delay times down to 1 ns or less).

These fast CMOS families have proliferated, with a head-spinning menagerie numbering into the dozens. Most of them operate over a range of supply voltages (e.g., 1.8–5 V for 74LVC), and in most cases the input logic voltage can swing beyond the positive supply (e.g., 74LVC is "5 V tolerant," regardless of supply voltage; see Figure 12.3). With greater integration of most of the needed logic into the large VLSI components, there is little reliance on discrete logic, and these "standard-logic" parts are used mostly as occasional glue. For that purpose they come in small packages, holding one or two gates, or a single flip-flop, with names like TinyLogic, Little Logic, MiniGate, or PicoGate.

These packages share with other surface-mount ICs the advantage of low-inductance ground (and power) pins, which reduce "ground bounce" (§10.8.3 and Figure 10.99) and other transient symptoms caused by fast edge rates driving the combined capacitance of wiring and load. Ground bounce had become a serious headache in the decade of the 1990s, with the new 74AC and 74ACT families, which combined fast full-swing (between +5 V and ground) logic transitions in traditional DIPs with corner power and ground pins. Some manufacturers (notably TI) addressed this problem by adding extra power and ground pins, and moving them to the center (creating new part numbers like 74AC11004, a DIP-20 repinned version of the DIP-14 '04 hex inverter); others created "quiet" logic families with controlled edge rates (e.g., the 74ACTQ family from FSC/NSC). The situation improved considerably with the move to lower supply voltages, low-inductance SMT packaging, and good printed-circuit layout practices (especially the use of dedicated power and ground planes in multilayer PCBs). And the use of low-voltage differential signaling (LVDS) for fast signals and clocks pretty much eliminated the problem altogether, because of both the balanced current changes, and their relatively small swing (∼0.4 V).

All CMOS families of standard logic (beginning with the original 4000A, and extending through HC, LVC, AUC, and a dozen others) have the pleasant characteristic of zero "static" (i.e., nothing happening) power dissipation, with typical quiescent currents less than a microamp. But CMOS does draw "dynamic" current when logic levels are switching, because of the combined effects of (a) transient rail-to-rail conduction of internal push–pull pairs during the middle of the logic swing and (b) dynamic current needed to charge and discharge internal and load capacitances. Dynamic supply currents are proportional to switching frequency and can rival bipolar logic as you reach maximum operating frequencies, as we discussed in Chapter 10 – see Figure 10.27. Note, however, that many VLSI functions implemented in CMOS (e.g., FPGAs and cPLDs, see Chapter 11) often have substantial quiescent current; this situation is changing, however, with a trend

[1] To see how this goes, consider the process of shrinking the linear feature size (channel length L and width W) on a CMOS IC by a factor of k ($k < 1$), while adjusting things to keep the electric field strengths constant; this is called "constant-field scaling." Then, with $L \propto k$ and $W \propto k$, constant-field scaling requires that $V_{DD} \propto k$; that makes the gate oxide insulation thickness scale as $t_{ox} \propto k$, which then makes the gate capacitance (which is proportional to LW/t_{ox}) go as $C_g \propto k$. A consequence of the geometry scaling is that the saturation drain current goes as $I_D \propto k$. Finally, with that drain current, a gate input can be driven through V_{DD} in a time $\tau \approx C_g V_{DD}/I_D$, which therefore scales as $\tau \propto k$. Voilà – the gate delay time scales as k; i.e., the speed increases proportionally to $1/k$. Even better, the power ($V_{DD}I_D$) decreases as $P \propto k^2$; and a reasonable figure-of-merit $1/P\tau$ (speed/power) goes as $1/k^3$. Perhaps this explains some of the semiconductor industry's relentless zeal always to move to the next-generation scaling "node" (shrink factor of $1/\sqrt{2}$).

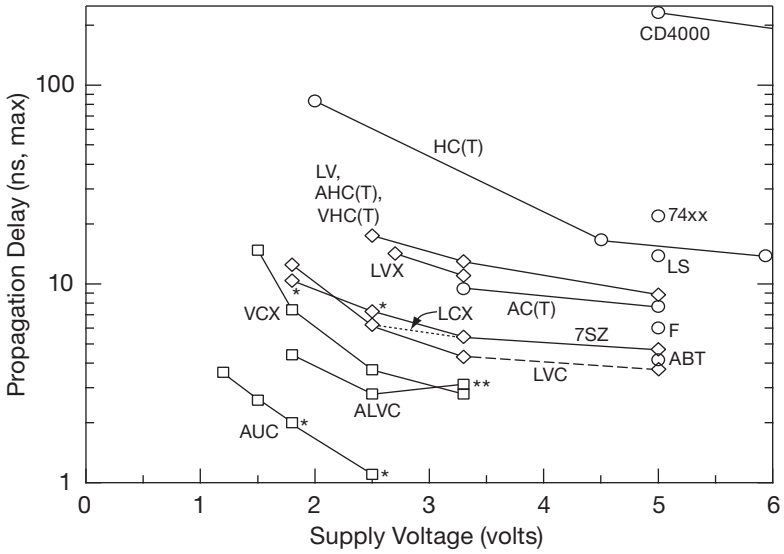


Figure 12.3. Gate speed versus supply voltage, for popular logic families. Maximum specified propagation delay ($t_{pd(max)}$) is shown for the standard supply voltages at which each family is specified. (As a rough guide, "typical" delays are in the range of 35–75% of $t_{pd(max)}$.) Open circles indicate families for which $V_{in(max)}$ is limited to V_{supply}, and data is for operation at 25°C; families with open diamonds have "5 V tolerant" inputs ($V_{in} \leq 5.5$ V, independent of supply voltage), and data is for operation over the "industrial" temperature range (-40°C to $+85$°C); families with open squares have "3.3 V tolerant" inputs ($V_{in} \leq 3.6$ V), data again over the industrial temperature range. Some families (e.g., LVC) have output-stage circuitry to ensure that the output does not load shared signal lines when unpowered. Plotted data is for load capacitance C_L=50 pF for 5 V and 3.3 V operation, 30 pF for 2.5 V and 1.8 V, and 15 pF for 1.5 V and below (except 50 pF for data points marked $**$, and 15 pF for $*$). The CMOS 4000-series logic runs to +15 V, at which $t_{pd(max)}$=70 ns. Only some LVC family members operate to +5V. Not shown are the speedy (and power-hungry) bipolar ECL families, operating at 5 V only: their maximum gate delays clock in at 0.6 ns (10E family), 0.44 ns (10EL), and 0.32 ns (10EP). Some families have evolved "enhanced" versions, for example LVC→LVCE, which operates down to 1.4 V and is 30% faster at low supply voltages. See also Figure 10.22 and §10.2.3.

toward micropower VLSI that is driven by battery-powered applications.

We wrap up our brief history with some recommendations.

- For simple logic circuits that are easy to assemble on a protoboard, and where blinding speed is not needed (nor desired), use 74HC or 74HCT logic (the latter for compatibility with either existing "TTL-level" signals, or with signals coming from low-voltage logic, e.g., powered from +3.3 V); you can substitute 74AC/74ACT/74ACTQ if you need the extra speed, but watch out for ground bounce.
- For low-voltage systems containing microcontrollers or other complex ICs, where some fast glue logic is needed, use a universal family like 74LVC, mindful that it is available only in surface-mount packages; these families are useful also for systems with several logic supply voltages.
- If you need to drive 5 V outputs (e.g., white LEDs, or solid-state relays) from a low-voltage system, use 74HCT
- for fast serial data and clock signals, use LVDS (or low-voltage PECL, "LVPECL") differential drivers, receivers, or SERDESs (serializer–deserializer).
- Choose the older 4000B/74C logic where the extended supply voltage range is needed and speed is unimportant (e.g., a portable device powered by an unregulated 9 V battery).
- Finally, use VLSI ICs (cPLDs, microcontrollers, ASSCs) rather than discrete logic – this reduces package count and wiring complexity and adds flexibility.

12.1.2 Input and output characteristics

Digital-logic families are designed so that the output from a chip can properly drive many inputs within the same logic family, powered from the same supply voltage. A typical fan-out capability is at least 10 loads, meaning that an output from a gate or flip-flop, for example, can be connected

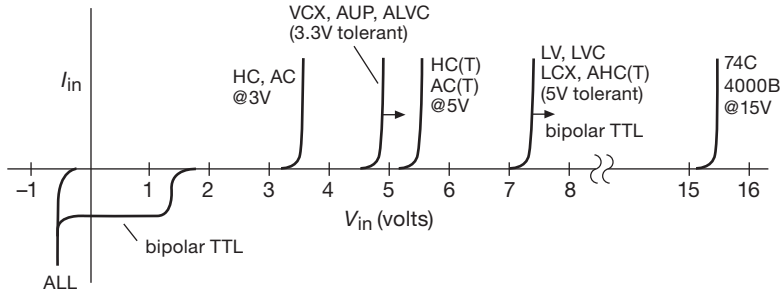


Figure 12.4. Logic gate input current versus input voltage. With the exception of bipolar "TTL," there is no static input current within the normal input-voltage range. All logic families include internal protective clamp diodes to ground. Some families (for example 74HC) clamp to the positive supply, and therefore draw input current when the input signal voltage exceeds a diode drop above V_+. More recent families (for example 74LVC or 74AUP) use internal zener-like protection, and permit inputs well above the supply voltage; these are called "5 V tolerant" or "3.3 V tolerant" (for example LVC and AUP, respectively), and permit such inputs even when unpowered (a situation in which a 74HC logic device would clamp at $+0.6V$ or, worse, cause the HIGH input to partially power the supply rail).

to 10 inputs and still perform within specifications.[2] In other words, in normal digital design practice you can get by without knowing anything about the electrical properties of the chips you're using, as long as your circuit consists only of digital logic driving more digital logic of the same type. In practice, this means that you don't often have to worry about what's actually going on at logic inputs and outputs.

However, as soon as you attempt to drive digital circuitry with externally generated signals, whether digital or analog, or whenever you use digital-logic outputs to drive other devices, you must face the realities of what it takes to drive a logic input and what a logic output can drive. Furthermore, when mixing logic families, or when you have logic running between different supply voltages, it is essential to understand the circuit properties of logic inputs and outputs. Interfacing between logic families is not an academic question. To take advantage of advanced VLSI chips, or special functions that are available in only one logic family, you must know how to mix logic types and voltages. In the next few subsections we will consider the circuit properties of logic inputs and outputs in detail, with examples of interfacing between logic families and between logic devices and the outside world.

A. Input characteristics

The graphs in Figures 12.4 and 12.5 show the important properties of digital-logic inputs: input current and output voltage (for an inverter) as functions of input voltage. We have extended the graphs to input voltages beyond the range normally encountered in digital circuits, because in interface situations the input signals may easily exceed the power-supply voltages. As the graphs imply, CMOS logic and bipolar TTL are normally operated with the negative supply pin connected to ground.

Input current (Figure 12.4)

Most logic these days is built with CMOS, where the inputs draw no current (except for leakage current, typically $10^{-5}\,\mu$A) for input voltages between ground and the supply voltage.[3] For voltages beyond the supply range, the input protection network looks like a clamping diode to ground, and either a diode to V_+ (e.g., the 74HCT family) or a zener-like clamp that permits input swings beyond the positive supply (e.g., the "5 V tolerant" 74LVC family); see Figures 12.3 and 12.4 for more details. Momentary currents greater than 20–50 mA through these diodes will damage or destroy the part, in some cases causing what's known as "SCR latchup" (§10.8.3B); you'll find such limits listed in the "Absolute Maximum Ratings" section of datasheets.

[2] For CMOS logic, which has zero dc input current, excessive loading merely slows down the transitions; in that sense the fan-out is "infinite." However, if you're driving bipolar logic, there's the additional issue of required dc input current (e.g., 1.6 mA for the 74LS family), which leads to a typical fan-out of 10.

[3] Note the peculiarity of the nearly obsolete bipolar TTL: an input sources a significant current (of order 0.1–1 mA) when held LOW, but draws only a small current when HIGH (typically a few microamperes, never more than 20 μA). To drive a TTL input, you must be able to sink a milliamp or so while holding the input below 0.4 V. Failure to understand this may lead to widespread circuit malfunction in interfacing situations!

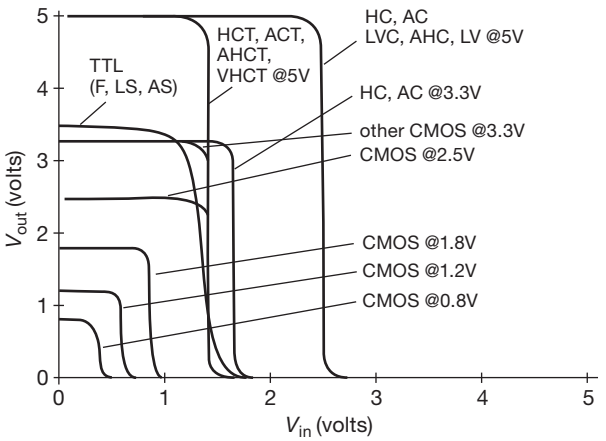


Figure 12.5. Output versus input ("transfer function") for logic inverters in commonly used families. In general, CMOS families that run at supply voltages of 2.5 V or less have their threshold at half the supply. At higher voltages most logic families (and many other more complex chips, such as microcontrollers and configurable logic) adhere to the "TTL" input specification, which guarantees that the threshold lies between 0.8 V and 2.0 V (and is typically 1.3–1.5 V). The HC/AC and 4000B families are exceptions, with their higher $V_+/2$ supply-tracking threshold.

Logic threshold (Figures 12.5 and 12.6)

The logic threshold voltage (the dividing line between logic LOW and logic HIGH inputs) depends on both the logic family, and on the supply voltage (for families where a range of supply voltages is permitted, as seen for example in Figure 12.3). The final authority is always the datasheet! But we can be of assistance here.

- Numerous digital devices adhere to what are known as "TTL thresholds," a legacy of the bipolar logic of the 1960s: $V < 0.8$ V is guaranteed to be interpreted as logic LOW, and $V > 2.0$ V is guaranteed to be interpreted as logic HIGH. (The voltages you send to such a logic device should stay outside those limits, to provide noise immunity; typically you'd provide a LOW level below 0.4 V, and a HIGH above 2.4 V.) The actual threshold voltage is typically around 1.35–1.5 V.[4] Common logic device families with TTL input thresholds (in addition to genuine bipolar TTL families such as 74F, 74LS, and 74AS) include the 74HCT, 74ACT, 74AHCT, and 74VHCT families; the "T" suffix signifies the family variant with (lowered) TTL thresholds, compared with the mid-supply ($V_+/2$) threshold of the non-T families (74HC, 74AC, 74AHC, 74VHC). Interestingly, many contemporary digital devices of high complexity, such as programmable logic (Chapter 11) and microcontrollers (Chapter 15) continue to adhere to both the TTL input threshold specification (LOW = 0.8 V or less, HIGH = 2.0 V or more), and the corresponding "TTL" output levels (LOW = 0.4 V or less, HIGH = 2.4 V or more).
- Logic inputs that are not of the "TTL-compatible" variety have their thresholds typically at mid-supply; that is, at $V_+/2$. This is true of the older CMOS families, for example 74HC and 74AC (which can operate with supply voltages from 2 V to 6 V) and the high-voltage 4000B/74C families (which can operate with supply voltages from 3 V to 18 V). It also generally holds for most newer low-voltage families like 74LVC and 74AUC. Note, however, that the actual threshold voltage for a logic part with a "mid-supply threshold" can vary considerably: the datasheet specifications typically permit a range from about one-third to two-thirds of V_+ (V_+ is often called V_{CC} or V_{DD}).

B. Output characteristics

The output circuit of CMOS logic devices almost invariably uses a pair of complementary MOSFETs, one ON and the other OFF (Figure 12.1). The output looks like a MOSFET r_{ON} to ground or to V_+ when it is within a volt or so of the respective rail, becoming something like a current source when you draw so much current that the output is forced more than a volt or two away from the supply rails. Typical values of r_{ON} are 30 Ω for 74HC(T) operating at +5 V, 12 Ω 74AC(T) operating at +5 V, 10 Ω/15 Ω (sinking/sourcing) for 74LVC operating at 3.3 V, and 200 Ω for 4000B operating at 15 V.[5]

The output circuit of bipolar-TTL devices (a disappearing species) uses an *npn* transistor switch to ground and an *npn* follower (or Darlington) to V_+ with a current-limiting resistor in its collector. One transistor is saturated and the other is off. As a result, a TTL device can sink a large current (8 mA for 74LS, 24 mA for 74F) to ground with a small (saturation) voltage drop, but when sourcing current

[4] We measured the threshold voltages for a sampling of "TTL threshold" inverters (by connecting the output back to the input), and here's what we found – 7404: 1.37 V; 74ACT04: 1.48 V; 74AS1004: 1.49 V; 74F04: 1.43 V; 74HCT04: 1.34 V; and 74LS04: 1.50 V.

[5] Watch out, though, for some "CMOS" digital functions (primarily older designs that operate from a +5 V supply, for example some cPLDs and microcontrollers) where the *p*-channel pullup switch is replaced with an *n*-channel source follower: for these devices the HIGH-state output does not reach the positive supply; instead it hovers around +3 V. You can recognize these parts from their datasheet's "DC Characteristics," which will say something like V_{OH}(min)=2.4 V, the telltale signs of an imitation legacy "TTL" output (where an *npn* follower forms the output pullup).

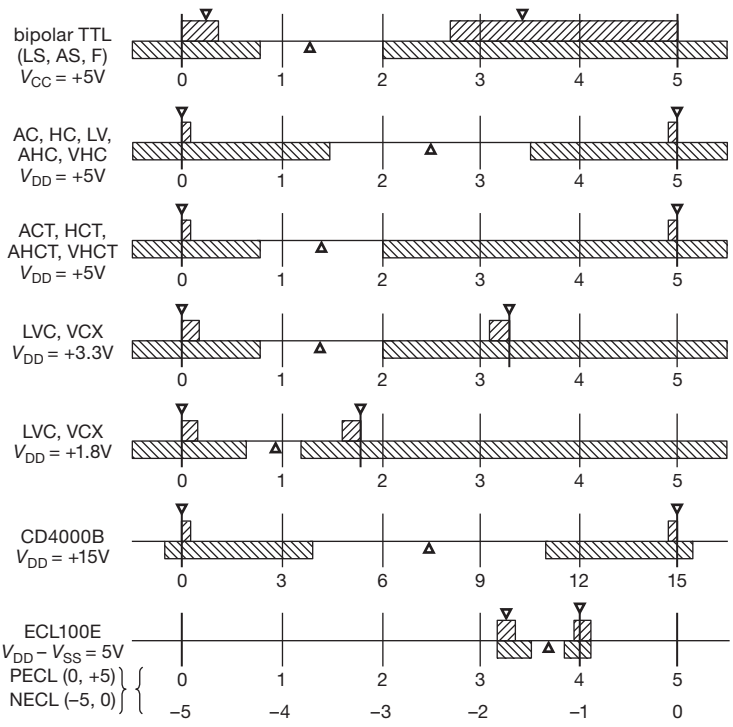


Figure 12.6. The range of voltages corresponding to the two logic states (HIGH and LOW) for popular families of digital logic. The shaded areas above the line show the specified range of output voltages within which a logic LOW or HIGH is guaranteed to fall, with the pair of arrows indicating typical output values (LOW, HIGH) encountered in practice. The shaded areas below the line show the range of input voltages guaranteed to be interpreted as LOW or HIGH, with the arrow indicating the typical *logic threshold* voltage, i.e., the dividing line between LOW and HIGH. In all cases a logic HIGH is more positive than a logic LOW. Table 10.1 and Figure 12.3 provide additional information about these families. See also Figure 12.135.

its output HIGH will be at least 1.5 V below the +5 V supply (see the 74AS and 74LS curves in Figure 12.7). The output circuit is designed to drive TTL inputs, or devices with "TTL input specifications" ($<$0.8 V guaranteed LOW, $>$2.0 V guaranteed HIGH) with a fan-out of 10.

In Figure 12.7 we have plotted the typical output voltage for both HIGH and LOW output states, against output current, for a selection of popular standard-logic families. To simplify the graph, output current is always drawn as positive. Note that true CMOS devices (that is, a complementary nMOS and pMOS push–pull switch pair) pull their outputs all the way to V_+ or ground, generating a full swing unless heavily loaded; so when driving only CMOS loads (zero dc current), the swing is fully rail-to-rail. Bipolar-TTL levels, by comparison, are typically 50–200 mV (LOW) or +3.5 volts (HIGH) when driving other TTL devices as loads. With a pullup resistor (discussed later), HIGH TTL outputs go all the way to +5 volts. We've also plotted two examples of MOSFET gate drivers, which accept a TTL-compatible logic-level input, and use a high-current CMOS output stage to generate an output swing between ground and a chosen positive V_{DD} supply; for the TC4420 series that can range from +4.5–18 V, whereas for the IXDD509 series the range is +4.5–30 V.[6]

In contemporary digital electronics we are increasingly exploiting the nifty capabilities of programmable logic devices (Chapter 11) and of microcontrollers (Chapter 15). So you really need to know what their outputs can do, in terms of driving external loads. Figure 12.8 shows the output drive characteristics of several popular PLDs (Altera MAX7000A, Lattice Mach 4000, Xilinx Coolrunner-II) and microcontrollers (Atmel ATmega, Microchip PIC16F, TI MSP430). These all use true CMOS

[6] Note, however, that MOSFET drivers do not operate at full logic speeds: although their timing is usually specified into the large capacitive loading characteristic of power MOSFETs, you're generally looking at propagation delays in the neighborhood of 10–25 ns or more (to be compared with $t_{pd}\sim$2 ns for standard-logic families like 74LVC).

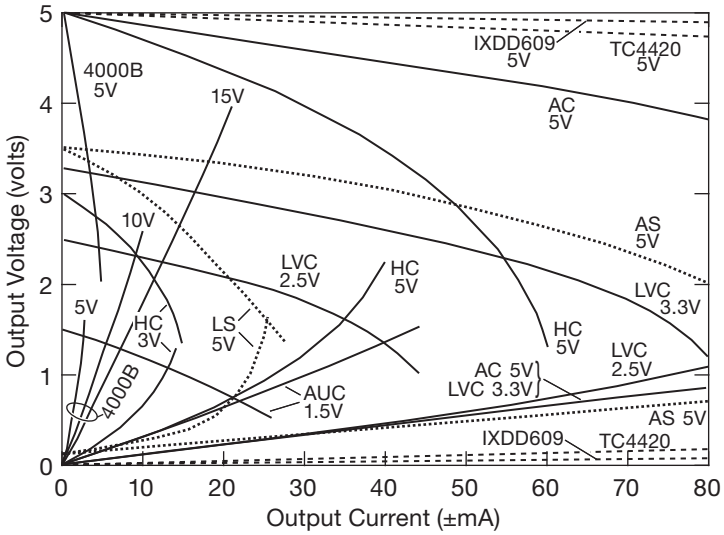


Figure 12.7. Logic gate output characteristics. The 74LS and 74AS families are 5 V bipolar TTL, with *npn* follower pullup, hence the ∼3.5 V HIGH output. All others are true CMOS, with rail-to-rail output swing. The TC4420/MCP1406 and IXDD609 are "MOSFET driver" ICs, with robust CMOS outputs that can source or sink up to 6 A and 9 A, respectively; they hardly even notice an 80 mA load.

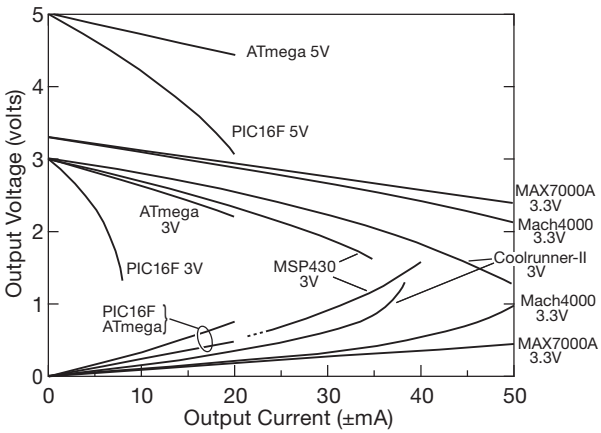


Figure 12.8. Output drive characteristics of selected PLDs and microcontrollers.

outputs with rail-to-rail voltage swing. As is evident from the graph, however, not all CMOS output stages are created equal.

12.1.3 Interfacing between logic families

It is important to know how to make different logic families talk to each other, because there are situations in which you must mix logic types, or portions of logic running from different supply voltages. In a typical situation you might want to use an output from a microcontroller running on +2.5 V to drive a 5 V single-gate inverter, so that the final 5 V full-swing output can drive a 5 V mechanical relay, or a solid-state relay (SSR), or a white LED. Or you might want to go the other way: a full-swing 5 V logic output needs to get to a low-voltage part running on 1.8 V.

The three things that can keep you from connecting any pair of logic chips together are (a) input logic-level incompatibility, (b) output drive capability, and (c) supply voltages. Rather than bore you with pages of rules and of explanations of what works and what doesn't, we've boiled down the interface problem to a simple set of recommended methods (Figure 12.9). Let's take a quick tour.

A. CMOS at same voltage

You can always make a direct logic connection between CMOS logic devices running from the same supply voltage. The output is full-swing, and therefore happily drives another CMOS device, regardless of the latter's particular threshold voltage.

B. "5 V logic" driving lower-voltage CMOS

CMOS running from 5 V can be connected directly to what's called "5 V-tolerant" logic (for example the 74LVC family) running at a lower supply voltage. As shown here, either CMOS (with its full-swing output), or "TTL" output-level devices (including true bipolar TTL, or ICs with nMOS follower pullups, either of which have a HIGH-state output of ∼3.5 V) satisfy the input voltage requirements

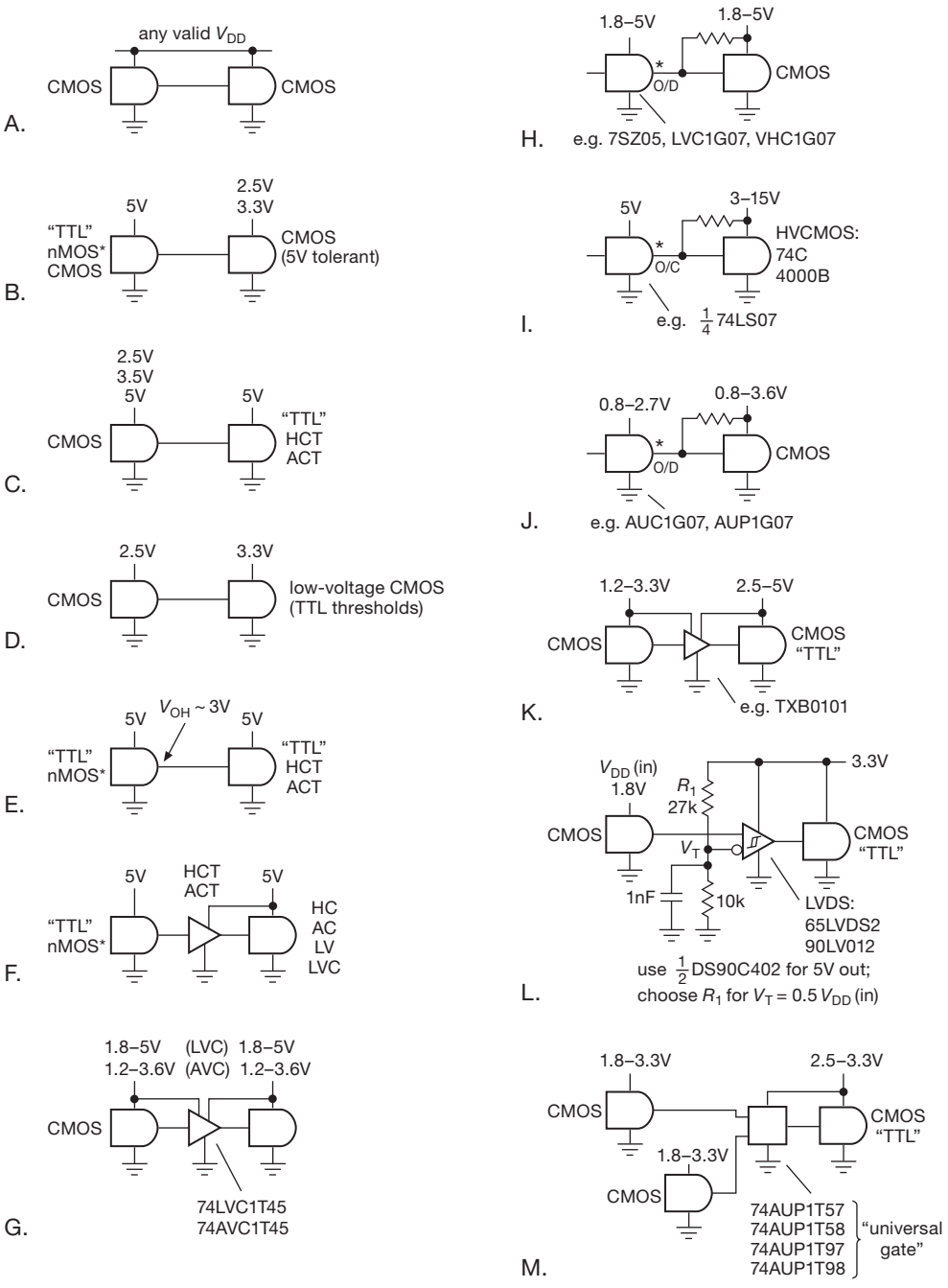


Figure 12.9. Logic family interconnections. See text for narrative.

of 5 V-tolerant CMOS running from supply voltages in the range of 2.5–3.3 V. There's also the 74LV1T level-translating family, which can do both "down translation" (as here) or "up translation" (as in C).

C. Lower-voltage CMOS driving 5 V logic

You can drive "TTL input" (reduced-threshold) 5 V logic directly from CMOS outputs that swing at least to 2.5 V, as shown. In addition to true bipolar-TTL parts, some

5 V CMOS families (such as 74HC, 74AC, 74AHC, 74VHC) include TTL threshold variants (74HCT, 74ACT, 74AHCT,[7] 74VHCT). There's also the 74LV1T level-translating family.

D. CMOS at 2.5 V driving CMOS at 3.3 V

Nearly all CMOS families that can operate at 3.3 V supply voltage have TTL-compatible input levels ($<$0.8 V guaranteed LOW, $>$2.0 V guaranteed HIGH), so it's safe to drive them from the full-swing output of CMOS powered from 2.5 V.

E. 5 V "TTL outputs" driving reduced-threshold 5 V logic

To drive logic powered from 5 V, those outputs – LOW close to 0 V, but HIGH only $\sim$3.5 V (and only guaranteed to be $>$2.4 V) – must be paired with TTL-compatible inputs; that limits you to true 5 V bipolar TTL (e.g., 74F), or to 5 V CMOS logic families (or more complex 5 V digital chips) with TTL-compatible inputs (74ACT, 74HCT, 74AHCT, 74VHCT).

F. 5 V "TTL outputs" driving incompatible 5 V logic

If you're stuck with normal threshold 5 V logic (i.e., threshold at $V_{DD}/2$, or $\sim$2.5 V), you can use a CMOS buffer or inverter with TTL thresholds (74HCT, etc) to convert the TTL swing to a 5 V full-swing signal. Note also that you can use instead a special level translator part like the 74LVC1T45 (Figure 12.9G).

G. Dual-supply level translator: 1.8–5 V and 1.2–3.6 V

There are some nice chips designed specifically for logic-level translation between a pair of supply voltages. The dual-supply 74LVC1T45 lets you translate between logic levels powered from 1.8 V to 5 V on either side (it is, in fact, bidirectional, controlled by a DIR pin input, just like the classic '245-style 8-bit bidirectional buffer). The lower-voltage 74AVC1T45 is similar, but operates from 1.2 V to 3.6 V on either side. You can also get these parts in duals (LVC2T45, AVC2T45). Note, however, that the LVC part, when operated at 5 V on its input side, has a "mid-supply" input threshold (guaranteed LOW $<$1.5 V, HIGH $>$3.5 V), and therefore cannot be driven from a TTL output level (with output HIGH only guaranteed to be $\geq$2.4 V). See also the TXB0101 dual-supply translator (Figure 12.9K).

[7] As an example, the 74AHCT1G125 is handy for converting a 3.3 V logic signal to a needed 5 V output swing; it comes in convenient SOT23-5 or SC70-5 packages, and costs less than $0.10.

H, I, J. Open-drain and open-collector

You can translate up or down in voltage level with an open-drain (or open-collector) buffer, though you pay a price in speed and quiescent current with the passive pullup resistor (the resistor value is a compromise – smaller is faster, but uses more power). Figure 12.9H shows translation between logic levels in the range 1.8–5 V on either side, using single gate '07 buffers in the wide voltage range logic families (7SZ, 74LVC, 74VHC), which support 5 V operation, and accept output pullup to 5 V. If you want a larger output swing, you can use the high-voltage open-collector 74LS07 (Figure 12.9I), which lets you pullup the output to +15 V. And if you need to translate from very low logic voltages, you can use a 74AUC1G07 or 74AUP1G07 (Figure 12.9J), which operates down to input logic levels as low as 0.8 V.

K. Low-voltage CMOS driving 2.5–5 V logic

The TXB0101 is another dual-supply translator (like the LVC/AVC1T45 in Figure 12.9G), but it has a couple of quirks. First, $V_{DD(B)}$ (the righthand supply in the figure) cannot be less than $V_{DD(A)}$. Second, it is bidirectional (like the 1T45), but it has no DIR control input; instead, it senses transitions on either side, responding by switching on the opposite port's CMOS drivers briefly, then maintaining that state weakly ($\sim$4k series output resistor) with positive feedback (useful for switch debouncing, Figure 12.16).

L. Very low voltage CMOS to 3.3 V or 5 V logic

Here's a nice trick: an LVDS receiver accepts a differential digital signal pair, within a common-mode range from 0 V to +2.4 V, and with guaranteed switching down to an input amplitude of 200 mV. The (single-ended) output is full-swing CMOS, powered from 2.5–3.3 V (e.g., 65LVDS2) or from 5 V (DS90C402, DS90C032). So you can trick it into working for you as a level translator, by supplying a reference dc level halfway between logic states (with a small bypass capacitor) to the unused input. This will work down to very low input logic swings, for example as little as 0.5 V, where none of the other level translators dare to tread. LVDS interface chips are *fast*, too, often specified for data rates of 400 Mbps, with typical propagation delays $<$2 ns.

M. Low-voltage CMOS translator with configurable logic

The elegant "universal" gates that we mentioned in Chapter 10 (§10.2.1A) can be used to perform some logic while translating across logic voltage domains. When used this way, the gate runs from a single supply in common with the logic on the output side (which can range from 2.5–3.3 V), accepting inputs from logic that can be powered variously

from 1.8–3.3 V. These gates have Schmitt-trigger inputs, with about 0.4 V hysteresis centered about the threshold at ∼0.7 V.

A. Dynamic incompatibility: slow edge rates

Here's a problem that sometimes crops up when you drive an input of fast digital logic with a digital signal that does not slew through the threshold fast enough: the abrupt switching output of the driven device can couple back to the input (through the ground or power-supply connection, or internally on the chip itself, or just capacitively), causing multiple output transitions, as seen in the measured traces of Figure 12.10. To encourage such unseemly behavior we used a 74LS05 open-collector (OC) inverter with a lazy 5k Ω pullup to +5 V, driving a frenetic 74AC04 gate.[8] You can clearly see the disruption of the leisurely input waveform, as the fast inverter furiously switches its output.

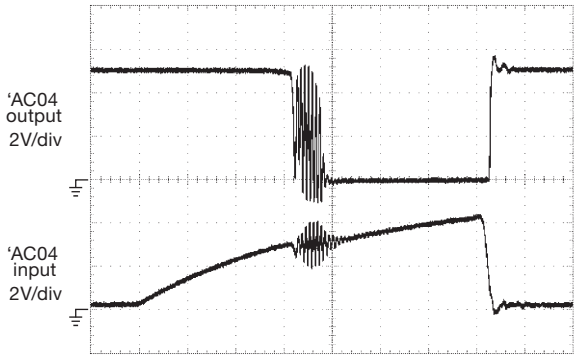


Figure 12.10. The relatively slow rising "edge" of a 'LS05 open-collector inverter with 5k Ω pullup (lower trace) drives a fast 'AC04 inverter, causing multiple output transitions. The trailing edge exhibits no bad behavior, because the open-collector falling edge is fast. Horizontal: 40 ns/div.

Multiple output transitions may be merely ugly; but this problem becomes devastating when you are driving edge-sensitive inputs (e.g., the clock input of flip-flops or counters). A flip-flop may fail to toggle; or a counter or shift register may clock several times at each edge. To illustrate this effect, we drove the clock input of a 74AC74 toggling flip-flop from the square wave output of a CD4001B NOR gate, with both powered from +5 V (and with a reset pulse applied to the flip-flop before the next clock). Figure 12.11 shows the messy result: sometimes the flip-flop toggles correctly, but sometimes it toggles twice in rapid succession.

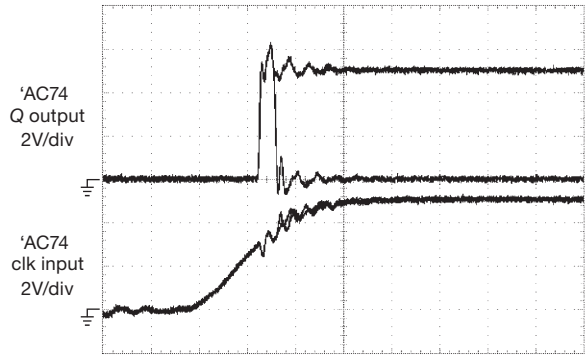


Figure 12.11. The slow rising edge of a CD4001B gate, loaded with 27 pF (a typical value of wiring and load capacitance), clocks a fast 'AC74 toggling flip-flop. The oscilloscope's persistence displays multiple clocking events, some more successful than others. Horizontal: 20 ns/div.

The lesson is clear: don't use slow edges to clock fast logic. It sometimes suffices to clean up the offending signal with a Schmitt-trigger inverter, available in most logic families as a '14 (e.g., 74LVC14). Figure 12.12 shows waveforms from the same setup as that of Figure 12.10, but with a 74AC14 (inverter with Schmitt) substituted for the 74AC04 (inverter without Schmitt). Better yet, stay away from slow edges completely.

You can get into the same kind of problems when you send digital signals between circuit boards, or between instruments, or over cables, an important set of topics we'll discuss presently (§12.9).

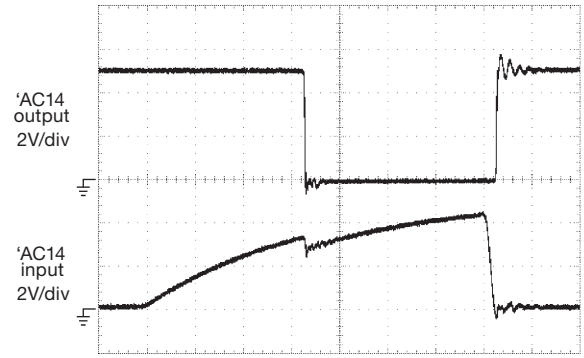


Figure 12.12. Same as Figure 12.10, but with the 'AC04 inverter replaced with an 'AC14 Schmitt-trigger inverter. The fast-switching output couples to the input, as before, but not enough to carry it back through the new (lower) threshold, thanks to the Schmitt's hysteresis. Horizontal: 40 ns/div.

[8] We've found that 74AC and 74ACT logic is fussy stuff, particularly in corner-pin power and ground incarnations packaged in through-hole DIP. Stay away from this stuff unless you need the speed; and then be sure to use a ground plane, keep ground leads short, and bypass close to the chip.

12.1.4 Driving digital logic inputs

A. Switches as input devices

It is easy to drive digital inputs from switches, keyboards, comparators, etc., if you keep in mind the input characteristics of the logic you're driving. The simplest way is to generate a valid logic level with a pullup or pull-down resistor (Figure 12.13). With CMOS logic, either is fine, because the inputs draw no current and the threshold is typically in the range of $0.3V_{DD}$ to $0.5V_{DD}$. It is usually more convenient to ground one side of the switch, but if the circuit is simplified by having a HIGH input when the switch is closed, the method with pull-down resistor will be perfectly OK. Be careful, though, with bipolar TTL: their inputs *source* a substantial current (e.g., a 74F family input sources 0.6 mA when held LOW), so it's best to use the configuration with a pullup resistor and the switch returned to ground.

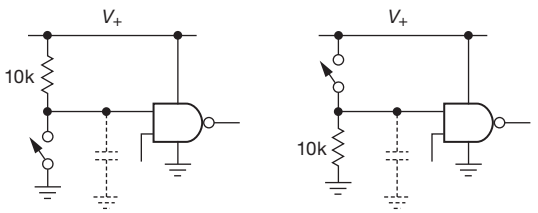


Figure 12.13. Mechanical switch to logic-level circuits (not debounced). If the switch is not close to the logic, it's common to see a small capacitor (∼100–1000 pF) used to suppress capacitively coupled noise.

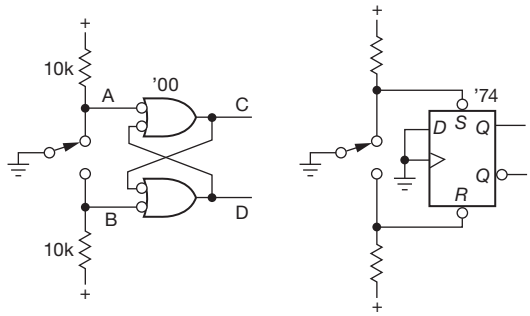


Figure 12.14. SR flip-flop switch debouncer, implemented either with cross-connected gates, or with a flip-flop with asynchronous SET and RESET inputs.

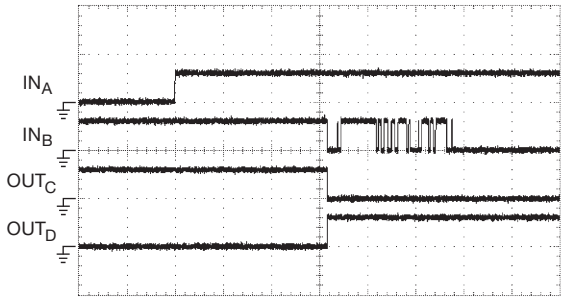


Figure 12.15. Waveforms from the debouncer of Figure 12.14, implemented with 3 V CMOS gates ('HC00) and a C&K 8121 SPDT (single-pole double-throw) pushbutton switch. Note the delay time from release of the *A* contact to the first closure of the *B* contact, as the switch's armature moves between the contacts. Vertical: 5 V/div; horizontal: 100 μs/div.

B. Switch bounce

As we remarked in Chapter 10, mechanical-switch contacts usually exhibit "bounce" after initial closure, with typical time scales of a millisecond. For physically large switches, the bounce can last as long as 50 ms. This can wreak havoc with circuits that are sensitive to changes of state, or "edges" (a flip-flop or counter would toggle many times if clocked directly from a switch input). In such cases it is essential to debounce the switch electronically. Here are a few methods:

- Use a pair of gates to make a jam-type *SR* flip-flop. Use pullups at the inputs to the debouncer, of course (Figures 12.14 and 12.15). Equivalently you can use a flip-flop with SET and CLEAR inputs (e.g., a '74); in that case, ground the clocking input.
- Use a noninverting buffer looped back into its input to make a latching "keeper" circuit, as in Figure 12.16. A buffer or noninverting gate (e.g., '1G34, '08, or '32) with its output looped back works fine. Timid circuit designers include a resistor in the feedback path (as shown) to limit the momentary transient current when the switch changes state; but, trust us, you can happily omit it.[9] The TXB0101 (§12.1.3, Figure 12.9K) is one of many "automatic" bidirectional level-shifter chips that hold their state, but can be overdriven by a change of state on either side.
- Use an *RC* slowdown network to drive a CMOS Schmitt trigger (Figures 12.17 and 12.18). The lowpass filter smooths the bouncy waveform so that the Schmitt-trigger gate makes only one transition. A 1 ms to 10 ms *RC* time constant is generally long enough. This method isn't

[9] It's always OK to override a logic output by shorting it to V_+ or ground, *providing the duration is kept short*. In this circuit there is no problem, because the output is forced for only a gate propagation delay, after which it contentedly holds itself in the new state.

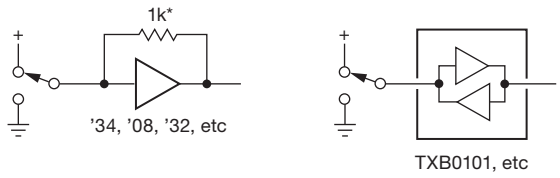


Figure 12.16. Switch debouncers using a "keeper" circuit that holds its logic state when the input is undriven.

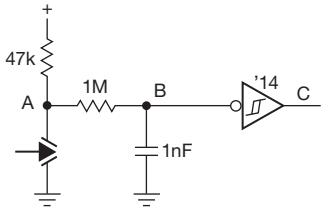


Figure 12.17. Debouncing an SPST (single-pole single-throw) switch with an *RC* smoothing circuit and a Schmitt trigger inverter.

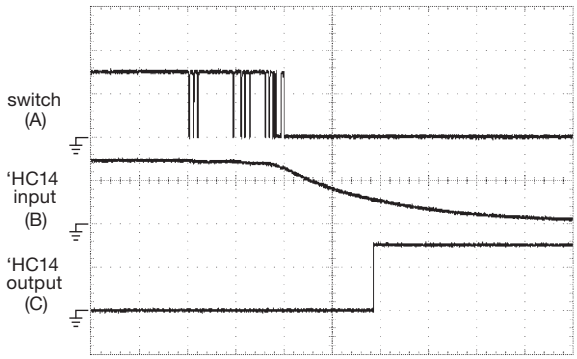


Figure 12.18. Waveforms from the debouncer of Figure 12.17, implemented with 3 V CMOS ('HC14) and a Microswitch 1PB13 pushbutton. Vertical: 2 V/div; Horizontal: 400 μs/div.

well suited to bipolar TTL because of the low driving impedance that TTL inputs require.

- Use a clocked circuit that samples the input level in a way that is not fooled by bounce. A simple way is to clock a *D*-type flip-flop with a period safely longer than the switch's bounce duration, e.g., with a clock frequency of 100 Hz (Figure 12.19). But there are dedicated debouncer chips, for example the MAX6816-8 (single, dual, and octal debouncers), which test for transition to a stable state for multiple clock periods (with internal oscillator, counters, and logic; effectively a digital lowpass filter) and generate a clean debounced output; they include internal pullup resistors, input protection to $\pm$25 V, and operation from 2.7 V to 5.5 V. You just tie your SPST switches from input to ground – no external components needed (Figure 12.19). A similar part is the MC14490 hex (6-section) debouncer, which belongs to the CMOS 4000B-series and can run with supply voltages from 3 V to 18 V; it includes internal pullups, but it requires one external capacitor to set the clock rate. Another alternative is the so-called "power-supply supervisory" chip, which is ordinarily used to detect brownout (undervoltage) and to generate a clean reset pulse on power restoration (or initial power-up). Many of these chips include a manual reset input, to which you can connect a pushbutton, thus hijacking them into service as a debouncer.
- Use a microcontroller, with programming to carry out "software debounce" (Figure 12.20). Most microcontrollers include internal pullups; and simple code (either interrupt driven, or polled) can look for a stable change of state. This is a favorite method used by circuit designers for any gadget that needs a microcontroller anyway.
- Use a device with built-in debouncer. Keyboard encoders, for instance, are designed with mechanical switches in mind as input devices, and they usually include debouncing circuitry. Another example is shown in Figure 12.20, namely a pushbutton-controlled "digital potentiometer" (an internal resistor string, with tap selection via MOSFET switches). Each press of a button increments or decrements the internal counter, so it's got to be bounce free. And it is.

A few general comments on switches as input devices: note that SPDT switches (sometimes called "form C") are necessary with the first two methods (SR flip-flop, "keeper" circuit), whereas the simpler SPST switches ("form A") can be used with the other methods. Keep in

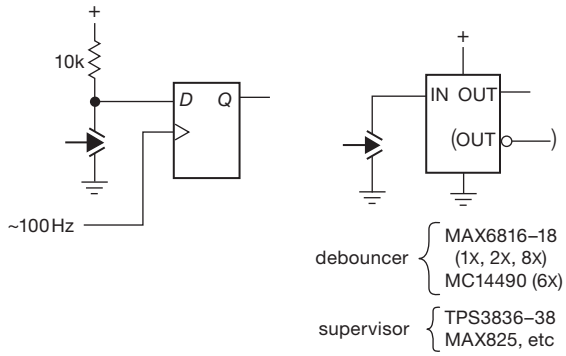


Figure 12.19. Clocked debouncers. Simplest is a slowly-clocked *D* flip-flop. There are better methods, embodied for example in special debouncer chips like the MAX6816-8 and MC14490, and in "supervisory" chips like the TPS3836-8.

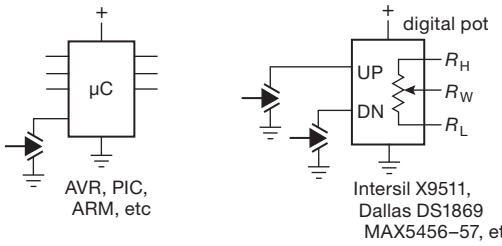


Figure 12.20. Debouncing with complex digital ICs: a microcontroller (μC) can use its user-written program (its *firmware*) to perform "software debounce"; and application-specific chips that accept pushbutton commands (like the digital potentiometers shown, with UP′ and DOWN′ inputs) often include internal pullup and debounce circuitry.

mind, also, that often it isn't necessary to debounce switch inputs, since they aren't always used to drive edge-sensitive circuitry. Another point: well-designed switches are usually "self-wiping" to maintain a clean contact surface (take one apart to see what that means), but it's a good idea to choose circuit values so that a current of at least a few milliamps flows through the switch contacts to clean them. With suitable choice of contact material (e.g., gold) and mechanical design, switches can be designed to avoid this "dry-switching" problem and will work properly even when switching zero current. More on this in Chapter *1x*.

12.1.5 Input protection

In these interfacing examples we've assumed that the signals being applied to the logic inputs are well behaved – that they do not have transient overvoltages or other destructive tendencies. That's not always the case. You can get into plenty of trouble with signals coming in from the outside, if you mount a connector on a box and casually hook it to the input of a logic gate (Figure 12.21A). A common source of transients is the static charge that accumulates readily in dry weather – those crackling sounds you hear when you take off a synthetic fleece jacket can elevate the body to a kilovolt or more. That voltage resides on the human capacitor (about 100 pF), and can take flight into a circuit input when you plug in a connector, or injudiciously touch components on a circuit board without first discharging yourself to ground.

The problem is well known in the electronics industry, and ICs are tested and rated for survival when subjected to electrostatic discharge. For this purpose the human body is modeled as shown in Figure 12.22. In the manner of the Physicist's "spherical cow," Electrical Engineers reduce all

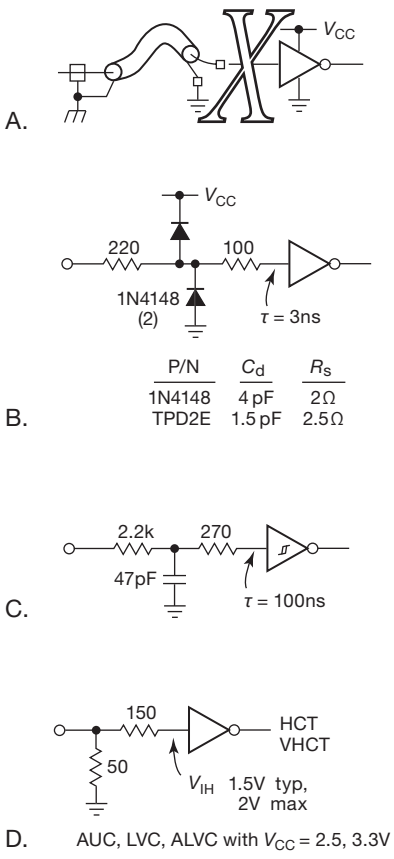


Figure 12.21. Protecting logic inputs from destructive transients: A. no protection; B. Diode clamps to V_{CC} and ground, or ESD protection array, with current-limiting series resistors; C. *RC* filtering plus Schmitt-trigger input; D. cable termination and series resistor, to dilute transient energy.

humans to a 100 pF capacitor in series with a 1.5 kΩ resistor,[10] and use that to test the robustness of their ICs. The usual charging voltages are 1–2.5 kV, but you see ICs rated up to 15 kV; for example, the MAX3232E RS-232 serial driver–receiver proclaims "ESD Protection for RS-232 Bus Pins – ±15 kV (HBM)." The figure shows the charged HBM (human body model) delivering a current pulse to the input of a logic-gate victim, which must sustain a transient current of an amp or more. That current is clamped to the supply rails, but the internal resistance of the clamp allows the input pin to go tens of volts beyond the rail (or below ground, for a negative transient), delivering a calculable energy to a very small area on the semiconductor die.

[10] JEDEC standard JESD22-A114D, dated 2006.

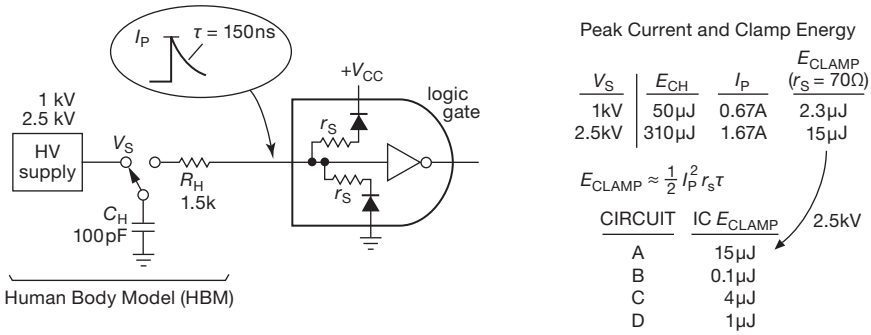


Figure 12.22. The Electrical Engineer's view of Biology: The "Human Body Model." ICs are tested and qualified at voltages of 1 kV, 2.5 kV, or more. The table lists estimated clamp energy values for the circuits in Figure 12.21. See also §3.5.4H.

ICs are rated to handle this. But plenty of damaged chips out there testify to the wisdom of adding external protection, especially if there's reason to expect exposure to high transient levels. Or inexperienced users. Look back at Figure 12.21. Circuit B is simple and effective. Ordinary diodes like the ubiquitous 1N4148 have more junction area than on-chip clamps, hence better clamping action; the upstream resistor limits the diode current, and the downstream resistor, fed from that clamped voltage, limits the chip's input current. You can get arrays of clamp diodes; these sometimes include a zener, as in Figure 12.23, for protecting differential logic like LVDS or RS-485.

Circuit C uses an *RC* transient filter to reduce the peak input current. To be effective, the time constant should be at least comparable to the HBM transient time scale, say 100 ns or so (and with an input resistance at least comparable to the HBM value of 1.5 k), and therefore should be followed by a Schmitt-trigger input. Circuit D takes a different approach: it combines a matched termination for 50 Ω coaxial cable (see Appendix H, and §12.10) with a larger series resistor, forming a "current divider." That dilutes the chip's transient input current by a factor of 4 (for the values shown), which is a 16× reduction of transient energy; and there's a further 2× to 5× reduction because the diluted energy is shared between the 150 Ω resistor and the chip's dynamic resistance during overvoltage. In Figure 12.22 we've estimated the delivered transient energy for each of these methods, compared with the 15μJ estimated value for a 2.5 kV HBM with no protection.

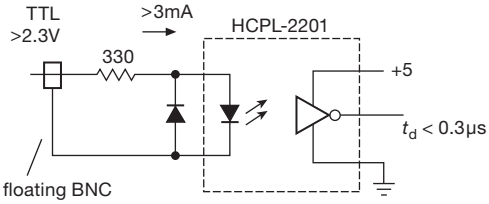


Figure 12.24. The ultimate in isolation: an optocoupler. This is more expensive than a diode clamp arrangement, but it's guaranteed to keep the bad stuff out.

Finally, for the ultimate in protection, use an optocoupler logic isolator, as in Figure 12.24. There's no galvanic connection at all (note the floating BNC connector), and these things can stand off potentials of many kilovolts (we'll see them later in the chapter, at §12.7). Note the current-limiting resistor and reverse protection diode; the latter is often omitted in error.

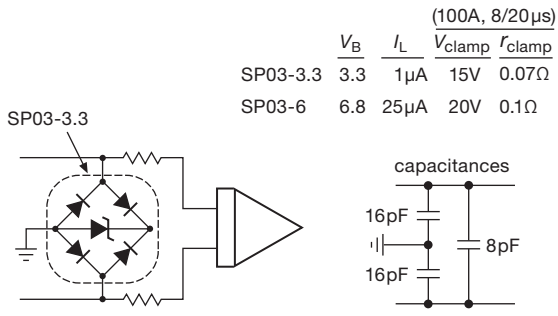


Figure 12.23. Diode protection arrays, like this one from Littelfuse, let you clamp single-ended or differential inputs, with very low clamp impedance. The internal zener sets the clamping voltage. You can get lower-current protection arrays with very low capacitance, for example the 5-volt CTLTVS5-4 from Central Semiconductor: it has a maximum 0.8 pF to ground (and 0.4 pF line-to-line) at zero volts, a peak current rating of 2.5 A, and it includes two protective bridges in a single SMT package.

12.1.6 Some comments about logic inputs

A. Pullups and pull-downs

Most contemporary digital logic is CMOS, with essentially zero input current. So even a weak pullup (or pull-down)

current is adequate to bring the input fully to V_+ (or to ground).[11] Watch out, though, for coupling of capacitive transients to such an input, for example from a panel switch with wiring that travels in proximity to signal-carrying lines. In such a case it's a good idea to add a small ($\sim$1 nF) bypass capacitor near the high-impedance logic input. In the case of complex digital ICs (microcontrollers, FPGAs, and other application-specific standard products) it's common to see internal pullups for inputs that may come from switches, so no pullup or pull-down resistor is needed (though you may want to add a small bypass capacitor to suppress coupled transients).

B. Input overdrive

Digital-logic inputs include protection against overvoltage, usually in the form of a clamp diode to ground, and either a clamp diode to V_+ or a zener-like clamp (for devices with input tolerance beyond their supply voltage). The "Absolute Maximum" datasheet ratings tell you the limits of what you can get away with (e.g., 74LVC devices specify $-0.5\,\text{V} < V_{\text{in}} < 5.5\,\text{V}$, regardless of supply voltage: they are "5 V tolerant."). However, it's often the input *current* that does the damage, which is duly noted in the same table: "The input and output negative voltage ratings may be exceeded if the input and output current ratings are observed." And the latter (input clamp current, output clamp current) are specified, in this case, as -50 mA, maximum. Although it's a nice gesture to keep your input drive voltages within specified bounds, it's OK to go beyond if you have some series impedance to limit the current, as we've shown in Figure 12.21.

C. Unused inputs

Unused inputs that affect the logic state of a chip (e.g., a RESET input of a flip-flop) must of course be tied HIGH or LOW, as appropriate. Perhaps less obvious, even inputs that have no effect (e.g., inputs of unused gate sections in the same package) should be tied HIGH or LOW (your choice), because an open input to a CMOS device can float up to the logic threshold, causing the output to go to half the supply voltage, with both MOS output transistors conducting, thus drawing considerable class-A current. This can result in excessive supply current, and can even lead to failure in devices with hefty output stages. It can also cause oscillation.

[11] However, be aware that bipolar-TTL devices are not so friendly: their inputs source significant current (up to a milliamp) in the LOW state, and sink a small (but nonzero) current (up to a few tens of microamps) in the HIGH state. Because of this asymmetry, external digital signals used as inputs will therefore nearly always have a pullup resistor and will pull LOW (sinking current) when active, a convenient arrangement because switches, etc., can use a common ground return. It also leads to greater noise immunity, since a line held near the bipolar V_+ of $+5$ V has 3 V of noise immunity, as compared with the $\sim$0.8 V of noise immunity of a line held near ground.

12.1.7 Driving digital logic from comparators or op-amps

Comparators (and sometimes op-amps), together with analog-to-digital converters, are the common input devices by which analog signals can interface into digital circuits (recall §4.3.2). If your circuit has a microcontroller ("μC," Chapter 15), then you can take advantage of built-in ADCs or comparators, which are common features of most μCs. But you'll sometimes want to go from a comparator (or op-amp) output directly into digital logic. This is not terribly difficult – but you've got to respect the allowable voltage input range of the driven logic. Let's look at some examples (for which it may be helpful to look ahead to Table 12.1 on page 812 and Table 12.2 on page 813, to keep score in the detailed discussion that follows).

A. Comparator driving logic

Figure 12.25 shows some common ways to connect comparator outputs to logic (and we'll have lots more to say about comparator interfacing in §12.3). The ever-popular and seriously inexpensive LM311 (and improved versions, like the LT1011) has a flexible open-collector output stage with a "ground" pin that sets the LOW state (which can be anywhere between V_+ and V_-); the pullup resistor, of course, sets the HIGH state, as shown. These are designated as output type "FL" in the tables. Some comparators, like the AD790, use internal active pullup, but you get to set the HIGH output state at a logic voltage pin V_{L}. Many high-speed comparators that operate from lower-voltage supplies simply use the V_+ voltage for their output HIGH, while still providing a "ground" pin – the LT1016 (shown) is an example; we like it because its design makes it particularly resistant to multiple transitions and oscillations. Note that this part has "TTL" output levels, namely ground and approximately +3.5 V (though guaranteed only to be $\geq$2.4 V and $\leq$5 V), and so it cannot be connected to 5 V-intolerant logic. Next is a large class of low-voltage "single-supply" comparators (marked "CM to neg rail" in the tables; e.g., the TLC3702) that simply swing their outputs between ground and V_+. Finally, single-supply comparators that can run from higher voltages and that are not going for break-neck speed (e.g., the classic LM393) are usually configured with open-collector (or open-drain) outputs, with external

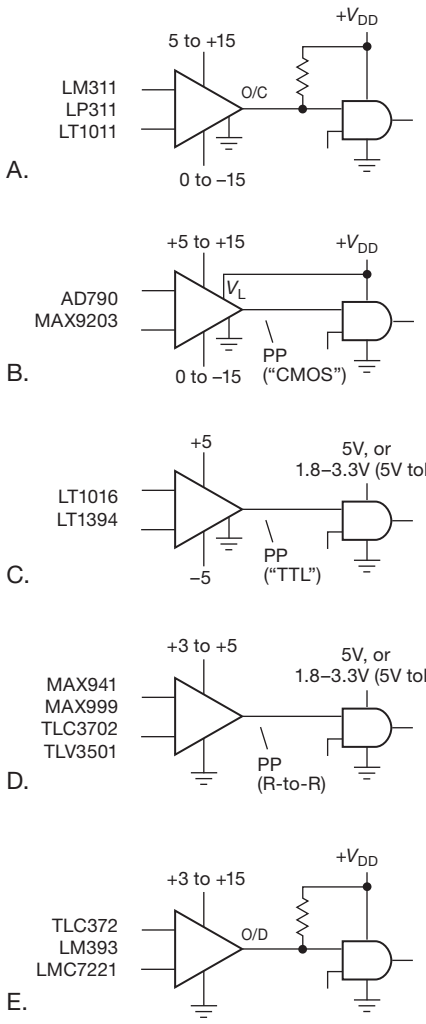


Figure 12.25. Driving digital logic from comparators.

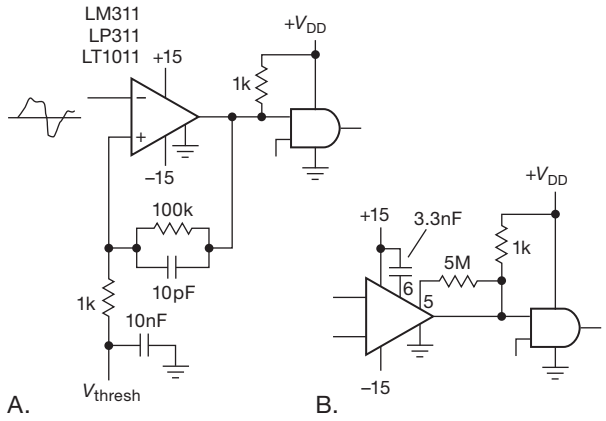


Figure 12.26. Threshold detector with hysteresis. A. Conventional circuit, for use with any comparator. B. Alternative method for the 311-type.

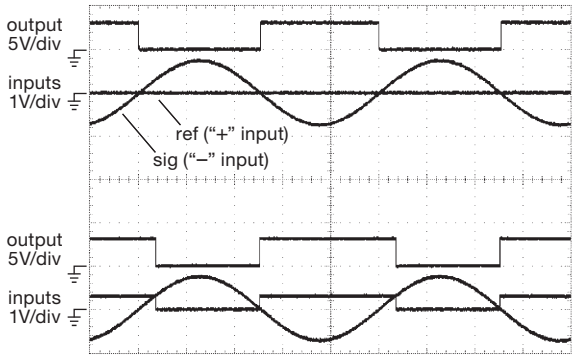


Figure 12.27. LM311 comparators driven with 1.5 V_{pp} 1 kHz sine wave, with output pullup to +3.3 V. Top: no hysteresis; bottom: 10% hysteresis (100k feedback, 11k to ground). Horizontal: 200 μs/div.

pullup resistor as shown; these are marked as output type "OC" or "OD" in the tables.

In Figure 12.25 we've ignored circuit configuration niceties like hysteresis, which we visited back in §4.3.2B. This is worth a reminder: Figure 12.26 shows the classic Schmitt-trigger threshold detector, configured with an amount of hysteresis equal to 1% of V_{DD}, and a small speed-up capacitor. In this circuit V_{thresh} must be provided from a low source impedance ($\ll$1k). That can be a serious disadvantage (although for slow or static V_{thresh} you can always use an op-amp buffer), in which case the alternative circuit shown may be just what you want. It takes advantage of the LM311's offset trim input terminals (pins 5 and 6) to produce both hysteresis (via the 5M resistor) and speed-up (via the 3.3 nF capacitor).[12] Figure 12.27 shows measured waveforms for an LM311 configured without and with 10% hysteresis. Note, in the latter case, the asymmetrical trigger points (and output waveform timing). Although it's not visible in the captured waveforms, you are asking for trouble when you send slow input waveforms to a comparator without hysteresis (trust us, we've been there!).

Finally, Figure 12.28 shows in detail the effect of a small speed-up capacitor in the feedback path of the Schmitt trigger. Without it, the positive feedback to the noninverting input is slowed by input and wiring capacitance, so an input signal with some low-level fast noise may produce multiple

[12] This trick, and variations on it, are nicely described in the LT1011 datasheet; there's some discussion, also, in National's extensive LM311 datasheet.

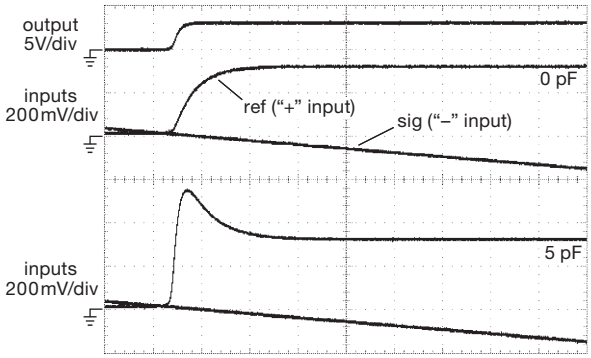


Figure 12.28. A small speed-up capacitor (bottom trace pair) compensates for the slowdown (middle trace pair) caused by input capacitance in the Schmitt trigger. Same circuit as Figure 12.27, but with expanded horizontal and vertical scales (to show detail of switching), and with 10 kHz sine wave input (so you can see the input signal's slope). Horizontal: 400 ns/div.

output transitions while the positive feedback is lollygagging along. The addition of very little feedback capacitance (here just 5 pF) fixes the problem (bottom trace pair). Don't overdo it, though – a large feedback capacitor generates a large hysteretic overshoot, with an undesirably long recovery time.

B. Op-amp driving logic

Op-amp driving logic – *what in heaven's name are you talking about?!* Well, sometimes you've used an op-amp section as a comparator, for example as a "battery-low" detector, or whatever. So its output swings between the rails, or nearly so (if it does not have a rail-to-rail output stage). All you want to do is get that output state into some digital logic. As with comparators, the only task is to make sure you respect the logic's input-voltage rules.

Figure 12.29 shows a few common situations. If the op-amp is running on a single low-voltage supply (in which case it likely has a rail-to-rail output stage) you can just connect it directly to logic that runs on that same V_+ voltage, or that is input tolerant to it. In the figure example, that could be 5 V logic, or 5 V-tolerant logic running on a lower supply (e.g., LV, LVC, or LVX: see Figure 12.3). If the op-amp has larger output swings, or swings both polarities, you need to limit it to logic bounds. One way is to interpose an nMOS inverter, as shown; alternatively you can use a passive clamp to the logic rail, combined with the logic's input protection negative clamp diode, as shown. We're not wild about this method, since it takes three components (and degrades the switching speed as well), but it does work. Whatever you do, be aware that the (analog and

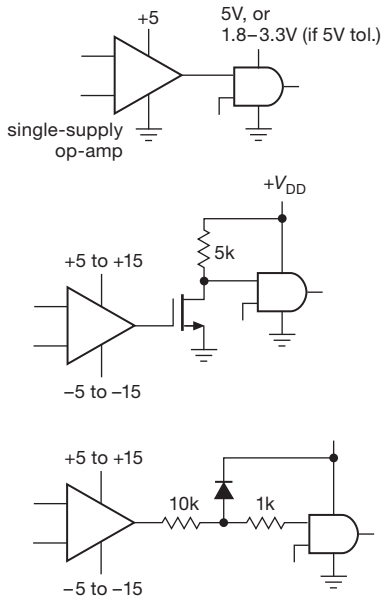


Figure 12.29. Driving digital logic from an op-amp's output.

slew-rate-limited) op-amp output transitions will be much slower than normal digital-logic transitions. That is, don't expect clean logic switching; these interfaces are merely a way to get the op-amp's state into some digital logic.

C. Clock inputs: hysteresis

A general comment about driving digital logic from op-amps: don't try to drive *clock* inputs from these op-amp interfaces – the transition times are too long, and you may get glitches as the input signal passes through the logic threshold voltage. If you intend to drive clocking inputs (of flip-flops, shift registers, counters, monostables, etc.), it is best to use a comparator with hysteresis (e.g., Figure 12.26), or to buffer the input with a gate (or other logic device) with Schmitt-trigger input (e.g., a '14 Schmitt-trigger inverter). The same comment goes for signals derived from transistor analog circuitry.

12.2 An aside: probing digital signals

We've had envious queries ("How do you guys get such nice traces? How do you get rid of all the wiggles?") about the clean digital 'scope waveforms in this book:[13] It's pretty easy, once you realize that you can't get away with the same lazy techniques that work OK at low frequencies.

[13] Take a look at some examples from this chapter, for example Figure 12.18 (slow) or 12.108 (fast).

In particular, the fast edges on digital signals produce probing artifacts (ringing, and soft edges) when you try to use a standard 10× passive 'scope probe with its 6″ grounding lead.[14] Figure 12.30 shows how the same logic waveform looks when probed three different ways.

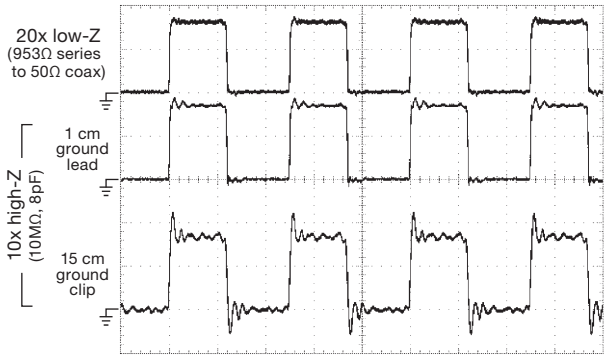


Figure 12.30. Logic waveform at 10 MHz from a 74AC14 inverter running at 3.3 V, as seen on a Tektronix TDS3044B 'scope when probed three different ways. Vertical: 2 V/div; horizontal: 40 ns/div.

This is a 10 MHz clock signal at the output of a 74AC14 inverter, wired up on a "solderless prototyping" board[15] (a dubious practice; but, hey, it works, most of the time anyway). We took the precaution of using an IC socket with integral SMT bypass capacitor, to put the best face on it. The bottom trace is business-as-usual, with a P6139A (500 MHz) 10× passive probe and 6″ ground lead. It's overshooting and ringing like crazy – can that be real, or is it an artifact of the probe's inductive ground path? You do considerably better by tossing the ground lead in the trash, removing the plastic sleeves, and using a little springy "ground tip contact." The middle trace shows the result: aha! Most of the ringing is gone. Better still, dump the 10× passive probe altogether (they don't come in speeds faster than 500 MHz anyway) and make your own by hooking a series resistor (we like 950 Ω) onto a length of skinny 50 Ω coax (we like RG-178); you temporarily solder the coax shield to a nearby ground, plug the other end into the 'scope (set for 50 Ω input) and voilà – a high-speed 20× probe![16]

[14] *Leads?* Yeah, sure. I'll just check with the boys down at the crime lab… they've got four more detectives working on the case… they got us working in *shifts!*

[15] For example, Global Specialties type UBS-100, or 3M type 923252.

[16] You can get these as a commercial product, the Keysight (Agilent) 54006A "6 GHz Passive Divider Probe Kit." It includes resistors for 10:1 and 20:1, and has a probe-tip capacitance of just 0.25 pF (as little as 0.1 pF, if you trim the tip). You can create greater ratios (thus higher loading resistance) by substituting other values of Caddock MG710-type resistors.

The top trace is about as clean as you can do, especially with through-hole (14-pin DIP) parts on a breadboard.

The homemade 50 Ω trick has the advantage of low cost, so you can do four traces easily; we used it for nearly all the digital 'scoping in this book. But because of its low input resistance, it is not useful for generalized circuit probing.

What happens with the more common configuration of surface-mount components on a printed-circuit board, and with much faster logic signals? Figure 12.31 shows what you see with four probing methods looking at the output of a 74AUC1G04 inverter, this time at 4 ns/div (and Figure 12.32 shows what the probes themselves look like).

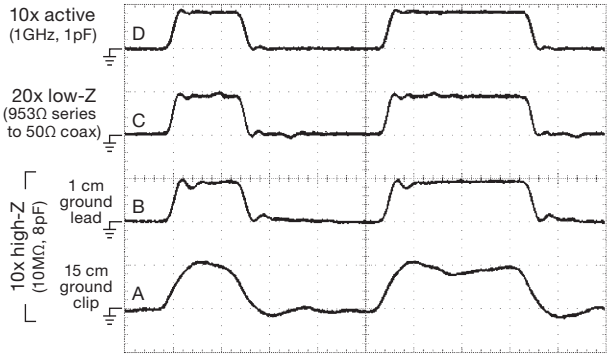


Figure 12.31. Logic waveform at a 6 ns clock rate from a 74AUC1G04 inverter running at 1.8 V, as seen on a Tektronix TDS3044B 'scope, when probed four different ways. The A–D labels match those in Figure 12.32. Vertical: 2 V/div; horizontal: 4 ns/div.

Here the passive-probe-with-ground-lead produces a mushy waveform (bottom trace) with moderate overshoot, considerably helped by using the shorter ground tip (next trace). The el-cheapo 20× trick looks better still. But best of all (if you've got the dough), is an "active probe" (an FET follower), with typical input capacitance less than 1 pF, and with speeds up into the high gigahertz (and prices to match). The top trace was made with a P6243 active probe (1 GHz bandwidth, less than – but not much less than – 1 kilodollar[17]).

12.3 Comparators

Comparators provide an important interface between analog (linear) input signals and the digital world, as we noted earlier in this chapter (§12.1.7A). In this section we would like to look at comparators in some detail, with emphasis on their output properties, their flexibility regarding power-supply voltages, and the care and feeding of input stages.

[17] But ∼8 dB less expensive on used auctions.

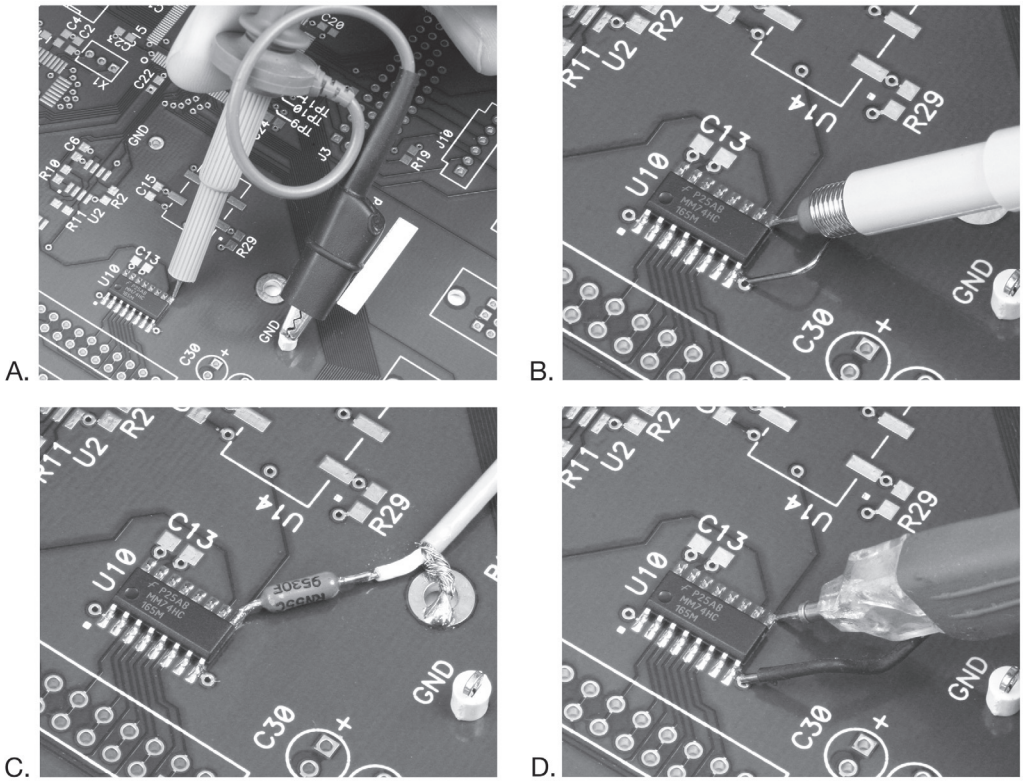


Figure 12.32. Probing digital signals: A. conventional 10× passive probe (Tektronix P6139A) with 6″ ground lead; B. short ground tip contact (Tek 016-1077-00) on 10× passive probe. C. Simple 20× passive "probe" for 50 Ω 'scope input: 953 Ω series resistor to coax; D. active probe (Tek P6243) with short ground tip. When probing fine-pitch ICs, it's a good idea to use a plastic guide (e.g., Tek "SureFoot" adapters) over the probe tip (not shown), so you don't short out adjacent contacts. The IC here is an SOIC-16, with 1.25 mm contact spacing.

Comparators were introduced briefly in §4.3.2A to illustrate the use of positive feedback (Schmitt trigger) and to show that special-purpose comparator ICs deliver considerably better performance than general-purpose op-amps used as comparators. These improvements (short delay times, high output slew rate, and relative immunity to large overdrive) come at the expense of the properties that make op-amps useful (in particular, careful control of phase shift versus frequency). Comparators are not frequency-compensated (§4.9) and cannot be used as linear amplifiers.

12.3.1 Outputs

We're used to op-amps, where the output can swing rail-to-rail (or nearly so), but where we usually stay in the linear region, deliberately avoiding saturation at the extremes of output swing. When the output is saturated, we're in trouble!

But comparators are different. Although the inputs are analog, the output is digital: it *lives* at the extremes. So what you care about is what the output does when LOW and when HIGH. As we've seen, the output may drive digital logic directly (Figure 12.25), in which case we need to bound its swing to that of the driven logic. Or we may want to drive an ON/OFF load, for example a relay (mechanical, or solid-state) or a bright LED, requiring plenty of output current, and perhaps powered from an external dc supply.

A. Output swing

Figure 12.33 shows the choices you've got to meet these various demands. In each case the comparator's analog circuitry is powered from a pair of supplies, V_+ and V_- (though for "single-supply" comparators, analogous to single-supply op-amps, the negative supply voltage V_- is ground). We'll have more to say soon about the input stage. What's interesting here are the output stages: one variety of comparators simply swing their output from rail to rail

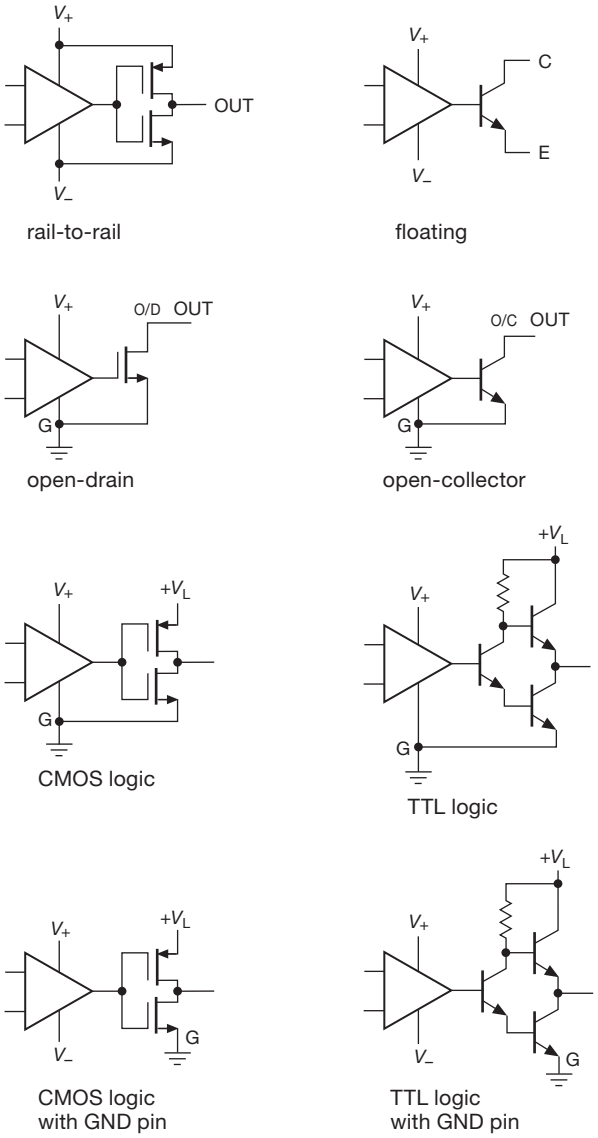


Figure 12.33. Don't let comparators confuse you: use this chart of simplified output stages, along with the listing in Tables 12.1 and 12.2.

("RR" in Tables 12.1 and 12.2), which is fine if that works for your application (e.g., if V_+ is +5 V, V_- is ground, and the output drives 5 V logic, or 5 V-tolerant logic; see Figure 12.25D). And the active pullup has the advantage of speed. But you may want to accommodate inputs that swing both sides of ground ("bipolarity") while driving digital logic (with its single positive supply), in which case you need to have V_- below ground; and so the right choice is either a comparator with floating output ("FL" in the tables) with resistive pullup (as in Figure 12.25A), or a comparator with separate GND and V_L (logic voltage) pins ("RR-G" and "TTL" in the tables), as shown in Figure 12.25B.

Finally, if you're interested in single-supply operation, with input signals only between ground and a positive supply V_+, you can use (a) logic-output types with active pullup to a logic voltage V_L ("RR-G" or "TTL" in the tables); or (b) if V_+ is a low logic voltage already, you can use a rail-to-rail comparator with V_- connected to ground (as mentioned above); or (c) you can use a comparator with open-drain or open-collector output ("OD" or "OC" in the tables), with pullup to a logic supply (as shown in Figure 12.25E). The open-collector style is nice for driving power loads, or loads connected to a high output voltage (e.g., the classic LM311 can sink up to 100 mA, and its output can be pulled up to +40 V); but passive pullup is slow (compared with active pullup), so you're better off with a logic-output type when driving digital logic, unless you don't care about speed.

B. Output current

Comparators vary widely in their output-current capability. When driving digital logic this doesn't much matter, but it's important when driving high current loads such as relays or LEDs. Figure 12.34 shows how it goes, for most of the open-collector and open-drain comparators in Table 12.1. You can see some interesting trends here: (a) comparators with MOSFET outputs (e.g., the TLC393) behave resistively (R_{on}) at low voltages, trending downward

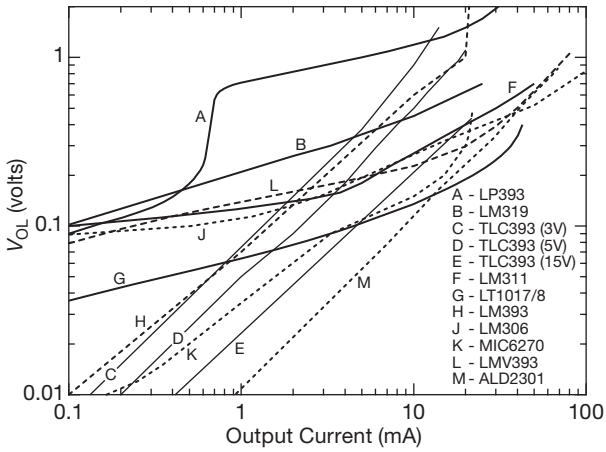


Figure 12.34. Saturated LOW output voltage versus sinking current, for a selection of open-collector and open-drain comparators. The popular TLC372-series is similar to the TLC393. Data are compiled from datasheet graphs, except for curve L and the low-current end of curves B and J, which were measured.

Table 12.1 Representative Comparators[a]

Type	t_d typ (ns)	V_{os} max (mV)	Input current typ (nA)	CM to neg rail?	CM to pos rail?	Supply voltage V_+ max (V)	Supply voltage V_- max (V)	Supply voltage Total min (V)	Supply voltage Total max (V)	Supply curr (mA)	Output type[b]	Comments
LM393	600	5	25	•	-	36	-	2	36	0.4	OC	jellybean dual
TLC372	650	5	0.005	•	-	18	-	3	18	0.15	OD	cmos dual; quad=374
TLC3702	2500	5	0.005	•	-	16	-	3	16	0.02	RR	cmos dual; quad=3704
LM311	200	3	60	-	-	30	−30	4.5	36	5	FL	strobe, popular; dual=2311
LT1016	10	3	5000	-	-	7	−7	5	14	25	TTL	fast, stable
AD8561	7	7	3000	•		7	−7	3.5	14	5.6	TTL	improved 1016; quad=8564
TLV3501	4.5	6.5	0.002	•	•	5.5	-	2.7	5.5	3.2	RR	fast; dual=3502

Notes: (a) See also the more extensive listing in Table 12.2. (b) See "Comparators, Outputs" in text; FL = floating *npn* output, collector and emitter output pins; OC = open collector; OD = open drain; RR = rail-to-rail; TTL = logic swing, separate V_L pin.

to the left with unit slope; (b) comparators with bipolar *npn* output stages (e.g., the LT1017/8) tend toward a finite saturation voltage; and then there is (c) the LP393, which has a bipolar-output stage that behaves like a Darlington at higher currents (with saturation around a V_{BE}, or ∼0.6 V), but becomes a simple grounded-emitter switch at low currents, which explains the weirdly gyrating curve.

The LM311's curve is worth an additional comment. The designers used a cute "antisaturation" circuit in the output stage (Figure 12.35): the series-connected V_{BE} drops of Q_{13} and Q_{14} are poised to rob base current from driver Q_{12} if Q_{15}'s collector gets too close to its emitter. If R_{11} were zero, this would happen just as Q_{15} saturates (figure out why!). But adding R_{11} causes this clamping action to occur when there's still about 20% of a V_{BE} across Q_{15}. This prevents deep saturation, which has two benefits: (a) it eliminates excessive turn-off delay caused by stored base charge in the output transistor; and (b) it reduces power consumption, by providing no more base drive to Q_{15} than is needed to bring it nearly to saturation, for whatever load it is driving.

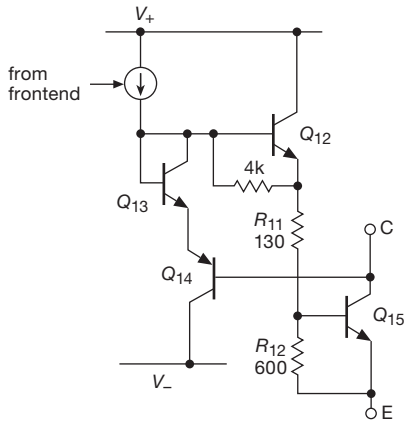


Figure 12.35. The output stage of the LM311 comparator incorporates a clever antisaturation circuit that limits base drive current when the output transistor is near saturation (V_{CE}≈100 mV). We've omitted here the output-stage current-limit circuit. Courtesy of Texas Instruments.

12.3.2 Inputs

A. Input common-mode range

Just as with op-amps, you've got to keep input voltages within the operating common-mode range. Comparators designed for low-voltage single-supply operation (in the range of 3–5 V; see Table 12.1) allow inputs to go to ground (or even a few tenths of a volt below), and some also work to the positive rail ("rail-to-rail" inputs). Examples are the TLC372/3702 and LMC7221/7211, respectively (with OD and RR versions listed in each pair). But these won't work with signals of both polarities (unless of course you tie the "ground" pin to a negative supply voltage, in which case you may be unhappy with an output that swings down to the negative supply). Instead you must use a dual-supply comparator (e.g., LM311, LT1016), many of which do not operate with inputs at or near the rails. Tables 12.1 and 12.2 include the common-mode operating range for the listed devices.

B. Offset voltage and trim

As with op-amps, garden-variety comparators have offset voltages in the millivolt range. You can do better with one of the "precision" comparators, from companies

Table 12.2 Comparators[a]

Type	Qty per pkg[h]	t_d typ (ns)	V_{os} max (mV)	Input current max (µA)	CM to neg rail?	CM to pos rail?	Cost qty25 ($US)	V_{diff} max (V)	Power supplies: Positive min (V)	Positive max (V)	Negative min (V)	Negative max (V)	Total min (V)	Total max (V)	Supply current[e] (mA)	Gain typ	Hyst (mV)	Bipolar sig	Offset trim	Q & Q* out	Shutdown	Latch?	V_{OH} (max, V)	Output type[g]	Packages: DIP	SOIC	SOT-23	Smaller	Comments
TLV3701	1,2,4	37000	5	250pA	•	•	1.39	rails	2.5	16	-	-	2.5	16	0.6µA	1000k	-	-	-	-	-	-	-	RR	14	8	5	-	TLV3441 for OD
LT1017	2	20000	1	0.015	-	-	2.92	40	1.1	40	-	-	1.1	40	0.03	500k	-	-	-	-	-	-	40	OC	8	8	-	-	micropower
LP339	4	10000	5	0.025	•	-	0.45	36	2	36	-	-	2	36	0.06	500k	-	-	-	-	-	-	30	OC	14	14	-	14	low-power '339
TLV3491	1,2,4	6500	15	10pA	•	•	0.97	rails	1.8	5.5	-	-	1.8	5.5	0.9µA		-	-	-	-	-	-	-	RR	8	14	5	8	fastest nanopower
LT1018	2	6000	1	0.075	-	-	2.79	40	1.1	40	-	-	1.1	40	0.11	2M	-	-	-	-	-	-	40	OC	8	8	-	-	low power
LM393	2	600	5	0.25	•	-	0.36	36	2	36	-	-	2	36	0.4	200k	-	-	-	-	-	-	30	OC	8	8	-	8	jellybean dual
LM339	4	600	5	0.25	•	-	0.36	36	2	36	-	-	2	36	0.8	200k	-	-	-	-	-	-	30	OC	14	14	-	14	jellybean quad
MIC6270	1	600	5	0.25	•	-	0.69	36	2	36	-	-	2	36	0.3	200k	-	-	-	-	-	-	36	OC	-	-	5	-	low cost
LM311	1,2	200	3	0.1	-	-	0.77	30	5	30	0	-30	4.5	36	5	200k	-	•	•	-	-	-	40	FL	8	8	-	-	strobe, popular; dual=2311
LP311	1	2000	7.5	0.1	-	-	0.95	30	3	30	0	-30	3	36	0.15	200k	-	•	•	-	-	-	40	FL	8	8	-	-	strobe, low-power 311
LT1011	1	150	0.5	0.025	-	-	2.13	36	3	36	0	-36	3	36	3.0	500k	-	•	•	-	-	-	50	FL	8	8	-	-	strobe, improved '311
LM319	2	80	4	0.5	-	-	0.99	5[b]	5	30	0	-30	4.5	36	6	40k	-	•	-	-	-	-	40	FL-G	14	14	-	-	
LM306	1	28	6.5	5	-	-	1.04	5[b]	10	15	-3	-15	-	30	6.6	40k	-	•	-	-	-	-	24	TTL,OC	8	8	-	-	strobe, 0.1A current sink
AD790	1	45	0.25	5	•	-	6.61	16.5	4.5	15	0	-15	4.5	36	10		0.4	•	-	•	-	•	-	RR-G	8	8	-	-	
LM361	1	14	5	30	-	-	2.21	5[b]	5	15	-6	-15	11	30	4.5	3k	-	•	-	•	-	-	7	TTL-G	14	14	-	-	strobe, aging, fastest 30V
LMC7221	1,2	4000	5	40fA[t]	•	•	1.15	rails	2.7	15	-	-	2.5	16	7µA	100k	-	-	-	-	-	-	16	OD	-	8	5	-	low pwr; RR=7211, dual=6772
TLC339	4,2	2500	5	5pA[t]	•	-	1.02	18	3	16	-	-	3	16	0.04		-	-	-	-	-	-	18	OD	14	14	-	14	cmos 339; dual=393
TLC3702	2,4	2500	5	5pA[t]	•	-	0.72	18	3	16	-	-	3	16	0.20		-	-	-	-	-	-	18	RR	8	8	-	8	cmos; quad=3704
TLC372	2,4	650	5	5pA[t]	•	-	0.67	18	3	18	-	-	3	18	0.15	200k	-	-	-	-	-	-	18	OD	8	8	-	8	cmos; quad=374
TLC352	2,4	650	5	5pA[t]	•	-	1.00	18	1.4	18	-	-	1.4	18	0.07		-	-	-	-	-	-	18	OD	8	8	-	-	low power, low voltage
ALD2301A	1	300	2	10pA[t]	•	-	4.04	rails	3	12	-	-	3	13.2	0.11	150k	-	-	-	-	-	-	13	OD	8	8	-	-	will sink 50mA (600mV)
TS861A	1,2,4	500	7	1pA[t]	•	•	1.19	rails	2.7	10	-	-	2.7	12	7µA		-	-	-	-	-	-	-	RR	-	8	5	8	µPwr; dual=862, quad=64
MAX9203	1	7	4	5	•	-	1.88	rails	4.75	12	0	-6	4.75	12	1.3		-	•	-	-	-	•	-	RR-G	-	8	8	-	Vdd pin; dual='02, quad='01
LT1016	1	10	3	10	-	-	4.08	5	4.5	7	0	-7	5	14	25	3k	-	•	-	•	-	•	-	TTL	8	8	-	-	fast, stable single +5
TL3016	1	7.6	3	10	-	-	1.76	5	4.5	7	0	-7	5	14	11		-	•	-	•	-	•	-	TTL	8	8	-	-	replace LT1016
LT1394	1	7	2.5	4.5	•	-	3.42	12	4.4	8	0	-8	4.4	12	6	1600	-	•	-	•	-	•	-	TTL	-	8	-	-	improved LT1016
AD8611	1	4	7	4	•	-	4.34	4	3	5.5	-	-3.5[c]	3	7	6		-	•	-	•	-	•	-	TTL	-	8	-	8	really fast 1016; dual=8612
MCP6541	1	4000	7	1pA[t]	•	•	0.40	rails	1.6	5.5	-	-	1.6	7	0.6µA		3.3	-	-	-	-	-	-	RR	8	8	5	-	µPwr; dual=42, quad=44
LMV339	4,2,1	400	7	0.25	•	-	0.94	5.5	2.7	5.5	-	-	2.7	5.5	0.14	50k	-	-	-	-	-	-	5.5	OC	-	8,14	5	8,14	dual=393, single=331
TLV3501	1,2	4.5	6.5	2pA[t]	•	•	2.97	rails	2.7	5.5	-	-	2.7	5.5	3.2		6	-	-	-	•	-	-	RR	-	8	6	-	fast; dual=3502
ADCMP371	1	2000	9	0.05	•	•	0.60	22	2.25	5.5	-	-	2	6	4µA		1	-	-	-	-	-	-	RR	-	-	-	5	lopwr; ref=356, OC(22V)=370
MAX941	1,2,4	80	2	0.3	•	•	2.60	rails[d]	2.7	6	-	-	2.5	6.5	0.43		•	-	-	-	•	•	-	RR	8	8	-	8	dual=942, quad=944
TS3021	1	38	6	0.16	•	•	0.92	5	1.8	5	-	-	1.8	5.5	0.07		-	-	-	-	-	-	-	RR	-	-	5	5	low voltage, low cost
LMV7219	1	7	6	0.95	•	-	2.06	rails	2.7	5.5	-	-	2.7	5.5	1.1		7.5	-	-	-	-	-	-	RR	-	-	5	5	low cost
MAX961	1	4.5	1.5	15	•	•	5.00	1.5[f]	2.7	5.5	-	-	2.7	6	7.2		3.5	-	-	•	•	•	-	RR	-	8	-	8	dual=963, quad=964
MAX999	1	4.5	1.5	15	•	•	5.64	1.5[f]	2.7	5.5	-	-	2.7	6	5		3.5	-	-	-	•	-	-	RR	-	-	5	-	shutdown=997
ADCMP600	1	3.5	5	5	•	•	3.09	rails	2.7	6	-	-	2.7	6	3.5	20k	2	-	-	-	-	-	-	RR	-	-	5	6,8	adj-hyst=601, shutdown=602
AD96685	1,2	2.5	2	10	-	-	4.77	5.5	4.8	5.2	-5	-6	10	13	15		-	•	-	•	-	•	-	ECL	-	16	-	-	dual=AD96687
ADCMP604	1	1.6	5	5	•	•	3.92	rails	2.5	5.5	-	-	2.5	5.5	15	2k	•	-	•	•	•	•	-	LVDS	-	-	-	6	hyst+shutdown+latch=605
ADCMP565	1	0.3	6	40	-	-	7.97	7	4.8	5.2	-5	-5.4	10	10.4	70	1k	adj	•	-	•	-	•	-	ECL	-	-	-	20	5GHz input bandwidth
ADCMP553	1	0.5	10	5	•	-	3.69	3	3.2	5.5	-	-	3	6	35	1k	-	-	-	•	-	•	-	PECL	-	-	-	8	dual=551,552, hyst=552
65LVDS34	2	4	-	20	•	•	1.84	rails	3	3.6	-	-	3	3.6	16		50	-	-	-	-	-	-	RR	-	8	-	-	LVDS rcvr; int term=LVDT34
65LVDS2	1	1.8	-	4	-	-	0.97	rails	2.4	3.6	-	-	2.4	3.6	5.5		•	-	-	-	-	-	-	RR	-	8	5	-	LVDS receiver
FIN1018	1	1.5	-	20	-	-	0.93	rails	2.4	3.6	-	-	2.4	3.6	5		•	-	-	-	-	-	-	RR	-	8	-	8	LVDS receiver
HMC874LC	1	0.1	5[t]	15[t]	-	-	50.00	1.8	3.3[n]		-3.3[n]		6.6[n]		50		1	•	-	•	-	•	-	PECL	-	-	-	16	clocked, 10GHz BW

Notes: (a) ordered by supply voltage groups, then by speed. (b) note V_{diff}(max) << Vsupply. (c) limited to 7V total analog supply voltage. (d) OK to rails, but 8.2kΩ for V_{diff} > 1V. (e) total, for listed part number. (f) back-to-back pairs of 2 clamp diodes, with 200Ω series resistor. (g) see "Comparators, Outputs" in text. (h) the first number corresponds to the listed part. (n) nominal. (t) typical.

like Analog Devices, Linear Technology, and Maxim; see Tables 12.1 and 12.2. A few comparators include external trim terminals; but, as with op-amps, an inexpensive trimmed comparator (e.g., an LM311) will have much larger tempco of V_{os} than an intrinsically accurate (i.e., precision) part. For example, the LT1011A "improved LM311" specifies a V_{os} tempco of $4\,\mu$V/°C, typ ($15\,\mu$V/°C max), whereas for the generic LM311 none of the half-dozen manufacturers even bothers to specify it. In applications where you care about accurate input thresholds, it's always best to avoid heavy loads at the comparator's output.

Here's an interesting riff on offset voltage: thermal gradients set up on the chip from dissipation in the output stage can degrade input offset voltage specifications. In particular, it is possible to have "motorboating" (a slow oscillation of the output state) take place for input signals near zero volts (differential), because the state-dependent heat generated at the output can cause the input to switch.

C. Input current

Here, also, familiarity with op-amps can lead to trouble if you assume (incorrectly) that the inputs present essentially infinite impedance, and draw no current. An important feature of comparator inputs is the bias current at the input terminals and the way it changes with differential input voltage. For a wake-up call, look at Figure 12.36, a plot of measured current at the two inputs of the ever-popular LM311.[18] *It isn't zero!*

What's going on here? Many comparators use bipolar transistors for their input stages, with input bias currents ranging from tens of nanoamps to tens of microamps. Because the input stage is just a high-gain differential amplifier, the bias current changes as the input signal takes the comparator through its threshold. In addition, internal protection circuitry may cause a larger change in bias current a few volts away from threshold.

To see how this works in detail, look at the LM311's input circuit, which we've drawn in simplified form in Figure 12.37. The input stage consists of current-biased *pnp* followers driving an *npn* differential amplifier; the followers have an impressive $\beta \approx 2000$, but even with that you've still got -35 nA of input current when the inputs are balanced. Of greater concern, the currents at the two inputs shift in opposite directions by roughly 10% when the inputs become unbalanced. This happens because the second-

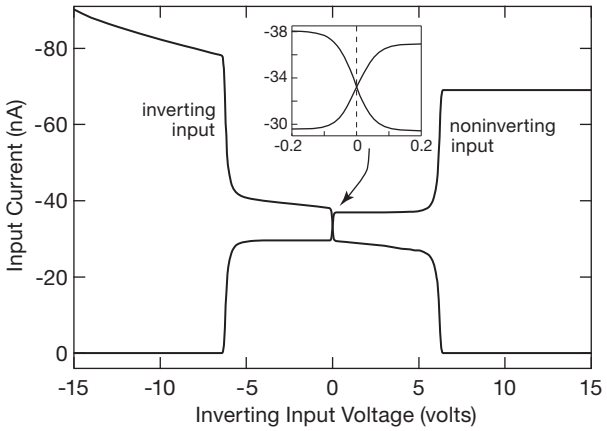


Figure 12.36. Measured input current of a specimen LM311N (NSC, with date code P134), with the noninverting input held at ground. The datasheet specifies an input current of 60 nA (typ), 100 nA (max).

stage differential amplifier transfers its operating current over to one side or the other, unbalancing its base current loading on the first stage. (The current "step" at zero volts differential is actually a smooth transition taking place over 100 mV or so, as seen in the expanded inset graph, and represents the voltage change necessary to switch the input differential amplifier stage fully from one state to the other.) So, unlike an op-amp (where feedback keeps the inputs balanced), a bipolar comparator's input current shifts at the input transition, which can cause trouble if the signal driving it is not of low source impedance.

For example, imagine you want to generate an output step when a slowly rising input signal (of finite source resistance) passes through zero volts. That's easy – you

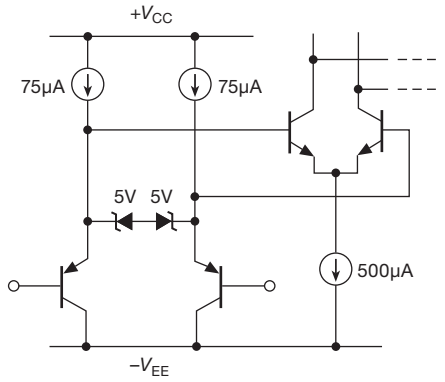


Figure 12.37. Simplified input stage of the classic LM311 bipolar analog comparator. Don't be alarmed by the 5 V zeners – the maximum allowed differential input is ±30 V.

[18] This is a true jellybean! A quick Internet search found at least a hundred 311 variants from a half-dozen manufacturers (Fairchild, NJR, National, ONsemi, ST, and TI), not even including "improved" 311s.

connect that signal to the inverting input, and you ground the noninverting input. Do this, and the output will likely exhibit fast multiple transitions as the input signal crosses through zero. The problem is that the drop in (negative) input current at zero volts causes the input voltage to reverse its rise, causing an extra transition; this continues several times, until the input signal has risen clear of the danger zone. Hysteresis (perhaps with a small speed-up capacitor) will generally cure this behavior, but it's helpful to understand its cause.

The graph in Figure 12.36 has some more surprises, namely the abrupt change in input currents when the differential input voltage reaches 6 V. This is caused by the symmetrical zener[19] clamp, included in the IC to prevent reverse base–emitter breakdown in the second-stage *npn* pair. Differential-input swings large enough to bring the clamp into conduction cause the *pnp* input transistor with the more negative input voltage to hog all the emitter current; so its base current doubles, and that of its twin drops to zero. The accurate graph of Figure 12.36 shows a feature you won't find in any of the official LM311 datasheets, namely the gradual increase in input current at large negative input voltages; this is evidently due to decreasing input-transistor beta at reduced V_{CE}. And a puzzle for the reader: why is this shape not mirrored in the noninverting input curve?

For comparator applications where extremely low input current is necessary, there are plenty of MOSFET-input comparators available, for example the TLC372, TLC3702, TLC393, and LMC7221. However, these are generally limited to a maximum total supply voltage of 16 V (compared with 36 V for "high-voltage" bipolar comparators); and, as with CMOS op-amps, they have poorer accuracy (V_{OS}) than precision bipolar comparators. In situations where the properties of a particular comparator are needed, but with lower input current, one solution is to add a matched-pair FET follower at the input.

D. Maximum differential-input voltage

Watch out for this one! Some comparators have a surprisingly limited differential input voltage range, as little as 5 V in some cases (e.g., the AD790, LM306, and LT1016), even though they can operate from a total supply voltage ($V_+ - V_-$) as high as 36 V. It may be necessary to use diode clamps to protect the inputs, because excessive differential input voltage will degrade beta, cause permanent input offset errors, and even destroy the base-emitter junctions of the input stage. General-purpose comparators that can operate from total supply voltages up to 36 V are generally better in this respect, with typical differential-input voltage ranges of ±30 V (e.g., the LM311, LM393, LT1011, etc.[20]).

E. Internal hysteresis

A little bit of hysteresis is usually a good thing. And some comparators (notably those intended for low-voltage single-supply operation) have a few millivolts of built-in hysteresis (see Tables 12.1 and 12.2). Some comparators (e.g., members of the ADCMP5xx and 6xx series from Analog Devices) permit adjustment of the amount of internal hysteresis.

12.3.3 Other parameters

A. Supply voltage

We've seen this already, because inputs have to stay in the "common-mode operating range," which at most extends slightly beyond the rails. Roughly speaking, there are three voltage ranges: (a) traditional bipolar comparators like the LM311 and LM393 can accept total supply voltages up to 36 V, and are now called "high-voltage" comparators; (b) a number of high-speed and CMOS comparators, for example the bipolar LT1016 and CMOS TLC/LMC parts, sit in a middle region, operating on total supply voltages up to 10–15 V; and (c) there has been a tremendous proliferation of "low-voltage" single-supply CMOS comparators like the LMV, TLV, and ADCMP600 series, which operate only to 6 V total supply. In the latter category there are some wickedly fast comparators (ADCMP572: 0.15 ns), and some micropower comparators (MCP6541: 0.6 μA typ; ISL28197: 0.8 μA typ). And there's pretty much everything in between.

B. Speed

It is convenient to think of a comparator as an ideal switching circuit for which any reversal in the differential-input voltage, however small, results in a sudden change at the output. In reality, a comparator behaves like an amplifier for small input signals, and the switching performance depends on the gain properties at high frequencies. As a result, a smaller input "overdrive" (i.e., more than enough signal to cause saturation at dc) causes a greater propagation delay and (often) a slower rise or fall time at the output. Comparator specifications usually include a graph of

[19] Implemented as a pair of back-to-back diode-connected transistors.

[20] These use integrated *pnp* input transistors, which tend to have high reverse emitter-base breakdown voltages, often more than 36 V (compared with typical *npn* breakdown around 6 V).

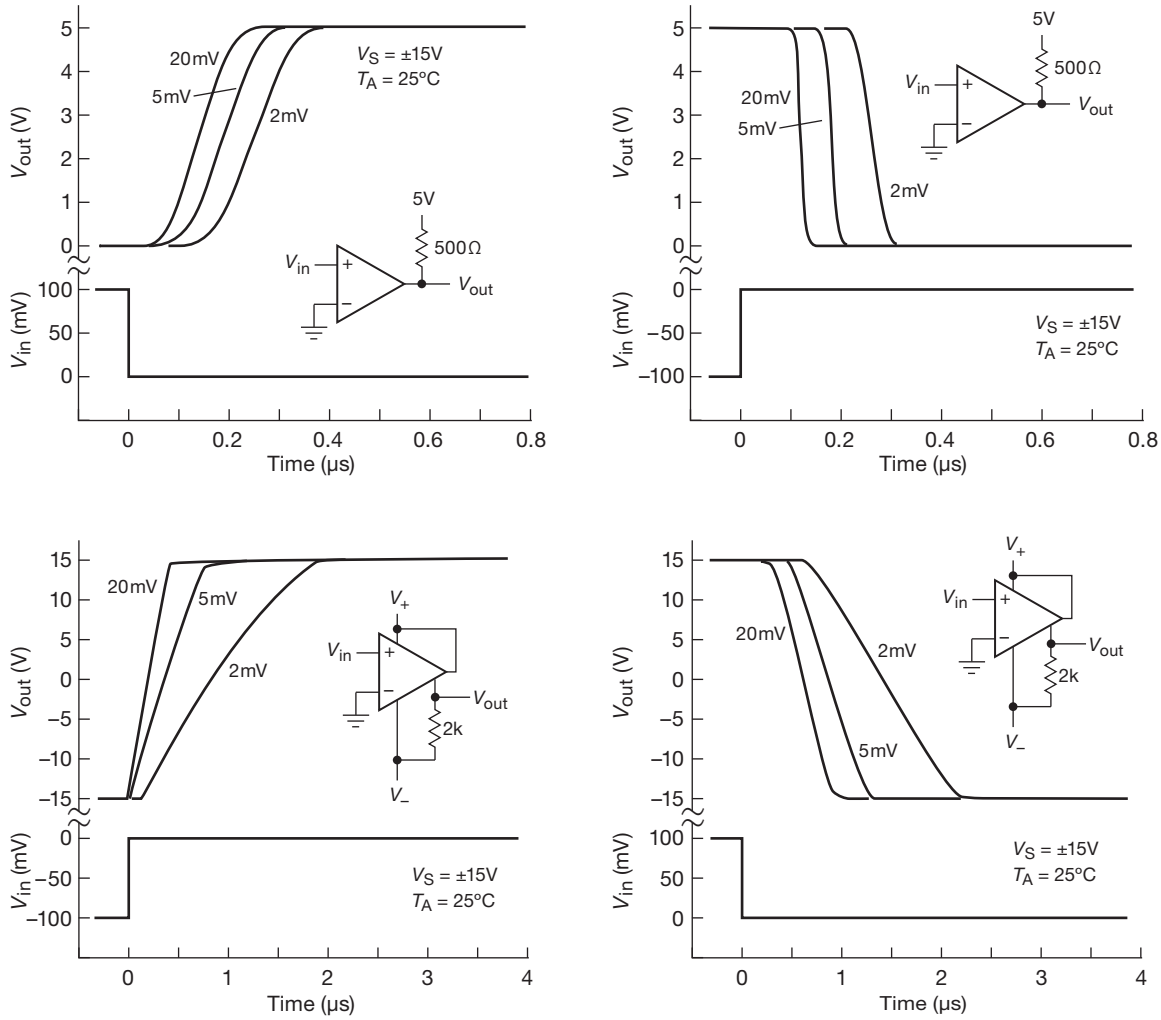


Figure 12.38. LM311 comparator response times for various input overdrives. Most comparators require considerable overdrive, 20 mV or more, for fast response. (Adapted from National Semiconductor Corp.)

"response times for various input overdrives." Figure 12.38 shows some for the LM311. Note particularly the reduced performance in the configuration in which the output transistor is used as a follower, i.e., with less gain. Increased input drive speeds things up because the amplifier's reduced gain at high frequencies is overcome by a larger signal. In addition, larger internal amplifier currents cause internal capacitances to charge faster.

The comparators in Tables 12.1 and 12.2 span a range of speeds (response time) from 0.3 ns to 300 μs – a ratio of a million to one! Haste makes waste – you pay the price in power dissipation (though low-voltage parts do much better), and in susceptibility to oscillations.

12.3.4 Other cautions

There are some general cautions concerning input circuits of comparators. Hysteresis (§4.3.2A) should be used whenever possible, because erratic switching is otherwise likely to result. To see why, imagine a comparator without hysteresis in which the differential-input voltage has just passed through zero volts, slewing relatively slowly, since it is an analog waveform. A mere 2 mV input differential causes the output to change state, with switching times of 50 ns or less. Suddenly you have 3000 mV fast digital-logic transitions in your system, with current pulses impressed on the power supplies, etc. It would be a miracle if some of these fast waveforms didn't couple into the input signal, at

least to the extent of a few millivolts, overcoming the 2 mV input differential and thus causing multiple transitions and oscillations. This is why generous amounts of hysteresis (including a small capacitor across the feedback resistor), combined with careful layout and bypassing, are generally required for making sensitive comparator circuits function well. It is usually a good idea to avoid driving comparator inputs directly from high-impedance signals; use an op-amp output instead. It is also a good idea to avoid high-speed comparators, which only aggravate these problems, if speed is not needed. Then, too, some comparators are more troublesome in this regard than others; we have had plenty of headaches using the otherwise admirable LM311.

12.4 Driving external digital loads from logic levels

It isn't hard to use a logic-level output signal (coming from something as simple as a gate or a flip-flop, or from a fancier device like an FPGA or microcontroller) to control on/off devices like lamps (LEDs), relays, displays, and even ac loads. In some situations you can drive such loads directly from the logic-level signal; but more often you've got to add a few components to make it work. A classic example of the latter might be switching a load that returns to a *negative* supply voltage.

12.4.1 Positive loads: direct drive

Loads that don't require too much current and that return to a low-voltage positive supply can often be driven directly from a logic output. Figure 12.39 shows some methods. Circuit A shows the standard method of driving LED indicator lamps from logic running from a 3–5 V supply. You choose the current limiting resistor to set the LED current: LEDs behave like a diode with a forward drop of 1.5–3.5 V (depending on the semiconductor material and emitted color; see Figure 2.8). Contemporary high-efficiency LEDs look pretty bright at just a few milliamps, which is easy work for all logic-family outputs (as well as more complex digital chips such as FPGAs and microcontrollers); so the logic output stays valid even while driving the LED load (see the listed V_{OL} values). One caution here: the forward drop of ∼3.5 V of GaN-based LEDs (blue, white, and "bright green") requires 5 V logic, whereas lower-voltage LEDs can be driven by 3.3 V or 5 V logic.

For historical reasons connected with the highly asymmetrical output properties of earlier bipolar and nMOS logic families, designers tend to prefer the current *sinking* connection of Figure 12.39A; but for contemporary CMOS logic families it's OK to connect an LED in the sourcing configuration of Figure 12.39B. The output sourcing is somewhat less muscular than sinking (see the listed V_{OH}), but it's plenty good enough for the job.

Some panel-mount LEDs come with built-in current limiting resistors, intended for direct 5 V voltage drive. This saves a resistor, but the selection is limited, and you may not be happy with the manufacturer's choice of operating current (e.g., 10–12 mA for the CML 5100H-LC-series, or the Dialight 558-series).

You can drive small mechanical relays in a similar manner, providing their coil operating voltage is low (there are plenty of 5 V dc units, from Coto, Omron, Panasonic, Tyco-P&B, and other suppliers; and you can get relays that operate on as little as 1.5 V, for example the TXS2 series from Panasonic), and their operating current is low enough (i.e., high coil resistance). Figures 12.39C–F show several examples. "Signal" relays like the TXS2 series are intended for switching low voltage and current, and they have gold-plated contacts for "dry switching" (see §*1x.6*); their coil current of 10–20 mA can be sunk by the logic families shown (and others as well). You can also get logic-drivable relays that can handle power switching, for example the Omron G5- and G6-series 5 V-coil relays shown in the figure. These can handle up to 5 A when switching 115 Vac (or even 240 Vac) power.

For relays (and other loads) that require somewhat higher drive voltages or currents, you can use logic devices with open-collector outputs intended for that kind of job (Figures 12.39G,H). The venerable 74LS07 is an OC hex inverter good for output swings to +30 V and (LOW-state, sinking) load currents to 40 mA. The equally crusty (and wildly popular) ULN2003 is a 7-section ("heptal"?) grounded-emitter Darlington with input resistors (i.e., a logic inverter with open-collector) that can swing to +50 V and sink up to 350 mA; its equally muscular close cousin (the 75468) can swing to +100 V.[21] And if you want to drive these kinds of loads with a comparator, the vintage LM311 or LM306 can handle those kinds of currents, though the open-collector output swing is limited to 40 V above the negative supply, and +24 V above ground, respectively.

When driving relays and other power loads from a microcontroller (Chapter 15), it's worth knowing about a class of serial-input power registers. These are elaborations of the '595 8-bit serial-in parallel-out logic shift register, but

[21] On a more modest scale you can use the 75451–4 dual open-collector gates (AND, NAND, OR, and NOR, respectively) in 8-pin packages for loads to 30 V and 300 mA.

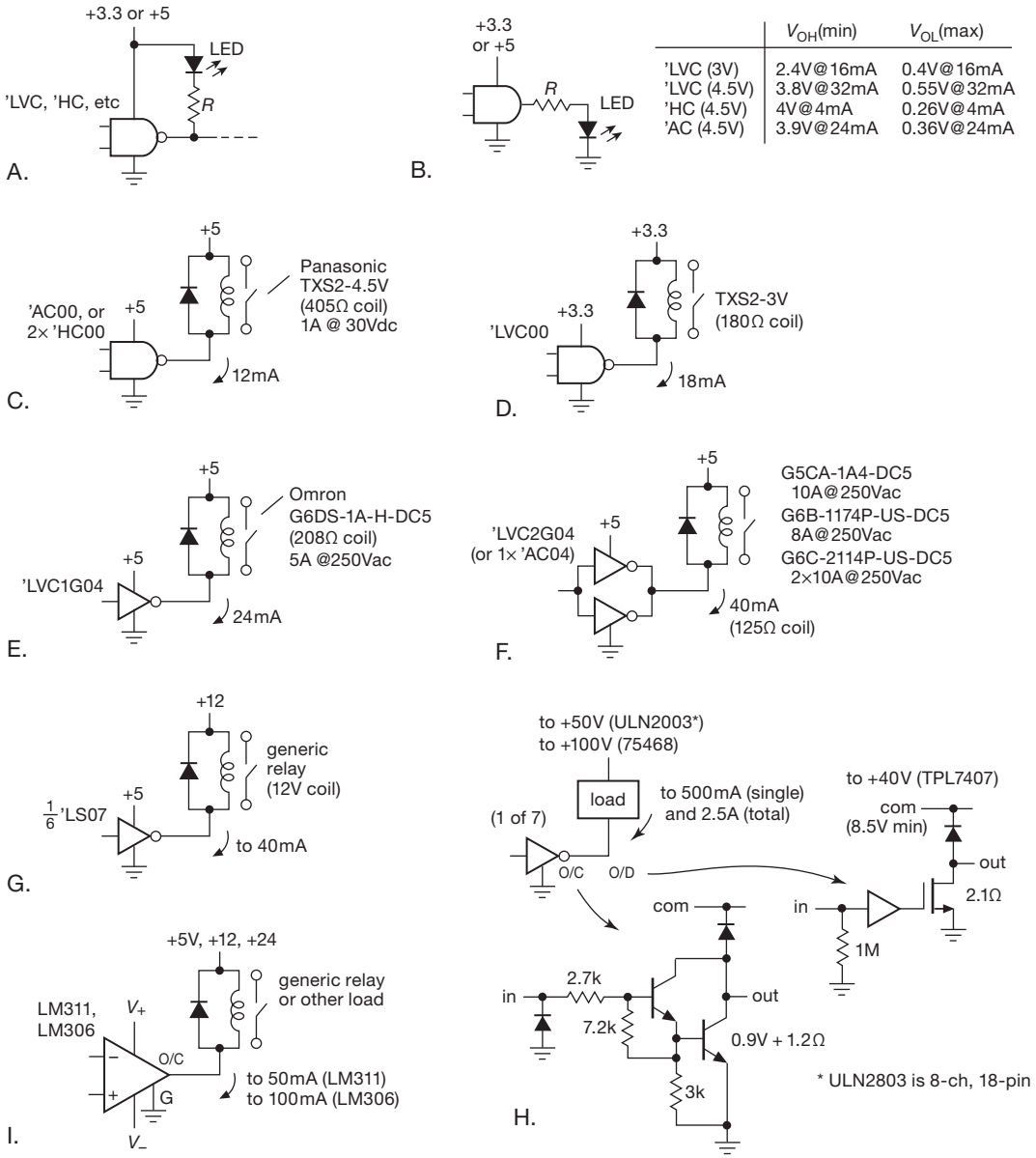
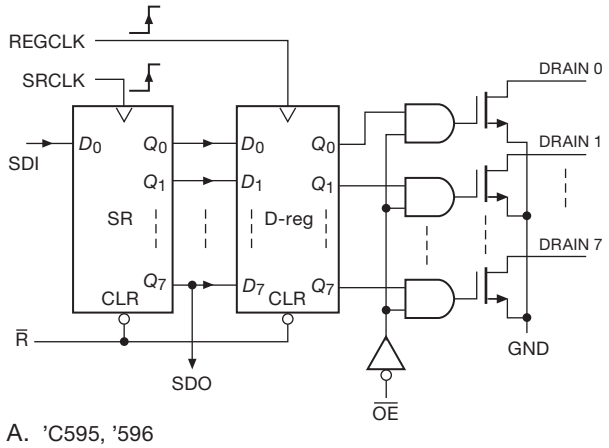


Figure 12.39. Driving loads directly from logic outputs. Relay coils can demand a lot of drive current; be sure your choice of driver IC can deliver.

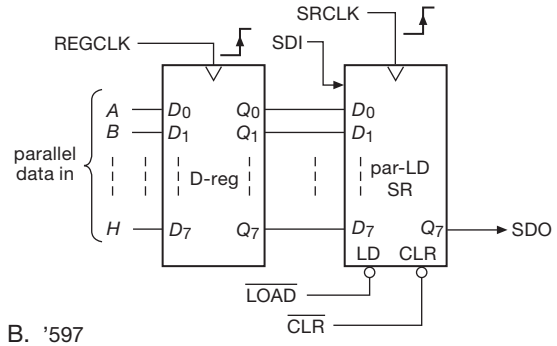
with open-drain outputs capable of sinking substantial currents, and with voltage ratings up to 50 V. Table 12.3 lists a nice selection of these, and Figure 12.40 shows what's inside these output devices, along with the companion '597 *input* register (i.e., parallel-in, serial-out). These are particularly helpful when you have a microcontroller with only a few I/O pins available because you can drive lots of power-hungry outputs (e.g., relays); and you can expand beyond eight outputs by chaining the SDO (serial data out) from one register to the SDI (serial data in) of the next. Figure 12.41 shows the basic idea.

A few points about relays in general, and these small easy-to-drive PCB-mounting relays in particular.

- Note that you don't ordinarily use a series resistor, because the coil's resistance sets the operating current at the rated operating voltage; add a series resistor, though,



A. ’C595, ’596



B. ’597

Figure 12.40. The ’595 and ’596 serial-input power registers accept a logic-level bit-serial input, clocked into an internal shift register; the contents can be latched in an output *D*-type register with substantial drive capability. The ’597 works the other way around, but accepts logic-level inputs only. See Table 12.3 for details.

Table 12.3 Power Logic Registers

Type[c]	Bits	Data comm[a]	V_O max (V)	I_O max (mA)	R_{DS} typ (Ω)	Cost qty25 ($US)	Reset	Prog Curr	Output Enable?	Output type[b]	Pkgs[d] DIP	Pkgs[d] SOIC
STP08CL596	8	SR	16	90	-	1.37	-	●	●	CS	(●)	●
STP08C596	16	SR	16	120	-	2.05	-	●	●	CS	(●)	●
TPIC6B259	8	AL	50	150	5	1.70	●	-	-	OD	●	●
74HC595	8	SR	V_{CC}	25	30	0.16	e	-	●	RR	●	●
TPIC6595	8	SR	45	250	1.3	2.50	e	-	●	OD	●	●
TPIC6B595	8	SR	50	150	5	1.41	e	-	●	OD	●	●
TPIC6C595	8	SR	33	100	7	1.09	●	-	●	OD	●	●
TPIC6C596	8	SR	33	100	7	0.98	●	-	●	OD	●	●
TPIC2810	8	I²C	40	210	5	2.15	f	-	●	OD	●	●
TPIC6B273	8	par	50	150	5	1.83	-	-	●	OD	(●)	●
TPIC6273	8	par	45	250	1.3	2.70	-	-	●	OD	●	●

Notes: (a) SR=shift register, AL=addressable latch. (b) CS=current sinking, adjust via ext resistor, range 15–90mA; OD=open drain; RR=rail-to-rail. (c) ‘596 types have registered data out. (d) “(●)” =no distributor stock. (e) resets SR, but not output latch. (f) power-on reset of SR and output latch.

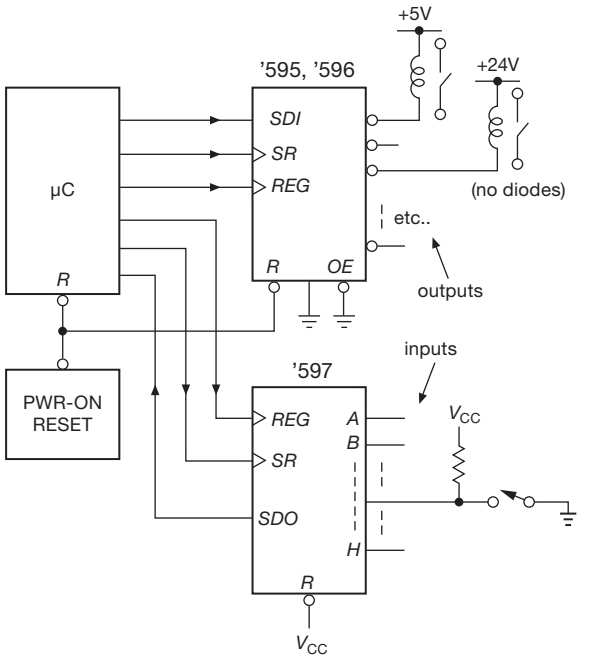


Figure 12.41. Driving relays and other power loads with a serial power register.

if you’re running from a higher supply voltage (e.g., a 12 V relay operating from a 15 V supply). In any case, be sure to include the diode to clamp the inductive spike.[22]

- Relays are always available in the normal “single-side stable” (also known as *non-latching*) configuration, in which the contacts are held in the energized position only while coil power is being applied. But you can also get “latching” relays, which remain in whichever state they have been sent after the coil drive is removed. There are two varieties of latching relays: “dual coil” (you drive one or the other, to set and reset the state), and “single coil” (you apply one polarity or the other, to set and reset). Latching relays are a good choice for something like a battery-powered lamp timer, because you need to

[22] Some driver parts include a diode, while other specifically give you an avalanche rating for the output MOSFET (with which to absorb the relay coil’s flyback spike). For example, the ’6B595 parts specify each output at 33 V and 30 mJ (some others allow 75 mJ), enough to handle the $E=LI^2/2$ inductive energy stored in a large coil. Always read the datasheet carefully.

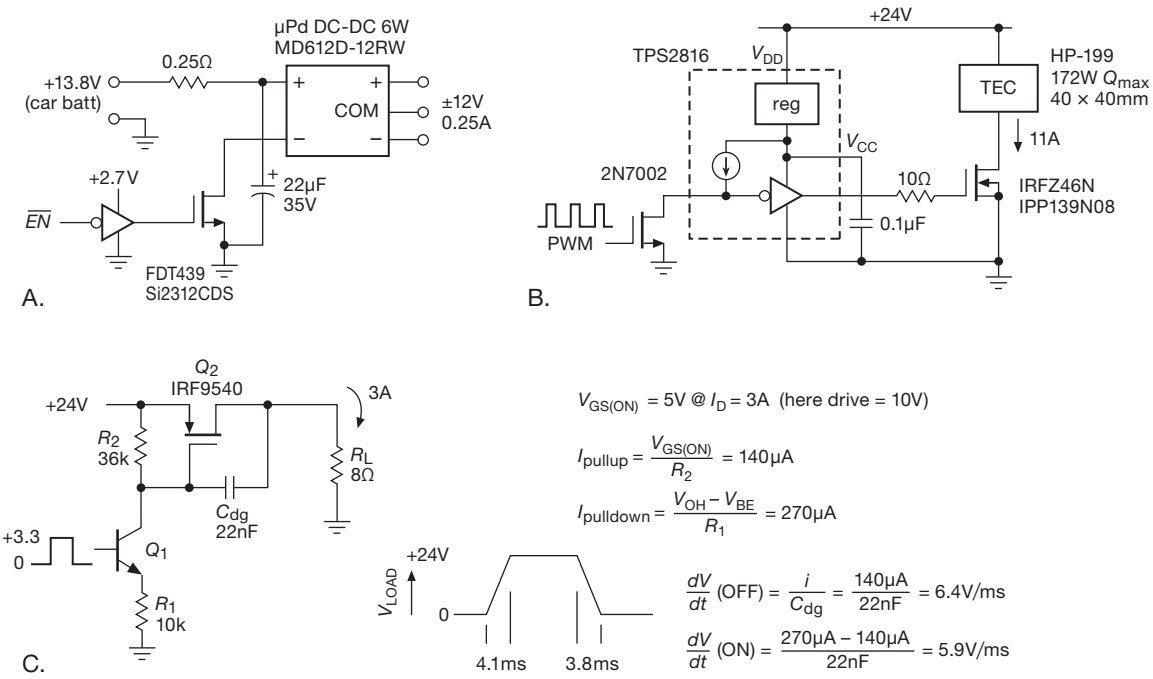


Figure 12.42. Some MOSFET switching examples: A. load with high inrush current (dc–dc converter); B. rapid (pulse-width modulation) switching of high-current thermoelectric module (the 2N7002 is needed because the TPS2816's internal pullup generates a 12 V input logic swing when powered from V_{DD}=24 V); C. high-side switching with controlled slew rate; the MOSFET Q_2 has full 10 V gate drive after a few milliseconds.

apply only a short energizing pulse (typically ∼10 ms minimum) to switch the load ON or OFF.[23]

- Single-coil latching relays evidently respect the polarity of the dc coil drive. But be aware that many of the normal (single-side stable) relays in these easy-to-drive PCB-mounting configurations also require that coil voltage be applied with correct polarity; it will be indicated on the datasheet, and usually with polarity markings on the relay body – and they mean it!

12.4.2 Positive loads: transistor assisted

With an external MOSFET or bipolar transistor you can drive just about anything. Look back at §3.5.3 (Figure 3.96) for some MOSFET drive circuits good for loads to hundred of volts and tens of amperes. MOSFETs (or insulated gate bipolar transistors, IGBTs; see §3.5.7) are the transistors of choice for such robust applications. They are also handy for "high-side" switching, as we saw in §3.5.6 (Figure 3.106).

Figure 12.42 shows some additional configurations. The challenge in the first circuit is to handle the relatively large inrush current at startup of the dc–dc converter (5 A peak, compared with 0.8 A while running at full power). The mighty FDT439, in its small SOT-223 case, guarantees a maximum R_{ON} of 0.08 Ω at V_{GS}=2.5 V (where its saturation drain current is around 20 A; see Table 3.4a). So it handles the inrush current with ease. The small input resistor and downstream bypass capacitor are included to filter switching noise and to isolate the turn-on transient to minimize disturbance of other electronic systems running from the car battery.

The challenge in circuit B is to do fast switching of a high power load (a thermoelectric cooling module), controlled by adjusting the duty cycle of its ON pulses (PWM, pulse-width modulation). Here we needed a hefty power MOSFET, and plenty of gate drive to switch the gate capacitance rapidly through its threshold (to minimize switching losses). The TPS2816 is a nice gate driver (see Table 3.8), with output HIGH voltage of +10 V (and peak current capability of ±2 A), and with a built-in regulator for

[23] We installed several Intermatic ST01C "Digital In-Wall Timers" to control our exterior home lighting. These puppies include enough brainpower to compensate for seasonal variations in dusk and dawn, and they happily switch 15 A 120 Vac loads, using only a single CR2 lithium battery. We could hardly wait to crack one open, and voilà – a latching relay!

operation from supply voltages to +40 V. The power MOSFETs shown are good to 30 A drain current, and come in power surface-mount packages (D^2Pak, TO-252, TO-262).

Finally, circuit C shows a high-side switch with controlled slew rate, to minimize production of transients. It's OK to switch power on time scales of milliseconds, as long as you don't try to do it at a high rate (as in PWM), and as long as the switch can handle the transient thermal pulse.[24] The slew rate is set by the gate drive current charging and discharging a "Miller capacitor" C_{dg}, here arranged for approximately equal net charging and discharging currents. The calculations in the figure say it all (five equations are worth a thousand words).

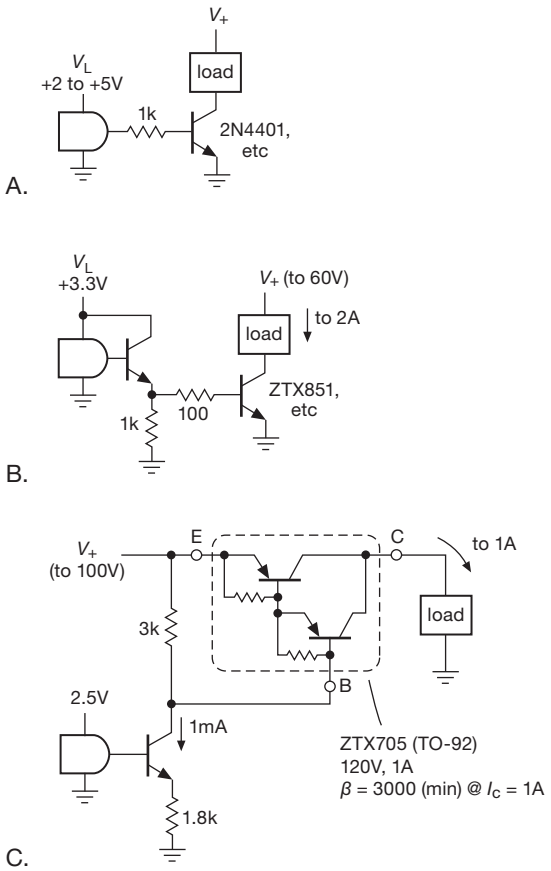


Figure 12.43. Bipolar transistors extend the voltage and current drive capability. Ditto for power MOSFETs – see Figures 3.96 and 12.42.

On a more modest scale you can use bipolar transistors (BJTs) for these same tasks, for example as shown in Figure 12.43. Something like a jellybean 2N4401[25] (about $0.06 in 100 qty) is good for up to +40 V and 500 mA; but beware – the minimum beta drops to 40 at that current, so you'll need to resize the series resistor to provide upward of 10 mA of base drive into the transistor's V_{BE}. And even then, the saturation voltage is not stunning: V_{CE}(sat)=0.75 V (max) at 500 mA with 15 mA of base drive (therefore 0.25 W transistor dissipation). A better choice for such currents is something like the ZTX851 from Zetex, which comes in a TO-92 variant that they call "E-line." It's good for 60 V and 5 A, minimum beta of 100 at 2 A, and V_{CE}(sat)=0.15 V (max) at 2 A with 50 mA of base drive. That's about the same dissipation as the 2N4401, but at four times the load current. To get the base drive you'd use an emitter follower driver, as in Figure 12.43B.

You can easily rig up a BJT as a high-side switch by using a logic-switched *npn* current sink to forward bias a high-side *pnp* switch (Figure 12.43C). Here we've used a Darlington from Zetex's excellent E-line series of BJTs, requiring only a milliamp of base drive to switch up to an amp of load current (where V_{CE}(sat)$\leq$0.75 V).

In these examples we've not worried about protecting the switches from fault conditions such as a shorted load. This should not be ignored; and we won't – see §12.4.4, coming up shortly.

12.4.3 Negative or ac loads

Figure 12.44 shows some methods by which a logic input can control loads that return to a negative supply rail; also shown is a common interface from logic to an ac power load. In circuits A and B a HIGH output state turns on the *pnp* transistor switch, pulling the collector into saturation at one diode drop above ground. In circuit A, the resistor (or gate output-current limit) sets the emitter current, and therefore the maximum collector (load) current, whereas in the more powerful circuit B, an *npn* follower is used as a buffer, and a diode in series with the output keeps the load from swinging above ground. In both cases the maximum load current is equal to the drive current to the *pnp* transistor's emitter. Circuit C requires a low-voltage negative bias supply voltage ($-V_{bias}$), but has the advantage of saturating cleanly to ground, and with a power MOSFET it can

[24] Datasheets specify a "transient thermal impedance" from junction to case, which for these transistors is about 0.5°C/W at millisecond time scales, fine for the 18 W (maximum) dissipation during switching. More on this in §*3x.13*.

[25] The 2N prefix is for the through-hole TO-92 package. For surface mount, the base part number becomes MMBT4401, with crazy suffixes for the different surface-mount packages (SOT-23, SOT-323/SC70, SOT-523, SOT-723).

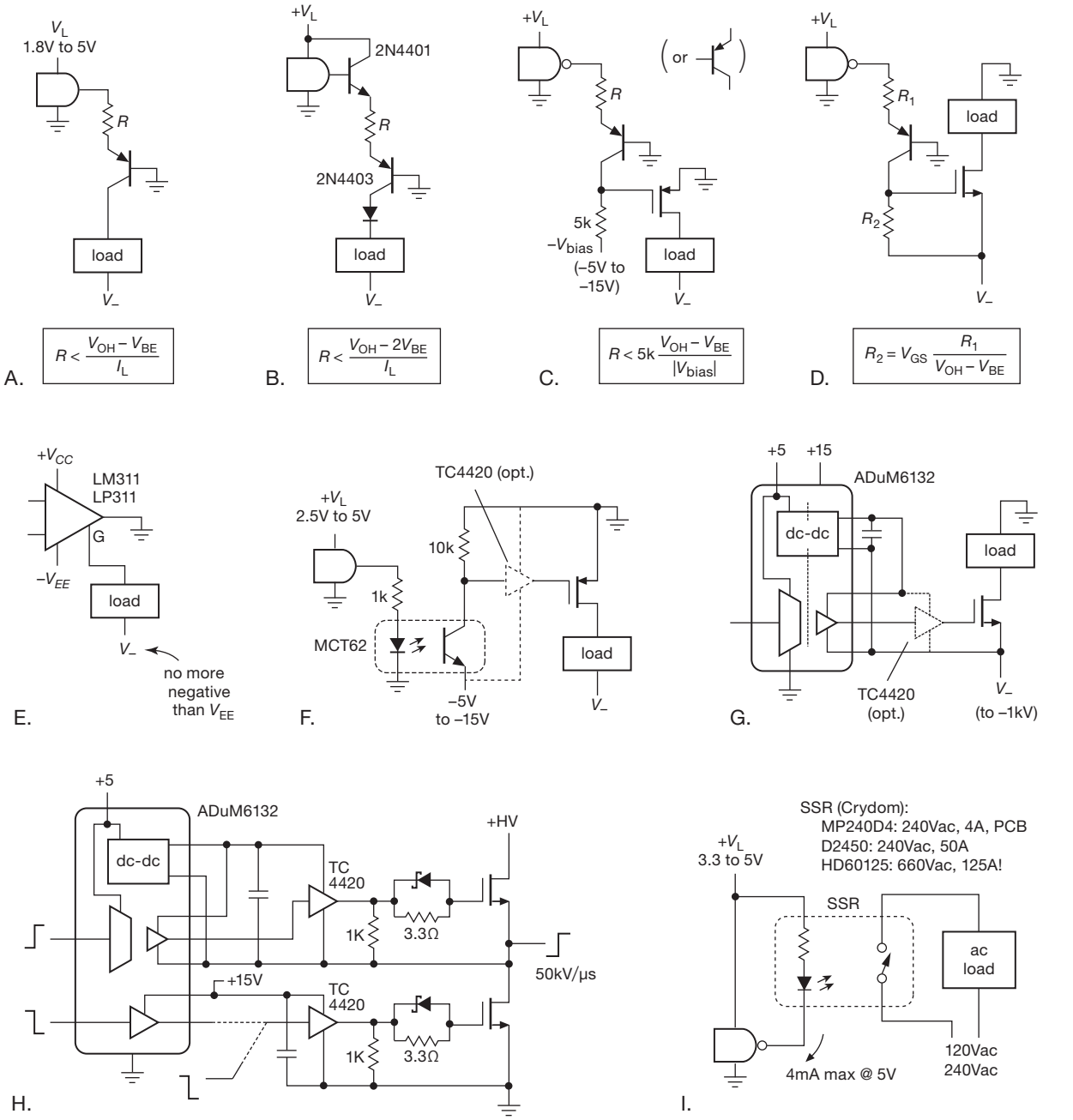


Figure 12.44. Driving negative and ac loads.

handle very large load voltages and currents, even when driven by low-voltage logic of minimal output drive current (but you may want to add a MOSFET gate driver IC, as in circuits F–H: its high current capability rail-to-rail output produces much faster switching into the MOSFETs large gate capacitance). Circuit D shows how to drive a ground-returned load to a negative voltage; this has the pleasant property of not requiring a separate low-voltage negative supply. Circuits like this do not provide protection against load faults; be sure to read §12.4.4, which deals with this important issue.

Comparators with a floating output (LM311 and

cousins) can drive modest negative-referred loads, as in circuit E; but the negative return cannot be more negative than the comparator's negative rail, and the current is limited to 50 mA. Circuit F shows how to use an optocoupler (see §12.7) to translate a positive logic output to a negative-going level that can drive a *p*-channel power MOSFET gate directly (or through a MOSFET gate driver, for faster switching; see Figure 3.97). Because *n*-channel MOSFETs are better performers (lower R_{ON} and available up to 1000 V rating, compared with −300 V for pMOS), it's nice to be able to use them whenever you can. Circuit G does this by generating a gate drive signal down at the V_- rail, using the elegant ADuM6132 logic isolator from Analog Devices. The latter uses tiny on-chip transformers to generate both an isolated (floating) dc supply voltage, and also to couple the input logic signal to a similarly isolated output. Once again, you can interpose a MOSFET gate driver, powered by the same isolated power.

As an aside, circuit H shows how to use the same ADuM6132 to generate a *high-side* gate drive, so that you can use *n*-channel power MOSFETs for both switches in a high-voltage push–pull output stage. This isolator chip is notable for its ability to work properly even when the isolated output is flying around at slew rates up to 50 kV/μs. Note that the high-side MOSFET, though it looks like a follower, is actually a switch, because its gate is either at the same voltage as its source (when OFF), or it is 15 V above the source (when ON).

Finally, for driving ac loads the easiest method is to use a "solid-state relay," as in circuit I. These are optically coupled triacs, SCRs, or IGBTs with logic-compatible input and 1–50 A (or more!) load-current capability when switching a 115 Vac (or more!) load. The low-current varieties are available in SMT and DIP packaging (e.g., the photoMOS series from NAiS Aromat, the MOSFET relays from Omron, and the PV series of "photovoltaic switches" from International Rectifier; see §12.7.5 and 12.7.6 for more choices), while the heftier ones come as a chassis-mounting rectangular block roughly 2 inches square.[26]

Alternatively, you can switch ac loads with an ordinary relay, energized from logic. However, be sure to check the specifications, because most small logic-driven relays cannot drive heavy ac loads, and you may have to use a MOSFET or logic relay to drive a second larger relay. Most solid-state relays use "zero-crossing" (or "zero-voltage") switching, which is actually a combination of zero-voltage turn-on and zero-current turn-off; it is a desirable feature that prevents spikes and noise from being impressed onto the powerline. Much of the "garbage" on the ac powerline comes from triac controllers that don't switch at zero crossings, e.g., phase-controlled dimmers used on lamps, thermostatic baths, motors, etc. As an alternative to the optical coupling used internally in circuit I, you sometimes see a pulse transformer used to couple triggering pulses to a triac or an SCR.

12.4.4 Protecting power switches

In these switch examples we've sidestepped an important topic: when driving power loads, you must worry about any of a variety of possible "fault conditions," for example a short-circuited load. This can happen more easily than you might expect, particularly with external loads wired through connectors and cables. Without some protective circuitry, the MOSFET is easily (and rapidly) destroyed, perhaps taking with it some additional circuitry. Let's take a closer look.

Figure 12.45 shows three versions of a pMOS high-

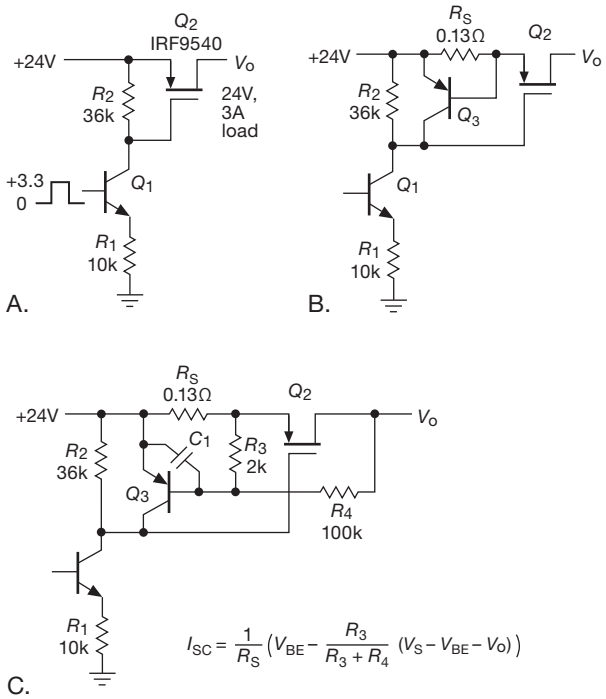


Figure 12.45. Current limiting the high-side switch: A. no protection; B. fixed 5 A current limit; C. foldback current limit.

[26] With high-current SSRs, the non-zero voltage drop can produce lots of power dissipation, requiring substantial heatsinks. For example, the 125 A HD60125 with its 1.7 V drop dissipates a bit over 200 W at full load! This is sobering: in spite of the no-moving-parts appeal of SSRs, you may decide that mechanical relays aren't so bad after all.

side power switch, intended for a ground-returned load that draws 3 A when powered by its rated +24 V. In all three circuits, *npn* transistor Q_1 converts the +3.3 V logic-level input into a 0.27 mA sinking current, which generates the ∼10 V negative-going gate drive for the *p*-channel power MOSFET Q_2. The IRF9540 has a maximum R_{ON} of 0.2 Ω at that gate drive, so there's at most 0.6 V drop at full load, or 1.8 W dissipation. You hardly need a heatsink.

What happens if the output is shorted? Circuit A has no protection, so the current is limited either by the capability of the +24 V supply or Q_2's saturation drain current, whichever is less. For the latter, the datasheet shows ∼50 A (at V_{GS}=−10 V and V_{DS}=24 V). For this 3 A application it's likely that the 24 V supply is less gutsy than that, perhaps good for 5–10 A. Taking the higher value, and using the datasheet's R_{ON}=200 mΩ at 25°C, we've got I^2R_{ON}∼20 W dissipation in Q_2 (rising to 30 W at 100°C), not a happy situation for a transistor that's ordinarily dissipating less than 2 W.

OK, you say, let's add *current limiting* (Circuit B). This is the usual circuit, with a current-sensing resistor R_S sized to generate a V_{BE} drop at the limiting current, thus bringing Q_3 into conduction, robbing gate drive and preventing further rise of current. It's best to set the limit current enough above the normal maximum load so that variations in V_{BE} with temperature don't cause premature limiting. Here the current limit is set at ∼5 A. The good news: we have a current limit. The bad news: it makes things worse for Q_2, whose dissipation into a short circuit rises to $I_{lim}V_{in}$=120 W.

OK, OK, you say, the problem with a simple current limit is that it permits a fault current at least equal to the maximum normal load current, and that with the full 24 V drop across the switch transistor. In other words, you'll always have at least 72 W dissipation (24 V times 3 A) into a short. So, let's rig up a better circuit that *reduces* the current limit when it sees the load dragging down the output; in other words, a protective circuit that permits full load current at rated output voltage, but reduced current at lower output voltage.

This is called *foldback current limiting*, and looks like circuit C. It behaves like the simple current limit when there's no significant drop across the switch; but if the output is held at ground (for instance), the voltage divider R_3R_4 creates about 0.5 V of forward bias across Q_3's base–emitter. So it takes only a bit over 1 A to bring Q_3 into conduction, limiting the current to that lower value and the dissipation to about 30 W. It's common to set the short-circuit current limit somewhere in the range of 25–35% of the full load current. Capacitor C_1 provides some delay before triggering current limit or foldback; a time constant of ∼1 ms guards against overzealous foldback while retaining protection.

Figure 12.46 shows the situation graphically. Among other things, you can see that the worst-case fault condition (in terms of transistor dissipation) occurs for a load that has a small resistance (rather than zero ohms). That's because the rising permitted load current more than compensates for the decreasing voltage drop across the switch. Even with this foldback scheme, we're once again faced with an enormous jump in switch dissipation: from 1.8 W (max) into a normal load, increasing to 34 W into a short, and 42 W into a 3 Ω fault-condition load. And foldback circuits have their disadvantages: if made sufficiently aggressive, they may prevent startup into a large capacitive load or other loads (like motors or dc–dc converters) that have a large inrush current.

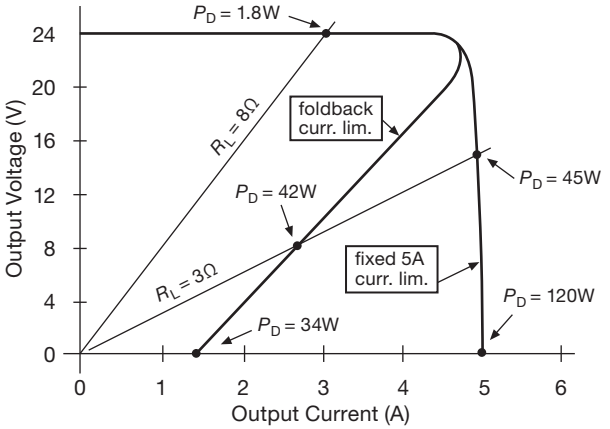


Figure 12.46. Foldback current limiting reduces the shorted-output switch dissipation by more than a factor of 3 compared with simple current limiting.

A. An easier way: protected switches

What to do? Foldback circuits can be made more precise,[27] for example by replacing Q_3 with a differential amplifier. And you can shape the foldback contour by including a zener in the feedback path. Then use a hefty heatsink, combined with temperature sensing to shut down the power when there's excessive heating, and you can make it all work reliably.

But there's a better way, in the form of "intelligent" pro-

[27] Our circuit suffers from uncertainties in V_{BE}, which is important: the short-circuit current is set by the *difference* between the actual V_{BE} and the 470 mV that the R_3R_4 divider creates during shorted output.

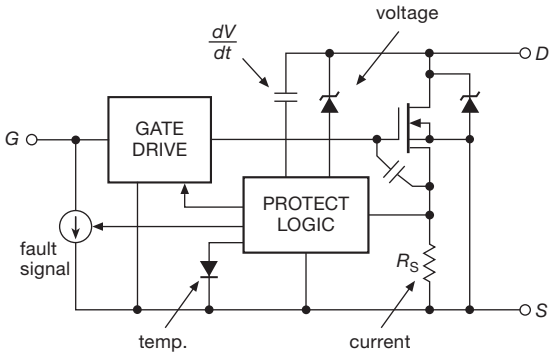


Figure 12.47. A protected MOSFET acts like an ordinary transistor, but it includes internal fault sensing and shutdown circuitry.

tected switches. One variety consists of *n*-channel MOSFETs, decorated with internal circuitry that senses overvoltage, overcurrent, and overheating, shutting down the gate drive accordingly (and providing an indication in the form of reduced gate input resistance). These devices come in standard 3-terminal transistor packages, but internally they have an integrated circuit watching over the power MOSFET (Figure 12.47). Table 12.4 lists a sampling of typical parts.

These are suitable for any application in which you would use a relatively low-voltage *n*-channel power MOSFET. For example, they would be fine for switching the low side of a load that returns to a positive supply, as in Figures 12.42A,B; or you could use one for switching a negative voltage to a ground-returned load, in the manner of Figure 12.44D.

For a switching application like ours, though, you'd need a protected *p-channel* MOSFET, a breed that appears not to exist. For this application there's another variety of protected switches, intended specifically for *high-side* switching: these use an *n*-channel power MOSFET, suitably protected, with an internal charge pump and level-shifting circuitry to drive the gate ∼10 V beyond the positive supply – see Figure 12.48. Once again there's sensing and protection against voltage, current, and temperature faults, sometimes responding also to undervoltage, reverse polarity, and loss of ground. So our problematic 3 A @ 24 V switching circuit becomes the thing shown in Figure 12.49: simple, inexpensive, and reliable.

Table 12.4 A Few Protected MOSFETS[a]

Type	V_{DS} max (V)	I_D max (A)	R_{DS} max (mΩ)	Q_g typ (nC)	Cost qty25 ($US)	Packages TO-220	TO-252	SOIC	SOT-223
BTS3207	42	0.6	500	-	0.60	-	-	-	●
VNN1VN04	40	1.7	250	5	0.62	-	3	●	●
VNN3VN04	40	3.5	120	8.5	1.08	-	3	●	●
IPS1041	36	4.5	100	-	1.92	-	3	d	●
BTS117	60	7	100	s	1.97	3	-	-	-
VNN7VN04	40	9	60	18	0.90	-	3	●	●
VNP10N07	70	10	100	30	1.36	3	-	-	-
VND14NV04	40	12	35	37	1.89	3	3	●	-
BTS133	60	21	50	s	1.27	3	-	-	-
BTS141	60	25	28	s	4.81	3	-	-	-
VNP35NV04	40	30	13	118	4.67	3	-	-	-

Notes: (a) All are n-channel types, with active current and overtemp limits; all require 5V min gate drive, and signal fault conditions with excess gate current; BTS types have slow *dV/dt*, while VN types are faster but have controlled *dI/dt*. (d) dual. (s) slew-rate limit 1V/μs.

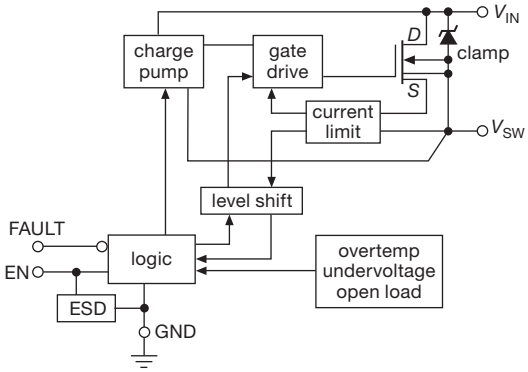


Figure 12.48. High-side intelligent switches have a charge pump and level-shifting circuits to drive the *n*-channel gate beyond the drain input supply; and they protectively monitor fault conditions such as overvoltage, overcurrent, and overtemp.

Table 12.5 lists the characteristics of selected high-side protected switches.[28] Within these there are two styles of logic input (Figure 12.50): one version accepts digital logic levels, relative to a ground pin; the other has no ground pin, instead requiring you to pull a current-sourcing pin to ground with a small external switch. This latter style provides a load-current indication, in the form of a "sense-output" pin that sources a current approximately proportional to load current.

[28] But a *caution*: the high-side intelligent-switch scene is strewn with discontinued parts. It's a healthy (but competitive) market sector (think automobiles), and new parts with improved performance (and pricing) are continually introduced. Table 12.5 gives a current list, and a sense of price and performance that's available; but you may have to look for comparable parts as these become consigned to the graveyard of obsoleted electronic components.

Table 12.5 Selected High-side Switches[a]

Type	Switches	V_{in} min (V)	V_{in} max (V)	I_o max (A)	R_{DS} typ (mΩ)	I_S typ (mA)	V_L min (V)	t_{ON} typ (ms)	Cost qty25 ($US)	Logic input?	Fault output	Active clamp?	Package	Comments
FDG6323L	1	2.5	8	0.6	550	b	1.5	0.01	0.35	●	●	-	SC-70-6	nMOS drvr + pMOS hi-side[b]
TPS22960	2	1.8	6	0.5	435	0.00	1.6	0.08	0.95	●	-	-	SOT-23-8	p-channel switch
FPF2110	1	1.8	8	0.4	160	0.08	1.8	0.03	0.99	●	●	-	SOT-23-5	p-channel switch
FPF2123[g]	1	1.8	8	1.5[c]	160	0.08	1.8	0.03	1.05	●	-	-	SOT-23-5	p-channel switch
MIC2514	1	3	14	1.5	900[d]	0.08	2.3	0.01	1.90	●	-	-	SOT-23-5	p-channel switch
STMPS2151	1	2.7	6	0.5	90	0.04	2.2	1	0.81	●	●	-	SOT-23-5	
AP2156	2	2.7	5.5	0.8	100	0.09	2.2	0.6	0.85	●	2	-	SOP-8	USB-port power
TPS2041	1	2.7	5.5	0.7	80	0.08	2.2	2.5	2.59	●	●	-	DIP-8	
BTS452	1	6	62	1.8	150	0.8	2.5	0.08	1.97	●	●	●	TO-252-4	
BTS410	1	4.7	65	2.7	190	1.0	2.5	0.10	3.15	●	●	●	TO-220-5	see also BTS462T
BTS611	2	5	43	2.3	200	4	4	0.20	3.17	●	1	●	TO-220-7	
IPS511	1	6	32	5	135	0.7	3.3	0.05	1.76	●	●	●	TO-220-5	
FPF2702	1	2.8	36	2[h]	88	0.09	0.8	2.7	2.10	●	●	-	SO-8	adjustable current limit
IPS6031	1	6	32	16	60	2.2	3.6	0.04	2.52	●	●	●	TO-220-5	
BUK202-50Y	1	5	50	20	28	2.2	3.3	0.14	4.16	●	●	●	TO-220-5	
BTS432	1	4.5	63	35	30	1.1	2.7	0.16	4.85	●	●	●	TO-220-5	
BTS6142	1	5.5	45	25	12	1.4	n	0.25	2.65	n	e	●	TO-252-5	$I_S = I_L$ / 10k (±20% at 30A)
BTS6133	1	5.5	38	33	10	1.4	n	0.25	3.97	n	e	●	TO-252-5	$I_S = I_L$ / 9.7k (±10% at 30A)
VN920	1	5.5	36	30	16	5e	3.6	0.10	3.10	●	●	●	TO-220-5	
BTS442	1	4.5	63	70	15	1.1	2.7	0.35[f]	5.37	●	●	●	TO-220-5	
IPS6011	1	6	35	60	14	2.2	3.3	0.07	4.40	●	●	●	TO-220-5	
BTS6144	1	5.5	30	37	9	2.2	n	0.30	4.77	n	e	●	TO-220-7	$I_S = I_L$ / 12.5k
BTS555	1	5.0	44	165	1.9	0.8	n	0.6[f]	5.80	n	e	●	TO-218-5	to 480A surge, $I_S = I_L$ / 30.2k

Notes: (a) all are *n*-channel switches, with charge pumps, unless marked; all have overcurrent and overtemperature protection. (b) transistor pair, not intelligent, add your own source and gate resistors. (c) adjustable 0.15-1.5A. (d) at 12V. (e) one pin signals load current and faults. (f) max. (g) shuts down after 10ms, retries every 160ms thereafter. (h) adjustable 0.4-2A. (n) Requires MOSFET closure to GND, sinking I_S.

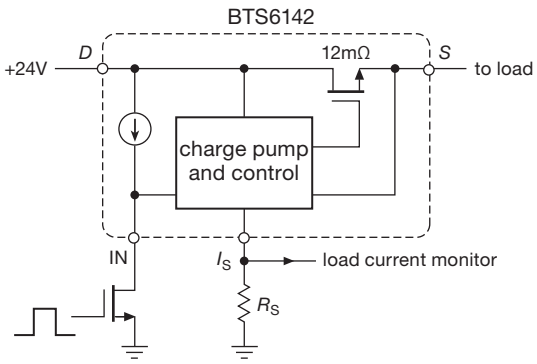


Figure 12.49. Problem solved! An intelligent high-side integrated switch provides a simple and reliable solution.

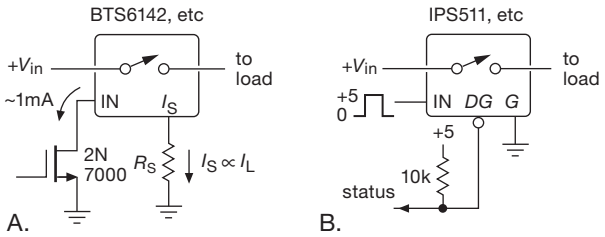


Figure 12.50. Two control schemes for high-side intelligent switches. A. No "ground" pin: enable by pulling input to ground, output current reported via proportional current-sourcing I_S pin. B. Logic-level voltage drive with respect to ground pin; active-LOW status output reports fault condition. This is the configuration shown in Figure 12.48.

12.4.5 nMOS LSI interfacing

Most LSI and VLSI circuits have true CMOS output drivers, with full rail-to-rail swings and with pretty much the same interfacing properties as the CMOS logic gates we just discussed. This is invariably true of ICs that operate from a supply voltage of +3.3 V or less. However, there are still useful ICs designed for 5 V supply operation that use a "totem-pole" *n*-channel MOSFET output stage (nMOS follower atop an nMOS switch; see Figure 12.51), thus producing a HIGH output voltage of only ~3.5 V, and

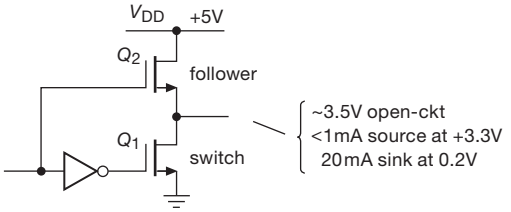


Figure 12.51. NMOS logic totem pole output circuit. The HIGH-state output voltage V_{OH} is around +3.5 V, with poor current-sourcing capability.

even that with hardly any sourcing current capability; you need to know about this quirky behavior. (The same goes for bipolar-TTL outputs, constructed with *npn* transistors in the same sort of totem pole.) While we're at it, we'll take a look also at a typical nMOS *input* stage, still widely used in ICs that can run over a range of supply voltages while maintaining an input logic threshold meeting the canonical TTL specifications (that is, anything less than +0.8 V is interpreted as LOW; anything above +2.0 V is HIGH).

A. nMOS and TTL outputs

The problem with an nMOS or TTL output is that its HIGH output, sitting anemically at ∼3.5 V, is inadequate to drive loads like LEDs or relays; and it cannot even legally drive the input of a 5 V 'HC-logic device (with its mid-supply threshold). Taking the example of the XC95xx family of Xilinx cPLDs trying to drive a bright white LED, Figure 12.52 shows graphically the problem: the dashed curves show the output HIGH situation, where the cPLD's sourcing current versus voltage drops to zero around 3.4 V. We can figure out what will happen if we use that to drive a white LED whose cathode ("−" terminal) is grounded, by plotting on the same graph the "load line" of the LED; we've done that for a 70 Ω series resistor, and for no resistor at all. Either way we're lucky to get 4 mA of LED current, and without any confidence in the result. Contrast that with the *sinking* configuration (cPLD output connected to LED cathode, anode returned to +5 V via a series resistor), where a logic LOW output's clean saturation to ground produces a robust and predictable ∼20 mA of drive with the series resistor (don't even *think* about omitting the resistor!). The results with a bipolar-TTL output structure are analogous (though with considerably poorer sourcing current).

The lesson is clear: when driving demanding loads directly from nMOS or TTL outputs, configure things so the output-LOW state does the heavy lifting.

If you insist on driving a load that returns to ground

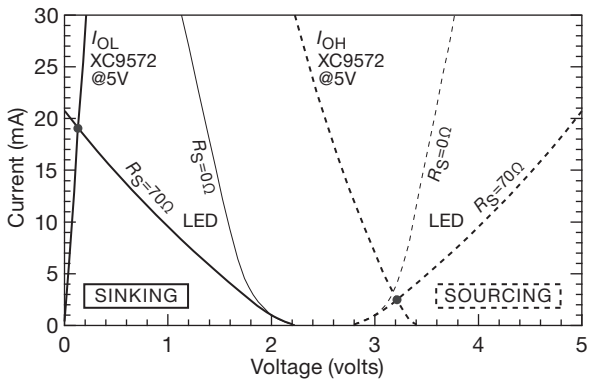


Figure 12.52. Driving a white LED with an nMOS output. The XC9500 5 V series of cPLDs provide good sinking all the way to ground, but weaker sourcing, and that only to ∼3 V. Here we've used load lines (see Appendix F) to estimate the LED current for both the sinking and (undesirable) sourcing configurations, in each case with and without a series current-limiting resistor. (See Figures 2.8 and *1x.48* for LED *VI* curves.)

from one of these wimpy outputs, you've got several choices. The *wrong* way is to say "hmmm, I'll just add a current-boosting emitter follower" (Figure 12.53). Nice try, but the additional V_{BE} drop only makes things worse. Figure 12.54 shows several ways that do work. In circuit A the nMOS output LOW sinks 2 mA, driving the *pnp* transistor into hard conduction; you can use a discrete resistor plus transistor, or a combination "digital transistor" (also known as a "prebiased transistor" or "bias–resistor transistor") like the DDTA123, good to 100 mA load current.[29] Circuit B substitutes a low-threshold *p*-channel MOSFET ($R_{ON}<100\,m\Omega$ at $V_{GS}=-2.5$ V), with a mild pullup resistor to ensure that output HIGH keeps the transistor OFF. Circuit C cheats a little, by interposing a CMOS inverter (or noninverting buffer) with TTL-compatible (therefore nMOS-compatible) input levels. The output can happily drive a 5 V load, sourcing (or sinking) tens of milliamps of current (see Figure 12.7). For more muscle, you can substitute a MOSFET gate driver like the ubiquitous TC4420-series (circuit D), which also lets you increase the output swing up to +18 V; these things hardly sweat at hundreds of milliamps source or sink (see Figure 12.7 again).

B. Weak CMOS outputs

Similar drive problems can crop up even with true CMOS rail-to-rail outputs. For example, look at the PIC16F sourcing (HIGH output) curve in Figure 12.8. We can use a

[29] Or the more than 300 alternatives in stock at DigiKey, from seven manufacturers.

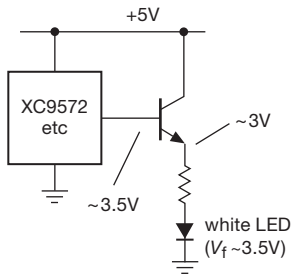


Figure 12.53. Don't do this! The low-hanging HIGH output voltage from an nMOS or TTL device is only made lower by the *npn* follower – it's a *black* LED.

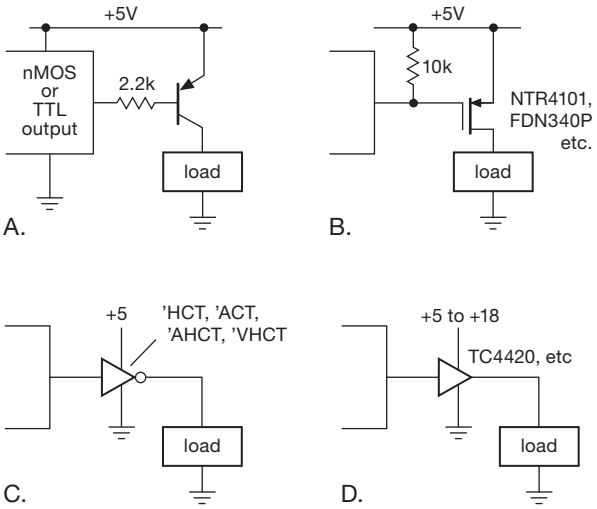


Figure 12.54. These nMOS logic outputs are driving loads that return to ground.

similar load-line graphical technique to see how such an output fares when driving a load like an *npn* switch with series base resistor (Figure 12.55), which you might use to switch on a 3 V relay coil (e.g., a Panasonic TXS2-3V, requiring 16.7 mA). The load lines are drawn for several choices of base resistor, allowing us to estimate the resulting base current from the intersection with the output-HIGH curve of the popular PIC10F microcontroller. Here a 2 kΩ series resistor produces about 1.7 mA of base drive, which (according to the transistor's datasheet) causes the collector to saturate at less than 50 mV with the relay coil as load. That's just fine – less than a milliwatt of transistor dissipation. But if you wanted to drive a heftier relay, for example the Omron G6RL-1A-3VDC, which has contacts rated at 8 A and 250 Vac, you'd have to contend with the 3 Vdc coil's resistance of 41 Ω, or 73 mA coil current. Now a 1 kΩ (or smaller) base resistor would be best: the

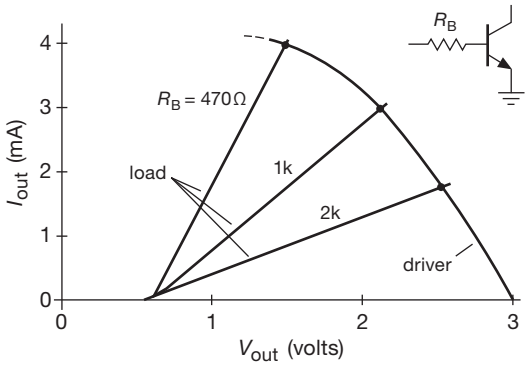


Figure 12.55. A weak CMOS output (here a PIC10F microcontroller, V_{DD}=3 V) can switch a "relay driver" transistor like the DRDNB16W (which includes both a 1k series base resistor and a clamp diode). Or you can use the resistorless DRDN005 with an external base resistor. Consider also an inexpensive 3-pin "digital transistor," which includes an integrated base resistor.

>3 mA base drive produces a collector saturation voltage of ∼100 mV, or 10 mW transistor dissipation.

C. nMOS inputs

You might think that nMOS is dead,[30] at the hands of the victorious CMOS. But you'd be wrong: many digital ICs that need to operate over a wide range of supply voltages use the simple input circuit of Figure 12.56. Q_1 is an inverter, and Q_2 is a small-geometry source follower providing pullup current (resistors take up too much space, so

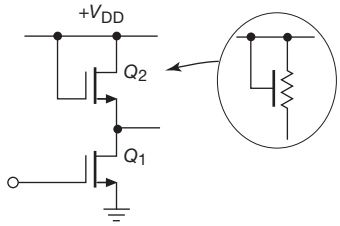


Figure 12.56. NMOS logic-input circuit.

MOSFETs are universally used as drain loads); the alternative symbol shown for Q_2 is widely used. The threshold voltage of the input transistor is in the range of 1–1.5 V, fully compatible with the long-standing "TTL input level" specification. A classic example of an IC that works this way is a MOSFET gate driver like the TC4420 (and its many cousins), which runs from a single +V_{DD} positive

[30] "She's not only merely dead, she's really most sincerely dead," as the Coroner proclaimed in *The Wizard of Oz*.

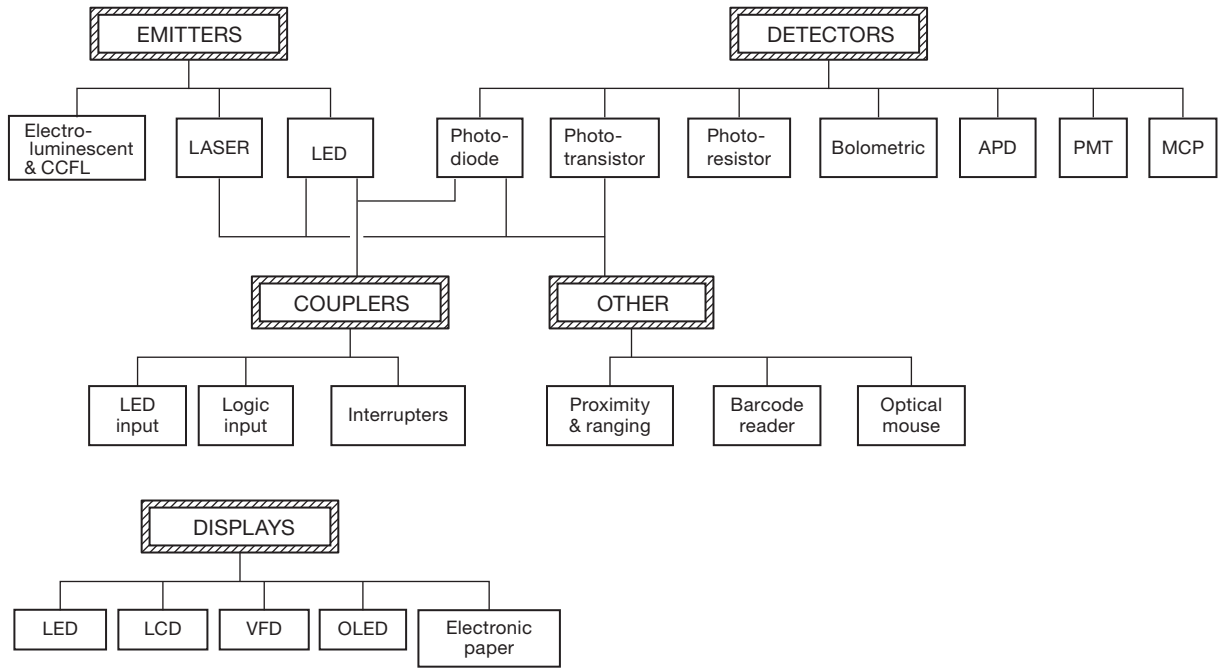


Figure 12.57. Optoelectronics family tree.

supply in the range of 4.5–18 V. Happily, such an input accepts pretty much any HIGH logic level you throw at it (including all the way to the elevated positive supply) while drawing essentially zero input current.

12.5 Optoelectronics: emitters

In the previous three chapters we have been using LED indicators and LED numeric display devices in various circuit contexts as we needed them. LEDs belong to the general area of *optoelectronics*, which includes displays based on other technologies as well, notably liquid crystals ("LCD") and gas discharge. It also includes optical electronics used for purposes other than indicators and displays: detectors (photodiodes and phototransistors), photomultipliers, array detectors such as "charge-coupled devices" (CCDs), light-coupled isolators ("opto-isolators"), solid-state relays, position and proximity sensors ("interrupters" and "reflective sensors"), diode lasers, image intensifiers, and a variety of components used in fiber optics.

Although we will continue to conjure up miscellaneous magic devices as we need them, this seems like a good place to pull together the area of optoelectronics, since it is related to the logic interface problems we have just been discussing.

To set the stage, we present a brief family tree of optoelectronics in Figure 12.57, and in the outline below. We've tried to make it inclusive, to provide perspective and orientation. And a family needs a *portrait*, so we gathered up a collection of optoelectronic goodies from our lab for Figures 12.58, 12.71, 12.80, 12.84, and 12.95. In the following sections we'll look at a subset of these devices, concentrating on the devices and techniques that are most important for day-to-day circuit and instrument design.

12.5.1 Indicators and LEDs

Electronic instruments look nicer, and are more fun to use, if they have pretty little colored lights on them. LEDs have replaced all earlier technologies (notably incandescent lamps) for this purpose. You can get red, yellow, green, blue, and white indicators, and you can get them in many packages, the most useful of which are (a) panel-mounting lights, and (b) PCB-mounting types. The catalogs present you with a bewildering variety of them, differing mostly in size, color, efficiency, and illumination angle. The latter deserves some explanation: a "flooded" (or "diffused") LED has some diffusing stuff mixed in, so the lamp looks uniformly bright over a range of viewing angles; that's usually best, but you pay a price in brightness.

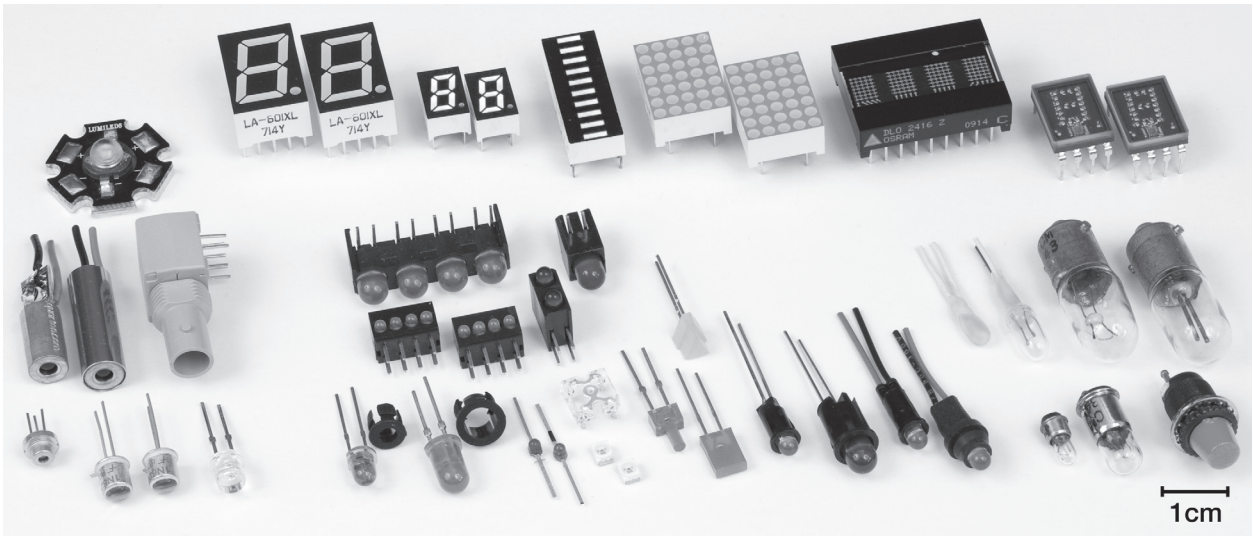


Figure 12.58. Optoelectronics: emitters and displays. At front center are visible LED indicators, in the popular 3 mm (T-1) and 5 mm (T-1$\frac{3}{4}$) panel-mount sizes (some with integral plastic mount), along with PCB-mount singles and arrays. To the left are three red lasers (bare, in 3-pin metal package; and with regulator-driver, in cylindrical housing), a few IR emitters, a fiber-optic ST-style transmitter, and a high-intensity white LED on an aluminum heat spreader. On the right are six legacy incandescent indicators and one NE-51 neon bulb. Along the rear are displays: 7-segment, bar graph, 5×7 dot matrix, 4-character dot matrix, and hexadecimal with latch–decoder–driver.

If the datasheet's "angle of half intensity" (or "viewing angle") specification is at least 90° (it may say "±45°"), and ideally 120° or more, it will look pretty good off-axis. Figure 12.59 shows a polar-plot comparison for LEDs with viewing angles of 30°, 60°, and 120°. These show relative intensity versus viewing angle, normalized to unit intensity on-axis (i.e., 0°). Without that normalization the plots for the 60° and 120° LEDs would shrink dramatically, as shown in the unnormalized plot of Figure 12.60.

An LED looks electrically like a diode, with a forward drop going from about 1.5 V (red) to 3.5 V (blue or white); they use semiconductors with a larger bandgap, and hence a larger forward drop, than silicon – see Figure 2.8 on page 76.[31] Panel-mount LEDs come mostly in 3 mm and 5 mm diameters (called T-1 and T-1$\frac{3}{4}$, respectively), sometimes with an integral mounting clip (see Figure 12.58). Typical flooded panel-type LED indicators look good at 4–10 mA forward current; on a board inside an instrument you can usually get away with 1 mA.

Figure 12.61 shows simple ways to drive small LED indicators. Basically you just need to provide a few milliamps of operating current through their forward voltage drop (V_F). That's usually just a series current-limiting resistor of value $R = (V_+ - V_F)/I_{LED}$, typically a few hundred ohms to a few kΩ. Some LED indicators come with internal current-limiting resistors (or even internal constant-

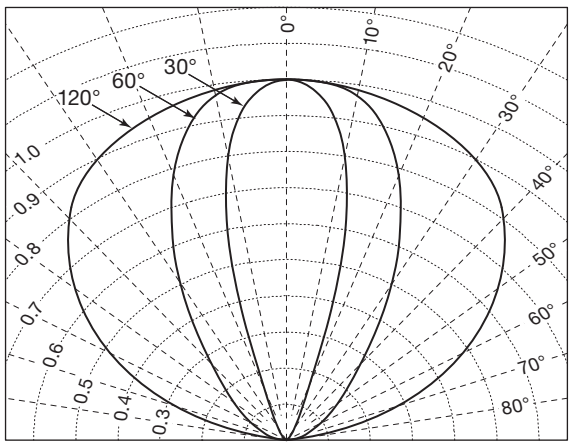


Figure 12.59. Intensity versus angle for three panel-mount LEDs with specified "viewing angles" (full-width to half-intensity) of 30°, 60°, and 120°, in each case "normalized" to unit intensity on-axis. *(After Vishay TLHx460, 520, and 640 series datasheets)*

[31] As an example, for the panel LEDs whose illumination patterns are plotted in Figure 12.59, the dependence of forward voltage V_F on wavelength is given approximately by $V_F(\text{volts}) \approx 1000/\lambda + 0.02 I_F$, where the wavelength λ is in nanometers, and the forward current I_F is in milliamps. The last term represents an effective series resistance of 20 Ω.

OPTOELECTRONICS FAMILY TREE

I. EMITTERS

LEDs (Light-Emitting Diodes) visible (red, yellow, green, blue, white) and infrared (IR); forward-biased diode, V_F ranges from ∼1–3.5 V, depending on color; panel-mount and PCB-mount; many configurations; available as displays

Laser diodes IR, red, and blue; fiber-optic transmitters, laser pointers, CD/DVD/Blu-ray players, barcode readers

Electroluminescent Night lights, "Indiglo"™ low-power backlight

II. DISPLAYS

LED-based 7-segment (numeric), dot-matrix (character), and "smart" (decoded–latched); organized as individual characters or as arrays ("sticks")

LCD-based (Liquid-Crystal Display) Bare LCD (standard or custom), or "smart" (decoded with memory; parallel and/or serial interface); character only, character plus configurable graphics, or full graphics; backlit transmissive, or "transflective"; varying quality (view angle and contrast)

VFD-based (Vacuum-Fluorescent Display) Smart LCD emulation, with superior readability; custom configurations for high-volume users

OLED (Organic LED) Inexpensive alternative to semiconductor LEDs; graphics, cellphone screens etc; larger sizes for best flat-screen TV

Electronic paper For example, E-Ink® microcapsule technology used in e-book readers; image retention at zero power except during erase–rewrite

III. DETECTORS

Photodiode *pn* (or PIN, positive–intrinsic–negative) diode junction; self-generated photocurrent into short ckt ("photovoltaic" mode), or when back-biased ("photoconductive" mode); fiber-optic receivers (speeds to gigabits/s); solar cells are large-area photodiodes

Arrays Linear (stripe); quad; proportional readout; full imaging array (CCD; CMOS)

Integrated Light → logic; light → voltage; light → current; light → frequency; synchronous detection pair

Phototransistor Transistor with base–emitter photodiode; higher current (factor of β), but slower; photo-Darlington even more so

Photoresistor Light-sensitive linear-resistive material (e.g., cadmium sulfide); slow response

Bolometric "Pyroelectric" material exhibiting large resistance change with temperature; motion detectors ("PIR", passive infrared)

APD (avalanche photodiode) High back-bias (∼100 V) multiplies the collected charge per photon; can be linear, or saturating ("Geiger mode")

PMT (photomultiplier) Vacuum-tube device with a photocathode and array of electron-multiplying (gain∼10^6) "dynodes"; operates at ∼1 kV

HAPD (hybrid APD) Vacuum-tube device, combination of photocathode and APD target; operates at >5 kV

Pixelated PMT Multi-anode PMT for coarse imaging; 4×4, 8×8 array

Microchannel plate Vacuum-tube device, combination of photocathode and electron-multiplying capillary array; an "imaging PMT"

IV. COUPLERS

LED-input LED→photodiode; LED→phototransistor; LED→photo-Darlington; LED→FET (via PV stack, "PV" = photovoltaic); LED→photoresistor; LED→SCR/triac (via PV stack): a "solid-state relay"; LED→logic output (active pullup, or open-collector)

Logic input Logic input→logic output

Interrupters With gap, or reflective

V. OTHER

Proximity and ranging Emitter plus position- or intensity-sensing detector: faucets, paper towel dispensers, LCD monitor power-down… and your iPhone (it needs to know that your cheek isn't trying to control the touch-screen)

Barcode reader

Optical mouse LED or laser emitter, plus smart detector

current circuits) – with these you omit the external resistor. For higher currents (large LEDs) use a transistor switch, as in Figures 12.61C–F.

We've listed some of our favorite panel LED indicators in Table 12.6. You can see them in the front row of Figure 12.58.

With the trend toward lower-voltage logic, along with the higher forward voltage of gallium nitride type LEDs

Table 12.6 Selected Panel-mount LEDs[a]

	LED only (bare wire leads)		**Integral mount, front entry**[b]			
			short bare leads		flying leads[c]	
Color	3mm (T-1)	5mm (T-1 3/4)	3mm	5mm	3mm	5mm
RED	TLHR4605	LX5093SRD/D	5111F1	5101H1	5110F1	5100H1
YELLOW	TLHY4405	LX5093LYD	5111F7	5101H7	5110F7	5100H7
GREEN	TLHG4605	TLHG6405	5111F5	5101H5	5110F5	5100H5
BLUE	LX3044USBD	LX5093USBD				

Notes: (a) all are diffused, without internal resistor. manufacturers: 51xx = CML; TLHxxx = Vishay; LXxxx = Lumex, with prefix SSL-. (b) Front-mounting: T-1 size take 5/32" (4.0 mm) hole, T-1 3/4 size take 1/4" (6.4 mm) hole. (c) Insulated, 24 gauge stranded, 15 cm long.

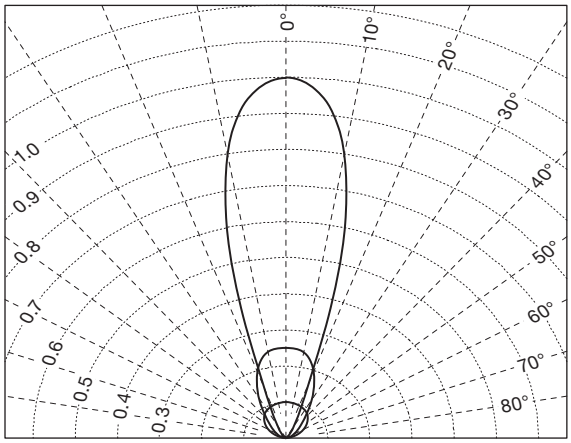


Figure 12.60. There's a price to pay in exchange for the pleasant appearance of a diffused ("flooded") LED, with its wide viewing angle. Here we've plotted intensity versus angle (relative to that of the 30° unit) for the same panel LEDs as in Figure 12.59, using datasheet values. It isn't quite as bad as it seems, though, owing to the eye's logarithmic-like sensitivity.

(used in blue, white, and bright-green LEDs – see Figures 2.8 and *1x.48*), you'll sometimes need a higher supply voltage. For example, +3.3 V is a popular supply voltage, but insufficient to drive a GaN-type LED. If you have a higher supply voltage (e.g., +5 V) available, use a transistor switch with pullup to that voltage (Figures 12.61E and F). If you don't, you'll need to generate the needed LED supply voltage. Figure 12.62 shows a few methods.

A favorite technique, particularly for driving several LEDs at once (e.g., a string of four or six white LEDs for backlighting) from a low supply voltage, is a nonisolated switching boost (step-up) converter, with feedback taken from a current-sensing resistor at the bottom of the LED string (Figure 12.62A). There are dozens of choices from many manufacturers; we've listed a few in the figure. These things will run on voltages as low as +1 V (i.e., a single cell), putting out tens of milliamps into 20 V or more, as needed to provide the LED operating current $I_{LED} = V_{ref}/R_{CS}$; and they're pretty efficient, typically 80% or more. Some variations include high-side sensing, internal current sensing (Figure 12.62B), linear intensity control, internal Schottky diode, and internal zener clamp.[32] Note well the latter: a zener clamp (D_Z in the figure) is *mandatory* if there's a way that the LED load might be disconnected; otherwise the output voltage runs away and destroys the IC.

You can get tiny individual LEDs, and arrays of LEDs – sticks of 2, 4, or 10 LEDs in a row – designed for PCB mounting. The latter are actually intended for linear "bar-graph" readouts. They come in upright or right-angle mounting. You can also get panel-mounting indicators with multiple colored LEDs in one uncolored package: red/green is inexpensive and common; you can also get red/blue/green and yellow/blue/green.[33] These make an impressive panel, with lights changing color to indicate good or bad conditions. When you turn on more than one LED you get "additive color," so, for example, you can generate *yellow* as red plus green, or *white* as red plus blue plus green. You can do this by switching each

[32] In addition to these *boost* (step-up) converters, you can get *buck* (step-down) converters for LED driving from a higher voltage supply; and you can get *buck–boost* converters that work with supply voltages higher or lower than the output voltage driving the LED load. You'll find more than a thousand varieties on the DigiKey website alone.

[33] Multicolor LEDs that include blue cost significantly more, and require the ∼3.5 V forward voltage characteristic of GaN-type LEDs.

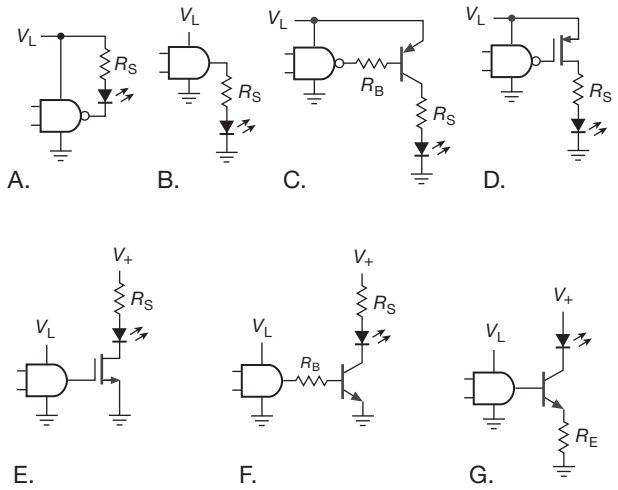


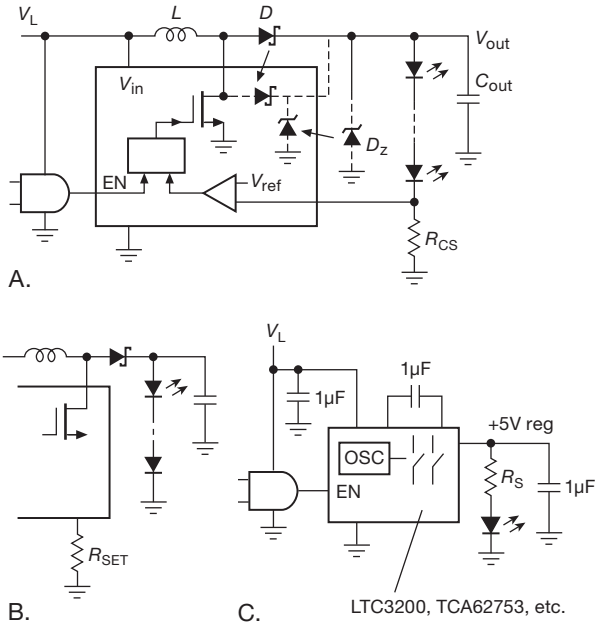
Figure 12.61. Driving LED indicators from logic – I: for LEDs with forward voltage V_F less than the logic supply voltage V_L, you can use the simple connections in circuits A and B to supply a few milliamps. For higher currents, use a transistor switch, as in circuits C–F (with V_+ tied to V_L). If you have a higher voltage available (e.g., +5 V) you can use circuits E–G to drive LEDs whose forward voltage is greater than the logic supply (e.g., a white LED from +2.5 V logic). Circuit G provides constant LED current, providing $V_+ \geq V_L + V_F$.

LED color fully ON or OFF; or you can produce intermediate levels of drive to each LED by varying the duty cycle (percentage of time ON) of a switching waveform (PWM). Multicolor LEDs come in both common-cathode (the − terminal) and common-anode (the + terminal) configurations; two-color LEDs also come as back-to-back diodes in a 2-pin package.

A. Example: "Superlamp"

Thanks to Shuji Nakamura's enabling breakthrough in GaN technology, white LEDs have reached impressive intensities and efficiencies, and they're becoming the preferred alternatives to incandescent and fluorescent lights for "area lighting" applications. We had a leftover Intel CPU heatsink, and we got to thinking, "'Hmmm, what hot thing can we stick onto *that*?" Nice bright white LEDs, of course!

Figure 12.63 shows the circuit: a CMOS 555 oscillator running at 28 kHz (chosen to be above audibility) and configured for a 0–100% duty cycle (see §7.1.3B) is used to switch a power MOSFET, generating a 12 V rectangular wave. This drives four strings of three white LEDs each (Philips Lumileds "Luxeon Star," mounted three-up on hexagonal metal heat spreaders) with 2 Ω current-limiting



part #	V_{in}[a] (V)	I_{out}[b] max (mA)	C_{IN}/C_{OUT} min (µF)	V_{ref} (V)	D_z max (V)	D	$ea (25pc)	notes
FAN5331	2.7–5.5	35	4.7[c]/1.6	1.22	20	ext	0.67	
TPS61041	1.8–6	30	4.7[c]/1	1.22	24	ext	1.10	
LT1937	2.5–10	20	1/0.22	0.095	24	ext	2.70	
LT3465	2.7–16	20	1/0.22	0.20	30[d]	int	3.00	lin.ctrl
LT3491	2.5–12	20	1/1	0.20[e]	27[d]	int	2.70	lin.ctrl
LT1932	1–10	30	2.2/1	int	32	ext	3.72	R_{set}

(a) V_{in} must be less than V_{out}. (b) at V_{in} (min). (c) typical. (d) internal. (e) R_S on high side, LED(s) return to GND.

Figure 12.62. Driving LED indicators from logic – II: there are dozens (maybe hundreds) of switching boost converters designed specifically for constant-current driving of strings of LEDs from low supply voltages, as in circuits A and B. For a single LED you can use an inductorless (charge-pump) regulated doubler (circuit C).

resistors, generating an ON current of ∼700 mA per string. A couple of design details. (a) The oscillator frequency is high enough to generate a modest amount of "switching loss" during the intervals when the MOSFET is transitioning between fully ON and fully OFF; adding a MOSFET gate buffer (like the TC4420) reduces the switching time, hence the loss, if that were a serious issue (it isn't). (b) The use of a current-limiting series resistor (rather than a current source) in each LED string has an interesting side effect – it cancels (approximately) the reduction of light output as the LED heats up (if anyone cares); that's because the LED's lower forward voltage (at higher temperatures)

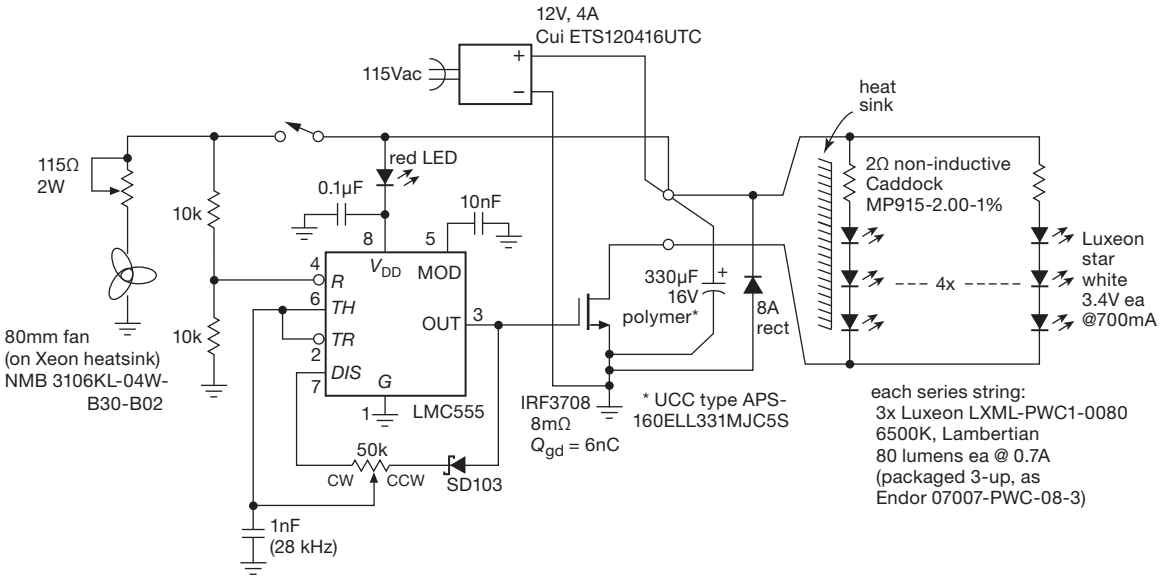


Figure 12.63. Our old friend, the 555, is at the heart of this bright desk lamp with 0–100% PWM dimming.

causes more drop across the resistors, hence a compensating rise in LED current. The compensation is nearly perfect for the values shown.

Visitors are amused by this off-the-beaten-path "desk lamp." One of them pointed out that our creation (Figure 12.64), though charming, is completely outclassed by the brilliant "light engines" from Enfis Limited; they make dense LED arrays (e.g., the "quattro mini") that put out as much as 5000 lumens of white light from 144 power LEDs packed into a 2 cm × 2 cm array (and their other colors put out as much as 24 *watts* of light from the same area[34]). We're loyal to our re-educated heatsink. And the ∼30 W of dissipated power is child's play for this Xeon heatsink, which was intended to handle four times as much in its regular day job; we substituted a much slower fan, and throttled it down to a bare whisper.

B. LED fast high-current driver

You can drive an LED with brief very high current pulses (far above its continuous rating), for example in a flashlamp replacement application, exploiting the LED's thermal mass to absorb the energy. For this purpose you can use a series resistor to set the current (Figures 12.62 and 12.63), but an active current regulator is better. An example is LTC's LT3743, showing off its prowess in Figure 12.65, where it is being instructed to make steps to 2 A and 20 A. Even though the PWM is running at 500 kHz, it's able to switch the LED current in just 2μs. Amazing... check it out.

12.5.2 Laser diodes

We make a brief digression on semiconductor diode lasers. These are the heart of the ubiquitous handheld laser pointers, objects once exotic and expensive, but now commonplace and ridiculously inexpensive ($5 or less for a red pointer). They are also ubiquitous in optical storage devices such as CD, DVD, and Blu-ray players and recorders, as well as in fiber optic transmitters. Low-power red lasers come as little modules with circuitry to run on 3–5 V (e.g., Quarton VLM-650-03-LPA); or, for the adventurous, you can get the bare laser diode in a metal 3-lead transistor package (see Figure 12.58). These include an integral monitor photodiode, which is used as feedback to regulate the light output via the applied drive current. The threshold current at which laser action starts is somewhat variable, generally in the neighborhood of 20–40 mA; above that threshold the light output rises rapidly. This behavior is shown in Figure 12.66, a 'scope trace of emitted intensity versus linearly ramping laser-diode current. The output beam emerges directly from a tiny spot on the

[34] At a somewhat more mundane level, high intensity white LEDs are replacing residential incandescent bulbs, floodlights, and spotlights, delivering comparable illumination at 1/5 the ac wallplug power, and lifetimes of 25,000+ hours; many of these are selling for around $10. And the automotive industry has taken notice, using LEDs even for headlights (check out the Lumileds Altilon series).

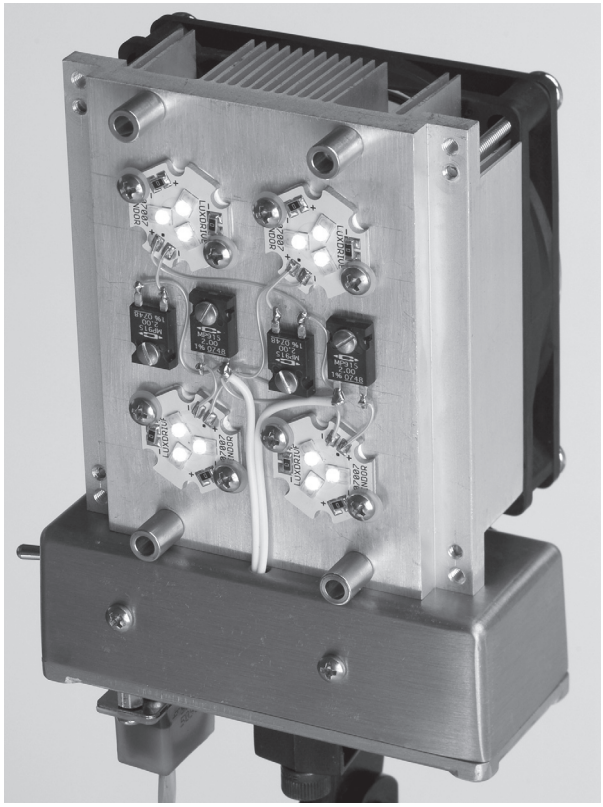


Figure 12.64. Computer CPU heatsink hijacked to make cool desk lamp. We cranked down the current so you can see the four sets of LED triples, along with the surface-mount ballasting resistors. At full brightness (1000 lumens) you can't look at this thing! The switching driver circuitry of Figure 12.63 lives in the box at bottom.

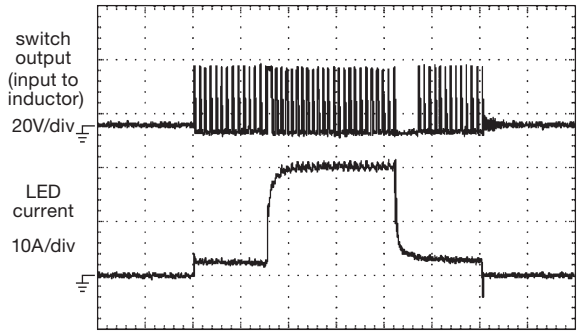


Figure 12.65. The LT3743 high-current LED driver uses cleverness to allow 2μs switching over a 3000:1 dimming range. Horizontal: 20μs/div. (adapted from LT3743 datasheet)

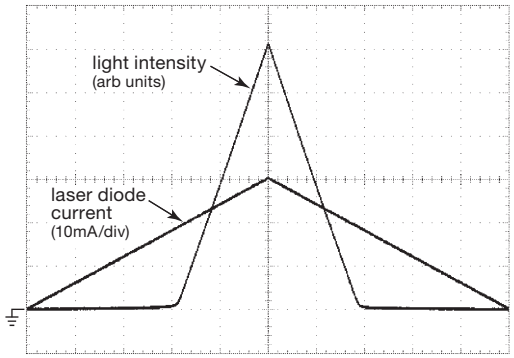


Figure 12.66. Diode laser output from a 0–30 mA triangular drive current. Horizontal: 100 μs/div.

light-emitting semiconductor (GaAlAs for infrared) chip, diverging with an angle of about 10–20°; it can be collimated with a lens to form a parallel beam or a very small focal spot.

It's easy to rig up a decent drive circuit, without even resorting to op-amps or special driver circuits. Just compare the monitor photodiode current with a threshold, and adjust the drive current accordingly. It's a good idea to include a current limit, so you don't blow out the diode; and you'll need to compensate the feedback loop to make it stable. Figure 12.67 shows a "good-enough" driver circuit: it converts the monitor current to a proportional voltage (switchable between two levels by Q_3), compared with a diode-compensated 1.2 V reference voltage. Q_5 drives the laser, with current limiting via Q_4. C_C and R_C compensate the loop, with final "cut-and-try" values as shown. This simple circuit runs fine from dc up to 1 Mb/s.

For such ON/OFF control you hardly need the whole loop business. But it's necessary if you want true *linear* control of laser intensity. Figure 12.68 shows waveforms for this same circuit, with an *npn* current sink attached at point X (replacing R_3, R_4, and Q_3). Here the programming current I_{prog} is a 200 kHz triangle wave with a 4:1 ratio of peak-to-valley current; the laser output is accurately proportional. Note that the laser drive current has a large offset (all three traces are positioned relative to the same ground).

This circuit is a simple example, and not intended for high performance. The real challenges come when you're trying to modulate a laser at hundreds of megabits per second or more. For this it's best to use an IC designed for fast laser modulation;[35] better yet, just buy the laser+driver module as a complete unit (see §12.8 on fiber optics for some examples).

[35] For example, the Maxim MAX3735-series, Micrel SY88722, or ADI ADN2870-series. The MAX3975-series and 3930-32-series go to 10 Gb/s; get their *Fiber Design Guide* at `www.maxim-ic.com/design_guides/`.

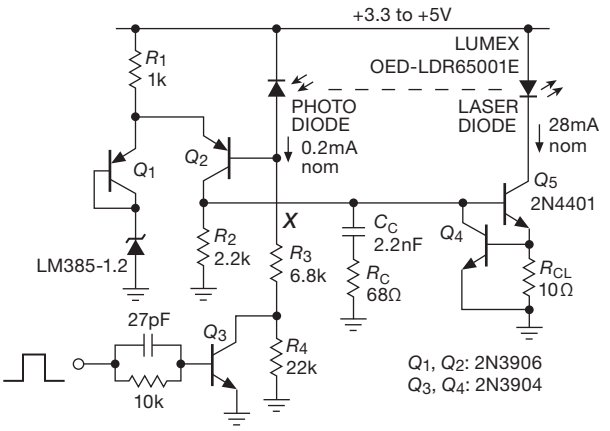


Figure 12.67. Simple laser drive circuit with monitor photodiode feedback and logic-level control.

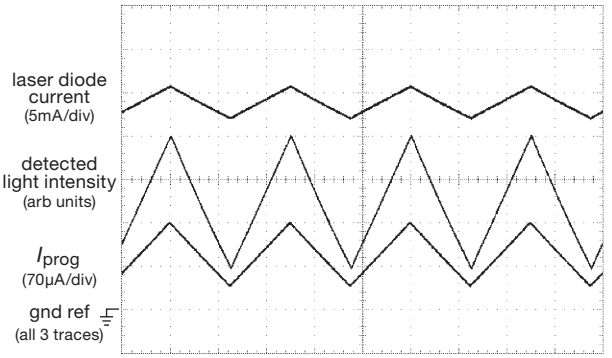


Figure 12.68. Linearizing the circuit of Figure 12.67 by replacing the components below "⊗" with an *npn* current sink: here a 200 kHz triangular programming current with 4:1 peak-to-valley ratio causes the offset laser current needed to produce an accurately proportional output intensity. Horizontal: 2 μs/div.

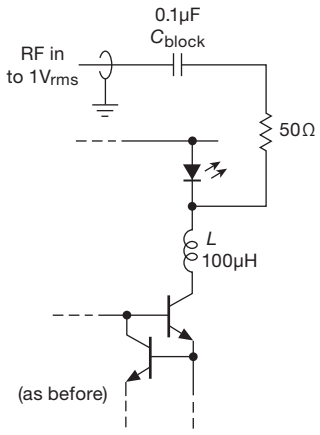


Figure 12.69. Adding a blocking capacitor, 50 Ω series resistor, and RF-isolating inductor to the laser drive circuit of Figure 12.67 provides a simple modulating input. We've used this circuit successfully with modulating frequencies up to ~1 GHz.

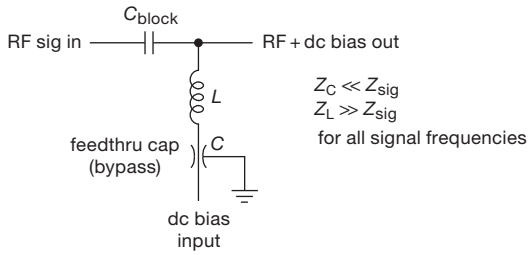


Figure 12.70. A "bias tee" lets you put a dc bias onto a signal (usually at radiofrequencies) that is coupled from input to output. Off-the-shelf bias tees are usually intended for line impedances of 50 Ω or 75 Ω, and they use careful inductor design for good performance over a wide frequency range (0.2 MHz to 12 GHz for the Minicircuits ZX85-12G, for example).

Figure 12.69 shows a simple alternative, if all you want is intensity modulation of a laser light source at some high frequency and you don't care much about linearity or response down to dc. The input signal, blocked by C_{block}, superposes a high-frequency variation of current at the laser diode, dc biased by the feedback circuit of Figure 12.67. Choose the blocking capacitor for the minimum modulating frequency (the value shown is good down to ~100 kHz), and be sure to bypass the supply rail effectively. The inductor L isolates the transistor's shunt capacitance, and should have a reactance greater than 50 Ω at all modulating frequencies. For good performance the inductor should be constructed from multiple series sections, cleverly designed (in the manner of a "bias tee," Figure 12.70) so that self-resonances do not produce low-impedance minima across the frequency range of interest.[36]

12.5.3 Displays

A *display* is an optoelectronic device that can show one or more numbers ("numeric" display); or the set of hexadecimal digits (0–9 and a–f, "hexadecimal display"); or any combination of letters, numbers, and punctuation ("alphanumeric display"); or most generally any graphic that can be represented by a dot matrix ("graphic display").

[36] Instead of suppressing resonances by dividing such an inductor into multiple sections, you can wind it in a *tapered* geometry; see, for example, the GL series of "Ultra Broad Band Inductors" from AVX. These have excellent performance from 1 MHz to 40 GHz.

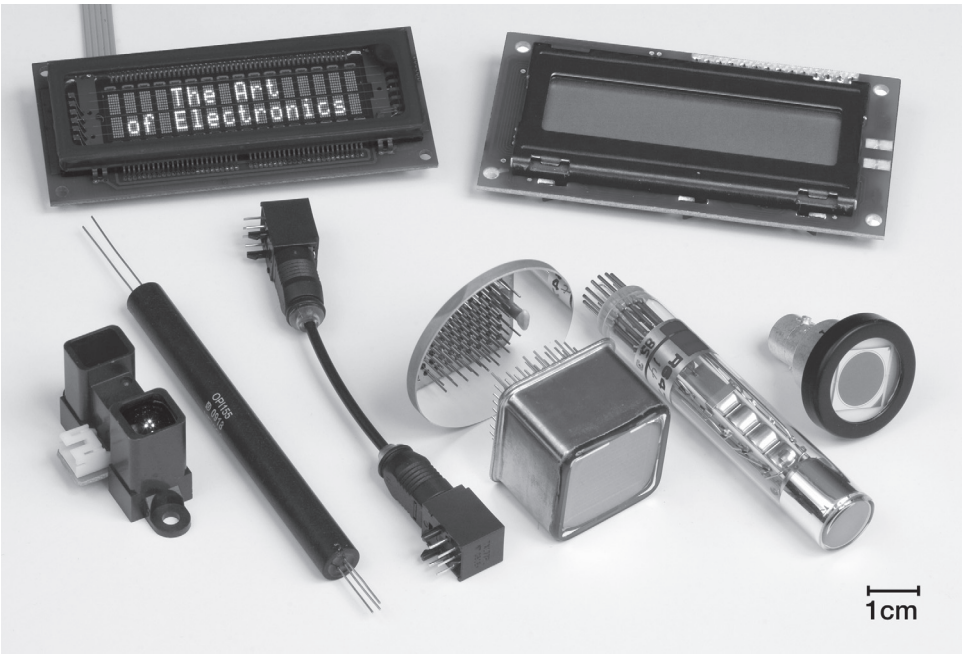


Figure 12.71. A collection of optoelectronic devices too large to fit in Figures 12.58, 12.80, and 12.84. At rear are VFD and LCD smart displays (Newhaven M0216SD-162SDARC-1 and Optrex DMC16207), the former with 3-wire serial interface. In front, left to right, are a ranging/proximity detector (Sharp GP2Y0D02YK0F), a 50 kV logic-output opto-isolator (Optek OPI155), a homemade TOSLINK isolator (Toshiba TOTX/TORX177PL), an 8×8 multianode photomultiplier (Hamamatsu R5900-00-M64), an end-on 13 mm PMT (Hamamatsu R647), and a 10 mm PIN-diode detector (OSI/UDT PIN-10D).

The dominant display technologies (in sizes relevant to electronic instruments) are LEDs, LCDs, VFDs (vacuum fluorescent displays), and OLEDs (organic LEDs). Figures 12.58 and 12.71 includes examples of all except OLED.

LED displays are bright, colorful, and available in large sizes; but they are power hungry, and not appropriate for graphics. LCD displays are quite popular – they are the monochrome yellowish or bluish backlit rectangular displays that commonly show one or two lines of 16 or 20 characters each. Depending on technology, backlit LCDs may be quite readable outdoors or in high ambient-lighting conditions ("transflective"), or they may look just terrible. LCDs intended for use without back-lighting (reflective) look pretty good. The clarity and off-axis readability depend considerably also on the liquid-crystal technology (active matrix versus passive matrix; "twisted nematic" versus "super-twisted nematic," and so on). Apart from the backlight, LCD displays can run on extremely low power (think of a digital wristwatch); and you can have them made with custom shapes and symbols. VFD displays are similar to their LCD cousins (and emulate their interface), but they are self-luminous; we think they look terrific.[37] VFD display prices are coming down nicely – for example, the 2-row ×16 character alphanumeric "smart" display in Figure 12.71 costs $30 in single quantities; a bare 1-row ×20 character stick costs $7 in single quantities. OLEDs are making inroads, particularly in small sizes.[38] They are likely to be the large-screen display technology of the future (currently dominated by LCD and plasma), when manufacturing costs decline sufficiently to make them competitive.

[37] Some engineers have complained that they generate RF interference, and they avoid them in proximity to sensitive circuits.

[38] We are particularly impressed by the stunning clarity of the small (∼1″) white OLED graphic modules (available from hobby suppliers like Adafruit, as well as from standard distributors like DigiKey, Mouser, and Newark); they can run from a single +3.3 V supply, drawing some 20 mA, and they communicate via SPI or I^2C serial buses. They cost less than $20 in unit quantities.

A. LED displays

Figure 12.72 shows the choices you have in single-character LED displays (and the same layouts are used in integrated multi-character displays). The original 7-segment display is the simplest and can display the digits 0–9 and the hex extension (A–F), albeit somewhat crudely (the hex letters are displayed as "AbcdEF"). You can get single-character 7-segment displays in many colors and sizes, and in "sticks" containing 2, 3, 4, or 8 characters (generally intended to be "multiplexed" – the characters displayed one at a time in rapid sequence). Single-character displays bring out leads for the 7 segments and the common electrode; the two flavors are thus "common cathode" and "common anode." Multiple-character sticks bring each character's common electrode out, but tie the corresponding segments together, which is what you want for multiplexing. However, if you want to display lots of characters, you're usually better off choosing a so-called "smart" display: these accept input codes for characters (or graphic symbols), and they do their decoding, multiplexing, and displaying internally. We'll get to them shortly, in §12.5.3A.

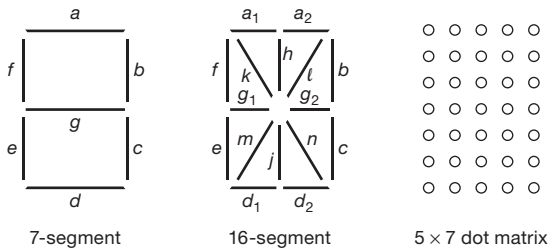


Figure 12.72. LED display layouts.

"Dumb" displays

Figure 12.73 shows how to drive a single-digit 7-segment common-cathode LED display. The 'HC4511 is a "BCD-to-7-segment latch–decoder–driver," able to source about 10 mA while holding its active outputs at +4.5 volts when powered from +5 V. The series resistors limit the segment drive current (to 2 mA or 4.7 mA, respectively, with 3 V or 5 V supply, assuming a typical LED drop of 1.5 V). Don't try to be clever and just stick a single resistor in the common cathode instead (why not?). You can get arrays of equal-value resistors in convenient SIPs (single in-line packages), in through-hole or surface-mount flavors.

You need only a single decoder–driver chip, even if you are displaying multiple digits, as long as you multiplex the display, i.e., illuminate only one displayed digit at a time. You do this by connecting the segment-driver outputs to all of the digits, then grounding the cathode of each digit in turn, simultaneously asserting the value to be displayed on decoder inputs D_0–D_3. Look back at Figure 10.90 (§10.6.2) to see how it's done in detail.

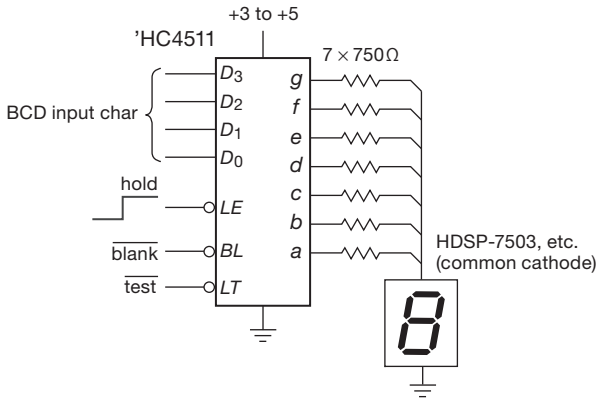


Figure 12.73. Driving a single 7-segment LED display from BCD (0–9) logic input.

Figures 12.74 and 12.75 show how to drive single-character LED displays in 16-segment or 5×7 configurations, respectively. Drivers for these displays generally assume you've got a microcontroller in your circuit somewhere, so they use serial input protocols such as SPI (serial peripheral interface; see §14.7.1). They include internal data registers, and current-source drivers programmed by a single external resistor. Note that the 16-segment driver in Figure 12.74 has no internal decoder – it just drives the segments you tell it to. So your microcontroller has to figure out which segments to drive; that's no big deal for these brainy parts, just be aware. The (smarter) 5×7 driver in

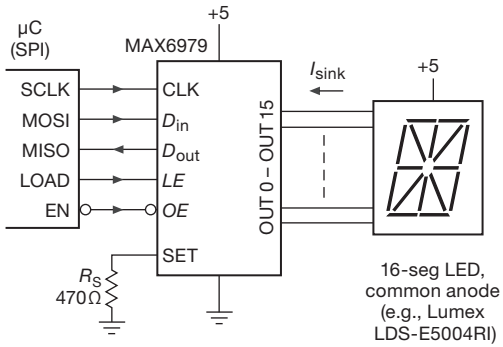


Figure 12.74. Driving a single 16-segment LED display from a serial (SPI) data input. There's no pre-stored character table in this particular driver chip; instead, you get to make up your own symbols, sending a HIGH or LOW bit for each segment.

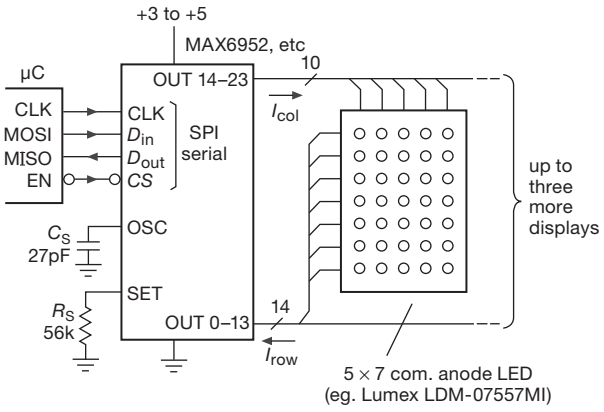


Figure 12.75. Driving up to four dot-matrix 5×7 LED displays from a serial data input (which can also control brightness and blinking). This driver has 104 predefined characters stored internally, and lets you create 24 more of your own.

Figure 12.75 *does* include an internal decoder with predefined character memory.

"Smart" displays

Except in high-volume applications that are cost-sensitive, it's usually a better idea to choose a display that integrates an array of characters (or graphics), with the decoders and drivers included. These come with LEDs, LCDs, VFDs, or OLEDs as the active display.

Single character LED The original smart display device is probably the single-character 5×7-dot LED, which delighted us back in the 1970s: you present it with a 4-bit code, and you can *see* (and latch) the result (Figure 12.76). These were made by HP and TI, and we liked them so much that we incorporated them in our electronic circuits lab course, confident that their ∼$15 single-piece price would drop nicely over time. It didn't work out that way, and these puppies (still made by Avago Technologies, the optoelectronics spinoff of HP) now cost double that price in single quantities.

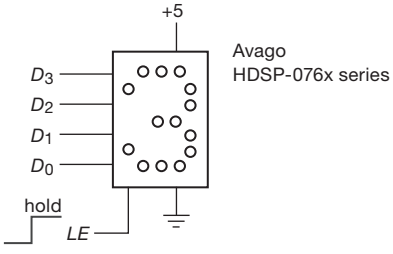


Figure 12.76. LED hexadecimal display with integral latch-decoder-driver. These are beautiful, and colorful, but expensive.

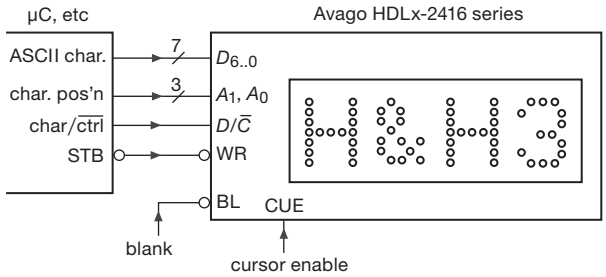


Figure 12.77. For half the price of Figure 12.76 you get a four-character smart LED display "stick." And you can display 128 ASCII characters with its built-in font ROM.

Several character LED Usually you want to display a few characters, at least: so there are handsome four-character smart display sticks (in the usual LED colors) that are as easy to use as the single-character display. Figure 12.77 shows one of these, originated by HP (now Avago) and available for about $15. These display a full character set (7-bit ASCII: upper-case and lower-case alphabets, numbers, and symbols), latched with a WR′ pulse and sent to the character position given by a 2-bit address (a simplified form of "parallel bus interface" that we'll learn about in Chapters 14 and 15). There are additional control functions (dimming, cursor), written by holding LOW the input D/C′ (which the datasheet oddly calls CU′, for cursor "select").

Many character LCD/VFD/OLED As we remarked earlier, the best way to display one or several lines of text is to use an intelligent (smarter than smart) LCD, VFD, or OLED "character" module (Figure 12.78). There's lots of competition,[39] with prices beginning at $10 or less (for a 16×2-character backlit LCD). They come in standard configurations of 1, 2, or 4 rows, with 16, 20, or 40 characters per row. They include built-in character sets, and many of them let you supplement that with a few (usually eight) custom characters. Alternatively, you can choose a full graphic version, consisting of a dot matrix (e.g., 64×260), which you program as a bitmap. That can be a chore if you want to display mostly text, so there are combination text-plus-graphics units that include a pre-stored character set in

[39] Here's an alphabetical list from our favorite suppliers (DigiKey, Mouser, Newark): 4D Systems, Adafruit, AND Displays, Batron, Displaytech, Electronic Assembly, Everbouquet, Hantronix, Lumex, Matrix Orbital, Microtips, Newhaven Display, Optrex, Powertip, Trident, Varitronix, and Vishay/Dale.

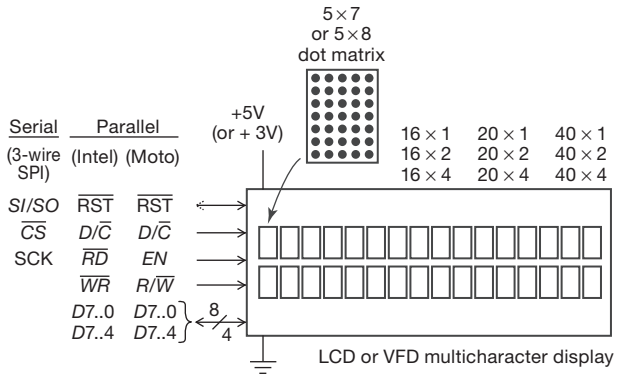


Figure 12.78. Better still, ride the popular wave with a smart LCD or VFD display stick. The character arrangements shown are standard. And some (as here) have both parallel and serial interfaces.

addition to full graphics capability. Examples of these, chosen from one company's offerings (Matrix Orbital, see Figure 12.79), are the LK162-12 (pre-stored text, plus eight user-defined symbols), the GLK24064-25 (full graphics, no built-in fonts), and GLK12232-25-SM (choice of prestored text or full set of user-defined symbols, plus full graphics).

To fully appreciate what these modules offer, it's important to understand that LCDs must be driven by an *ac* waveform; otherwise their liquid guts are ruined. So LCD driver chips usually have some way to generate a square-wave segment drive, synchronized to the LCD backplane waveform. An example is the 'HC4543, the LCD cousin of the 'HC4511 LED 7-segment latch–decoder–driver. Another complication, of course, is the need to continually drive the large array of dots with the pattern to be displayed. Add to that the need for an interface that lets you change individual displayed characters on command, or to scroll the display, or advance a cursor, and so on, and you can understand how complicated things can get. Fortunately, the manufacturers understand this, and therefore they provide complete displays that are more than intelligent; they're positively at the genius level.

These displays are invariably used in conjunction with a microcontroller, which communicates via a simple interface. The original LCD smart display sticks use a simple parallel interface, with 8 bits of data (either the character or control information, depending on a data/control′ line: D/C′ in the figure, but traditionally called "RS") and a few control lines.[40] Most now offer a 3-wire or 4-wire serial bus alternative, or provide both options (jumper selectable)

[40] They allow a "nybble" mode, in which 8-bit data is sent as two successive 4-bit quantities.

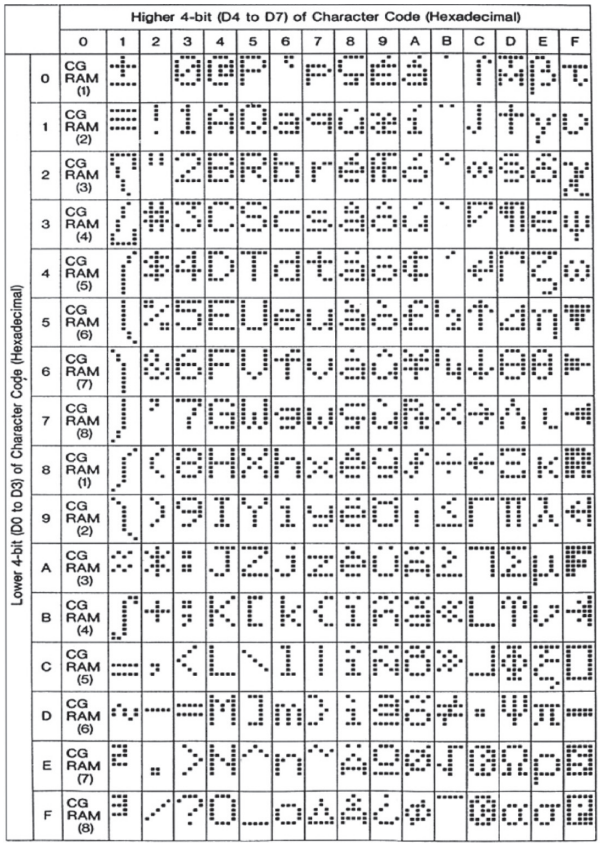


Figure 12.79. Display codes for an LCD alphanumeric "smart display" (Matrix Orbital LK-series) with a standard character set plus eight user-configurable characters (CG RAM). Reproduced with permission of Matrix Orbital.

on the same display, as in Figure 12.78 (which even offers a choice of "Intel" or "Motorola" parallel protocols, §14.3).

We like these displays a lot; you can see both LCD and VFD types in the photograph (Figure 12.71), the latter expressing fluorescent joy in the completion of this weighty tome.

12.6 Optoelectronics: detectors

Figure 12.80 shows a selection of detectors, most of which are variations on a *photodiode* or *phototransistor*. We've seen these earlier, in the op-amp chapter (§4.3.1C), where we showed how to use a simple op-amp current-to-voltage ("transimpedance") circuit to convert the photocurrent to a proportional output voltage. In §8.11 we dealt in detail with the design of exceptionally fast and quiet photodiode amplifiers. And in §*4x.3* there's a discussion of the thorny problem of stability in transimpedance amplifiers, taking

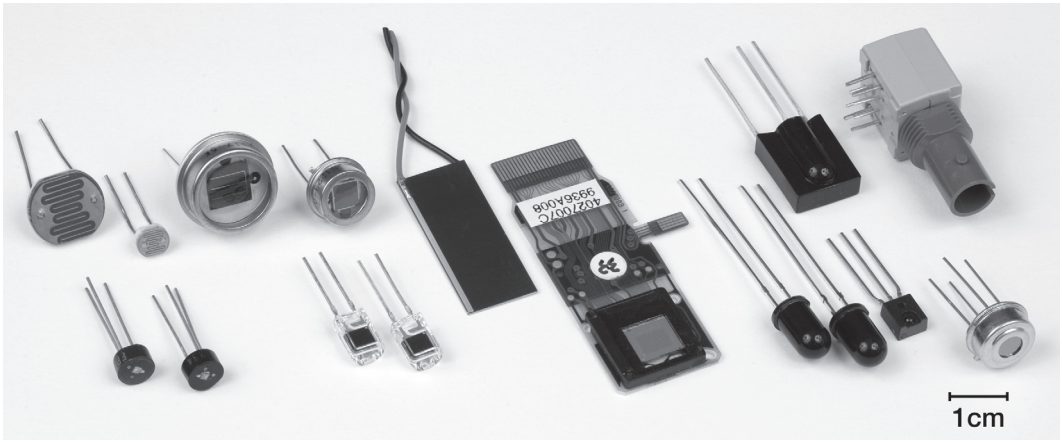


Figure 12.80. Optoelectronics: detectors. At left rear are a pair of cadmium sulfide photoresistive sensors, and a pair of photodiodes (GaAs, silicon) in hermetically sealed metal cases. Below them are plastic-case photodiodes (top- and side-looking), and to their right is a bare photodiode and a small CMOS image sensor (with attached flexible circuit connector) of the sort used in webcams and cellphones. The four opaque black objects toward the right are IR-detectors (filtered to eliminate interfering visible light); the small square one generates a logic-level output, and the large square one is used as a frequency selective (30–56 kHz) "clicker" receiver in consumer audio/video gear. At top right is an ST-style fiber-optic receiver, and below it is a pyroelectric thermal-IR sensor used in PIR motion detectors (featured in the geek movie classic "Sneakers").

into consideration the destabilizing effects of input capacitance.

Soon we'll see the ubiquitous use of photodetectors in the context of optocouplers (also called opto-isolators); and a bit later we'll see them again in connection with fiber optics. Here we simply summarize the earlier photodiode and phototransistor circuits (Figure 12.81), along with some nice parts you can get that integrate a photodiode with additional circuitry to produce either a digital-logic output, or a proportional output in the form of a voltage, a current, or a frequency.

This is a good place to say something also about *photomultipliers*: these sensitive and fast light detectors (two are shown in Figure 12.71) use a cascade of electron-multiplying electrodes (dynodes) to convert a single photoelectron (released, with about 20% probability, when a photon of light strikes the sensitive photocathode) into a fast ($\sim$ns) pulse of 10^5–10^6 electrons. The resultant current pulse has thus been amplified sufficiently before reaching any circuitry, so that amplifier noise is not a problem: a million electrons in a nanosecond is nearly 0.2 mA!

12.6.1 Photodiodes and phototransistors

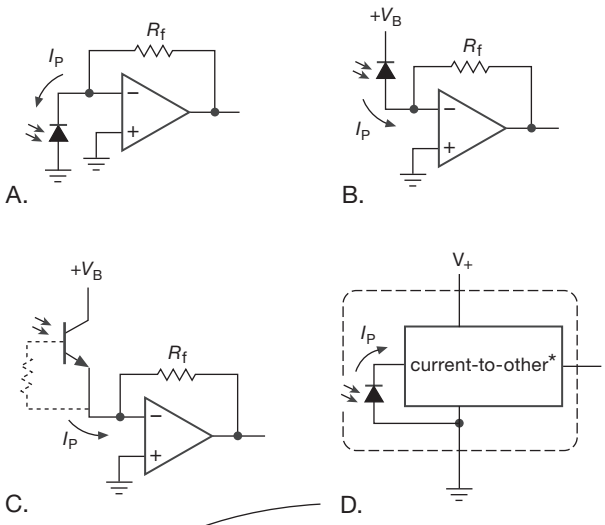
Figure 12.81 reviews the standard ways that photodiode and phototransistors are used. In circuit A the photodiode operates in *photovoltaic mode*; that is, generating a photocurrent into a short circuit. The transimpedance amplifier generates a positive output voltage ($V_{\text{out}} = R_{\text{f}} I_{\text{P}}$), so you can operate from a single positive supply. You might think the op-amp is unnecessary – after all, Ohm's law gives you the same answer – but you'd be wrong! First, the diode's self-generated photocurrent drops to zero if allowed to develop a diode drop of forward voltage, so you'd have to use a small resistive load, such that the maximum output voltage (with maximum light input) is less than $\sim$0.5 V; and second, the circuit would be slower, set by the time constant of the diode's capacitance and load resistance.

In circuit B the photodiode is back-biased (*photoconductive* mode), which increases the speed by reducing the diode's capacitance and by providing an electric field to sweep away charge. So this circuit is faster; but it's also noisier, and the diode's leakage current limits the low light-level performance. Capacitance can be a real problem in these circuits, especially when the photodiode sits at the far end of a connecting coax cable (§§8.11 and *4x.3*).

Circuit C uses a phototransistor, effectively boosting the photodiode current by the transistor's beta (the back-biased collector–base junction acts as a photodiode, whose photocurrent is multiplied by beta). So a small phototransistor under ordinary room illumination typically gives you currents in the $\sim 100\mu$A range, compared with $\sim 1\mu$A for photodiodes.[41] This configuration is the slowest of all, but

[41] Experimental data: we put an FPT110 phototransistor (lower left in

it can be speeded up by adding a base resistor. That produces a threshold, however, because the photodiode current must develop a diode drop across the base resistor in order to get the current amplification factor of beta. Pushing this further, you can get photo-*Darlington* detectors: lots more gain, but lots slower. Finally, Figure 12.81D shows some integrated possibilities, inexpensive and nicely engineered to save you time and pain.



*Function	Example Part Number, and (mfgr)
$E_V \rightarrow$ logic	QSE156-159 (FSC)
$E_V \rightarrow$ frequency	MLX75304, TSL230-series (Melexis, TAOS)
$E_V \rightarrow$ current	TPS851 (Toshiba)
$E_V \rightarrow$ voltage	OPT101, PNA4603, TSL250-2 etc. (TI, Panasonic, TAOS)
$E_V \rightarrow$ serial digital	TSL2560-series (TAOS)

Figure 12.81. Photodiode and phototransistor circuits. A. Photovoltaic mode. B. Photoconductive mode. C. Phototransistor. D. Integrated light-to-other modules.

converts the energetic photon into a visible light pulse). There is ample sophistication, if you want to get really serious about these things; but the basic configurations are shown in Figures 12.82 and 12.83.

Simplest is the "photon-counting" configuration (Figure 12.82), in which the current pulse at the anode generates a (negative) fast pulse across the 50 Ω load resistor, amplified by a wideband amplifier of the sort used in RF applications (the UPC2710 costs about $1 and works fine: we've got a thousand of them running right now).

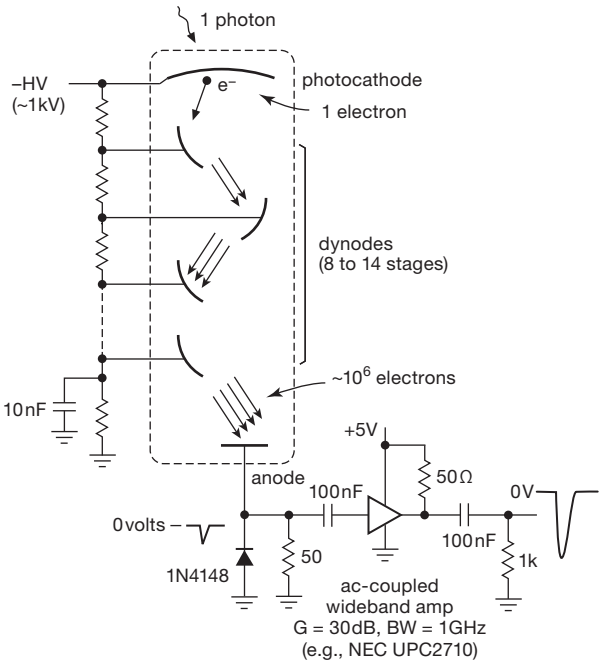


Figure 12.82. Photomultiplier "pulse counting" circuit. The cascade of electron-multiplying dynodes produces a ~ns pulse of $\sim 10^6$ electrons for each detected photon of light, which develops a ~mV negative pulse across 50 Ω.

12.6.2 Photomultipliers

Photomultipliers are favorite detectors for photon-counting applications, low light-level detection and integration, and X-ray or gamma ray detection (via a scintillator, which

If you want instead to detect accurately very low light levels, and don't need nanosecond speed, you probably want a bandwidth-limited "integrating" amplifier like the one shown in Figure 12.83.[42] Here the excellent (and nearly unique) OPA656 wideband low-I_B (JFET) op-amp converts the anode current to a voltage waveform. Even though we're aiming for only a few megahertz of bandwidth, we need a fast op-amp, as explained in §§8.11 and *4x.3*. For reasons of bandwidth and noise, it's especially

Figure 12.80) on our lab bench and measured its photodiode current as I_{CB}=0.4μA, compared with its phototransistor current (with 10 V bias) of I_{CE}=60μA. In hazy sunlight (an August day in Boston) the photodiode current was 250 times higher (100μA). Although the datasheet for this device doesn't tell you the active area, we measured it as 0.7 mm^2. That works out to an efficiency of about 5%, if you were to use the photodiode as a solar cell.

[42] This is similar to the amplifier designed for our colleague Lene Hau, and was used in her pioneering "slow-light" experiment in 1999: 17 m/s (38 mph), slower than a speeding bicycle!

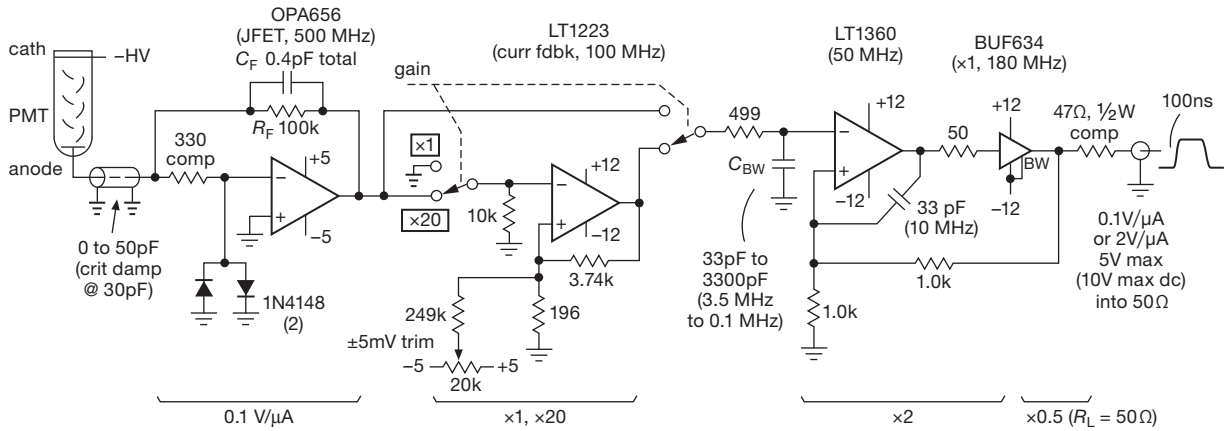


Figure 12.83. Low noise photomultiplier amplifier, with 3.5 MHz bandwidth and 2V/μA gain. See also the discussion in §8.11.

important to keep the input capacitance to a minimum, and to choose the stabilizing feedback capacitance C_F carefully.

The rest of the circuit is pretty straightforward: the series input resistor and clamp diodes protect the op-amp from input spikes, caused, for example, by PMT breakdown (it uses a kilovolt supply). The second stage is an optional $\times 20$ gain boost, for which we used a current-mode op-amp. These have the interesting property that their closed-loop bandwidth is largely independent of closed-loop gain: the ratio of resistors sets the gain (as with the familiar "voltage-mode" op-amp), but the bandwidth is determined by the value of feedback resistor alone. For example, this particular op-amp has 100 MHz bandwidth with a 1k feedback resistor and G_{CL}=1, dropping modestly to 60 MHz for G_{CL}=30; by contrast, a voltage-feedback op-amp's gain would decrease inversely with closed-loop bandwidth. Here we chose a larger feedback resistor (3.74k), deliberately limiting the bandwidth to about 15 MHz, because that's all we need (and extra bandwidth just invites trouble).

This op-amp specifies ± 3 mV (max) offset voltage, which, when combined with the ± 2 mV (max) offset of the input stage, could produce as much as 100 mV of output offset; so we added an offset trim. There's a simple *RC* bandwidth-limiting lowpass on the way to the third stage; this could be elaborated into a switched set of 3 dB breakpoints, if desired. Note however that the maximum bandwidth of $\sim$3.5 MHz is limited by the input-stage rolloff: BW=$(1/2\pi)R_F C_F$. Finally, the output stage consists of a wideband unity-gain power buffer (good to ± 250 mA) enclosed within a gain-of-2 noninverting feedback stage. You need the op-amp, because the buffer alone does not have internal feedback to discipline its output: its offset is specified as ± 100 mV (max). But you have to admire the brute – it's fast, and it's powerful.

12.7 Optocouplers and relays

An LED emitter, combined with a photodetector in close proximity, forms a very useful object known variously as an *optocoupler*, *opto-isolator*, or *photocoupler* (Figure 12.84). In a nutshell, optocouplers let you send digital (and sometimes analog) signals between circuits with separate grounds. This "galvanic isolation" is a good way to prevent ground loops in equipment that drives a remote load. It is essential in circuits that interact with the ac power mains. For example, you might want to turn a heater on and off from a digital signal provided by a microprocessor; in this case you would probably use a solid-state relay, which consists of an LED coupled to a high-current triac or SCR. Most ac-operated switching supplies (e.g., those used in computers, telecommunications, and instrumentation) use optocouplers for the isolated feedback path (see for example Figure 9.83 in §9.8). Similarly, designers of high-voltage power supplies sometimes use optocouplers to get a signal up to a circuit floating at high voltage.

Even in less exotic situations you can take advantage of opto-isolators. For example, an opto-FET lets you switch analog signals with essentially no charge injection (apart from effects of the fractional-pF isolation capacitance); the same goes for sample-and-hold circuits and integrators. Opto-isolators can keep you out of trouble when driving industrial current loops, hammer drivers, etc. Finally, the galvanic isolation of opto-isolators comes in handy in high-precision or low-level circuits: for example, it is difficult to take full advantage of a 16-bit analog-to-digital converter,

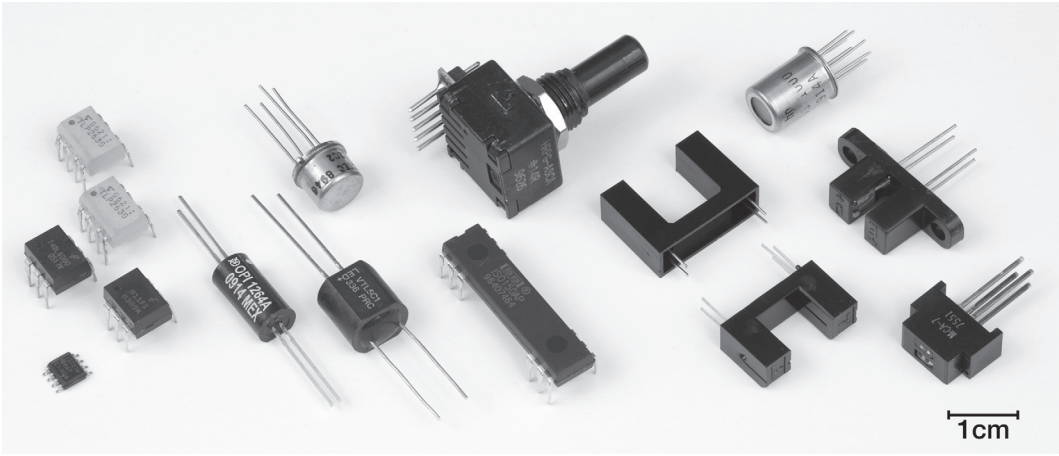


Figure 12.84. Optoelectronics: couplers and interrupters. The five ICs at left are optocouplers, with LED–photodiode pairs and (in some cases) logic inputs and outputs. They are good to several kilovolts or so of isolation, whereas the adjacent cylindrical object with whiskers is rated at 10 kV (and the stretch version in Figure 12.71 is good to 50 kV). The oval package (and metal can above it) are LED–photoresistor couplers, which use a CdS resistive sensor like the ones at top left in Figure 12.80. The stretched IC at center (ISO150) is also a digital isolator, but it uses a *capacitive* coupler; it doesn't belong at this party, but it's too elegant to omit! The three gapped objects to its right are opto-interrupters, and the boxy thing without a gap is a reflective sensor (as is the metal-can barcode reader at top). The panel control at top center is an optical incremental encoder, generating 120 quadrature cycles per turn.

because the digital output signals (and noise on the digital ground to which you connect the converter's output) get back into the analog front end. You can extricate yourself from "noise city" with optical isolation of the digital half. Optocouplers typically provide 2500 volts (rms) isolation, $10^{12}\,\Omega$ insulation resistance, and less than a picofarad coupling between input and output.

There are many varieties of optocouplers, the choice depending on the intended application – for example, the coupling of analog signals, or digital-logic signals, or ac power switching. In the sections below we've sorted these into seven categories, illustrated with examples of some of the more popular (or more interesting) parts currently on the market. The choice of categories is somewhat arbitrary, but it makes sense to us. They are (in order of appearance): I. Phototransistor output; II. Logic output; III. Gate driver output; IV. Analog-oriented; V. Solid-state relay with transistor output; VI. Solid-state relay with triac/SCR output; and VII. ac-input optocouplers.

12.7.1 I: Phototransistor output optocouplers

Figure 12.85 shows an assortment of optocouplers with bipolar transistor output. These are primarily intended for digital logic-level coupling (though it is possible to exploit the configuration in Figure 12.85C to make an approximately linear coupler; see Figures 12.88 and 12.89). The earliest (and simplest) is typified by the 4N35, an LED–phototransistor pair with 40% (min) current transfer ratio (CTR) as a phototransistor, and sluggish 5μs turn-off time (t_{OFF}) into a $100\,\Omega$ load. Circuit A shows how to use it: a gate output and pullup resistor generate current-limited 8 mA drive, and a relatively large output-side collector resistor guarantees saturated switching between logic levels. Note the use of a Schmitt-trigger inverter, a good idea here because of the long switching times. You can get LED-phototransistor pairs with CTRs of 100% or more (e.g., the jellybean CNY17-4, with CTR=160% min), and you can get LED–photo-Darlingtons, as shown; they're even slower than phototransistors! To get improved speed, the manufacturers sometimes use separate photodiode and transistor, as shown in the 6N136 and 6N139 optotransistor and opto-Darlingtons. With optocouplers that provide access to the base, you can add a resistor from base to emitter to improve the speed (configurations B and F); however, this produces a threshold effect (as shown alongside circuit F), because the phototransistor doesn't begin to conduct until the photodiode current is large enough to produce a V_{BE} across the external base resistor. In digital applications the threshold can be useful, but in analog applications it is an undesirable nonlinearity.

12.7.2 II: Logic-output optocouplers

These preceding optocouplers are nice, but somewhat annoying to use because you have to supply discrete

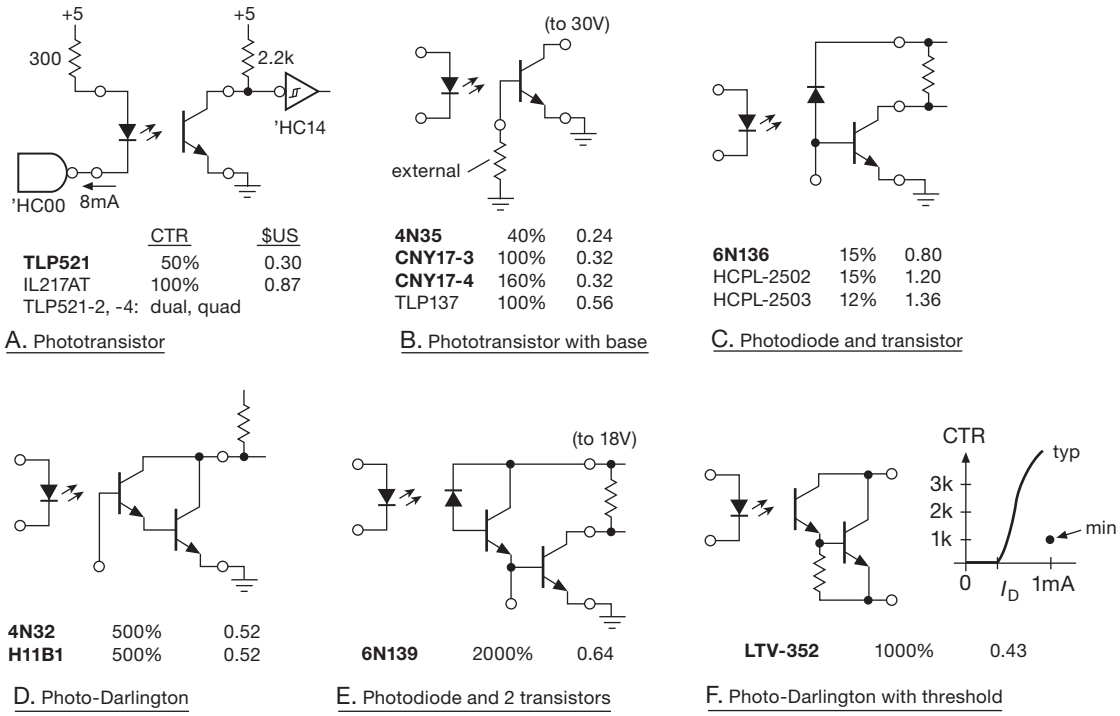


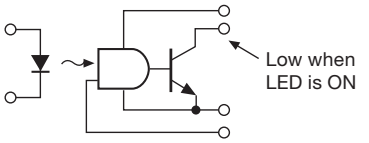
Figure 12.85. Optocouplers-I: Phototransistor output. Parts in **bold** are "jellybeans" – inexpensive and ubiquitous, though not necessarily the best performers.

components at both input and output. Furthermore, the current needed to drive the LED may exceed the drive capability of some logic families; and the passive pullup at the output side suffers from slow switching and mediocre noise immunity. To remedy these deficiencies the silicon wizards bring us "logic" optocouplers (Figure 12.86). The 6N137 and friends (configurations A and B) go halfway, with bare LED at the input, but buffered logic at the output. You still need plenty of input current (for the 6N137 it's specified as 6.3 mA, min, to guarantee output switching), but you get clean logic swings at the output (albeit open-collector), and speeds to 10 Mb/s. Note that you must supply +5 volts to the internal receive-side output-logic circuitry. Logic couplers with reduced drive are listed; among these are the classic (and curiously named) H11L1 and H11N1, descendants of the pioneering optoelectronic offerings from General Electric, happily enjoying great popularity well past their 30th anniversary. But note the cost penalty, particularly for high speed combined with low drive current.

As long as you have internal logic circuitry at the output, why not provide honest active pullup? Why not indeed. That's configuration B, among which you'll find improved versions, some faster, some with impressive isolation slewing rates, and some with three-state outputs; but note the "improved" pricing!

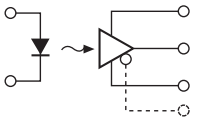
The optocouplers in configuration C bring us to the promised land (where, however, the real estate values are somewhat elevated): these accept logic-level inputs, and produce logic-level outputs with active pullup. Because of the internal logic circuitry at both input and output, both sides of the chip require logic supply voltages. Some varieties (e.g., the ACPL-772L) will happily run on either 3 V or 5 V supplies at both ends, in any combination. These couplers are pretty fast, up to 50 Mb/s.

We've listed also three isolating logic couplers that perform similarly, but that use capacitive or transformer-based isolation techniques in place of light. They are faster, but note one complication: their isolating methods are all *ac-coupled*, using short pulses to transfer changes of state across the gap. That is, they are not intrinsically "dc correct," and they may exhibit artifacts such as timing-delay skew, or require an initializing signal to force the output to a known state.



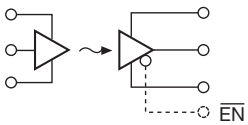
	I_{LED}	EN	V_{CC}	Mbps	$US
6N137	6.3mA	H	5	10	0.71
HCPL-2601	6	H	5	10	1.04
HCPL-061	3	H	5	10	1.90
H11L1	2	—	3–15	1	0.58
H11N1	3	—	4–16	5	4.02
HCPL-2300	0.75	—	5	8	6.40
ACPL-W60L	5	—	3–5	15	1.85
ACPL-P456	10	—	5–20	1	2.09

A. LED-to-logic (open-collector)



	I_{LED}	EN	V_{out} (LED ON)	V_{CC}	Mbps	$US
HCPL2200						
FOD2200	2.2mA	L	H	5–20	2.5	2.62
TLP2200						
HCPL-2201	2	—	H	5–20	5	2.00
HCPL-2400	4	L	L	5	40	3.92
ACPL-4800 (30kV/μs!)	6	—	H	5–20	3	1.85

B. LED-to-logic (active pullup)



	EN	V_{CC}	Mbps	$US	coupling
HCPL-0721	—	5	25	4.07	opto
FOD0721				2.67	
PC412S	—	5	25	2.31	opto
ACPL-772L	—	3–5	25	2.68	opto
HCPL-7723	—	5	50	7.00	opto
HCPL-0900	L	3–5	100	5.30	xfmr
ISO721,722	–, L	3–5	150	2.80	cap
IL710	L	3–5	150	3.30	GMR

C. logic-to-logic (active pullup)

Figure 12.86. Optocouplers II: Logic output.

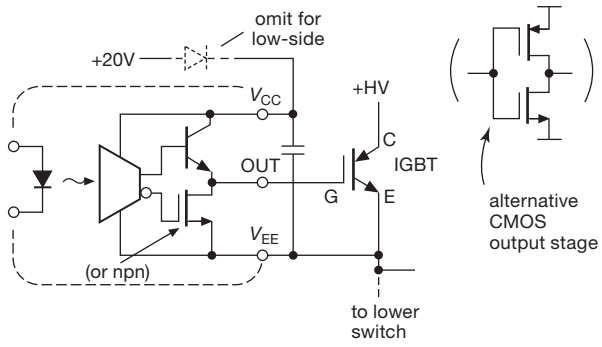
12.7.3 III: Gate driver optocouplers

Isolated optical coupling lets you float the output, relative to the input, at voltages up to the isolation rating of a few kilovolts (limited also by a maximum slewing specification, called "common-mode transient immunity," typically a few kilovolts per microsecond). We know what you're thinking now – whatever can such a high-voltage and swiftly moving load *be*?! Well, one important application that needs both isolation and fast slewing is "high-side drive" of MOSFETs or IGBTs (the latter favored for high-voltage power switching, see §3.5.7), where you have a push–pull pair going between ground and a high-voltage positive rail (see Figures 9.73C and D, and §*9x.10*). And what you need is an isolator that can deliver a full gate drive of +10 V or more, relative to the MOSFET source (or IGBT emitter) of the high-side transistor, which is acting like an nMOS follower. Because it's a follower, rather than a switch, the latter is "flying" along with the output.

Figure 12.87 shows the basic gate-driver optocoupler. In block diagram form you might mistake it for a simple logic coupler; but its output stage consists of a robust push–pull driver that can source and sink currents of an amp or more, and operate from an output supply rail up to ∼30 V. When the coupler's output goes HIGH, it switches on the IGBT, which promptly flies up to the positive high-voltage rail at a kilovolt or more, carrying the gate driver output along with it. That's why you need both high-voltage isolation and common-mode transient tolerance.

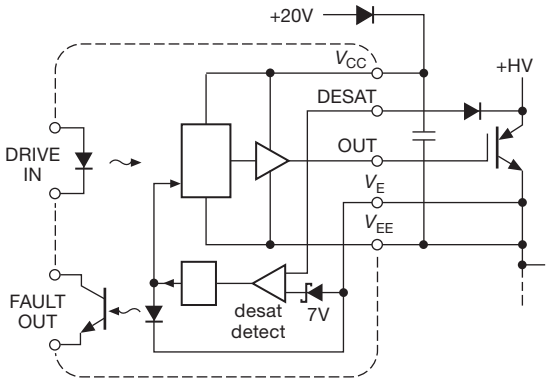
Of course, the ∼20 V supply for the isolator's output side has to fly also! Sounds like a serious problem – but there's a terribly clever solution, which exploits the fact that the output is switching between ground and +HV: if you hook a high-voltage diode to the coupler's V_{CC} from a plain ordinary +20 V supply (relative to circuit ground), as shown in Figure 12.87A, it will conduct during times when the output is LOW, charging up the bypass capacitor. Just make the latter large enough to keep the coupler's output powered during the periodic HIGH states, and you're done; that's easy enough, because the coupler output-stage quiescent current is only a few milliamps. This is basically a high-side "charge pump," sometimes called a "bootstrap supply." You can use one of these couplers to drive the low side also; in that case just power it directly from the +20 V, omitting the diode.

High-voltage switching is fraught with peril, and is not a recommended activity for the faint of heart: a momentary output short can annihilate things in a hurry. You need to protect against such "faults," for example with current-limiting circuitry. But even then you can destroy the IGBTs or MOSFETS, because a load short puts the full supply voltage across the high-side IGBT while it is running at that current limit. What you need to do is sense when the IGBT's gate is being driven, but its output has not gone promptly into voltage saturation. Happily, there



	I_O	max V_{CC}	$US	
PC924	0.5A	35V	1.04	
ACPL-P302	0.4	30	1.26	
HCPL-314J	0.4	30	2.37	dual
FOD3184	3	30	1.14	
HCPL-3120	2	30	3.10	

A. Push-pull gate driver



	I_O	max V_{CC}	$US	
HCPL-316J	1.5A	35V	4.92	logic in
ACPL-332J	2A	33V	5.78	

B. Gate driver with fault shutdown

Figure 12.87. Optocouplers III: Gate driver output.

are improved gate-driver optocouplers with "desaturation" fault detection, as in Figure 12.87B: internal circuitry looks at the drop across the driven IGBT, and shuts off its gate drive during a non-saturating fault.[43] It also sends a fault indication back across the isolation gap, as shown.

[43] There's time to do this: an IGBT's thermal mass provides about 10 μs at its maximum conduction current, with full rated voltage across it, before damage occurs.

12.7.4 IV: Analog-oriented optocouplers

So far we've seen only *switching* applications of optocouplers, where linearity is of little concern. But sometimes you need to isolate *analog* circuits. One method, of course, is to use a pair of converters, converting the analog quantity to a digital bitstream, coupling that via logic optocouplers, then converting back to analog. But there are analog-oriented optocouplers that do the job directly.

Figure 12.88 shows most of these. The classic H11F1 (another winner from the pioneers at GE, this one going back to 1979) is an opto-FET, in which the LED drive current affects the FET in the same way that a gate drive voltage would (we're not exactly sure what's inside this part). So increasing levels of LED drive increase the saturation current (that is, channel current at channel voltages greater than a few tenths of a volt or so), reaching about 1 mA for an LED drive of 25 mA. Notice that the output is completely symmetrical, and works up to ± 30 V across the output terminals. And, as with ordinary FETs, the output terminals look approximately resistive for small voltages across the channel; here, however, the R_{ON} value is set by the LED drive current, and varies from $>300\,\text{M}\Omega$ (with no LED drive) down to about $100\,\Omega$ (with 16 mA LED drive). Once again this property is symmetrical both sides of zero volts, but it's not particularly linear beyond about ± 50 mV, as seen in the graph of Figure 12.88A.

For really good linearity you can use an opto-photoresistor (configuration B), which is an LED illuminating a CdS photoresistor. The sensors are slow, and exhibit memory effects; but the output terminals behave like very linear resistors, with linearities of $\sim 0.01\%$ for voltage swings up to ± 1 V.[44] This would make an excellent amplitude limiter for a low-distortion Wien Bridge oscillator like the one in Figure 7.22.

There's a small class of "video optocouplers," which rely on the inherent linearity of LED intensity versus drive current (except at low currents), and the good linearity of photodiode current versus illumination. Figure 12.88C

[44] Sadly, the continued availability of these nice devices is threatened by legislated restrictions on hazardous substances (RoHS): cadmium is not a nice material, and worries about its toxicity put it on the no-no list. Ironically, while the milligrams of cadmium in a CdS photocell are a target of RoHS legislation, such potential exposure (from *eating* the things?) is dwarfed by the general population's exposure to cadmium via fossil-fuel burning and cigarette smoking. And, while we're whining, how about the vast quantities of lead that are permitted in automobile batteries? These weigh 50 pounds or more, most of it lead, and about 100 million are manufactured annually: that's more than a megaton of lead!

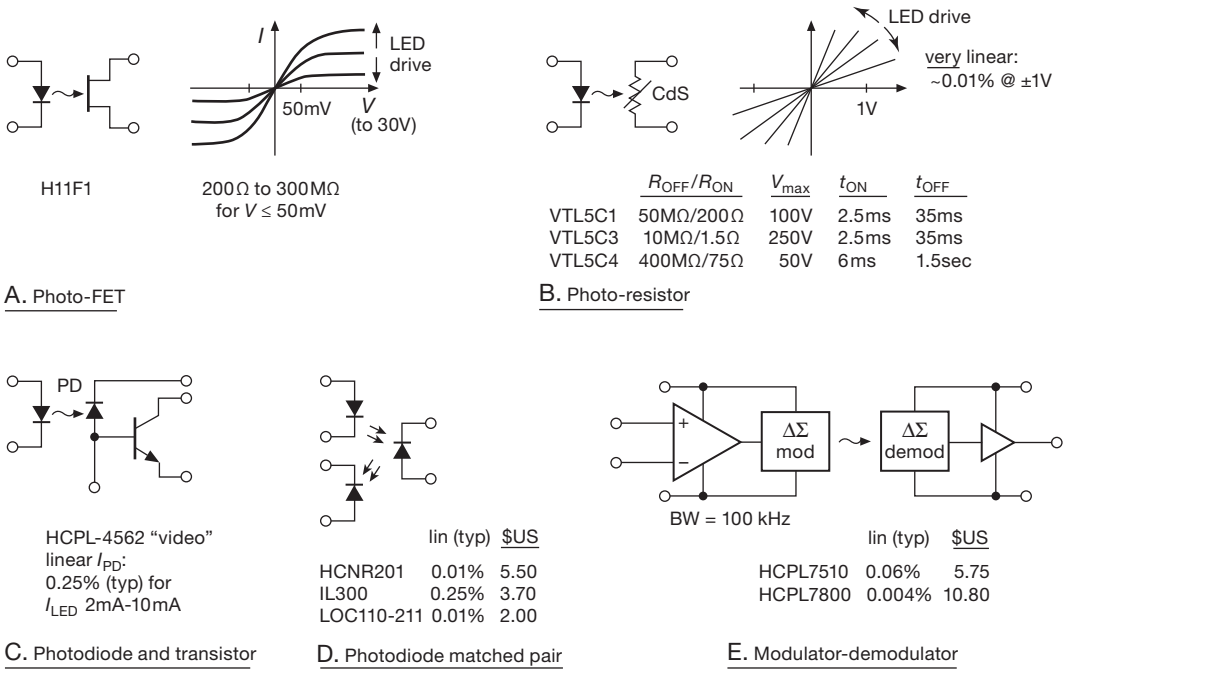


Figure 12.88. Optocouplers IV: Analog-oriented output.

shows on example, with a claimed bandwidth of 17 MHz when used in the circuit of Figure 12.89. Another way to achieve reasonable linearity is to package a matched pair of photodiodes with an LED (Figure 12.88D), then use one of them to provide feedback on the driving side (Figure 12.90); the far-side photodiode will then exhibit linearity limited only by the degree of matching. The devices listed in Figure 12.88D achieve linearities of better than 1% in such a configuration.

Finally, there's an interesting class of linear optocouplers that integrate an A-to-D converter and a D-to-A converter, with digital coupling. These use "delta–sigma" conversion (sometimes called "1-bit" conversion), which we discuss later in §13.9. Figure 12.88E shows the scheme. These are very linear, but the delta–sigma process leads to significant output noise (∼30 mVrms, compared with 3 V full-scale output), and some signal delay (∼5 μs). These parts have found widespread use in half-bridge three-phase variable-frequency-drive motor-power systems, where they measure the current in each leg. As a result they generally have full-scale ranges of only 300 mV.

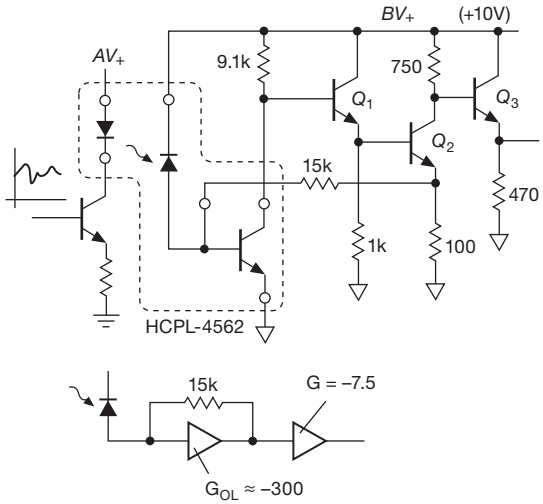


Figure 12.89. Using the HCPL-4562 "video optocoupler" in the datasheet's recommended transresistance configuration (∼17 MHz bandwidth to −3 dB).

12.7.5 V: Solid-state relays (transistor output)

Turning back again to couplers whose output is a switch, we have the class of "solid-state relays" (SSRs; Figures 12.91 and 12.92). These are characterized by isolated two-terminal outputs that are either open (nonconducting) or closed (conducting), depending on the state of input LED drive, and therefore can be thought of as substitutes for electromechanical relays (see §*1x.6*). They do not

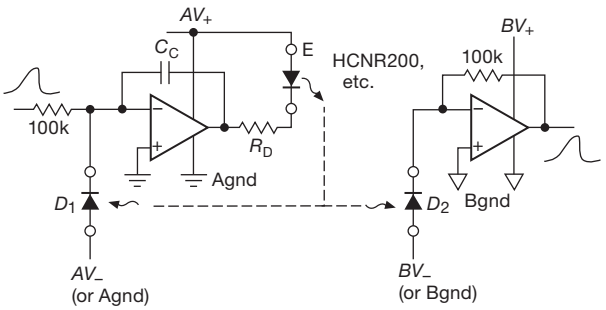


Figure 12.90. Linearizing an optocoupler with photodiode matched pair.

require any external source of supply voltage on the output side – they're "just a switch." One class of SSRs uses *thyristors* (SCRs and triacs) as the output switch; these devices, once triggered into conduction, remain conducting until the current is removed, and so they are suitable only for ac loads (where the current goes through zero twice per cycle). We treat these below, after looking at SSRs with MOSFETs as the output switches, which is the class of SSRs that is suitable for dc loads (and, as we'll see, these can also switch ac loads when configured as a series pair of MOSFETs).

To bring the output transistor(s) into conduction, MOSFET-output SSRs use a series string of a dozen or more photodiodes to generate the needed gate voltage. This "photovoltaic stack" is illuminated by the LED, generating 5–10 V of gate bias. It can supply only a few microamps to the gate, whose capacitance causes the turn-on and turn-off times to be typically in the range of 0.1–5 ms. You don't have to crack an SSR open to play with this portion of the relay, though, because you can buy it à la carte (Figure 12.91A). The datasheets for these devices don't tell you much about what's inside; but it's evident from the fast t_{OFF} specifications that most of them use an auxiliary bit of circuitry to discharge the gate for fast turn-off. This could be a *pnp* transistor that is pulled into conduction when the PV stack's output current ceases, perhaps aided by an SCR (indicated by dashed lines).

Most MOSFET SSRs use a series-connected pair of *n*-channel FETs, as shown in Figure 12.91D, driven by a photovoltaic stack.[45] For an ac load you use the top and bottom (drain) terminals. When the relay is OFF, one transistor or the other is acting as the open switch, depending on polarity; you need the series pair, because otherwise the reverse (body) diode would conduct. When the relay is ON, both transistors act as switches, characterized by an R_{ON}.[46]

You can of course use this same connection for a dc load, but you do better by using the transistors in parallel (connecting the drains together as shown), which reduces R_{ON} by a factor of 4 (at the cost of increased output capacitance, if that matters to you). Note that enhancement-mode MOSFETs give you a "normally OFF" (form A) relay, whereas depletion-mode MOSFETs (with the gates connected to the negative end of the PV stack) make a "normally ON" (form B) relay.

Keep in mind that these "relays" come in an enormous range of current capabilities, and that the small ones can be used effectively for low-level switching: parts like the AQY221N3 or NEC/CEL PS7801-1A, for example, have an R_{ON} of 10 Ω or less while presenting a mere picofarad of output capacitance. Unlike CMOS analog switches (§3.4.1A), they have *zero* charge injection (apart from the isolation capacitance, about 1 pF for most parts, but a mere ∼0.3 pF for the PS7801A) – you may want to use them even when you don't need the isolation. For example, you might use one of these in a sample-and-hold or integrator circuit.

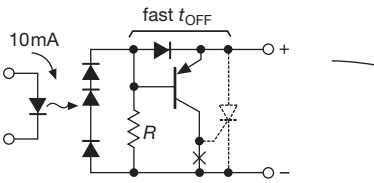
There are a couple of oddball FET relays listed in Figures 12.91B and C: the H11F1 (featured earlier as "analog oriented") is both fast (∼15 μs) and symmetrical (good to ±30 V when OFF, ±100 mA when ON). We're not sure how they do it. And the LH1514 has an interesting configuration, intended for balanced ac signals: it uses a "T-switch" arrangement, with a pair of normally-open series switches on each line, bridged across with a normally-closed switch (a depletion-mode MOSFET, driven from the same PV stack). This results in good signal attenuation when OFF, even for signals at radiofrequencies (65 dB at 1 MHz).

12.7.6 VI: Solid-state relays (triac/SCR output)

For powerline ac switching, it's common to use a solid-state relay with a triac or a pair of SCRs (collectively *thyristors*) as the switching device. The low-current SSRs shown in Figures 12.92A and B are used primarily as trigger devices to activate a high-current thyristor, as shown in Figure 12.93. But the big gorilla in the room is the integrated SSR, which includes the optocoupler and (usually)

[45] Some SSRs, intended for dc loads only, use only a single MOSFET; examples are the AQV100- and AQZ100-series from Panasonic.

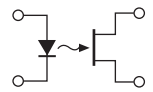
[46] In §3.5.6B we suggested a general-purpose roll-your-own version of this SSR circuit (Figure 3.107). The advantage of providing your own MOSFETs is that you can select low-resistance, high-current parts (to over 1kA pulsed, 100 A continuous), or common low-cost parts like an IRF640 (200 V, 12 A, 0.25 Ω), or amazing high-voltage parts (to 4.5 kV); a handy table is provided there.



typ, @10mA

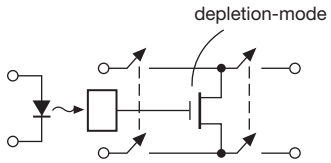
	V_{OC}	I_{SC}	t_{ON}	t_{OFF}	$US
PVI5033	8.5V	5µA	6ms	0.4ms	5.05 dual
TLP190	8	20	0.2	1	1.53
APV1122	8.7	14	0.4	0.1	2.00
ASSR-V622	7	20	0.3	0.03	3.00
TLP3924	40	6	—	—	—

A. Photovoltaic stack output



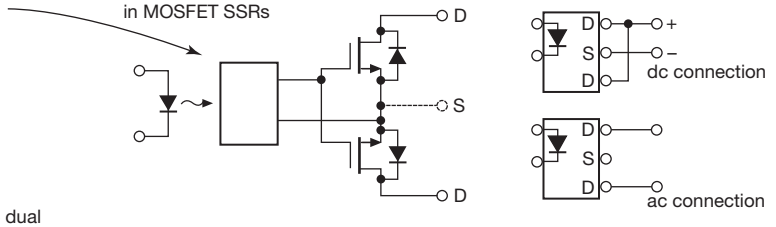
	V_{OFF}	I_{ON}	C_{OUT}	$US
H11F1	±30V	±100mA	15pF	1.50
	R_{ON}	≤200Ω @ I_L = 16mA		
	R_{ON}	≥300M @ I_L = 0		

B. FET output



	V_{OFF}	I_{ON}	I_{LED}	C_{OUT}
LH1514	±15V	±100mA	3mA	20pF
	OFF Isolation (with R_L=50Ω):			
	65dB @ 1 MHz			
	30dB @ 50 MHz			

C. Balanced T-switch



	V_{OFF} (V)	ac I_{ON} (mA)	$R_{ON}^{h,m}$ ac (Ω)	no "S" term[d]	C_{OUT} (@10V) pF	$US[w]	comments[a]
form A (N.O.)							
AQY221N	40	120	12.5	•	2.5[y]	7.07	RF relay
CPC1117N[b]	60	150	16	•	17	1.38	
AQV212	60	550	2.5	–	40	4.66	
PVDZ172	60	1400	0.25[g]	–	220	5.27	DC only
PVA1354N	100	375	5	•	22	4.43	
AQV227	200	50	50	–	4.5	5.74	RF relay
AQV257	200	250	4	–	70	5.92	soft on/off
PVA3055	300	50	160	•	4.5	6.99	
PVA3324	300	130	24	•	23	5.22	200nV offset
LCA110	350	120	35	–	26[x]	1.68	
AQV210[d]	350	130	25	–	8	3.58	
LAA110	350	120	35	•	25[x]	3.62	this is a dual
AQV216[d]	600	50	120	–	10	4.93	
AQV259	1000	30	200	–	30	6.93	
AQV258	1500	20	500	–	-	6.60	
form B (N.C.)[z]							
LH1501	350	150	25	–	35[x]	2.97	
CPC1130	350	120	30	•	25[x]	2.43	
LCB110	350	120	35	–	25	2.25	
AQV414[d]	400	120	50	–	10	4.66	
form C (N.O. & N.C.)							
LCC120	250	170	20	•	50[x]	4.72	can wire as SPDT

Notes: (a) all devices specified at 5mA LED drive current, except CP1117N. (b) 2mA LED drive current. (d) no source terminal pin; can use for ac or dc, but without the benefit of low the dc R_{ON} you get with the "dc connection." (g) single MOSFET, no ac-connection mode available. (h) ac-mode resistance; 4 times lower with dc connection (if available). (m) max. (w) qty 25. (x) at 50V. (y) at 0V (z) ON when unpowered.

D. Solid-state relay – MOSFET output

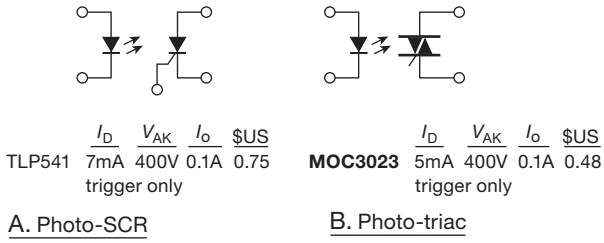
Figure 12.91. Optocouplers V: "dc solid-state relay" (transistor output).

zero-voltage-switching (ZVS) trigger circuit, along with a substantial output triac or SCR pair. This is good: when driving ac loads, it's best to switch on the load during a zero crossing of the ac waveform, to avoid putting spikes onto the powerline; and the triac or SCR inherently switches off at zero current.[47] So we've got "ZVS/ZCS."

Figure 12.92C lists just a few of the hundreds of available types. High-current SSRs aren't cheap, but they're delightfully easy to use. The larger ones (10 A or more) come in a panel-mount "brick" package (about $1.75'' \times 2.25'' \times 1''$, intended for heatsinking), whereas the smaller ones come in several sizes of PCB-mounting "single in-line" packages (SIPs).[48]

[47] An exception is when switching powerline ac across a transformer primary: there ZVS is the *worst* case, because the full unipolarity half-cycle applied voltage brings the core closest to (or into) saturation, with enormous peak currents. Ideally you'd want to switch the ac close to its *peak* voltage. We encountered this effect in our lab, when powering a 20 A Variac (§1.9.5D) – about half the time when you switched it on, the 20 A circuit breaker on the wall would snap out (even though the Variac was unloaded); you had to be lucky and catch the ac waveform near a peak to successfully switch it on.

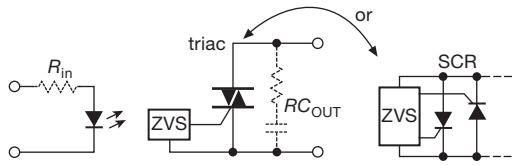
[48] But see our warning footnote in §12.4.3 about heatsinks. Be sure to



	I_D	V_{AK}	I_o	$US
TLP541	7mA	400V	0.1A	0.75
	trigger only			

	I_D	V_{AK}	I_o	$US
MOC3023	5mA	400V	0.1A	0.48
	trigger only			

A. Photo-SCR B. Photo-triac



	I_D	V_{AK}	I_o	$US	R_{in}	R_{out}	package
MOC3043	5mA	400V	0.1A	0.91	—	—	DIP
AQG22105	10	240	2A	5.82	300Ω	✓	SIP mod.
G3M-203P	10	240	3A	12.65	300Ω	—	SIP mod.
MP240D4	3	240	4A	21.00	1.5k	—	SIP mod.
CWD2450S	10	240	50A	42.00	1k	✓	brick
PRGD24150	15	240	150A	~100	200Ω	✓	brick

C. Solid-State Relay (ZVS)

Figure 12.92. Optocouplers VI: "ac solid-state relay" (triac/SCR output).

12.7.7 VII: ac-input optocouplers

Finally, there is a class of optocouplers intended for ac input drive (Figure 12.94). Some use a pair of LEDs back-to-back, coupled to a phototransistor or photo-Darlington. The output phototransistor conducts according to the magnitude of LED current, with typical current-transfer ratios of 20–100%; this is useful for detecting zero crossings of the ac powerline.

Input–output modules

There is a category of opto-isolating "input–output modules" used in industrial settings: The *input* modules monitor an ac or dc signal (with a selection of voltage ratings, up to powerline voltage), producing an isolated low-voltage open-collector logic output to go to a computer or other industrial controller; the *output* modules use a low-voltage logic signal from a computer or controller to switch an ac or dc load, typically at powerline voltages. In other words, the output modules are SSRs with thyristor or transistor outputs (ac and dc, respectively), and the input modules are ac-input optocouplers with open-collector outputs. These

consult the datasheet for the ON voltage drop when designing an SSR into a system.

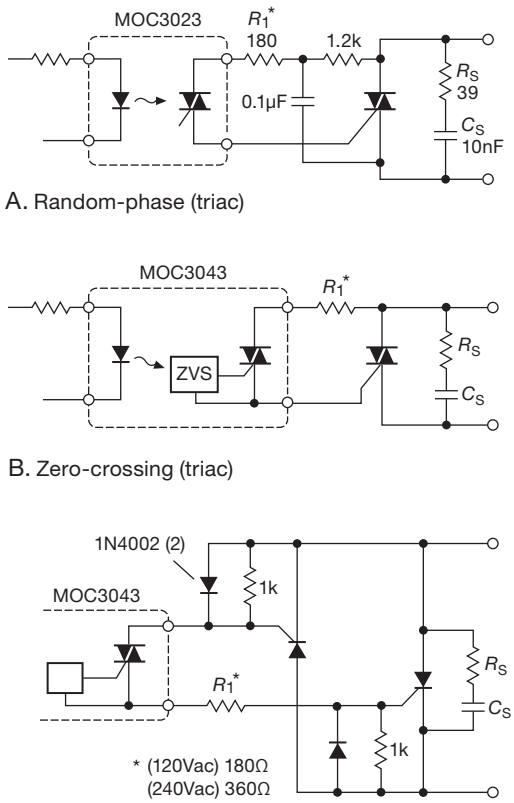
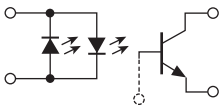


Figure 12.93. A small opto-triac triggers a large triac or SCR pair. Choose resistor R_1 according to the ac line voltage.

latter, therefore, are an easy way to use an ac powerline input to create an isolated logic-level swing (or higher, because the O/C output typically can go to +30 V), as shown in the figure. Note that these puppies include an internal *RC* filter, so the output indicates the presence or absence of an ac waveform, but does not capture the individual cycles. Input–output modules come in several standard upright module package sizes (e.g., $1.7'' \times 1.25'' \times 0.6''$ and $1.7'' \times 1'' \times 0.4''$), with a hold-down screw-fastener arrangement, and (often) an indicator LED on top.

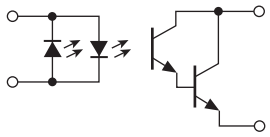
12.7.8 Interrupters

You can use an LED–phototransistor pair to sense proximity or motion. An "optical interrupter" consists of an LED coupled to a phototransistor across an open slot. It can sense the presence of an opaque strip, for example, or the rotation of a slotted disk. An alternative form has the LED and photodetector looking in the same direction, and



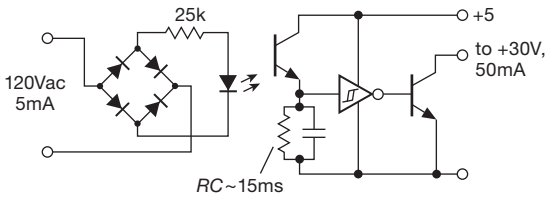
	CTR	$US	
LTV-814	20%	0.26	
ILD252	100%	3.00	dual
ACPL-824	20%	0.46	dual
ACPL-844	20%	0.93	quad
IL252	100%	1.28	w. base
MOC256	20%	0.77	w. base

A. AC to phototransistor



LTV-8141 600% $0.47

B. AC to Darlington



Crydom M-IAC5
Omron G3TC-IAC5
Tyco IAC-5

C. AC "input module"

Figure 12.94. Optocouplers VII: ac input.

it senses the presence of a reflective object nearby (most of the time, anyway!). As with optocouplers, you can get these with a simple phototransistor on the receive side, or you can get them with logic-level outputs (either open-collector, or active pullup). You can see some examples in Figure 12.84. Optical interrupters are used in mechanical devices such as printers, to sense the end of travel of the moving assembly. Optical interrupters can have problems when ambient light levels are high. A nice trick in such situations is to use *synchronous detection* (§8.14.1), making the detector selectively sensitive to the frequency at which the emitter is driven. Hamamatsu offers a nice selection (their S4282/89-, S6809/46/86-, and S7136-series) of detectors with built-in preamp and signal processing electronics. You can get optical "rotary encoders" that generate a quadrature pulse train (two outputs, 90° out of phase) as the shaft is rotated; there's an example of one in Figure 12.84. These provide a nice alternative to resistive panel controls (potentiometers) – see §15.5.

In any application where you're considering an optical interrupter or reflective sensor, take a look at Hall-effect sensors (not shown) as an alternative; they use solid-state magnetic-field sensors to indicate proximity. They're commonly used in applications such as automobile ignition systems (as an alternative to mechanical breaker points), antilock brakes (sensing wheel rotation), and brushless dc motors.

12.8 Optoelectronics: fiber-optic digital links

Fiber-optic transmission of digital signals provides a convenient galvanically isolated link, capable of carrying digital communications at rates to 10 Gb/s, over distances to 10 km,[49] without any susceptibility to interference, even in the most electromagnetically noisy environments (factory floor, automobile, mountaintop observatory). Although the sophisticated user may want to design 10 Gb Ethernet fiber-optic circuits, we're mostly interested here in rather more humble goals, for example, simply linking some instruments together over distances of 10 m (or perhaps 1 km), with data rates of megabits per second (or perhaps with fast Ethernet, at 100 Mb/s). Let's see what off-the-shelf components will do the job.

12.8.1 TOSLINK

A very simple and inexpensive short-range digital-fiber link is provided by the TOSLINK™ family of transmitter–receiver pairs (known generically as "EIAJ optical," "JIS F05," "ADAT optical," or "Digital Audio Optical Cable"), see Figure 12.95. The TOSLINK standard was originated by Toshiba,[50] and is widely used for digital-audio interconnections, for example between audio and video components. It is one of the two "digital-audio" jacks you see on the rear of DVD and Blu-ray players, sometimes la-

[49] The speed record at time of publication is 1 petabit/s (10^{15}b/s) over a 50 km length of fiber optic cable, achieved by a collaboration of NTT (Japan), Fujikara Ltd., Hokkaido University, and the Technical University of Denmark. The "cable" is a 12-core bundle, each fiber of which carries 84.5 Tb/s by multiplexing upon 222 separate wavelengths a signal rate of 380 Gb/s per wavelength (the latter achieved by modulating eight separate carriers, each with polarization multiplexed 32QAM).

[50] **Caution**: we've found current availability largely through second-source suppliers such as Comoss, Sys Concept, and FiberFin; the parts used for the examples in this section appear to be unavailable from distributors, and may have been discontinued by Toshiba.

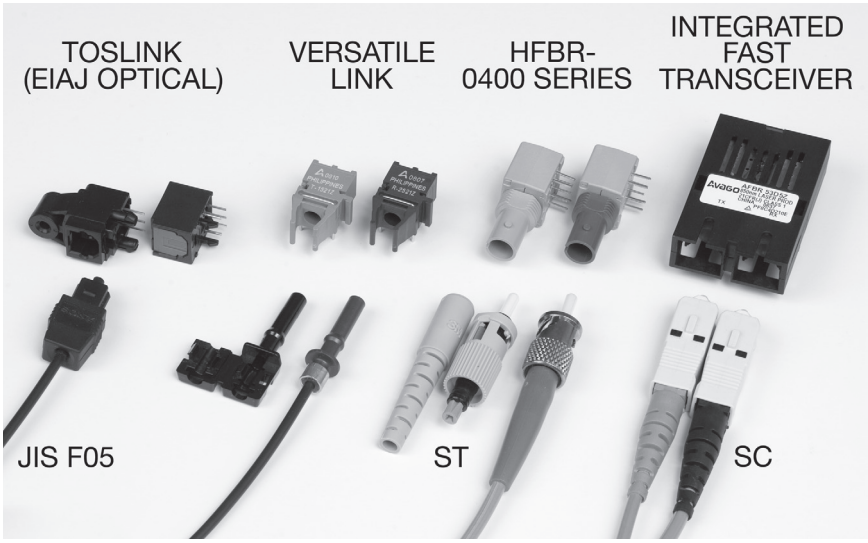


Figure 12.95. Some popular fiber-optic formats. The TOSLINK devices, widely used in consumer audio, are here shown in a shuttered PC-mount version (right), and a panel-mount unshuttered version (left). The other connector formats can be field terminated (bare connectors are shown for ST and Versatile Link), but it's easier to buy preterminated fiber cables.

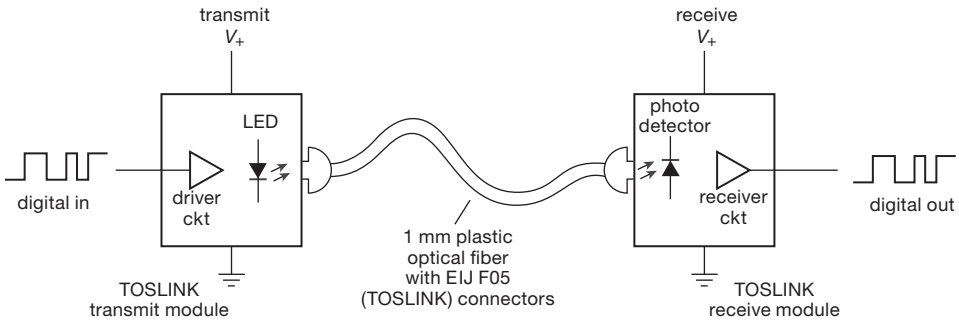


Figure 12.96. Inexpensive PC-mounting TOSLINK-style fiber-optic modules include all drive and receive electronics. They accept and reconstruct standard logic levels, at rates to 15 Mb/s over ranges to 10 m.

beled "digital optical" (the other being a coaxial electrical audio socket that is physically identical to the common "RCA" jack, usually placed alongside the TOSLINK connector, and that carries, electrically, the same digital stream as the optical port). TOSLINK uses visible red LEDs, operating at 650 nm; it's easy to see when it's working, and which end is the transmitter (unlike the case with invisible infrared fiber modules).

The charm of TOSLINK is its simplicity and low cost: the Toshiba TOTX147 and TORX147 are a typical transmitter–receiver pair, costing about $1 at each end (in small quantities). They run on a 2.7–3.6 V logic supply, with the transmitter accepting logic-level inputs, and the receiver replicating the logic levels at its output. That is, all the logic interface circuitry is built in; all you do is connect it up (Figure 12.96). They are intended for short-range links, operating up to 15 Mb/s over a distance of 5 meters or less. You use an inexpensive 1 mm plastic fiber (APF, for all-plastic fiber, or POF, for plastic optical fiber), which you can get as a 2.2 mm (or thicker) sheathed cable with TOSLINK jacks at each end; we've found such "patchcords" (e.g., at Amazon.com) for as little as a dollar or less, and are standard equipment at audio–video supply stores. Figure 12.97 shows logic input and output from a TOSLINK pair running at +3.3 V, driven with a 15 Mb/s data stream, and connected with an 8-foot fiber patch-cord. You can get 5 V versions as well, e.g., Toshiba part numbers TOTX/TORX177 (analogous Sharp part numbers are

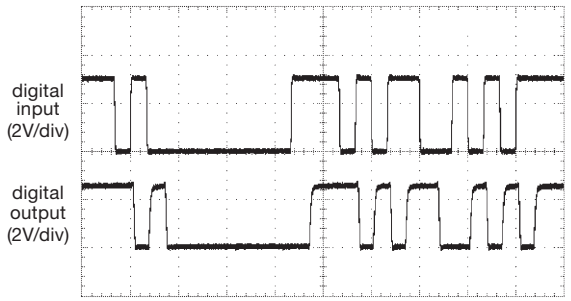


Figure 12.97. Transmitted and received data, sent through 8 feet of 1 mm plastic fiber at 15 Mb/s with TOSLINK digital-audio components (Toshiba TOTX141FPT, Sharp GP1FA352RZ). Horizontal scale: 200 ns/div.

GP1FA351TZ/RZ and GP1FA551TZ/RZ, for 3 V and 5 V transmitter–receiver pairs).

One drawback of these digital-audio TOSLINK components is that the receiver bandwidth does not extend down to dc, and is specified to operate properly only with a minimum data rate of 0.1 Mb/s.[51] Note that the transmitter end is dc-coupled; the problem is at the receiver end, where the optical signal's switching amplitude is used to establish the receiver's threshold. This is done to minimize pulse-width distortion: if a *fixed* threshold were used (as is common with other fiber protocols, for example the Versatile Link devices below), the receiver's output would show a widening or narrowing of the reconstructed data bits, according to the amplitude of the received optical signal (i.e., dependent on length of fiber and other losses).

If you need response down to dc, you can get true dc-coupled TOSLINK-style receivers, with generally improved performance: for example, the TOTX197 and TORX198 pair of "general-purpose" (as contrasted with "digital-audio") modules operate from dc to 6 Mb/s over a distance of 40 meters or less, using the same plastic-fiber cable and connector. The TOTX/TORX1350 series specify data rates of dc to 10 Mb/s over distances to 100 meters, again with all-plastic fiber (APF). And, by using a lower-loss plastic-clad glass fiber (PCF, with a fiber diameter of 0.2 mm) the range is extended to 1 km, for example, with the TOTX/TORX196 pair. As in life generally, there are tradeoffs: these devices exhibit greater pulse-width distortion (typically ± 55 ns, compared with ± 15 ns for the audio-style devices[52] with their adaptive threshold circuitry); also, you pay about twenty times as much for them ($20–25 or so, in small quantities).

A. Connectorless plastic-fiber components

If you dislike connectors, you can get connectorless fiber transmitters and receivers from Industrial Fiber Optics, Inc. These accept 1 mm plastic fiber, in a PCB-mount cinch-nut collet housing; you just cut the fiber, insert, and twist the thing tight. They make a range of power and speed grades, up to 155 Mbps; pretty colors (red, green, blue, and infrared) too! While you're shopping, you might pick up one of their nifty "Infrared Detection Cards" (part # IF-850052), with which you can tell if an IR LED (or laser) is working. You hold it in front of the emitter, and you see an orange-yellow spot on its target area.[53]

12.8.2 Versatile Link

The Versatile Link (VL) series of optical-fiber transmit and receive modules was introduced by Hewlett–Packard (then spun off as Agilent, and finally Avago) around 1990, and continues in wide use. It runs through the same 1 mm plastic fiber as TOSLINK, but with a somewhat different connector format, namely a cylindrical snap-in connector that engages a pair of plastic pincers (see the photograph, Figure 12.95). These modules use a similar visible red wavelength (660 nm), and come in several models, most usefully a 5 Mb/s dc-coupled pair (HFBR-1521Z/2521Z) that works out to 20 m distance, and a 40 kb/s dc-coupled pair (HFBR-1523Z/2523Z) that's good to 100 m. The receive end provides a logic-level output (using an external or internal pullup resistor); but the transmit end is a bare LED, so you have to provide a current-limiting resistor and a saturated switch to ground, or the equivalent (Figure 12.98). This provides some flexibility in transmitted optical level, but does require extra components. Both transmit and receive modules cost less than $10 in small quantities. You can get the optical cable with connectors attached; or you can roll your own, which turns out to be simple and quick, using the "crimpless" (i.e., no tooling required) HFBR-4531Z series of connectors; they cost about $0.50 apiece, in small quantities.

For greater link speeds you can get the faster HFBR-

[51] It is not required that the data stream have balanced quantities of 1s and 0s, just that there be transitions at frequent-enough intervals.

[52] There has been some grumbling in the audiophile community about jitter in this interconnection technology. Although this can be addressed with PLL clock regeneration or sample-rate conversion, it is an issue that does not arise with electrical (coaxial) interconnection of digital audio devices.

[53] Puzzle for the reader: converting a lower-energy photon into one of higher energy would appear to violate energy conservation. Figure it out!

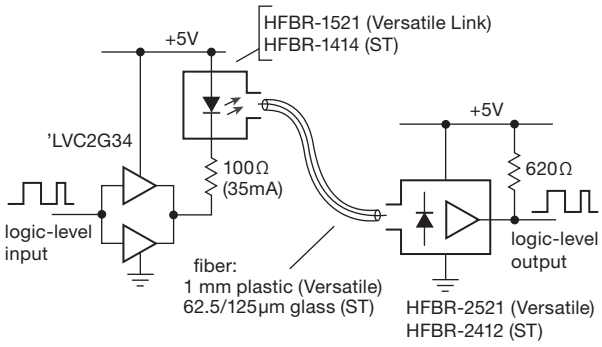


Figure 12.98. Typical drive and receive circuit for the Avago dc-coupled 5 Mb/s "Versatile Link" and ST-connectored fiber-optic series. For 3.3 V logic levels use 50 Ω and +3.3 V on the driver side, and pullup to +3.3 V on the receiver side. Operating currents shown here are suitable for lengths to 10 m (Versatile Link: plastic fiber) or 1 km (ST: glass fiber); they can be altered, depending on link length and maximum data rate (see Avago datasheets and App Note 1035).

1527Z/2526Z modules, which use the same fiber and connector geometry, and which permit signal rates to 125 Mb/s (but not dc-coupled). These, and analogous devices (e.g., the TODX2402 or TOTX/RX1701 in the TOSLINK series; or the HFBR-1424/2426 in the 820 nm ST/SC-connector devices from Avago that use all-silica graded-index fiber, see below) are popular as fiber-based fast serial links, for example to carry Ethernet signaling, or serialized data between a SERDES (serializer–deserializer) pair.

12.8.3 ST/SC glass-fiber modules

For many years we've used the HFBR-14xx/24xx series of fiber transmitters and receivers from Avago. These are more expensive (about $15 in small quantity) than the plastic-fiber devices above, but by using graded-index silica-clad silica fiber (sometimes called "ASF," all-silica fiber, or "AGF," all-glass fiber) at 820 nm wavelength, they work happily even through a kilometer or more of fiber. An added bonus is the robustness of the fiber cables, which, with their very thin glass core/cladding (62.5 μm/125 μm, in the most common variety) and tough jacket, can withstand plenty of bending and pulling without damage.

We use the ST connector style, which is a locking bayonet (see Figure 12.95). You can buy preterminated fiber cables as simplex (one fiber, ST termination at each end), or duplex (also called "zip cord": a pair of fibers, each with its own ST connector at each end), from companies like Tyco/AMP, 3M, and Amphenol, for a few dollars per meter. If you're willing to spend a few hundred dollars on a termination kit, and suffer a bit, you can even put the connectors on by yourself; look at connectors and tooling from companies like Tyco/AMP and 3M.

As with the Versatile Link components, you can get dc-coupled low-speed receivers, for example, the HFBR-2412Z, which goes to 5 Mb/s at distances to 2 km. Its open-collector output requires only a pullup resistor to generate the reconstructed digital logic levels. For high speeds use instead the HFBR-2416Z, which is good to 155 Mb/s at distances to 0.6 km. The latter provides a high-bandwidth "analog" output from its internal PIN-diode detector and preamp, which you ac-couple to external amplifier–comparator circuitry to generate the fast LVDS or ECL reconstructed digital stream. For either receiver you would use the HFBR-1414Z transmitter, which, like the analogous Versatile Link component, looks electrically like a bare LED; you have to provide the requisite drive, either with a transistor switch and resistor, or with a hefty logic gate like a 'LVC2Q34 (see Avago App Note 1123 and App Brief 78).

12.8.4 Fully integrated high-speed fiber-transceiver modules

Why not integrate *all* the logic-level interfacing within the fiber-optic module itself? Indeed. With widespread use of duplex fiber to carry fast serial data – for example as optical fast Ethernet (125 Mb/s), or firewire (rates to 250 Mb/s), or simply between a pair of SERDES parallel–serial chips – there are now plenty of easy-to-use fiber-optic-transceiver ("FOT") modules. These devices have both transmitter and receiver optical modules and connectors (usually in duplex SC or ST format, for glass fiber; or as the duplex "SMI" connector for plastic fiber), along with driver and receiver circuitry using fast differential signaling (usually 5 V or 3.3 V PECL); see Figure 12.99 (and the portrait in Figure 12.95).

Contemporary examples are the Avago AFBR-5xxx-series, which come in duplex SC or ST connector styles, and which can handle 100 Mb/s data rates for fast Ethernet (100Base-FX) or ATM (asynchronous transfer mode); you connect the transceiver's PECL ports directly to the corresponding inputs or outputs on an Ethernet, firewire, or ATM "PHY" (physical layer IC). A typical member of this family, the AFBR-5803, good to 125 Mb/s, currently costs about $30; the faster AFBR-53D5 can handle gigabit Ethernet and costs about $60. You can use a fast fiber transceiver to connect to a SERDES like the Cypress CY7C924 or HDMP-1636, to link a pair of widely separated parallel ports, with data rates of 20–100 Mbytes/s.

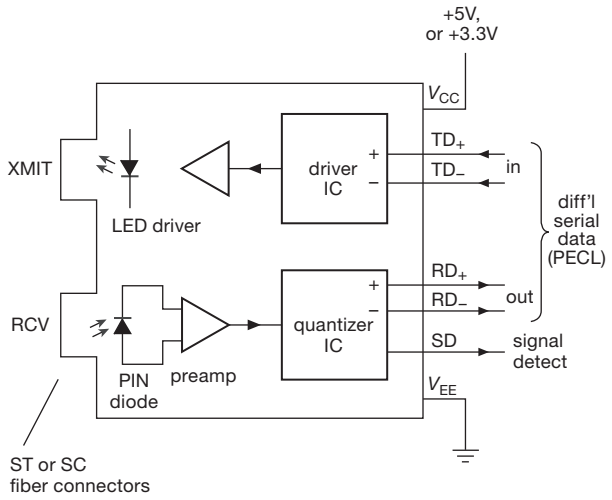


Figure 12.99. High-speed integrated fiber-optic transceivers like the Avago AFBR-5xxx-series, in the industry standard 1×9 SIP package, include all driver and receiver circuitry, to interface directly via 3 V or 5 V serial differential PECL.

For short links, fast transceivers that mate with plastic fiber are available, for example the Toshiba TODX2402 (PECL in–out, duplex SMI, rates to 250 Mb/s; about $25 in small quantity).

12.9 Digital signals and long wires

Special problems arise when you try to send digital signals through cables or between instruments. Effects such as capacitive loading of the fast signals, common-mode interference pickup, and "transmission-line" effects (reflections from impedance mismatching; see Appendix H) become important, and special techniques and interface ICs are often necessary to ensure reliable transmission of digital signals. Some of these problems arise even on a single circuit board, so a knowledge of digital transmission techniques is generally handy. We begin by considering on-card problems. Then we go on to consider the problems that arise when signals are sent between cards, on data buses, and finally between instruments via twisted-pair or coaxial cables.

12.9.1 On-board interconnections

A. Output-stage current transient

The push–pull output circuit for logic ICs consists of a pair of transistors going from V_+ to ground. As we remarked earlier (§3.4.4B and 10.8.3B), when the output changes state there is a brief interval during which both transistors are ON; during that time, a pulse of "shoot-through" current flows from V_+ to ground, putting a short negative-going spike on the V_+ line and a short positive spike on the ground line. The situation is shown in Figure 12.100. Suppose that IC_1 makes a transition, with a momentary large current from +5 to ground along the paths as indicated; with 74F*xx* or 74AC(T)*xx* circuits the current might reach 100 mA. This current, in combination with the inductance of the ground and V_+ leads, causes short voltage spikes relative to the reference point, as shown. These spikes may be only 5 ns to 20 ns long, but they can cause plenty of trouble: suppose that IC_2, an innocent bystander located near the offending chip, has a steady LOW output that drives IC_3 located some distance away. The positive spike at IC_2's ground line appears also at its output, and, if it is large enough, it gets interpreted by IC_3 as a short HIGH spike. Thus, at IC_3, some distance away from the troublemaker IC_1, a full-size bona fide logic-output pulse appears, ready to mess up an otherwise well-behaved circuit. It doesn't take very much to toggle or reset a flip-flop, and this sort of ground-current spike can do the job nicely.

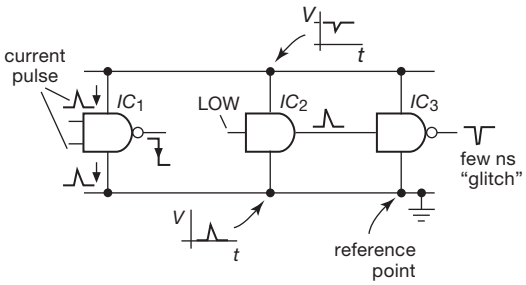


Figure 12.100. Ground-current noise, also known as *ground bounce*.

The best therapy for this situation consists of (a) using hefty ground lines throughout the circuit, or preferably a "ground plane" (one side of a double-sided PC board) or an inner layer of a multilayer PCB, and (b) using bypass capacitors liberally throughout the circuit. Large ground lines mean smaller current-induced spikes (lower inductance and resistance), and bypass capacitors from V_+ to ground sprinkled throughout the circuit mean that current spikes travel over only short paths, with the reduced inductance resulting in much smaller spikes (the capacitor acts as a local voltage source, since its voltage does not change appreciably during the brief current spikes). It is best to use a 0.1μF ceramic capacitor near each IC, although one capacitor for every two or three ICs may suffice. In addition, a few larger electrolytic capacitors ($\sim$10μF or so) scattered throughout the circuit for energy storage and

resonance damping[54] is a good idea. We can hardly emphasize enough the importance of bypass capacitors from power-supply lines to ground in any circuit, digital or linear. They help make the supply lines low-impedance voltage sources at high frequencies, and they prevent signal coupling between circuits via the power supply. Unbypassed power-supply lines can cause peculiar circuit behavior, oscillations, and headaches. *Don't do it!*

B. Spikes caused by driving capacitive loads

Even with the supplies bypassed, your problems aren't over, as we mentioned earlier in §10.8.3A. Figure 12.101 shows why. A digital output sees the stray wiring capacitance and the input capacitance of the chip it drives (5–10 pF, typically) as part of its overall load. To make a fast transition between states it must sink or source a large current into such a load, according to $I = C(dV/dt)$. For instance, consider a 74LVC*xx* chip in a +3.3 V logic system, driving a total load capacitance of 25 pF (equivalent to three or four logic loads connected with short leads). With typical output rise and fall times of ∼2 ns, the current during the logic transition is 40 mA. This current returns through ground (HIGH-to-LOW transition) or the +3.3 V line (LOW-to-HIGH transition), producing those cute little spikes at the receiving end, as before. To get an idea of the effects of such current transients, consider the fact that wiring inductance is roughly 5 nH/cm. An inch of ground wire carrying this logic transition current would have a spike of $V = L(dI/dt) = 0.2$ V. And if the chip happened to be an octal buffer, with simultaneous transitions on a half dozen outputs, the ground spike would be over a volt; refer back to Figure 10.99. A similar (though generally smaller) ground spike is generated near the driven chip, where the drive-current spikes return to ground through the input capacitance of the driven device.

In a synchronous system, with a number of devices making output transitions simultaneously, the noise-spike situation can become so serious that the circuit will not function reliably. This is especially true in a large printed-circuit card, with long interconnections. The circuit may fail only occasionally, when a whole group of data lines unluckily happen to make a simultaneous HIGH-to-LOW transition, generating a momentary very large ground current.

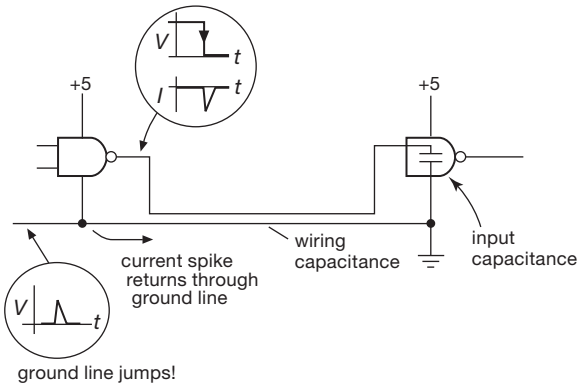


Figure 12.101. Capacitive-load ground current noise.

This kind of pattern sensitivity is characteristic of noise-induced error and is a good reason for running extensive memory tests on microprocessor systems (where you typically have 16 or 32 data lines and 32 address lines bouncing around in crazy patterns).

The best design approach is to use an internal ground-plane layer on a multilayer circuit board, or at least a perpendicular "gridded" arrangement of grounds on both sides of a simpler two-sided board. Copious use of bypass capacitors is mandatory. These problems have been mitigated considerably by measures such as (a) low-inductance surface-mount packages for discrete logic devices, (b) the use of multiple ground pins on complex logic devices,[55] (c) the nearly universal adoption of multilayer PC boards with dedicated power and ground planes, and with extravagant use of SMT chip bypass capacitors, and (d) controlled edge-rate chip designs (e.g., 74ACQ, 74ACTQ, or Gunning Transceiver Logic[56]) and redundant "center-pin" power–ground pinouts for situations in which fast logic chips will be used with less favorable layouts (i.e., through-hole packages, two-layer PCBs, etc.).

Because of these noise problems, it is generally best not to use a faster logic family than you need.[57] That's why we use 74HC(T) logic, rather than 74AC(T), for general-purpose use in our electronics course (where students put their circuits together using "solderless breadboards").

[54] A circuit with multiple 0.1 μF ceramic bypass capacitors, connected by inductive supply-bus wiring, is susceptible to ringing and even oscillation, owing to the multiple high-Q resonances; these are often in the 5–20 MHz range. They can be effectively damped by the equivalent series resistance (ESR) of one or more paralleled electrolytic capacitors.

[55] For example, the "Virtex-5" FPGA that we're using in our lab has 197 redundant ground pins; the larger "FF1760 package" version has 322!

[56] Read Fairchild's App Note AN-1072 to see how it works. GTL is a single-ended bus-driving logic family with reduced swing (<1 V) and controlled edge rates.

[57] This advice goes for *analog* circuits, too: don't use a 100 MHz op-amp, or a 2 ns comparator, when you don't need the speed.

12.9.2 Intercard connections

With logic signals going between circuit cards, the opportunities for trouble multiply rapidly. There is greater wiring capacitance, as well as longer ground paths through cables, connectors, card extenders, etc., so the ground spikes induced by drive currents during logic transitions are generally larger and more troublesome. It is best to avoid sending clocking signals with large fan-out between boards, if possible, and the ground connections to the individual cards should be robust. For fast signals (characteristic times of order a few nanoseconds, or less) the interconnections must be treated as constant-impedance *transmission lines* (see §12.10.1 and Appendix H), which may be single-ended (coaxial cable) or differential (twisted pair). We'll have much more to say about this soon (§12.10).

If clocking signals are sent between boards, it is important to use a gate (for single-ended signals) or a differential receiver (for differential signals such as LVDS) as an input buffer on each board. In some cases it may be best to use line driver and receiver chips, as we will discuss shortly. In any case, it is best to try to keep critical circuits together on one card, where you can control the inductance of the ground paths and keep wiring capacitance at a minimum. Fast signals (edge times of order 1 ns or less), and especially clocking signals, that go between the circuits on a single card will often be routed as "stripline" or "microstrip" transmission lines (§*1x.1.3*).[58] This can take the form of a single-ended trace above a ground plane (microstrip) or between ground planes (stripline); or it can be a differential pair, with two traces side-by-side or stacked vertically. Such constant-impedance routes will be terminated in their characteristic impedance, commonly 50 Ω (single ended) or 100 Ω (differential), either with "back termination" at the driving end, or with far-end termination, or both. The problems you'll encounter in sending fast signals around through several cards should not be underestimated; they can turn out to be the major headache of an entire project!

12.10 Driving Cables

You can't run digital signals from one instrument to another by just stringing a single conductor between them, because such an arrangement is prone to pickup of interference (as well as generating interference of its own), and also serious degradation of the digital signals themselves. Instead, digital signals are generally piped through coaxial cables, twisted pairs, flat ribbon cables (sometimes with ground plane or shield), multiwire bundled cables, and, increasingly, fiber-optic cables.[59] Let's look at some of the methods used to send digital signals between boxes of electronics, since these methods constitute an important part of digital interfacing. In most cases there are special-purpose driver–receiver chips available to make your job easier.

12.10.1 Coaxial cable

If you've never dealt with fast signals going through cables, you're in for a surprise.

A. The wrong way

Here's a typical mistake that we see again and again: you've got some digital signals coming out of a "Digital I/O" interface card, for example the PCI-6509 from the popular series of data acquisition products made by National Instruments. This puppy plugs into a computer motherboard's PCI slot, and gives you 96 bits of bidirectional digital I/O, grouped in 12 bytes, each byte of which can be configured as input or output. As an output, each bit generates full-swing 5 V CMOS logic levels (i.e., 0 V and +5 V), with plenty of drive capability (24 mA sink or source), enough to drive easily loads like solid-state relays, small mechanical relays, bright LEDs, and the like.[60]

The mistake is to connect this digital output to a length of coax, and expect it to arrive safely at the far end, as in Figure 12.102A. The thinking goes like this: we've got at least ±24 mA of drive, which should be able to drive the ∼200 pF of a 2-meter length of coax (C=100 pF/m) just fine; after all, $I = C\,dV/dt$ predicts a rise time of ∼20 ns, assuming a typical switching output current of ∼40 mA into the capacitive load. So, what's the problem?

The problem is that we have to treat a coax cable as a transmission line, rather than as the low-frequency approximation of a lumped capacitance, when dealing with signals that are changing on a time scale comparable with (or shorter than) the round-trip delay time of the cable. If you try to use the logic output to drive the cable directly, you'll get a messy waveform at the far end, with overshoots and polarity reversals, producing incorrect recovery of the waveform (and even destruction of the far-end gate). But you can magically fix the problem by simply adding a 50 Ω

[58] This is mandatory, for example, with the fast logic families known as ECL-100K, ECL-100E, ECL-100EL, and ECL-100EP.

[59] Coaxial cables (affectionately called "coax") and twisted pairs are examples of *transmission lines*, discussed in detail in Appendix H.

[60] Look up the 74LVC4245A, if you want to see the specs on the voltage-translating (3 V ↔ 5 V) octal transceivers used in this product.

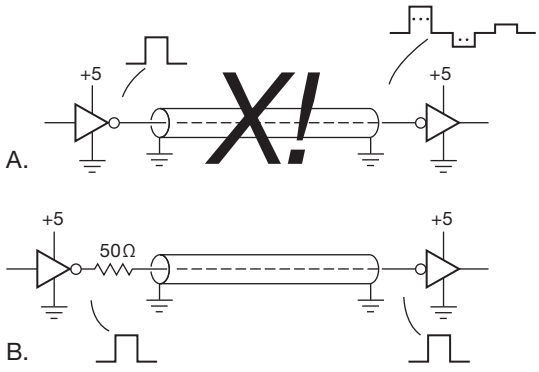


Figure 12.102. A. Driving a length of cable with a logic output produces a mangled waveform of overshoots and polarity reversals. B. Adding a ∼50 Ω series resistor effects a magic cure.

series resistor at the driving end (Figure 12.102B): the far-end waveform becomes a good replica of the driving waveform.

Let's look at this situation in more detail, beginning with the "wrong" way, then progressing through three configurations that solve the problem, each of which have advantages and disadvantages. We'll wind up with the magic fix of Figure 12.102B, known as "series termination," which is well suited to digital-logic signals.[61]

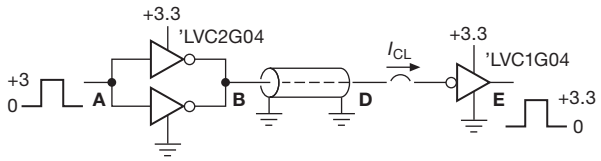


Figure 12.103. Test circuit that attempts to send digital logic signals directly through a length of cable whose far end is "unterminated". The disastrous results can be seen in Figures 12.104–12.106. *Don't do this!*

To illustrate the problem, we hooked up the circuit of Figure 12.103 and ran a pulse pattern through it. You can see the measured results in Figures 12.104–12.106. In Figure 12.104 the far end has been left unconnected: the first transition reaches the far end 12 ns later, where it bounces off the open end (with unchanged polarity), producing an output voltage nearly double the step size; things get messy as the signals bounce back and forth, reversed in polarity with each bounce off the source end, and decaying slowly with each bounce, but always adding the new source signal transitions. The signal at the far end looks terrible – and it

[61] But not to *every* kind of signal: in the world of RF and video the method of "double-ended termination" is used universally.

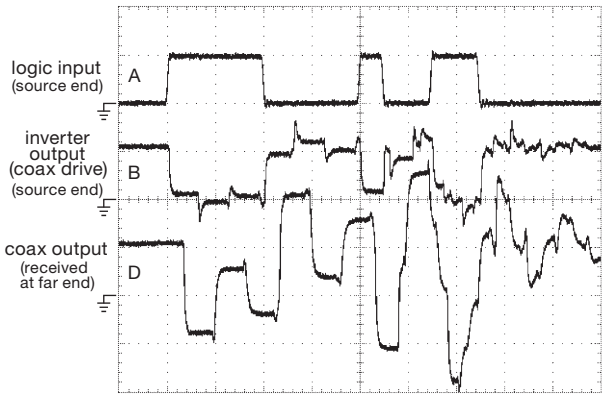


Figure 12.104. Waveforms seen in the circuit of Figure 12.103, when driven with a pulse pattern clocked at 20 ns, and with the far end of the 8 foot (2.4 m) RG-58 coax cable unconnected. The 3.3 V logic-level drive to the cable produces far-end swings of nearly 15 V peak-to-peak! Horizontal: 40 ns/div; Vertical: 3 V/div.

swings to +8 V and −6 V, even though we are driving the cable with 0 V and +3.3 V only. Imagine what will happen when we connect a logic inverter at the far end!

You needn't imagine. In Figure 12.105 we've attached the far end inverter, running also from +3.3 V. Its input protection diodes clamp the renegade waveform, limiting negative swings to roughly a diode drop, but allowing occasional positive swings to +8 V (the 'LVC1G04 has 5 V–compliant inputs, regardless of supply voltage). It's an ugly situation, and it's not surprising that the recovered logic output has some false transitions.

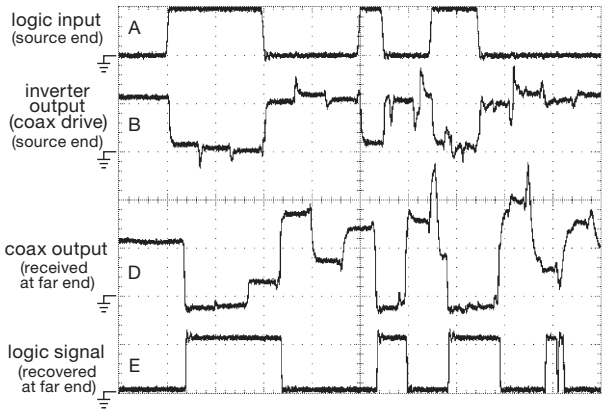


Figure 12.105. Same as Figure 12.104, but with the 'LVC1G04 inverter connected at the far end. The clamping effect of the inverter's input protection diodes reduces the swing at the coax output. This unhealthy situation produces some false transitions at the output; it can also destroy the output inverter.

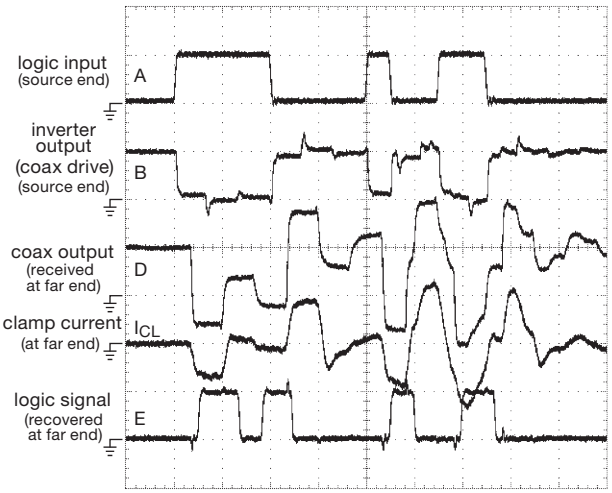


Figure 12.106. Same as Figure 12.105, but with 5 V logic ('LVC2G04 drivers, 'HCT04 far-end receiver). Note the substantial clamping currents, and the error-prone signal recovery. Horizontal: 40 ns/div; vertical: 5 V/div and 20 mA/div.

Turning now to the original scenario – a 2 m cable connecting a digital I/O card to some instrument – the driving signal comes from a hefty 5 V logic output, and the receiver (within some commercial instrument) is likely to be something like an 'HCT04 (5 V logic, with TTL threshold ∼1.4 V). Figure 12.106 shows what happens. It's not a pretty picture: once again the flailing far-end signal is partially clamped, but reaches peaks of +10 V and −5 V, with corresponding clamp currents of ±25 mA. This exceeds the "absolute maximum" input clamp current specification of ±20 mA.[62] And the recovered output is a mess. It's pretty clear that you just cannot hook things up this way.

12.10.2 The right way – I: Far-end termination

The solution is to *terminate* the cable in its *characteristic impedance* $\mathbf{Z}_0$ (see Appendix H), which for most coaxial cables is 50 Ω (resistive).[63] There are three different ways this can be arranged, namely far-end termination, double termination (both ends), and near-end termination ("series termination," "source termination," or "back termination").

Far-end termination is the easiest to understand; see Figure 12.107. The *amazing fact* about transmission lines (explained in Appendix H) is that the addition of a simple resistor of value $R = Z_0$ at the far end suppresses all reflections and, furthermore, makes the cable input look like a pure resistance equal to R. Amazing, because *all capacitance disappears*.

The good news is that this solves the problem. The bad news is that Z_0 is uncomfortably low, usually 50 Ω, which requires rather high driver output currents (20 mA per volt of drive). But you can do it, with hefty logic chips, typically several in parallel, operating with modest logic supply voltages (for example, +3.3 V or less).[64]

Figure 12.107 shows one way to do this, using several sections of 'LVC04 or 'AC04 in parallel to drive the near end of a length of 50 Ω coax that is terminated with a 50 Ω resistor at the far end. The 'LVC2G04 is fully specified at ±24 mA output current and 3 V supply; the corresponding figure for the 'AC04 is ±12 mA. So we're pushing things just a bit here, demanding +60 mA of source current. (No worry, however, if we replaced the single 50 Ω termination with a 100 Ω–100 Ω divider at the far end. That would require ±30 mA of drive, for which we could probably get away with a single 'LVC1G04.)

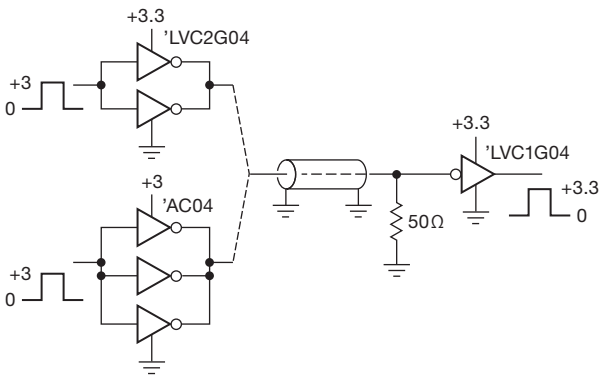


Figure 12.107. Digital logic levels driving end-terminated 50 Ω coax. Apart from losses, the received signals are full-swing replicas of the driving signal.

But it works: Figures 12.108 and 12.109 show the nice waveforms you get, in this case with a 10 m (33 foot) length, operating at the same 20 ns clocking rate. Note the clean waveforms throughout (with a small bit of ringing evident with the 'AC04 driver, probably caused by

[62] For which the datasheet warns "Stresses beyond those listed under 'absolute maximum ratings' may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under 'recommended operating conditions' is not implied."

[63] A big exception is the video community, which has chosen 75 Ω (typified by RG-59 coax); and in pulse circuits you occasionally see 93 Ω (RG-62).

[64] A nice trick that reduces the needed drive current by a factor of 2 is to use a pair of 100 Ω resistors at the far end, configured as a divider between the supply and ground.

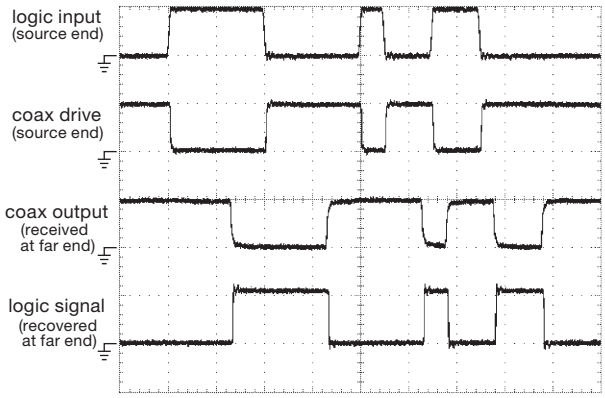


Figure 12.108. Waveforms from the circuit of Figure 12.107 ('LVC2G04 driver), with the same 20 ns clocked pulse pattern as in Figures 12.104–12.106, but with a 10 m length of coax. Horizontal: 40 ns/div; vertical: 3 V/div.

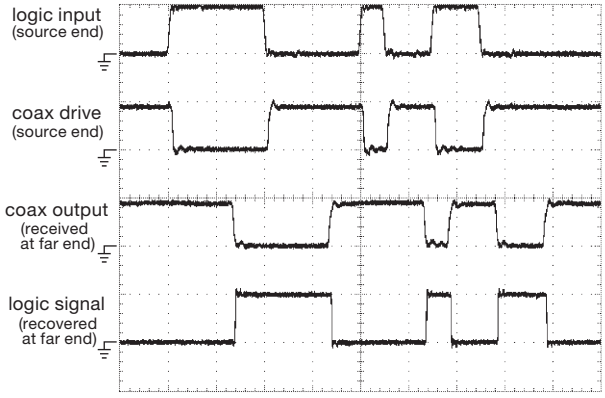


Figure 12.109. Same as Figure 12.108, but with 'AC04 driver.

the greater inductances in power and ground leads of its DIP-14 package, as compared with the compact SOT23-6 surface-mount package of the 'LVC2G04).

If you're worried about the substantial drive current needed to push full rail-to-rail logic-level swings into a 50 Ω load, and if you are not operating at maximum logic speeds, you can substitute a "MOSFET low-side driver" IC for the paralleled logic inverters we used. For example, the venerable TC4420-series of MOSFET drivers accept logic-level inputs, and generate a robust rail-to-rail output swing between ground and the supply voltage, which can range from +4.5 V to +18 V. The peak output current ranges from 1.5 A to 9 A in this series – no problem driving 50 Ω! These are dual drivers, available in many package styles (including DIP) and in inverting or noninverting flavors; they cost about a dollar in small quantity (see Table 3.8 on page 218). There are literally hundreds of MOSFET driver chips available, with some very nice choices from companies like Fairchild, IXYS, Microchip, ST, and TI, among others; most are happy to operate from 5 V supplies (where they have reduced output-drive capability, but they're still pretty muscular; the 9 A IXDD609, for example, can source or sink 2 A when powered from a 5 V supply). Don't be discouraged by the rather relaxed speed specifications on datasheets (e.g., rise and fall times of 20 ns or more), because these are usually specified into the horrendous capacitive loads (1000–10,000 pF, aided and abetted by the Miller effect) of switching power MOSFETs. You'll do much better when driving a 50 Ω far-end terminated coax, which presents a pure resistive load to the driver. For example, the IXYS IXDD509 9 A noninverting driver specifies ∼25 ns rise and fall, but that's into 10,000 pF! From the curves later in the datasheet, however, you'll discover rise and fall times of 4 ns or less for supply voltages anywhere between +5 V and +30 V.[65]

A. The right way – II: Double-ended termination

The high currents needed to drive end-terminated 50 Ω coax can be remedied somewhat by adding a series resistance at the cable's input, equal in value to its characteristic impedance (i.e., 50 Ω; see Figure 12.110). Then the driver sees a load of 100 Ω (the series resistor plus the 50 Ω seen at the cable's input). That's sometimes called "double-ended termination." It has an additional advantage, namely that any signals reflected from the far end are swallowed by the input resistor, which acts effectively as a termination for signals traveling backward.

It works, but now the output-signal amplitude is half that of the driver, because the cable's input resistance forms a voltage divider with the series resistor at the input. That's why we used 5 V logic at the input, combined with inverters at the output whose thresholds are in the range of 1.2–1.4 V: either the TTL threshold 'ACT04 (see Figure 12.111), or logic running from a lower +3 V supply (see Figure 12.112). The 'ACT04 specifies input logic levels of $<$0.8 V and $>$2.0 V; the corresponding figures for the 'LVC1G04 (powered from +3.0 V) are $<$0.8 V and $>$1.7 V.

[65] In spite of the good rise and fall times, there tend to be substantial delay times, regardless of load capacitance, at low supply voltages; for this IXYS part, the delay times are of order 30 ns with a +5 V supply, dropping to half that at V_S=10 V. From a practical point of view, this limits operation to speeds of a few megahertz. A more serious problem at high frequencies is the rising rail-to-rail supply current, and consequent power dissipation, at frequencies above a megahertz that are due to the high internal capacitance of the large output MOS devices. Some parts show plots of no-load supply current versus frequency in their datasheets.

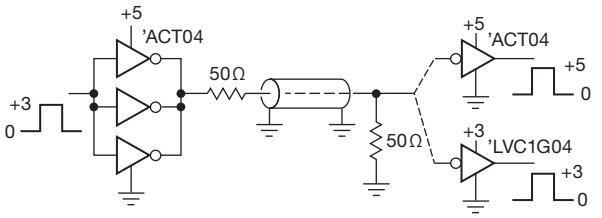


Figure 12.110. Digital logic driving double-end terminated 50 Ω coax. The received signal amplitude is half that of the driver logic output, therefore ~2.5 V in this example.

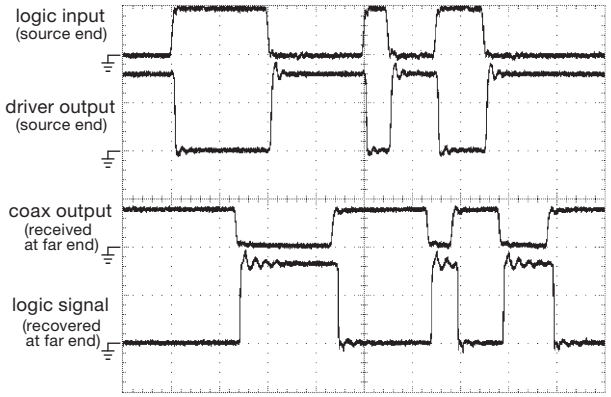


Figure 12.111. Waveforms from the circuit of Figure 12.110 ('ACT04 receiver), with the same 20 ns clocked pulse pattern as in Figures 12.104–12.109, and with a 10 m length of coax. Horizontal: 40 ns/div; vertical: 3 V/div.

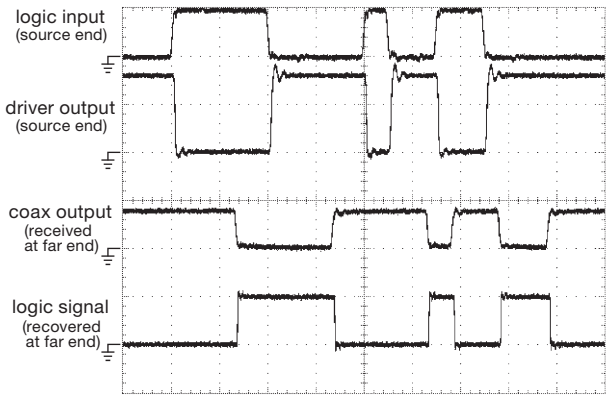


Figure 12.112. Same as Figure 12.111, but with 'LVC1G04 receiver.

The method of choice, for RF and video Because double-ended termination results in a received signal that is half the amplitude of the unloaded driver, it is not well suited for *digital logic* applications like this. It's better to use series termination (next subsection). Lest we leave the wrong impression, however, we note that the use of double-ended termination is *universal* in the RF and video worlds: all such signal sources are built with an output impedance equal to that of the cable impedance (50 Ω for RF, 75 Ω for video), and the far end of every cable is terminated in that same resistance. And the factor-of-two amplitude reduction is addressed by arranging the open-circuit amplitude of every signal source to be exactly twice what is ultimately needed at the (properly terminated) far end. You see this in signal and function generators – the output amplitude, as measured with a 'scope probe, is double what you've set it for (because it assumes you've attached a 50 Ω load resistor, or a cable with 50 Ω far-end termination). And in the video world you'll find plenty of "video buffer amplifiers" that provide a gain of exactly ×2, to compensate for the corresponding loss when driving a terminated cable. Figure 12.113 shows an example from LTC: their LT6553 is a triple video buffer (three independent amplifiers, to handle color analog video), with an internally set gain of 2, intended for driving 75 Ω video cable, as shown. It has impressive bandwidth (650 MHz) and slew rate (2500 V/μs), and can drive a ±3.5 V swing into the 150 Ω load it sees in this circuit, when powered from ±5 V supplies. Their companion LT6554 has the same specs, but with unity gain.

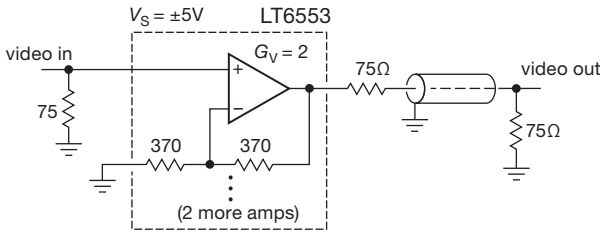


Figure 12.113. A video buffer amplifier drives a 75 Ω load through 75 Ω coax. The voltage gain of G_V=2 compensates for the ×2 signal attenuation caused by the series resistor at the amplifier's output.

B. The right way – III: Series termination

To summarize the previous methods: end termination with direct drive requires lots of drive current, but it delivers the full driver signal at the far end. Double-ended termination reduces the drive current, but attenuates the driver signal amplitude by a factor of 2.

There's a third way, which captures the best of each: use a series resistor at the source end and *no* termination at the far end. This is sometimes called "series termination" or "back termination" (Figure 12.114). It exploits a property of open-ended transmission lines, namely that there is a

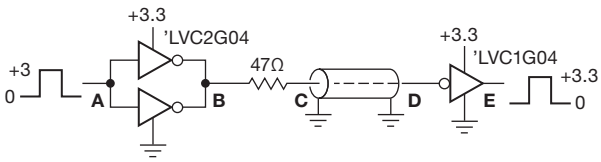


Figure 12.114. Series termination presents a load of twice the cable's characteristic impedance (therefore 100 Ω), while delivering the full driver amplitude at the far end. We recommend this method for logic outputs on instrument panels.

full-amplitude reflection from the far end, of the same sign as the incident signal, thus producing an output amplitude of twice the incident amplitude. But, because the incident amplitude is half the driver's output (owing to the voltage divider formed by the series resistor and the cable's input impedance), the net result is an output just equal to that of the driver. Voilà – full output swing, without the need to drive the cable's low impedance. And we don't get the ugly situation of signals bouncing back and forth (as in Figures 12.105 and 12.106), because the series resistor acts as a proper end termination for the backward-going signals. Instead you get one reflection off the far end, which gets swallowed at the source end, with the production of some steppy waveforms at the near end (Figure 12.115).

Series termination is the method of choice for piping logic signals through cable. This technique has the pleasant properties of (a) presenting a (less severe) load impedance equal to twice the cable's characteristic impedance, and (b) requiring *no* continuing drive current after a logic-level change has propagated down and back. To expand on this latter point: immediately after a step change, current flows from the driving gate, $I = V_{CC}/2Z_0$, or 50 mA for 5-volt logic driving a 50 Ω cable; but this ceases after a round-trip propagation delay. We have +50 mA for low-to-high steps, and −50 mA for high-to-low steps. Often we'll estimate the R_{OUT} of the driving gate and reduce the value of the added source resistor accordingly. But if the total source resistance is too low (less than 50 Ω), the returning echo will overshoot V_{CC} (for low-to-high steps) or undershoot ground (for high-to-low steps) for a time equal to a propagation delay. If the overshoot is high enough (i.e., more than 10% for a 5 V supply) it will cause current to flow in the gate's output clamp diode.

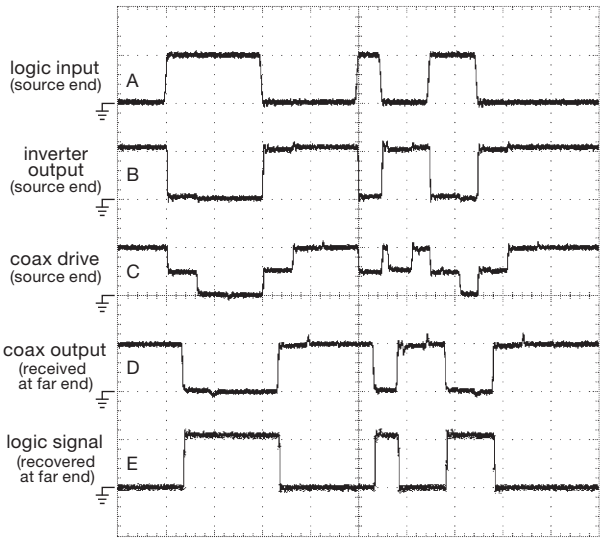


Figure 12.115. Waveforms from the circuit of Figure 12.114, with the same 20 ns clocked pulse pattern as in Figures 12.104–12.112, and with an 8-foot (2.4 m) length of coax. Horizontal: 40 ns/div; vertical: 3 V/div.

The logic family known as "AUC Little Logic" (one of TI's low-voltage logic families, one or two gates in 5-lead or 6-lead SMT packages) has an unusual output structure that is well suited for driving PCB traces or lengths of coax. Its output impedance is a reasonable approximation to a series termination, so you can drive a 50 Ω transmission line directly from the logic gate output, without any series resistor. The family is optimized for a +1.8 V supply voltage. As described in the Application Report[66] the output stage consists of three paralleled inverters, such that the driving impedance changes during a logic transition: it starts low (for high drive current), then becomes an approximate match to the transmission line, suppressing ringing or reflections. Also, there is no clamp diode from the output to the positive supply, so the output is "3.6 V tolerant" and is not damaged by back reflections from the open-ended transmission line.

Although they suggest cable or trace lengths only to 15 cm or so, we found that these devices work well for substantially longer runs of cable. Figure 12.116 shows the signals when driving a 30 cm length of 50 Ω cable (RG-316 "skinny" coax) with 100 MHz NRZ[67] data (twice the rate used in Figures 12.104–12.112 and 12.115). How far can you go with this driver, and at these speeds? Figure 12.117 shows the logic input pattern, and the (open-ended) coax output (points "A" and "D" in Figure 12.114), when driving lengths of 50 Ω coax[68] up to 5 m long. Looks pretty good, to us!

[66] *Application of the Texas Instruments AUC Sub-1-V Little Logic Devices*, SCEA027A (September 2002).

[67] "Non-return to zero," a fancy name for just sending each bit value, as a logic level, for one clock period.

[68] RG-141 for 500 cm, RG-316 for the rest.

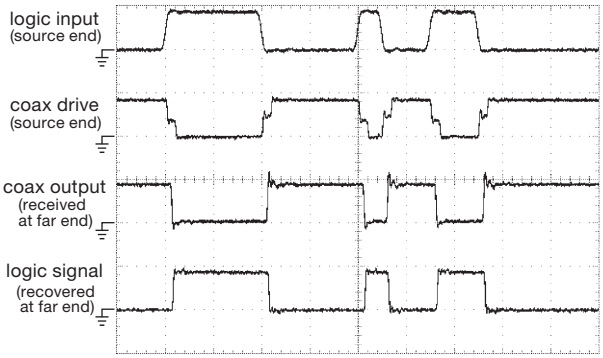


Figure 12.116. Waveforms from a 74AUC1G04 inverter operating at 1.8 V and directly driving a 30 cm length of coax, with the same pulse pattern as earlier, but at a 10 ns clock rate. The four traces (top to bottom) correspond to points A, C, D, and E in Figure 12.114, but with no resistor. The approximate half-step at the coax input confirms a drive impedance close to 50 Ω. Horizontal: 20 ns/div; vertical: 2 V/div.

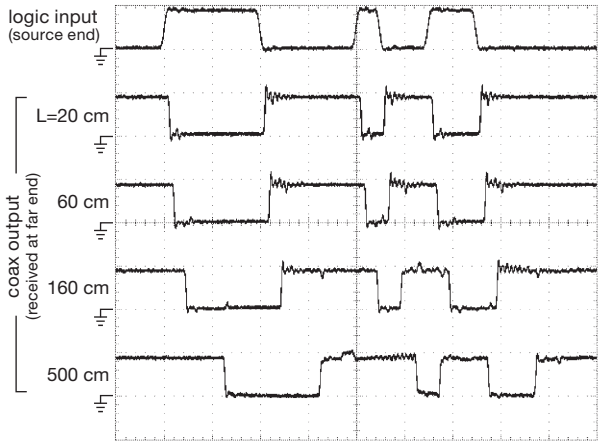


Figure 12.117. Same setup as Figure 12.116, showing received waveform at far end of coax cables with the indicated lengths. Horizontal: 20 ns/div; vertical: 2 V/div.

C. Driver pre-emphasis and receiver equalization

The measured waveforms above look pretty good. But we're not really pushing the limits, at these data rates of 50–100 Mbps NRZ. Things get pretty messy when you try to send upward of a few hundred megabits per second, because the coax cable itself becomes lossy and attenuates higher frequencies (see Appendix H). For example, the popular RG-58A (used for everybody's BNC patch cords) attenuates about 10 dB per 30 m at 500 MHz. You do better, of course, with cables of lower loss (e.g., RG-8 attenuates about 5 dB per 30 m). But these are bulkier – RG-8 cable is 10 mm in diameter, double that of RG-58.

Here's what you can do instead: at either the driver end or the receiver end (or both) you can compensate for the cable's loss (and phase shift) at higher frequencies. This is called *pre-emphasis* and *equalization*, respectively. Figures 12.130–12.132 later in the chapter show how it goes, in the case of a differential twisted-pair cable or printed-circuit differential stripline. For a (single-ended) coax line you could use a chip set like NSC's DS15BA101 and DS15EA101; the latter is a receiver with adaptive equalizer, which can apply up to 35 dB of boost at 750 MHz.[69] You can think of this trick as analogous to "turning up the treble," in an audio system, though here we need to worry about phase as well. It works quite well, owing to the very good signal-to-noise ratio of these large signal swings running through shielded (or twisted-pair) transmission lines. We'll see this next, in connection with LVDS signaling on differential pairs.

12.10.3 Differential-pair cable

There's another way to convey digital signals on cables, namely to use differential signaling, most often by way of a twisted-pair cable.[70] A common example of the latter is "Cat-5" (or Cat-6) Ethernet cable, which is an unshielded cable containing four independent twisted pairs, with characteristic impedances of 100 Ω. Some advantages of differential signaling are the suppression of common-mode interference and ground noise, the ability to use smaller signal swings (hence smaller drive currents),[71] and the large reduction of both radiated noise and signal-rate ground-current fluctuations (from the balanced differential swings, as well as the smaller amplitudes). Two popular differential signaling standards are RS-422 and LVDS. They differ in several ways. Roughly speaking, RS-422 is used for

[69] This chip, like other adaptive cable equalizers that are widely used in digital video systems (conforming to 75 Ω coaxial video serial digital interface standards with names like SMTE 259M, 292M, 344M, or 424M), is meant to be ac-coupled, and works only down to some minimum frequency of order 150 Mbps. See National Semiconductor's App Note AN-1909.

[70] Which may be shielded twisted pair ("STP") or unshielded twisted pair ("UTP"). It need not be twisted, however: you'll see adjacent wire pairs on a flat ribbon cable used for differential signaling, as well as parallel trace pairs ("microstrip," "stripline") on a printed circuit board.

[71] Because the signal is differential, it does not require accurate alignment of thresholds (as, for example, with the fussy single-ended 10K/100K ECL logic, whose ∼0.8 V swing mandates careful temperature compensation); the differential signal pair need stay only in the common-mode range of the receiver, e.g., 0 V to +2.4 V for LVDS, with its ∼0.35 V differential swing.

data rates up to a few megabits per second, and cable runs up to a kilometer. It uses differential *voltage* drive, and is common in industrial control applications. By contrast, LVDS is used for data rates to several gigabits per second, over distances to several meters. It uses differential *current* drive, and is common in short-range high-rate applications such as backplanes (e.g., PCIe[72]) and transmission of serial data (e.g., SATA, Firewire). Both are point-to-point connections, but provide multidrop variants (RS-422 → RS-485; LVDS → M-LVDS).

A. RS-422 and RS-485

These comprise a popular industrial data bus, improving on the venerable RS-232 data link standards (see discussion at §§12.10.4 and §14.7.8), and substantially extending the latter's capabilities, see Figure 12.134. The RS-422 and RS-485 standards[73] specify signal properties used in a differential voltage-drive arrangement like that shown in Figure 12.118. The differential outputs typically swing most of the way between ground and the +5 V rail, although the specification permits differential-output swings as small as ±2 V or as large as ±10 V. The receiver must respond to differential inputs as small as ±0.2 V, over a common-mode range of −7 V to +7 V (−7 V to +12 V for RS-485). The characteristic impedance of differential twisted pair cables is typically 100–120 Ω, so you've got to terminate the far end with that value resistor. You often see both ends terminated (as in Figure 12.118), which is not necessary (or desirable) with unidirectional RS-422, but required with the bidirectional (or multipoint) variant (RS-485).

Figure 12.119 shows measured signals traversing one of the four twisted pairs in a 140 m length of Cat-6 Ethernet cable, with a, by now, familiar bit pattern clocked at 10 MHz (i.e., an NRZ rate of 10 Mbps). Although this is ten times faster than allowed by the RS-422 specification (Figure 12.134), the differential nature of the received signal allows clean logic recovery nonetheless; note the propagation delay of ∼700 ns (mostly caused by the 4.7 ns/m signal velocity in the cable).

Differential signaling provides impressive immunity to common-mode interference. The latter may arise from nearby signal-carrying wires or from radiated signals (from intentional emitters like radio/TV, wireless networks, etc.);

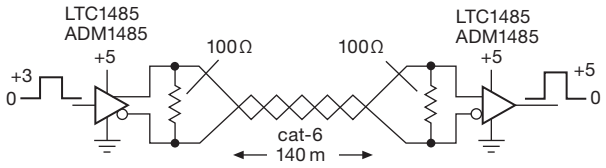


Figure 12.118. Differential voltage signaling with RS-422/485 over a length of unshielded twisted pair, using transceiver chips with 3-state outputs (each chip has a separately enabled Tx and Rx, sharing a common differential pair).

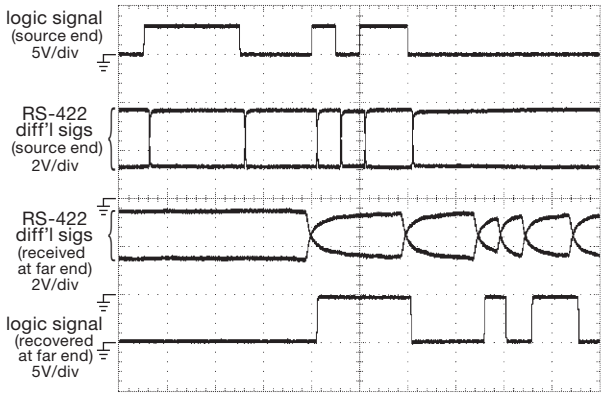


Figure 12.119. Waveforms from the circuit of Figure 12.118, with a 100 ns clocked pulse pattern, and a 140 m length of Cat-6 network cable. Horizontal: 200 ns/div.

it's also common to see a difference in ground potentials between equipment that is plugged into different power outlets. We have seen as much as a volt or two of 60 Hz ac between instruments in the same room!

We hooked up the circuit in Figure 12.120 to illustrate common-mode rejection in RS-422 signaling. The pseudo-random noise generator adds a ∼15 Vpp noise voltage into the (floating) ground of the signal source, as seen from the destination end. The results are shown in Figure 12.121: both the 5 V logic signal and the RS-485 differential pair signals are hopelessly noised up, reaching peak levels of −7 V and +12 V (the specified receiver common-mode range) as seen from the destination end. But the magic of common-mode rejection recovers the original logic intact. Note that the noise is very much "in-band," with wiggles on the time scale of the digital data. Although the figure captures a single-shot data burst, the result is robust – we ran it many times, with never an error.

Slew rate limiting If you don't need the speed, it's wise to choose driver ICs of lower slew rate because you get less signal coupling from signals on adjacent pairs. RS-422 and

[72] Shorthand for "PCI Express," itself shorthand for "Peripheral Component Interconnect Express."

[73] Officially known as ANSI TIA/EIA-422 and TIA/EIA-485; however, most engineers continue to use the original designations "RS-422" and "RS-485," or, more loosely, they'll just say "422" or "485." In context it's clear what they mean.

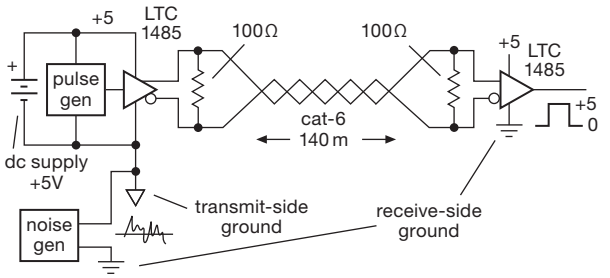


Figure 12.120. Torture test for RS-422 common-mode noise rejection. We floated the driver-side circuitry, and drove its "ground" with band-limited noise of ∼15 Vpp amplitude, relative to far-side ground.

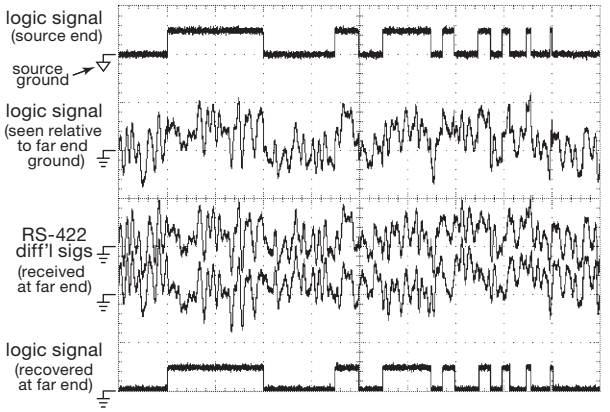


Figure 12.121. Waveforms from the circuit of Figure 12.120. As seen from the far end, the input logic signal is buried in the noise, extending from −7 V to +12 V. The individual differential signals are likewise a mess, but the recovered logic output is error free. The signal pattern includes pulses with widths from 200 μs down to 5 μs. Horizontal: 100 μs/div; vertical: 10 V/div.

RS-485 drivers are available with a selection of slew rates, for example the Maxim MAX3293-95 series, with specified rates of 250 kbps, 2.5 Mbps, or 20 Mbps. Other examples include the MAX481–489- and 1481-1487-series, the LTC2856–2858-series, and the 65ALS176 versus the 75ALS176B.

RS-422 versus RS-485 RS-485 is basically RS-422 with some additions that make it possible to have several drivers share a single signaling pair: this requires that the drivers have an ENABLE input, so they can be put into a high-impedance (noninterfering) state, analogous to the use of three-state drivers on a shared (single-ended) data bus line.[74] RS-485 interface chips (e.g., the classic 75ALS176, or LTC1485) usually combine a transmit–receive pair, sharing the same differential signal lines ("half duplex"),[75] with complementary ENABLE pins (called DE and RE′); such a *transceiver* chip looks like Figure 12.122.

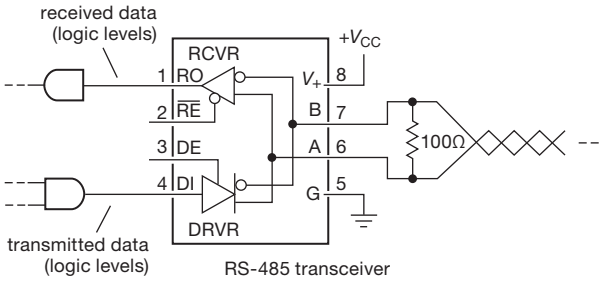


Figure 12.122. An RS-485 *transceiver* combines driver and receiver in one package, sharing a single twisted pair. Separate ENABLE pins (usually tied together) permit operation in either direction. The pin numbering shown is an industry standard.

Along with the ENABLE feature that is necessary for multiple drivers (called "multipoint"), the RS-485 specification beefs up some other RS-422 specifications: (a) it expands the receiver common-mode input range (−7 V to +12 V), thereby allowing symmetrical common-mode swings (up to ±7 V) around the traditional power-supply and signaling levels of 0 V and +5 V; (b) it lowers the minimum load resistance (to 54 Ω), which is necessary because a driver at either end requires end-termination at the other end (thus a 100–120 Ω at *both* ends). The RS-485 specification also (c) expands the allowable common-mode range that can be applied to a driver *output* (in the disabled state), from a diode drop beyond the rails (RS-422) to the full −7 V to +12 V range of an RS-485 input. This is necessary, of course, because the driver and receiver are both connected permanently to the signal lines in a typical bidirectional (or multipoint) arrangement; if the driver's output clamped to ground or the positive rail, it would defeat the receiver's wider common-mode range.

Most RS-485 transceivers satisfy also the narrower RS-422 specifications, so you might as well just use one of the hundreds of available RS-485 interface chips (from manufacturers like Analog Devices, Intersil, LTC, Maxim, or TI), even if you are sending data in only one direction; simply enable what you are using at each end. Some common choices are the 'ALS176 (and its many imitators –

[74] Good references: Analog Devices AN-960 and AN-727, National Semiconductor AN-759, Maxim AN-3776, and TI SLLA112.

[75] An interface chip that brings RS-485 driver and receiver pairs out separately is called "full duplex."

65LBC176, 65HVD1176, etc.), the 75176 and 75ALS180 classics, the LTC1480/5, and the ADM1485.

Isolated RS-422/485. Scenario: a "factory floor," with a host of automated machine tools and with products moving along on conveyor belts (think of a scene from the ever-informative "How It's Made" series on cable channels like Discovery Channel or Science Channel). Sensors send digital information through cables that snake along on overhead cable trays, and wind up at a central computer control; the same cables, or different ones, carry commands back to actuators. This centrally controlled network choreographs factory activities, ultimately directed toward the practical (and profitable) goal of… making stuff. It's common to see these signals piped around as differential RS-485 or a variant like the Process Field Bus (PROFIBUS®) that uses a similar "physical-layer" signal.[76]

These activities may be spread out over distances of hundreds of feet, and may go between buildings. Let's face it, with all that heavy machinery thumping along, you're likely to see common-mode transients that exceed even the "generous" specifications of RS-485.[77] The solution here is to use *isolated* interface chips, in which the RS-485 signals have their own independent ground, galvanically isolated from the ground of the logic signals. This requires a second dc power source, of course, floating with respect to the logic signal ground.

There are many isolated RS-485 interface chips to choose from. The clever designers use various tricks to make fast digital signals cross a gap that can withstand a kilovolt or so. For example, the LTC1535 (or MXL1535) "isolated RS485 Transceiver" uses small capacitors to couple the (modulated) digital signals (Figure 12.123). It also helpfully includes a high-frequency (∼420 kHz) oscillator, whose output you can use with a transformer to isolate and rectify to make the isolated dc for the RS-485 signal side. The ISO15/35 from TI is a less-expensive alternative, which omits the oscillator; you've got to provide isolated dc from scratch (either from an isolated dc–dc converter, or an ac-line-powered dc supply).[78]

Another approach is to couple with small (chip-scale) transformers. This technique is used in the ADM2485 series of isolated transceiver from Analog Devices, which also provides an oscillator output that can be used for generating the isolated dc (like the LTC1535). Or, you can add the popular MAX845 or MAX253 chips, which generate a pair of complementary square wave outputs at ∼0.75 MHz, suitable for directly driving a small isolation power transformer, good to a half-watt or so of isolated dc. The electronics industry makes it easy to use this part: major transformer companies even offer "MAX845 transformers."

An interesting technique is used in the IL3485 from NVE Corp, namely the "giant magnetoresistive effect" (GMR, widely used in hard-disk drives to sense the magnetically stored bits on the rotating platters). Instead of conventional transformer coupling, where a secondary winding senses flux *changes*, the NVE part uses GMR to sense the field directly.

Finally, isolated transceivers like the MAX1480/90 use optocouplers for the digital signals. And, to make your life really easy, they include an internal transformer to generate isolated dc power. These hybrid ICs include also the diodes and capacitor to complete the power circuit, so all you need to provide is a +5 V logic-side supply. That's the good news; the bad news is that these puppies will set you back twenty bucks (in small quantities), compared with $5–$8 for the capacitor- or transformer-coupled chips.

Ethernet PHY When thinking about moving digital data across isolation barriers, don't forget about the optical couplers and fibers earlier in this chapter, and also about isolation via pulse transformers, as used, for example, in local-area networks such as Ethernet. Figure 12.124 shows the physical layer ("PHY") of an Ethernet link, with pulse isolation transformers (which everyone just calls the "magnetics."). They provide excellent rejection of common-mode interference (and of course of differences in potential at the ends), using both a transformer and a common-mode

[76] PROFIBUS transceivers have higher maximum data rates, typically 30 Mbps or 40 Mbps, and generally meet the (less-rigorous) specifications of RS-422 and RS-485; check out parts like the 65ALS1176, ADM1486, ISO1176, or ISL4486. Some other industrial buses that use RS-485 signaling include BITBUS, Data Highway (DH-485), INTERBUS-S, Measurement Bus (DIN 66348), Optomux, P-NET, and Series 90 (SNP).

[77] As eloquently expressed in the MAX1480 datasheet, "The RS-422/485 standard is specified for cable lengths up to 4000 feet. When approaching or exceeding the specified maximum cable length, a ground-potential difference of several tens of volts can easily develop. This difference can be either DC, AC, at powerline frequency, or any imaginable noise or impulse waveform. It is typically very low impedance so that if a connection between the two grounds is attempted, very large currents may flow. These currents are by their nature unstable and unpredictable. In addition, they may cause noise to be injected into sensitive instrumentation and, in severe cases, might actually cause physical damage to such equipment." Amen.

[78] You sometimes see industrial RS-485 hookups with bused dc (+24 V or +48 V "telecom dc") bundled in the same cable as the RS-485 signal pair(s); hang an isolated dc–dc converter onto that, at each transceiver node, to power the RS-485 signal side of a transceiver like the ISO15/35.

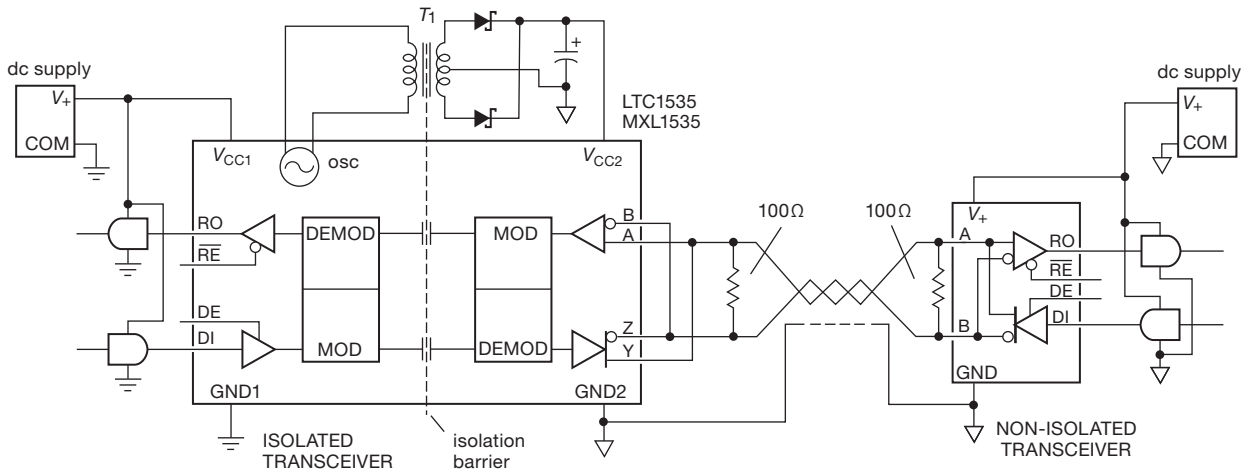


Figure 12.123. An "isolated" RS-485 transceiver galvanically separates the circuitry connected to the cable pair from the logic-level circuitry. The far-end ground ("GND2," triangular symbol) is brought back to the isolated transceiver, but is never connected to the local ground ("GND1," normal symbol). Such an arrangement prevents ground loops, and can accommodate common-mode and ground offset potentials of hundreds of volts. Additional isolated transceivers can be connected ("multidrop") along the span, referenced to the same GND2, each with its own isolated RS-485-side power supply; such midspan "stubs" must be kept short in length, and without 100 Ω terminating resistors.

choke, as shown.[79] If you've got a microprocessor in your system already, why invent your own isolation hardware, when Ethernet works just fine? And, if you need to go greater distances, you can use a media converter to carry Ethernet signals on fiber. Check out the offerings from manufacturers like Allied Telesis.

B. LVDS

In contrast to RS-422, which is intended for modest data rates (10 Mbps and below) over relatively long cables (to 1 km), the LVDS (low-voltage differential signaling, also known as RS-644) standard is aimed at much higher data rates (to 1 Gbps and above) over shorter cables (to ∼10 m) or even shorter circuit-board trace runs. Rather than driving the wire pair with crisscrossing *voltages* of a few volts amplitude, LVDS switches *currents*: an LVDS driver sinks and sources 3.5 mA (nominal) into a wire pair that is terminated at the far end in its characteristic impedance (usually 100 Ω). That produces a differential voltage at the far end of ±350 mV. The driver is obliged to maintain a common-mode voltage, nominally +1.2 V; so the receiver sees crisscrossing voltages of approximately +1.0 V and +1.4 V. The relatively low common-mode voltage was chosen deliberately to accommodate both driver and receiver chips operating at low supply voltages. That's important because digital chips are moving steadily to lower voltages, and the

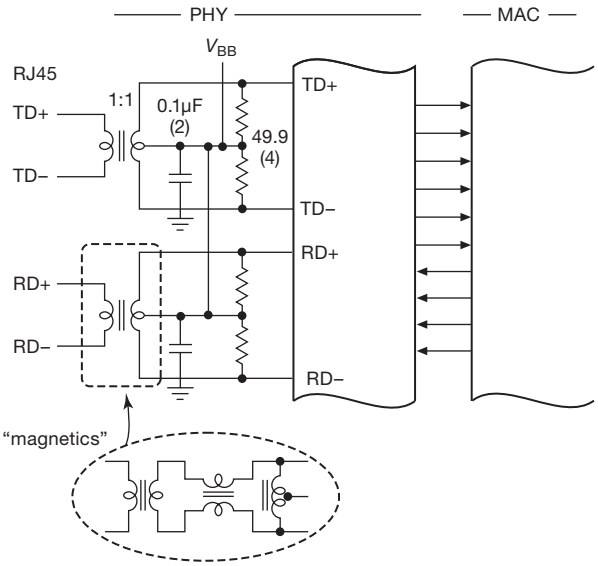


Figure 12.124. Ethernet uses transformer coupling and common-mode chokes for robust isolation of its differential signaling. Closest to the wiring is the physical layer ("PHY"), followed by the media-access control layer ("MAC," as in "MAC address").

[79] You get additional robustness by adding surge-protecting components, notably the Transient Blocking Units (TBU™) from Bourns. These are little 2-terminal elements that go in series with the signal leads from the transformer: they act like a low resistance, up to a critical current, at which point they go into a high resistance state.

LVDS interface is often incorporated right into a complex chip that wants to run at low voltages.

We hooked up the circuit in Figure 12.125 (analogous to Figures 12.120 and 12.121) to illustrate LVDS's rejection of common-mode interference. We injected a trapezoidal waveform at ∼30 MHz into unused pairs of a network cable, and cranked up the amplitude until the received signal reached the specified common-mode receiver limits of 0 V and +2.4 V. The results are shown in Figure 12.126: You can see the two received signal voltages weaving busily past each other, at the command of the input signal (delayed by that speed-of-light nuisance, of course), cleanly recovered by the LVDS receiver.

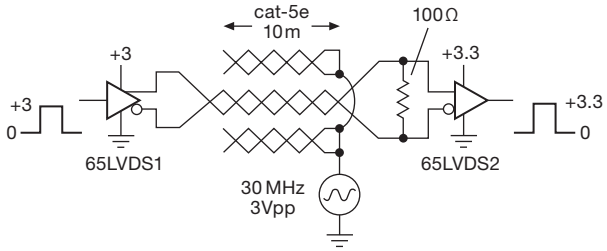


Figure 12.125. Torture test for LVDS common-mode interference rejection. We applied a ∼30 MHz signal to two unused pairs at the far end of a 10 m length of Cat-5e network cable, while transmitting a pulse sequence (clocked at 20 ns) through a different pair.

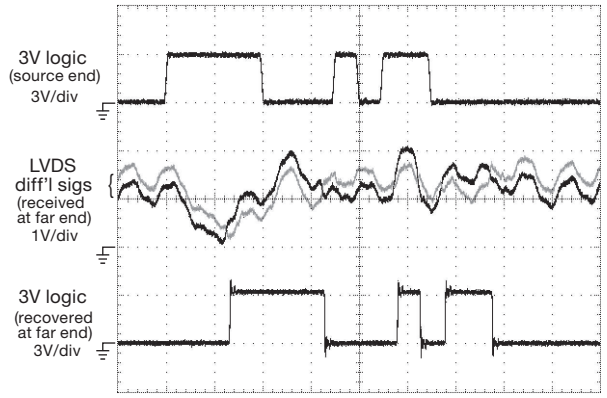


Figure 12.126. Waveforms from the circuit of Figure 12.125. The injected signal adds up to ±1 V of common-mode interference to the ∼400 mV differential 50 Mbps received signal. Horizontal: 40 ns/div.

To illustrate the problem of conveying digital signals between instruments whose grounds differ by a small amount of 60 Hz ac, we rigged up the circuit of Figure 12.127. We "floated" the source device's ground, analogous to the arrangement earlier with RS-422 (Figure 12.120), then drove it with a 2 Vpp sinewave. The results are shown in Figure 12.128, where we've clocked down the data rate so you can see the powerline frequency. Note that LVDS permits only 2 Vpp of added common-mode signal, compared with 14 Vpp for RS-422/485; but with LVDS you get lots of speed, and compatibility with low-voltage logic.[80]

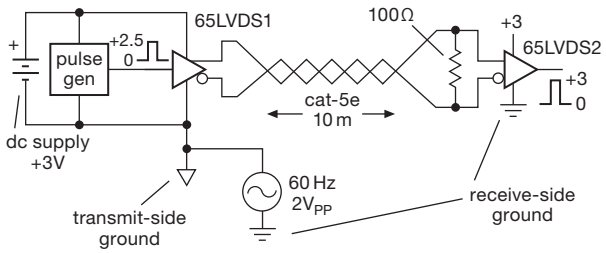


Figure 12.127. Powerline common-mode test setup, using a floating LVDS signal source (complete with pulse pattern generator and power supply).

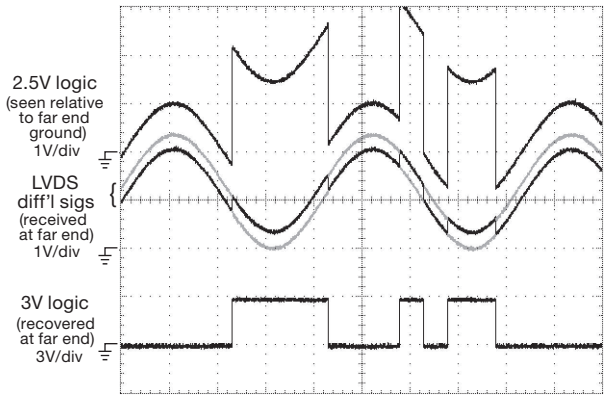


Figure 12.128. Waveforms from the circuit of Figure 12.127. The ∼400 mV differential signal is superposed on the added 2 Vpp 60 Hz sinewave, reaching the specified receiver common-mode limits of 0 V and +2.4 V. Horizontal: 4 ms/div.

The LVDS protocol is widely used in serializer-deserializer links (see also §12.8.4, §12.10.5, and §14.7), in which a fast serial link connects a pair of separated parallel registers. At each end you think you're talking to a parallel port (and you are!), but in between the data travels as serial bits. A common data width is 10 bits, which lets you send a byte plus two extra bits to signify whatever you want (a new byte, or the beginning of a new "frame"

[80] You can get LVDS receivers with extended common-mode capability, for example the 65LVDS34, which specifies operating common-mode range of −4 V to +5 V (they do it with a resistive input divider, so R_{in}=250 kΩ).

of bytes). Figure 12.129 shows how it goes, in this case with a SERDES pair with a relatively relaxed speed specification: the transmit clock, which times the entry of 10-bit data symbols, can be in the range of 16–40 MHz. The serialized data bits that go over on the LVDS link are at ten times that rate, i.e., up to 400 Mbps.[81] You need only one LVDS differential pair, because the deserializer performs "clock recovery" from the transmitted bitstream. In the figure we've omitted some additional details having to do with modes and synchronization.

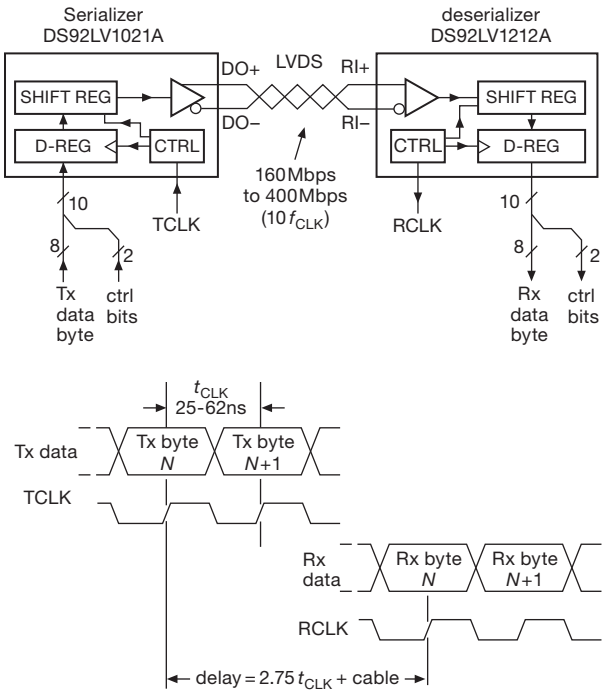


Figure 12.129. A serializer-deserializer pair lets you clock parallel data into a register at the transmit end; it appears, magically, as parallel data at the receive end. There are a few clock cycles of delay, but nothing to write home about.

LVDS signal conditioning As we remarked earlier (§12.10.2C), the use of transmit pre-emphasis and receive equalization can compensate for the frequency-dependent loss of the connecting cable. These *signal-conditioning* techniques reduced greatly the effects of data-dependent jitter, which is also known as *intersymbol interference*, or ISI.

This ISI effect is worth a few words of explanation: because of the lowpass filtering effect of the losses in cables or printed-circuit traces, particularly when operating at very high bitrates (gigabits per second), the initial received signal voltage at the beginning of each bit cell depends on the previous bit (or bits), and so the time to cross the threshold will vary somewhat, depending on the previous bit(s). Intersymbol interference plagues high-speed communication in all forms. You can see this effect in the 1.5 Gbps recovered LVDS waveforms in Figure 12.132, in which the 'scope traces, triggered by the clean clock signal, show the scatter of signal levels on the data line carrying a pseudo-random data pattern. This is called an "eye diagram;" it is a standard tool for visualizing noise and jitter in assessing signal quality in a clocked data stream.

The use of pre-emphasis and equalization can significantly extend bitrates and transmission distances, as we remarked earlier in §12.10.2C in connection with coaxial cables. Figure 12.130 shows the scheme, and Figure 12.131 shows how pre-emphasis boosts the voltage levels following each transition; and when combined with receiver equalization (high-frequency boost), the result is a clean replica of the original input signal. Such signal conditioning is used in various driver and receiver ICs, for example the DS25BR100/200/400 and DS25CP102 series from NSC; see their App Note AN-1957 for details and illustrative waveforms.

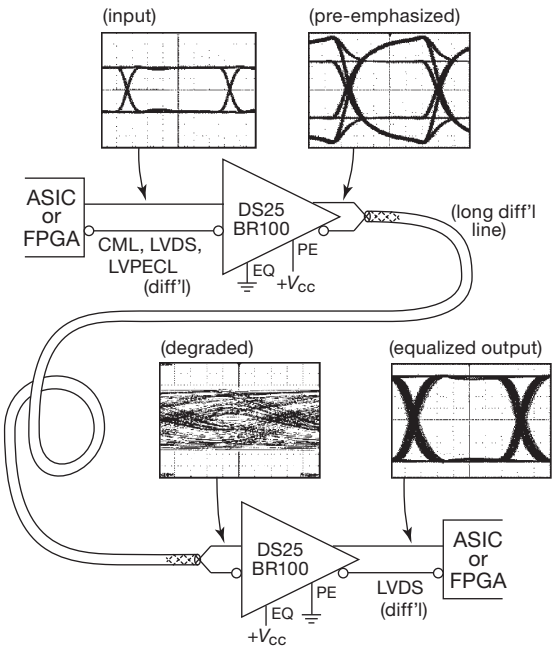


Figure 12.130. Driver pre-emphasis, receiver equalization, or both can compensate for cable losses when transmitting at high data rates over lossy media.

[81] The serial bitrate is actually 12 times the clock rate, because the serializer adds two clocking bits.

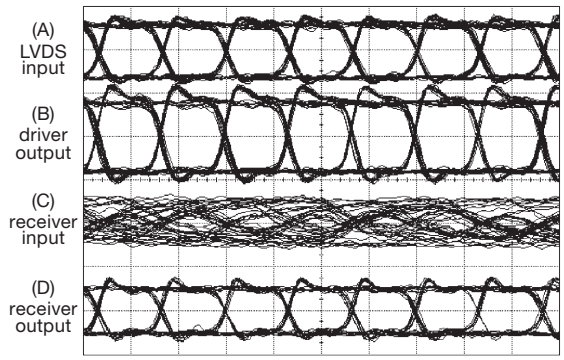


Figure 12.131. Waveforms from a 1.5 Gbps pseudorandom LVDS signal traversing a 2.5 m length of differential stripline on a printed circuit board. A. Input to DS25BR120 driver; B. Driver output, with pre-emphasis; C. Input to DS25BR110 receiver; D. Receiver output, with pre-emphasis and equalization. Vertical: 500 mV/div; horizontal: 500 ps/div.

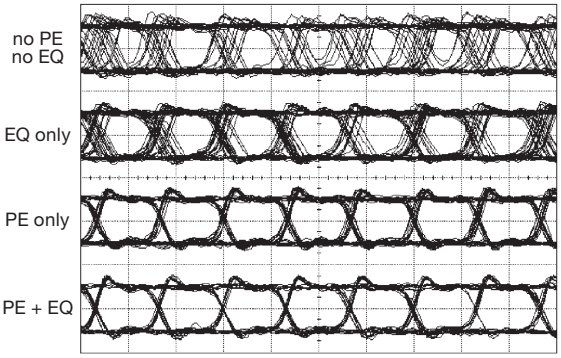


Figure 12.132. Receiver output waveforms ("eye diagrams") from the same setup as Figure 12.131, showing the effect of varying amounts of driver pre-emphasis and receiver equalization. The bottom trace represents PE=+9 dB and EQ=+8 dB at 1.5 GHz. Vertical: 500 mV/div; horizontal: 500 ps/div.

12.10.4 RS-232

This is a signaling format that goes back to the 1960s, intended originally for low-speed (<19.2 kbps) serial links between alphanumeric terminals (e.g., the legendary DEC VT-100) and computers. RS-232 has been revised several times and is now known officially as EIA232; although RS-232 ports are still seen on some computers and instruments, the standard is considered old-fashioned, and arguably headed for the trashcan. It's still with us, however (and perhaps always will be[82]). Although not part of the standard, the data is usually sent as asynchronous 8-bit serial data bytes, with a synchronizing START bit, and one or two STOP bits (these extra bits allow the receiver to resynchronize after each byte; see §14.7.8). The usual bit rates (also not specified in the RS-232 standard itself) are power-of-2 multiples of 300 bps (thus 300, 600, 1200, 2400, 4800, 9600, and 19,200 bps, supplemented with an interleaved set consisting of 14.4 kbps, 28.8 kbps, 57.6 kbps, and 115.2 kbps). Thus, for example, when you set up a serial port for "9600 8N1," you are sending groups of 8 bits, with one START and one STOP bit, and no parity, at 9600 bits/s. Note that the synchronizing bits (START and STOP) are included in the overall bitrate measure; for this example, then, you're sending a payload of 960 bytes/s.

What the standard *does* specify are the signaling voltages, load resistance and capacitance, and slew rates, along with the connector pinouts. We'll see RS-232 again, in the context of computer communications, in Chapter 14. But here we stick to the use of RS-232, at the physical level (hence the term "PHY"), as a way to drive cables with digital data. The RS-232 voltage levels are bipolarity, with legal driver output voltages of either -5 V to -15 V (logical 1, also known as "MARK"), or +5 V to +15 V (logical 0, also known as "SPACE"). RS-232 drivers and receivers are inverting, so MARK corresponds to logic HIGH at the input of a driver or at the output of a receiver (see Figure 12.135). As indicated in the figure, RS-232 is single ended; and, because of the relatively low signaling rates, the cable is unterminated. That's a terrible thing to do, normally, because rapidly changing signals (i.e., on a time scale shorter than the round-trip signal travel time) reflect from the open end, as we've seen. RS-232 finesses that problem by specifying a maximum slew rate (30 V/μs), which is slow enough for typical cable runs of fifteen meters (the original specified maximum, later replaced with a maximum load capacitance) or less. To be in compliance with the standards, the load capacitance must be 2500 pF or less;[83] and the load resistance is a nominal 5kΩ ($\pm$2kΩ). RS-232 drivers must withstand a continuous short to ground or to any dc voltage in the range $\pm$25 V.

[82] It's hard to completely dismiss RS-232, when so many test instruments are happily working well with it, as one of their several interface choices. The move toward USB is being stolen by Ethernet instead (or other replacements), leaving an RS-232 connector as the easy-to-implement port. Unlike more advanced serial interfaces like USB and Firewire, RS-232 does not require any "smarts" – there's no initialization or negotiation needed. Partly for this reason, most microcontrollers include an easy-to-use serial port controller. And for connection to contemporary laptop computers and the like, you can get USB-to-RS-232 adapter pods, for example from FTDI.

[83] Which corresponds to 50 m of Cat-5 twisted pair (50 pF/m), or 25 m of 50 Ω coax (100 pF/m).

So, RS-232 rumbles along, slowly. The good news is that it can ride along in a multiwire unshielded cable, which need not act as a well-behaved transmission line; the limited slew rate minimizes both crosstalk and reflections. The bad news is that the driver needs both positive and negative supply voltages of at least ±5 V. The original driver–receiver chips (1488/1489, and their CMOS follow-ons DS14C88/89 and MC145406) required such dual supplies (±9 V, nominal), which is asking a lot in a digital system (like a computer motherboard) that runs on positive supply voltages only. Maxim was the first to introduce RS-232 drivers with an on-chip charge pump (flying capacitor) voltage doubler and voltage inverter, to generate ±10 V from a single +5 V; they named it (naturally) the MAX232. There are now dozens of such chips (some with the capacitors included in the package, e.g., the MAX203 or LT1039), spanning a range of maximum speed, power consumption, supply voltage, number of drivers and receivers in one package, and so on. As an example, the MAX3232E is a dual transceiver (two drivers, two receivers) that runs from a single +3 V to +5.5 V supply, requires four external 0.1 μF capacitors, and is guaranteed to meet RS-232 specifications to 120 kbps; it comes in five different packages (SMT and DIP). The -E suffix designates enhanced robustness to electrostatic discharge, namely ±15 kV ESD-protected, specified with the Human Body Model (HBM, recall §12.1.5) of a 100 pF charged capacitor in series with a 1.5kΩ resistor.

The RS-232 interface specification includes a number of additional control signals, intended for hardware "flow control" when a terminal is connected to a computer (see §14.7.8); these have names like Data Terminal Ready (DTR), Data Set Ready (DSR), Request to Send (RTS), and Clear to Send (CTS). You can ignore all of this if you simply want to use an RS-232 interface chip pair to send low-speed digital data over a wire connection. In fact, these are often ignored even in computer serial ports: you can simply use the transmit data (TD) and receive data (RD) lines, plus ground, and do the flow control with software (more on this in Chapter 14). Note, however, the curious (and confusing) official naming of the actual signal lines TD and RD: you would expect that TD is the output from a driver, which should be connected to RD of the (distant) receiver. Not so! In RS-232 nomenclature, a device is either Data Communications Equipment (DCE) or Data Terminal Equipment (DTE); the signal sent by the latter is called TD *at both ends of the link!* (and likewise for RD). Go figure. In practice, most engineers ignore this confusion; they call an outgoing signal TD and an incoming signal RD, at whichever end it's happening.

Figure 12.133 shows received RS-232 signals and the recovered +3.3 V logic-level digital data. We used a pair of MAX3232E dual RS-232 transceivers, powered from +3 V, and driven by a random byte generator[84] emitting standard serial-formatted data at 115.2 kbaud. The 'scope capture, triggering on the transition from negative STOP bit to positive START bit, shows many such bytes superposed, so the data bits attain both values. The recovered logic signal is clean after 10 m of twisted pair (approximately its rated limit), but after 140 m there is significant spread in the recovered data's edge timings, caused by intersymbol interference (§12.10.3B) in the slow-transitioning RS-232 signal. Here, the slow transition time is comparable to the time for one bit (1 "UI," or unit interval), causing the time to cross the 0 V threshold to vary somewhat, according to the preceding bit values. You can see this effect in the long cable data in Figure 12.133, where the received cable signal falls short of its asymptotic voltage levels (approximately ±4.5 V) by as much as a volt.[85]

When sending digital signals on wires over substantial distances, or in electrically noisy environments, there

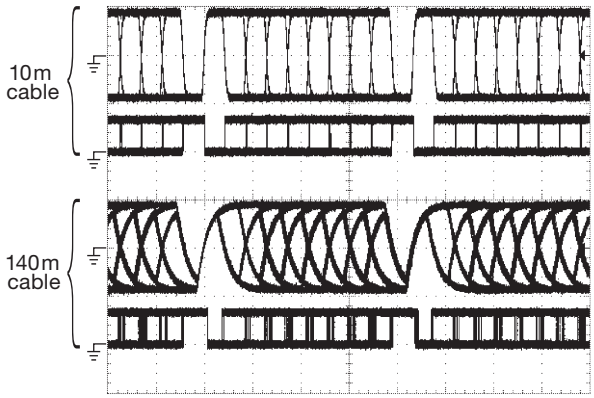


Figure 12.133. RS-232 signaling uses single-ended voltage drive, alternating between positive and negative voltage levels. These 'scope captures show received random bytes ("8N1") at 115 kbaud (top trace of each pair) after transmission through two lengths of Cat-5e/6 network cable; the drivers and receivers are inverting, as can be seen in the recovered logic signal (bottom trace of each pair). The marginal performance with the longer cable is consistent with Figure 12.134's limit of ∼10 kbaud for this cable length. Driver/receiver: MAX3232E at +3 V. Horizontal: 20 μs/div; vertical: 5 V/div.

[84] The one used to generate the ∼250 MB of random bytes on the *Numerical Recipes* CD-ROM.

[85] Users of RS-232 often use slower data rates, the most popular of which is 9600 baud, at which these effects are insignificant.

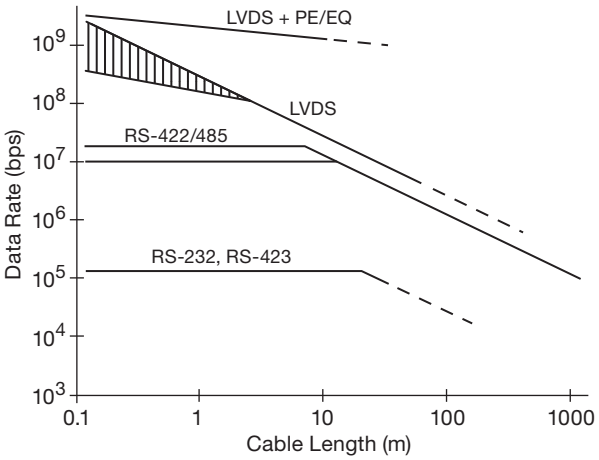


Figure 12.134. Approximate data rate limits versus cable length, for several signaling protocols. Your actual mileage may vary, depending on cable quality and interference environment. Note the speed improvement gained with pre-emphasis and equalization ("PE/EQ," see Figures 12.130–12.132).

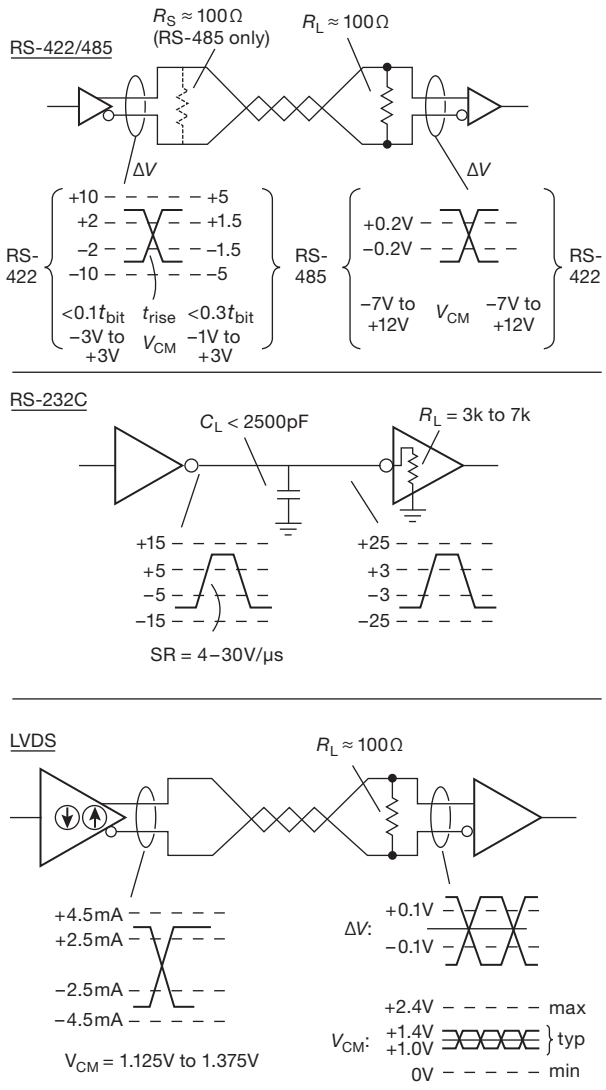


Figure 12.135. Allowable driver and receiver signal levels for LVDS, RS-422/485, and RS-232 signaling. Except for the bipolarity (and single-ended) RS-232, the output voltages from the drivers are of positive polarity only. LVDS outputs are *current-mode* drive (which are converted to a differential voltage signal at the cable's terminating resistance), at a common-mode voltage of +1.25 V; the others are *voltage-mode* outputs. All of the receivers respond to voltages, whether single ended (RS-232) or differential (LVDS, RS-422/485).

are always the problems of ground currents and of injected noise. Galvanic isolation is the best solution. Although RS-232 does not receive as much attention as RS-422/485, there is at least one good isolated driver, namely the ADM3251E from Analog Devices. It uses internal transformer coupling (with a modulator–demodulator pair) for the transmit and receive data, and an additional transformer (plus rectifier and regulator) to generate the isolated dc; the only external components are five 0.1 μF for the (internal) charge pumps.

One final comment: we've been talking mostly about the RS-232 *physical layer* (the actual voltages on the cable), with its awkward bipolarity signaling, as a simple tool for direct transmission of digital data. In the real world, RS-232 is ordinarily used in conjunction with asynchronous serial data sources, for example between a host computer's "serial COM port" and a device such as a modem or programming pod. In that role, you need to use driver and receiver interface chips (like the MAX3232) at each end, to translate between the logic-level signals and the RS-232 signals. Although we're less than enthusiastic about RS-232 signals, we believe that the use of the simple asynchronous serial data protocol *without conversion to RS-232 voltage levels* will continue to be useful. That's because it is the last of the uncomplicated serial interface standards – later serial protocols such as USB, Firewire, and SATA require substantial brainpower to negotiate and operate the link. Most microcontrollers include one or more serial ports (called UARTs, or COM ports), which are easy to use, and which can talk to any computer via a serial-to-USB converter such as the popular FTDI TTL-232R-3V3. This handy device plugs into a USB host at one end, and gives you a serial port with +3.3 V logic levels (for direct connection to a microcontroller, or whatever) at the other end.

12.10.5 Wrapup

In contemporary practice LVDS is quite popular, owing to its combination of high speed (to 3 Gbps and beyond), low emission of interference, low power, and compatibility across low-voltage logic families.[86] It's used widely for high-speed signal transport and distribution on circuit boards and backplanes, as well as through short runs ($\lesssim$10 m). You can get discrete drivers and receivers in small packages (e.g., 65LVDS1/2), and many complex ICs include LVDS driver/receivers for serialized data, for example a SERDES pair like the DS92LV1023/1224, or a complex FPGA like the Xilinx Spartan-3 or Altera Stratix series.

RS-422/485 is used for longer runs, often in industrial environments, where the data rates are lower (to 10 Mbps) but the common-mode noise is higher. And RS-232 survives, in spite of frequent predictions of its demise, for simple data links at low rates. For applications where the signal must be isolated from interference (both incoming and outgoing) you'll see shielded cables, either as shielded twisted pairs for differential signaling, or as coax cables for single-ended (or analog) signaling. It's also common to see galvanically isolated RS-485 interface chips, employing electrostatic (via capacitors), magnetic (via transformers, or GMR), or optical (via LEDs) techniques to bring the data across a several-kilovolt isolation barrier. Finally, fiber optics (§12.8) provides a totally noise-immune and galvanically isolated digital link, capable of very high data rates and long runs, in exchange for higher costs in driver–receiver components.

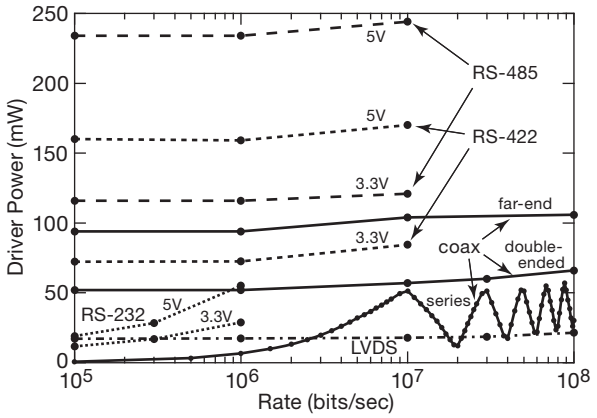


Figure 12.136. Measured driver power versus bitrate, for several single-ended (coax) and differential (twisted-pair) cable-driving configurations. All tests used a 10 m length of RG-58A coax or Cat-5e network cable, driven with alternating 1s and 0s (i.e., a square wave whose frequency is half the bit rate). The far-end termination resistance was 50 Ω for coax (except open circuit for "series" terminated), and 100 Ω for twisted pair (except 5 kΩ for RS-232) with another 100 Ω at the source end for RS-485. Note the effect of reflections from the mismatched end termination (5 kΩ) with RS-232, particularly evident here because of the single-frequency square-wave drive.

Figure 12.134 is a rough guide to speeds and lengths of LVDS, RS-422/485, and RS-232 signaling. Figure 12.135 summarizes the signal characteristics of these link standards. And Figure 12.136 compares their driver power requirements (i.e., the driver chip's supply voltage times current), along with that of 50 Ω coax.[87]

[86] There are several closely related protocols, namely PECL (positive emitter-coupled logic), LVPECL (low-voltage PECL), and CML (current-mode logic). The latter is used, for example, in digital-video links such as DVI and HDMI.

[87] The driver ICs we used for these power measurements were: coax – 74LVC2G04 (both sections in parallel); LVDS – 65LVDS1; RS-232 – MAX3232E; RS-422/485 – LTC1485 (5 V) or LTC1480 (3.3 V).

Review of Chapter 12

An A-to-S summary of what we have learned in Chapter 12. This summary reviews basic principles, facts, and application advice in Chapter 12.

¶A. Logic Interconnections.
The subject of this chapter is the interconnection of digital logic signals and logic devices to... *everything*, where "everything" includes (a) other logic devices, (b) input sources (switches, optoelectronics, cables), and (c) output devices (dc and ac power loads, optoelectonics, cables). So it's a long chapter, rich with multiple themes. Here we try to organize these diverse topics into manageable paragraphs.

¶B. Logic Families.
§12.1.1. Contemporary digital logic is owned by CMOS, with the minor exception of some emitter-coupled logic families (ECL, PECL, and LVPECL) and some BiCMOS (ABT, BCT). CMOS inputs draw no current, and their logic thresholds are usually close to mid-supply, except for the -T suffix types (HCT, ACT, AHCT, VHCT) where the threshold conforms to that of legacy bipolar TTL ($\sim$1.4 V); see Figure 12.5.

The 74HC[T] logic is good for easy prototyping and general-purpose use, and is available in DIP or SOIC; substitute 74AC[T] or 74LV for applications needing higher speed. For lower-voltage applications at moderate speeds, use 74LVC or 74VCX (available in SOIC and mini-logic packages only). For operation up to 15 V you can use 4000B logic, but this stuff is *slow*. Finally, for differential signaling use LVDS (or LVPECL, if specified) drivers and receivers. See Figures 12.2 and 12.3 for logic-family speed versus power, and speed versus supply voltage.

¶C. Input and Output Characteristics.
§§12.1.2A and 12.1.2B. CMOS *inputs* draw no current when $0V \leq V_{in} \leq V_{DD}$ (and even for inputs a few tenths of a volt beyond). For most logic families, the input protection diodes clamp to ground and V_{DD}, thus causing substantial input current for input voltages as much as a volt beyond V_{DD} or below GND; this can cause SCR latchup. However, some families use zener-like clamping, allowing input swings well beyond the rail (and even when unpowered); examples include 5V-tolerant 74LVC, and 3.3V-tolerant 74AUP. See Figure 12.4.

CMOS *outputs* swing to the rails, looking resistive for small load currents (Figure 12.7). The output current capability varies widely among families; within a family it increases with increasing supply voltage. See Figure 12.6 for a summary of thresholds, and valid input and output voltage ranges. Note that the outputs of (essentially obsolete) bipolar TTL logic (and also of digital ICs with nMOS outputs) do not saturate to the supply rail; see §12.4.5A.

¶D. Interfacing between Logic Families.
You can mix logic types as long as you respect logic-level input requirements. At the basic level this means that (a) you can always make a direct connection between logic running at the same voltage; (b) logic running from a higher voltage (e.g., +5 V) can drive lower-voltage logic if the latter's input is "tolerant"; and (c) lower-voltage logic can drive higher-voltage logic if the latter has "TTL thresholds" and the former is powered from at least +2.5 V. With dozens of logic families, most capable of operating over a range of supply voltages, you've got a lot of possibilities. Most are dealt with in §12.1.3, and presented pictorially in Figure 12.9.

¶E. Driving Digital Logic Inputs.
§12.1.4. A mechanical switch with pullup (§12.1.4A) generates the right logic levels, but with *bounce*. That may be OK for some applications; but for clean edges you need a debouncer, illustrated in a half-dozen variations in §12.1.4B. A logic input can come instead from a *comparator* (a subject summarized in ¶G below, and treated in detail in §12.3), whose output may have active pullup to the same V_{DD} as the driven logic, or it may come from an "open-drain" (or open-collector) terminal; in the latter case you need an external pullup to V_{DD}. Figure 12.25 shows representative circuit configurations.

When driving logic inputs from signals of any type, be careful not to overdrive the input (whether from a signal source of greater swing, or from transmission-line effects – see the egregious example in Figures 12.103 and 12.104); and do not leave unused logic inputs floating.

Contemporary logic devices are engineered to withstand substantial static-electricity insults (defined in terms of the human-body model, HBM, of 100 pF in series with 1.5kΩ, §12.1.5), but it's best not to tempt fate; be careful to discharge yourself, use antistatic materials, etc. See also the earlier discussion in §3.5.4H.

¶F. Driving External Loads from Logic Outputs.
§12.4. You can drive small loads (LEDs, SSRs, small mechanical relays) directly from logic outputs (§12.4.1 and Figure 12.39), taking care to respect the output drive capability (voltage swing, current), and, for a mechanical relay, adding a diode clamp across its coil. As shown in the figure, you can use an open-collector driver (ULN2003, 75468) to

accommodate voltages and currents as high as 100 V and 350 mA respectively. There are analogous drivers for use with serial data (from a microcontroller), see Figures 12.40 and 12.41, and the listing in Table 12.3 on page 819.

For driving heftier loads you can append an external transistor or power-driving module. For driving *positive* loads see §12.4.2, and Figures 12.42 and 12.43. For driving *negative* or *ac* loads see §12.4.3 and Figure 12.44. Serious power switching requires *fault protection*, see §12.4.4 and Figures 12.45–12.48.

¶G. Comparators.

A comparator is a high-gain uncompensated differential amplifier, used to signal which of two analog voltage inputs is greater. It is an important interface between analog signals and the digital world. Comparators were introduced in §4.3.2, and treated in detail in §12.3 (with listings in Tables 12.1 on page 812 and 12.2 on page 813).

As with op-amps, comparators come in a range of speeds ($<$1 ns to tens of μs), supply voltages (total supply voltages from as little as 1.1 V to as much as 40 V), input offset voltages (0.25 mV to 10 mV), input bias currents (1 pA to $>10\,\mu$A), input common-mode voltage ranges (to both rails, one rail, or neither), and quiescent currents ($<1\,\mu$A to tens of mA). But, unlike an op-amp, which is normally operated in the linear region where the (analog) output voltage avoids saturation, a comparator's output is digital; it lives at the extremes. It may be used to drive logic, or it may drive an ON/OFF load like a relay or an LED. To accommodate various loads, comparator output stages come in a half-dozen variations (Figure 12.33), including (a) rail-to-rail (like an op-amp); (b) open-drain or open-collector; (c) logic-level with auxiliary logic-rail pin V_L; (d) logic-level with both V_L and GND pins; and (e) floating-transistor output stage.

Comparator miscellany: comparators are usually configured with some hysteresis (Schmitt trigger) to prevent multiple transitions and oscillation. Some comparators have quite limited differential input voltage ranges (as little as a few volts). For comparators with BJT input stages, the input currents may undergo an abrupt step at zero differential voltage (e.g., Figure 12.36). The delay and switching time of a comparator depends on the input overdrive (Figure 12.38).

¶H. Optoelectronics.

§12.5. Continuing the theme of "logic-to-everything," we must include among the latter *humans*, with senses that delight in visual presentations.[88] There's much richness here, including emitters, indicators, displays, detectors, and couplers, as illustrated in the photographs in Figures 12.58 (page 830), 12.71 (page 837), 12.80 (page 841), 12.84 (page 844), and 12.95 (page 853). To remind you of the breadth of optoelectronics, here is a condensed listing of devices (expanded in ¶¶I–M below) in the family tree:

Emitters: LED; Laser diode; Electroluminescent

Displays: LED-based; LCD-based; VFD-based; OLED; E-ink

Detectors: Photodiode; Phototransistor; Photoresistor; Bolometer; APD; PMT; HAPD; Microchannel plate

Couplers: LED-input or logic-input, with transistor-, thyristor-, or logic-output

Other: Opto-interrupter, proximity and ranging; Barcode reader; Optical mouse.

¶I. LED Indicators.

These have diode-like *I–V* curves, but with greater (and color-dependent) forward voltage drops (Figure 2.8). They come in many form factors (panel mount, PCB mount, arrays, 7-segment displays), sizes, viewing angles, and colors (§12.5.1). You can drive them from a logic signal of adequate swing, with a current-limiting series resistor (mindful that blue, bright-green, and white LEDs have forward drops of 3 V or more), or you can add an external transistor for higher voltage or current (see Figure 12.61). For serious illuminations purposes you're better off using a current-sensing dc-dc switching converter (Figure 12.62). By contrast, to drive *laser diodes* you use feedback from the integrated monitor photodiode to set the drive current (see Figure 12.67 and associated text).

¶J. Displays.

A popular device for numeric or hexadecimal data is the 7-segment (or 16-segment) *display* (§12.5.3A), available in multicharacter sticks of both "dumb" and "smart" varieties (see Figures 10.90 and 12.77, respectively). Once you've bought into the idea of a multicharacter display device, you should consider the multicharacter (and multiline) LCD smart display (Figure 12.78) and the nicer-looking (and compatible) VFD display, seen in Figures 12.71 and 15.25.

¶K. Detectors.

In the detector arena, *photodiodes* convert light to a proportional photocurrent, and can be operated in either of

[88] As a colleague once remarked, as we were admiring the sunset while walking along the beach, "nice graphics."

two modes – *photovoltaic* (self-generating, into a short circuit or virtual ground); or *photoconductive* (back-biased) – see Figure 12.81. The latter is faster, but suffers from leakage current and increased noise. A *phototransistor*, which operates only in photoconductive mode, combines a photodiode and transistor, for greater gain, but with reduced speed; similarly (and more so) for a *photo-Darlington*. The *avalanche photodiode* exploits the phenomenon of avalanche multiplication in a photodiode biased close to breakdown; each detected photon causes multiple electrons to be collected, thus providing higher gain (in linear operation), or (if biased still further) a full-sized "Geiger-mode" pulse for each detected photon. A different class of detector is the cadmium sulfide *photoresistive* sensor, which behaves like a linear resistance ($I \propto V$) that depends on illumination; CdS sensors are slow, and they are becoming a rarity owing to RoHS regulations (cadmium is not good for you), but they are useful in applications where you want a light-controlled linear resistance (see for example Figure 7.21). Finally, a still-popular old-school photodetector is the *photomultiplier* tube (PMT, §12.6.2), in which a photoelectron is accelerated and collides with successive electron-multiplying dynodes to produce a cascade of some 10^6 electrons at a collecting anode (Figures 12.82 and 12.83).

¶L. Optocouplers and Solid-state Relays.

Optocouplers (also known as opto-isolators, or photocouplers) consist of an opto-emitter combined with a detector in an opaque package (§12.7). They are used to convey digital (and sometimes analog) signals between circuits with separate grounds; such *galvanic isolation* prevents ground loops in sensitive circuits, it permits safe switching of ac powerline circuits, and it allows communication and control for circuits operating at a high voltage.

All optocouplers use an LED at the *input end*, in most cases simply providing the anode (+) and cathode (−) terminals (so you have to limit the driving current with an external resistor, see Figure 12.85A); some optocouplers include current limiting (e.g., SSRs, Figure 12.92C), while some others (e.g., high-speed logic-to-logic couplers, Figure 12.86C) accept a logic-level input signal. At the *output end* there are many configurations: transistor or Darlington (Figure 12.85), digital logic with open-collector or active pullup (Figure 12.86), active-pullup MOSFET-driver (Figure 12.87), ac-output SSR (Figure 12.92), and dc-output SSR in many flavors (Figure 12.91). There are also analog-oriented optocouplers (Figures 12.88, 12.89, and 12.90), ac-input optocouplers (Figure 12.94), and a whole range of industrial *input/output modules* (§12.7.7).

¶M. Opto-interrupters, Proximity Sensors, and Angle Encoders.

An *opto-interrupter* is an LED-detector pair with an open gap (Figure 12.84), widely used to sense end-of-travel in mechanical devices. An optical *proximity sensor* bounces a beam off an external object; the *distance measuring sensor* is a variant that incorporates a position-sensitive detector that uses parallax (triangulation) to measure object distance (to $\sim$1 m or so). Optical *shaft angle encoders* measure shaft position; they come in incremental (quadrature square-wave) and absolute versions, with resolutions of 32 to 128 counts per rotation (for panel controls) to 30,000 or more for high-resolution shaft encoders.

¶N. Fiber Optics.

Light travels happily, and unimpeded, through glass or plastic fiber. For long-haul high-datarate communications you use single-mode glass fiber (with rates to 100 Gbps or more per fiber); but for more modest data communications you can use multimode fibers, ranging from the simplest TOSLINK 1 mm plastic fiber to the very popular 62.5/125 μm graded-index glass fiber. Transmitter and receiver modules are widely available (§12.8.1–12.8.3), as well as high speed transceiver modules that include all necessary driver/receiver electronics (§12.8.4).

¶O. Digital Signals and Long Wires.

Fully 25% of the chapter (§12.9) is devoted to the problem of sending digital signals somewhere else, where the *somewhere* may be on the same PCB, or through a backplane, or through a cable to a remote electronic device. As simple as these may seem, numerous pitfalls await the unwary: for example, we were surprised (but should not have been) when encountering errors while sending parallel digital data of only modest speed (HC logic) through just 4″ of flat ribbon cable. Read the following paragraphs, to avoid membership in the club of the unwary.

¶P. Short Links.

For the *short links* that stay on a PCB, the distributed capacitive loading and consequent ground-current spikes cause logic glitches (§12.9.1A), requiring liberal power-supply bypassing (which is always salutory) and low-inductance ground plane layouts. The problem is less severe in synchronous systems; but it's more severe with fast logic, where interconnections may require constant-impedance transmission-line layouts (see §*1x.1.4*). These problems are more pronounced for signals moving between circuit boards or along backplanes, where you may need to use line driver and line receiver ICs, perhaps in

combination with controlled impedance traces that are properly terminated; the latter is standard practice with LVDS (differential) signaling, as seen for example in Figure 12.127.

¶Q. Cable Runs.

Longer signal runs require *cable* (§12.10), usually *coaxial* (e.g., the ubiquitous RG58 BNC patch cords), multiwire *ribbon*, *twisted pair* (e.g., the comparably ubiquitous cat-5 or cat-6 network cable), or *fiber optic* (generally recognizable by their brightly colored svelte profile). With the exception of "slow" waveforms,[89] signal cables must be treated as *transmission lines* (Appendix H), with their *characteristic impedance*[90] Z_0. For coax, Z_0 is usually 50 Ω (but 75 Ω for video), while twisted-pair is usually 100 Ω. The significance of Z_0 is that *a signal applied to a cable, terminated with a load resistance equal to its characteristic impedance, is completely absorbed by the load*; it does not reflect. No other load impedance can make that statement.

¶R. Terminating Digital Coax.

So you can (a) put a 50 Ω resistor across the far end of a coax line (a "terminated line," or "matched line") and drive it with your digital signal (which sees a pure 50 Ω resistive load – the capacitance disappears!), as in Figure 12.107; or (b) drive such a terminated line with a signal of the same impedance, as in Figure 12.110, noting that the output signal amplitude is reduced by ×2; or (c) keep the matched source resistor but omit the load resistor, as in Figure 12.107, to preserve the full output amplitude. In the latter case there *is* a reflection of substantial amplitude, which however is hospitably absorbed by the source resistor – see the highly educational waveforms in Figure 12.115. See §12.10.1 for pretty pictures, and plenty of detail.

¶S. Differential Cable.

Differential signals are treated similarly, with a termination connected across the pair. For long hauls (to ∼1 km) and modest speeds (10 kbps–10 Mbps) the long-lived RS-422/485 protocol is popular, whereas the LVDS protocol is used for higher datarates (10 Mbps–1 Gbps) over shorter distances; see the speed limit contours in Figure 12.134. To achieve the highest datarates over significant lengths of cable one commonly uses driver *pre-emphasis* and receiver *equalization*, see Figures 12.130–12.132 and associated discussion.

[89] Those whose transition times are much longer than the round-trip travel time, $t=2L\sqrt{\varepsilon}/c$, where the factor $\sqrt{\varepsilon}$ accounts for the slower-than-light "velocity factor" of wave propagation in the cable.

[90] Which should really be called characteristic *resistance*.